systems, the system designer should recognize the interacting effects of attitude control system loop gain, capability of the attitude control system to compensate for disturbances, accuracy of attitude control, and control system bandwidth.

3-Axis Stabilization. Different types of active control systems have different key parameters and algorithms. For systems in which spacecraft rates will be kept small, 3-axis control can frequently be decoupled into three independent axes. In this simplest form, each axis of a spacecraft attitude control system can be represented as a double-integrator plant (i.e., $1/Is^2$ in the s-domain) and may be controlled by a *proportional-derivative (PD) controller*, which has control torque

$$T_C = K_P\theta_E + K_D\omega_E \tag{19-9}$$

Here, θ_E is the attitude error angle, and ω_E is the attitude rate error. The most basic design parameter in each axis is its *proportional gain*, K_P. (K_P is also often called a *position gain*.) This gain represents the amount of control torque we want to result from a unit of attitude error, and so it has units of torque divided by angle, e.g., newton-meters per radian. The proportional gain is selected by the designer and must be high enough to provide the required control accuracy in the presence of disturbances, which can be guaranteed by setting $K_P > T_D/\theta_{E,max}$, where T_D is the peak disturbance torque, and $\theta_{E,max}$ is the allowable attitude error. Note that this error will remain for as long as the disturbance torque remains; the error that remains in the presence of disturbances is called *steady-state error*.

The value of K_P also largely determines the attitude control system bandwidth, which is a measure of the range of errors the system can correct. The mathematical definition of *bandwidth* requires a bit more explanation than we want to go into here, but it can be approximated as

$$\omega_n \approx \left(K_p / I\right)^{1/2} \tag{19-10}$$

where I is the spacecraft moment of inertia. The bandwidth defines the frequency of disturbances and of commanded motions at which control authority begins to diminish. Attitude control and disturbance rejection are effective from 0 frequency (i.e., constant or d.c. inputs) up to the bandwidth. *Speed of response* is approximately the reciprocal of bandwidth. Note from Chap. 16, proportional gain is inversely proportional to allowable error, and bandwidth is proportional to the square root of the proportional gain. Therefore, high accuracy implies high K_P and high bandwidth. However, there are practical limits on bandwidth. If the bandwidth is high enough that it includes bending frequencies for the spacecraft structure, then control-structure interactions must be carefully considered. If the bandwidth is higher than the maximum sampling rate of critical attitude sensors, some care must be taken to avoid excessive attitude responses to the slow updates. Finally, high bandwidth control systems may be more likely to have adverse reactions to inaccurate or lagging system inputs; this condition is referred to as having insufficient stability margin.

There are design techniques to address these concerns. One such technique is to modify the system response to increase stability while keeping the same attitude accuracy by increasing attitude rate *damping*, which is controlled by the *derivative gain*, or *rate gain*, K_D. For a given value of K_P, increasing K_D will slow down the response of the control system, but it will not diminish the accuracy in the end, provided the disturbance environment or slewing requirements do not require very high speed of response. This technique is often used to improve stability margins in systems where the final accuracy is more important than the amount of time it takes to achieve that accuracy. Note that, though there are multiple methods for selecting K_P and K_D, such as linear-quadratic regulation (LQR) and other state-space methods, the system response generally must still be understood in terms of bandwidth and damping to achieve confidence that the control gains will provide both stability and adequate performance.

Sometimes the problem facing a designer is that stability margin requirements are not difficult to meet, but accuracy requirements are very difficult. A common technique for improving on the performance of a PD is to add an integrator to the feedback loop; this is called a *proportional-integral-derivative controller (PID)*. Recall that in a regular PD controller, a steady-state error of some kind will remain as long as there are disturbance torques to create it. The integrator causes the correcting control torque to gradually increase over time to remove that steady-state error completely. If the disturbance torque changes magnitude or direction, the PID controller will take some time to adjust, but it will track with no steady-state error. This improved performance generally comes at the price of reduced stability margins, so the stability and performance requirements must always be carefully balanced.

In addressing control-structure interactions, the control designer must keep in mind that, if bending causes apparent attitude errors, the control system may aggravate the bending motion, eventually affecting control stability and perhaps damaging the bending structures. Though there are control design techniques to deal with these potential problems, the control designer will often start by recommending a minimum frequency for structural bending modes that would allow flexible effects to be neglected. This minimum frequency is generally an order of magnitude greater than the bandwidth needed to achieve required accuracy. Then, structural engineers will attempt to honor this restriction in their design. For further discussion of how structural flexibility interacts with feedback control systems, see Sec. 3.12 of Agrawal [1986].

Spin Stabilization and Momentum Bias. The fundamental concept in spin stabilization is the nutation frequency of the vehicle. For a spinning body, the inertial nutation frequency, ω_{ni} is

Table 19-15◀, Fig. 19-8◀, Eq. 19-10

$$\omega_{ni} = \omega_S I_S / I_T \tag{19-11}$$

where ω_S is the spin frequency, I_S is the spin axis of inertia, and I_T is the transverse axis inertia. For a momentum-bias vehicle with a non-rotating body and a momentum wheel (or a dual-spin vehicle with a non-rotating platform), the nutation frequency is

$$\omega_{ni} = h / I_T \tag{19-12}$$

where h is the angular momentum of the spinning body (or bodies). (Note that this is really the same equation as for a spinning body, for which $h = \omega_S I_S$.) Thus, spacecraft with large inertias and small wheels have small nutation frequencies (i.e., long periods).

Attempting to control the vehicle with a bandwidth faster than the nutation frequency causes it to act more like a 3-axis stabilized vehicle, that is, the stiffness of the spin is effectively reduced. In general, we attempt to control near the nutation frequency or slower, with correspondingly small torques for given attitude errors. Because we generally wish to avoid changing the spin rate with control torques, the torque is applied perpendicularly to the spin. We can then relate the achieved angular rate, ω, to the applied torque by simplifying Euler's equations to only the gyroscopic torque component:

$$\mathbf{T} = \omega \times \mathbf{H} = \omega \mathbf{H} \tag{19-13}$$

where **H** is the system angular momentum. A lower limit on control bandwidth is usually provided by the orbit rate ω_O, which for a circular orbit, is

$$\omega_O = (\mu / r^3)^2 \tag{19-14}$$

where $\mu = 3.986 \times 10^{14} \text{m}^3/\text{s}^2$ for Earth orbits and r is the orbit radius.

Example of Magnetic Momentum Management. Managing the long-term buildup of angular momentum in the orbiting spacecraft often drives the ADCS requirements. As would be expected for an element so central to ADCS design, there are many different methods for keeping momentum to levels that do not degrade data or equipment. We present here a simple, common, and relatively inexpensive algorithm for commanding magnetic torquers using only magnetometer data. It functions as an excellent active rate damping controller, and after the spacecraft's rate becomes very low, it keeps the angular momentum at manageable levels by canceling it out on an orbit-averaged basis. Often called B-dot, this algorithm is based on back-differencing magnetometer data to estimate the time derivative of the magnetic field (or "Bdot"). For rate damping purposes, we assume the B-field is changing in the body due only to spacecraft rotation, which is true for all but rotation rates on the order of orbital rate. Specifically,

$$\mathbf{Bdot} = \mathbf{B_{k+1}} - \mathbf{B_k} = -\omega \times \mathbf{B} \tag{19-15}$$

The spacecraft dipole vector, **D**, is set equal to a gain multiplied by the magnetic field:

$$\mathbf{D} = -k\,\mathbf{Bdot} = k\omega \times \mathbf{B} \tag{19-16}$$

Then, when the magnetic field acts on that vector, it produces a control torque

$$\mathbf{T_c} = \mathbf{D} \times \mathbf{B} \tag{19-17}$$

that is always in a direction that slows down the spacecraft rotation except when ω is aligned with **B** (in vector terminology, $\omega \cdot \mathbf{T_c} \le 0$). Once the spacecraft is slowed down, the algorithm will attempt to follow the magnetic field vector, its success determined by the strength of the gain, k, that is chosen. However, whether it follows the field doesn't matter: The goal of slowing the spacecraft and then maintaining a slow momentum has been achieved. Flatley, Morgenstern, Reth and Bauer [1997] did not invent the B-dot controller, but their paper nicely explains in clear English and equations how something so simple works so well.

Algorithms for Attitude Determination. A full discussion of attitude determination algorithms requires a dedicated reference such as Wertz [1978]. We will highlight only some of the basic concepts.

The basic algorithms for determination depend on the coordinate frames of interest and the geometry and parameterization of the measurements. Useful coordinate frames can include the individual sensor frames, the local vertical frame, and an inertial frame, such as Earth-centered inertial (ECI). The geometry of the measurements is different for different sensors, as discussed in Sec. 19.1.4, and they are generally parameterized as sequential rotations about the axes of a frame, called *Euler angles* (e.g., roll, pitch, yaw), or as attitude *quaternions*, which are unit vectors with four elements that define a rotational axis in a frame, called the eigenaxis, and the amount of rotation about the eigenaxis. Inertial reference units (with their supporting software) and star sensors are well suited to quaternions, whereas Earth-pointing vehicles often use a local-vertical set of Euler angles, much like aircraft.

Simple spacecraft may use the sensor output directly for control, whereas more complex vehicles or those with higher accuracy requirements employ some form of averaging, smoothing or Kalman filtering of the data. The exact algorithms depend on the vehicle properties, orbit, and sensors used.

FireSat II Algorithms. For our momentum-bias FireSat II example, control separates into pitch-axis control using torque commands to the momentum wheel and roll-yaw control using current commands to the magnetic torquers. The pitch-wheel desaturation commands must also be fed (at a slower rate) to the torquers. The pitch-wheel control is straightforward, using PD control and, optionally, *integral control*, in which commands are augmented by a small torque proportional to the integral of the attitude error. The roll-yaw control design starts by using the linearized nutation dynamics of the system, and is complicated by the directional limitations of electromagnetic torque (the achievable torque is perpendicular to the instantaneous Earth magnetic field).

Table 19-15◀, Fig. 19-8◀, Eq. 19-17

Table 19-16. ADCS Vendors. Typical suppliers for ADCS components. An up-to-date version of this table can be found on the book website.

Company	Sun Sensors	Earth (horizon) Sensors	Magnetometers	Star Sensors	Gyro/IMU	GPS	Momentum/ Reaction Wheels	CMG	Magnetic Torquers	Thrusters
Adcole Corporation	X									
Aerojet				X						X
Ball Aerospace and Technologies Corp.			X							
Billingsley Aerospace and Defense	X						X			X
Bradford Engineering	X			X						
Comtech AeroAstro					X	X		X		X
EADS Astrium		X		X						
EADS SODERN				X						
EMS Technologies, Inc.	X	X		X	X					
Finmeccanica (incl. SELEX Galileo)						X				
ITT Aerospace						X				
General Dynamics			X		X		X		X	
Goodrich (incl. Ithaco)							X	X		
Honeywell Space Systems (incl. Allied)	X			X						
Jena Optronik					X		X	X		
L-3 Space & Navigation					X	X				
Kearfott Guidance and Navigation Corp.			X							
MEDA					X					X
Micro Aerospace Solutions									X	
Microcosm, Inc.						X	X			
NASA Goddard Space Flight Center					X					X
Northrop Grumman (incl Litton)				X						
Ørsted—DTU	X	X								
Optical Energy Technologies							X			
Rockwell Collins Deutschland (incl. Teldix)		X								
Servo Corp. of America				X						
StarVision Technologies										
Surrey Satellite Technologies—US LLC	X		X	X	X	X	X	X	X	X
Systron Donner Inertial					X					
Terma				X						
Watson Industries, Inc.			X		X					

The nadir-oriented control system may use an Earth-referenced, aircraft-like Euler angle set (i.e., roll, pitch, and yaw), although quaternions may simplify calculations during off-nominal pointing; quaternion calculations generally require fewer arithmetic steps, they simplify tracking commands, and they have no singularities. The horizon sensors directly read two of the angles of interest, pitch and roll. Yaw needs to be measured directly from Sun position (during orbit day) or from the magnetometer readings (using a stored model of the Earth's field plus an orbit ephemeris), or inferred from the roll-yaw coupling described earlier. MEMS gyro data can be used to enhance the attitude estimate, but they can have large biases that change with temperature, so it may be best not to have to rely on them for our accuracy requirements. The magnetic field and Sun information require

Table 19-16, Fig. 19-8◀, Eq. 19-17◀

an uplinked set of orbit parameters, and increase the computational requirements of the subsystem. Overall, meeting the 0.1 deg yaw requirement when the Sun is not visible will be the toughest challenge facing the ADCS designer, and planning to coast through the dark periods without direct yaw control may be most appropriate.

SCS Algorithms. A 3-axis PD controller will be sufficient for our control accuracy needs. In this control algorithm, an error angle and error rate is calculated for each axis, and a control torque calculated based on those. That calculation done for the three axes gives a control torque vector, which is then distributed among the reaction wheels. If one of the wheels is asked to issue a control torque that exceeds its capability, then the spacecraft would have a total torque vector that is not in the same direction as the requested vector. For our mission, that is not too much of a danger, since we don't have any elements that would be damaged by Sun exposure or the like. Still, it is a simple matter to scale all 3 vector components by the same factor that would make the largest of them equal to its limit while preserving the direction of the torque vector. Note that this is safe for a PD controller, but that stability problems can result if the same tactic is used for a PID controller without taking some care to prevent problems.

The data from the star camera and the rate gyros in each SCS satellite can be combined in any of a number of ways. We will simply trust the star camera attitude as accurate whenever it gives us data it indicates is valid, and we will use the gyros to provide direct rate measurements and to propagate the attitude solution by integrating the rate over the sampling time if the star camera fails to provide a valid attitude. This method is not the most accurate, but generally, more accuracy comes at the expense of more complexity, which then costs money and time in flight software development and testing. This is a good lesson to wrap up this section: Though it can be tempting to always reach for your best, most expensive tool, it is better engineering to try to get by with the cheapest system that will meet your requirements with appropriate margin, as discussed in Chap. 13.

19.2 Spacecraft Trajectory Navigation and Control Systems

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The previous section covered various aspects regarding the determination and control of a spacecraft's rotational motion. This section covers the corresponding aspects of a spacecraft's translational motion, utilizing the terminology defined below.

Trajectory refers to the path traveled by a spacecraft. *Orbit* refers to a special type of trajectory that forms a closed circular or elliptical path. Trajectories that do not fall under the classification of an orbit have open paths, e.g., a hyperbolic path of a planetary fly-by, or a launch/reentry path that intersects the surface of the Earth or other celestial body. Although a distinction is made here in the context of navigation and translational control, the terms orbit and trajectory are sometimes used interchangeably, as in Chap. 9. *Navigation* is the determination of the spacecraft's current translational states (position and velocity), relative to a specified reference frame, as a function of time. *Trajectory control* is the determination and execution of the translational commands (thruster firings) required to drive the current states towards the desired states which may be determined by a guidance system.

The purpose of this section is to help answer the following questions: As a function of the system used, how accurately can the user know the spacecraft's translational states (position and velocity)? Similarly, how accurately can the user control the spacecraft's trajectory? What are the mass and power requirements of the navigation and control systems to achieve the corresponding functions? Section 19.2.1 outlines the system definition process. Section 19.2.2 then discusses the various options available for navigation systems, and Sec. 19.2.3 describes the control options for orbit maintenance and making changes to trajectory parameters. Some design considerations for trajectory navigation and control systems are presented in Sec. 19.2.4.

19.2.1 System Definition Process

Table 19-17 summarizes the process for defining the trajectory navigation and control functions. Each of the steps is described below.

Step 1. Define top-level functions. The first step is to determine the key mission objectives that require navigation data or trajectory control. Examples include a repeating ground track orbit, a geostationary slot, or interplanetary flight.

Step 2. Estimate navigation accuracy requirements. Mapping and pointing trades (see Sec. 8.6) typically provide preliminary estimates of the required navigation accuracy. In most cases, the pointing and mapping requirements are satisfied by trading between position and attitude accuracy, so navigation accuracy requirements are generally performed in conjunction with attitude accuracy requirements.

Step 3. Determine where navigation data is needed. A data flow analysis helps determine where the data comes from and where it is needed. For a simple low-cost satellite mission where a single ground station performs all of the data evaluation, system control, and mission planning, the best solution is probably to do the navigation at the ground station. At the other extreme would be a satellite communicating with many distributed users, each of whom needs to know the satellite position or derivative information such as the ground look-point location. In this case, the navigation data is probably needed on the spacecraft, although we could also navigate from the primary ground station and distribute the results to other users. In some cases, navigation data may be needed by the spacecraft itself for payload functions such as target selection and/or for control functions such as orbit maintenance.

Table 19-16◀, Fig. 19-8◀, Eq. 19-17◀

Table 19-17. Process for Defining the Trajectory Navigation and Control Functions.

Step	Principal Issues	Discussed
Define top-level functions and requirements for trajectory navigation and control	How do the key mission objectives depend on trajectory navigation and control functions?	Secs. 3.5, 4.4, 6.1, 10.1
Estimate navigation (position) accuracy requirements	What are the pointing/mapping trades?	Sec. 8.6
Determine where navigation data is needed	Is it needed only at ground station or at multiple user sites, for mission planning and execution? Is it needed on board the spacecraft for payload and/or control functions?	Sec. 4.2.1
Conduct trade for autonomous vs. ground-based navigation	Is there a need for real-time navigation data? Do the potential operational benefits justify the risk associated with a non-traditional approach?	Secs. 4.2.1, 19.2.4
Select navigation method	Which method best satisfies the mission requirements?	Sec. 19.2.2
Determine whether trajectory control is needed	What are the mission requirements for orbit maintenance and/or change?	Chap. 10, Sec. 19.2.3
If yes, conduct trade for autonomous vs. ground-based trajectory control	Do the potential operational benefits justify the risk associated with a non-traditional approach?	Secs. 4.2.2, 19.2.3
Define requirements for trajectory navigation and control systems	State requirements in terms of **what** is needed rather than **how** it is done	Sec. 19.2.1

Step 4. Conduct autonomous navigation trade. Ground-based navigation is the traditional approach. Decide if the potential operational benefits justify the risk (technical and/or programmatic) associated with a non-traditional approach. Autonomous navigation may be preferred if real-time navigation data is needed, either on board the satellite or by the end user.

Step 5. Select navigation method. Section 19.2.2 summarizes the various options available for spacecraft navigation, along with advantages and disadvantages. Based on the mission objectives, typical navigation methods can be ground-based, space-based, semi-autonomous, or fully autonomous.

Step 6. Determine need for trajectory control. At the system design level, decide if there is a need to maintain or change the orbit. If trajectory control is not needed, eliminating the propulsion subsystem can save some weight/cost. On the other hand, maintaining the orbit may significantly extend mission life and thereby reduce the cost per year.

Step 7. Conduct autonomous control trade. If trajectory control is required, decide whether to do so from the ground or autonomously on board the spacecraft. Ground-based trajectory control is the traditional approach. Decide if the potential operational benefits justify the risk (technical and/or programmatic) associated with a non-traditional approach. A key question is the complexity and frequency of the trajectory control actions that need to be performed. For example, ground-based control may be preferred if the mission calls for infrequent but complex maneuvers.

Step 8. Define system requirements. Top-level system requirements are defined in terms of what capability is needed rather than how it is implemented. Detailed trade studies, done later in the design cycle, will help select the best trajectory navigation and control systems that meet these requirements.

Table 19-17, Fig. 19-8◀, Eq. 19-17◀

19.2.2 Navigation

Spacecraft navigation pertains to the knowledge of the vehicle's translational states: components of position and velocity. For closed trajectories, i.e., orbits, the 6 translational states can also be expressed in terms of the 6 orbital elements. In principle, knowledge of the state vector at any one time allows us to predict the position and velocity of the spacecraft at other times.

Spacecraft navigation data can be obtained from onboard sensors such as accelerometers and star trackers, or from external tracking/surveillance sensors such as radar or optical devices. *Tracking* is the process of updating an object's location by first predicting where it will be and then observing it. Tracking sensors are used when high-precision data are needed about a specific object. *Surveillance* is the process of searching for objects whose location may or may not be known. These sensors typically "stare" across an area of the sky and register objects passing through their field of view. They can observe multiple objects simultaneously, and continually add to the quality of the tracking catalog by ensuring that the data are accurate and up-to-date.

Radar sensors provide range and range-rate measurements, and may additionally provide azimuth and elevation angles. Rates of azimuth and elevation angles may also be available but can be quite noisy and hence may not be useful. Range, azimuth, and elevation data provide position vectors. Velocity vectors can be determined from position-vector rates (directly measured or computed from successive position measurements) or by other means such as Doppler shifts.

Optical sensors, unlike radar sensors, cannot provide range information. Hence the state vector has to be determined from angles-only observations of right ascension and declination from time-tagged images of the satellite's star background by comparing with star catalogs. Iteration or additional processing are required to com-

pensate for the lack of direct range information. Navigation data can be obtained from three pairs of right ascension and declination angles using various techniques [Vallado, 2007].

When a radar/optical sensor detects an object, it makes a series of observations as the object moves along its trajectory. Most modern sensors make hundreds or thousands of observations during each pass of the space object. A typical pass interval is ~15 min for low Earth orbit (LEO) and ~10 hrs for deep space orbits. Using dense measurements (~100 observations per pass) from a small number of sensors often provides higher quality navigation data than using sparse measurements from many sensors.

There are many techniques available for determining position and velocity vectors using various types of measurements: range and range-rate (three simultaneous observations from three separate sensors); range-only (six independent time-tagged observations); three time-tagged position vectors (range, elevation, and azimuth); two position vectors and time-of-flight [Vallado, 2007]. The sensor-relative position and velocity data are then transformed to the Earth-centered inertial (ECI) frame. The transformation requires that the sensor site location and velocity be known accurately, and this requirement poses some challenges for satellite-based sensors. Combining multiple data types/sources can reduce the error in the estimated state vectors. Range data is preferred, but angular data can be useful too.

State vector (position and velocity) data forms the basis of orbit determination, which involves estimating the spacecraft's orbital elements and accounting for perturbations to its natural orbit. The overall process of orbit determination requires estimation (which is linked with initial orbit determination), prediction, and uncertainty estimates. The process of orbit determination iteratively solves for a description of a satellite's orbit in terms of a state vector at an epoch, based upon observations and measurements.

The observation update interval for a spacecraft should be timed so that the next observation occurs before trajectory propagation theory would introduce enough error to keep the next sensor from re-acquiring the target, or when the most stringent navigation accuracy requirement for the satellite would not be met. Satellite state vector update rates are typically once per day for LEO to once per week for geosynchronous Earth orbits (GEO).

19.2.2.1 Ground-based Navigation Systems

Ground-based navigation systems have sensors at one or more locations on the surface of the Earth. It is the traditional way to obtain navigation data. These systems can be categorized as active or passive.

Active ground-based navigation systems interact with the spacecraft being tracked via telemetry signals. A ranging signal is sent to the spacecraft on the uplink, and a response signal from an on-board transponder is received on the downlink. The signal transit time and Doppler shift are used to compute position and velocity relative to the ground station. This type of navigation data is generally collected at the spacecraft operator's ground station.

Passive ground-based navigation systems utilize sensors that do not interact with the object being tracked. They simply use reflected energy: radio waves for radar sensors and visible-spectrum light waves for optical sensors.

Satellite position accuracies from ground-based navigation systems depend on sensor accuracy, amount of data, and the spacecraft's altitude. However, position accuracies typically range from a few kilometers in LEO (much better for laser ranging) to tens of kilometers in GEO. The principal ground-based navigation systems are described below.

Space Surveillance Network (SSN). Operated by the US military, this is the world's most comprehensive space surveillance system. The SSN is a network of radars and optical sensors located at 25+ sites around the northern hemisphere. It can track space objects as small as 10 cm in low Earth orbit. SSN sensors take a combined total of roughly 500,000 observations in a day. Data from the SSN, fused with data from some other sources, is used to maintain the active space catalog that in 2010 contained around 15,000 objects of which only about 1,000 were operational satellites. A key function of the SSN is to catalog and track space debris to support analyses of close approaches (conjunctions) that may pose a threat to operational spacecraft. Spacecraft operators can exchange space situational awareness data with the SSN operator through the Commercial and Foreign Entities (CFE) program.

There are three types of radars in the SSN, the most common being phased-array radars that can track objects up to an altitude of a few thousand kilometers. There are also conventional dish-type radars that can track objects as small as ~25 cm out to the GEO belt (about 40,000 km). Finally, there is the Space Fence which is a system of multi-static radars (3 transmitter sites and 6 receiving stations) spread out across the continental US at a latitude of roughly 33 deg north. This component of the SSN performs surveillance in the strict sense of the word because it does not need to be tasked. The existing VHF space fence tracks objects that pass through its sensor field out to an altitude of about 30,000 km. The new S-band space fence, planned for initial operational capability in 2015 and full capability in 2020, will be able to track even smaller objects. It will use multiple globally positioned radars which will allow for the detection of much smaller space objects than the current system due to their geographic separation and higher radio frequency. With the capability to perform 750,000 observations per day and track over 100,000 objects, the new Space Fence will significantly reduce coverage gaps and significantly improve space situational awareness in low and medium-altitude Earth orbits [James, 2010].

The SSN also has optical telescopes that passively collect light reflected off space objects and can track

Table 19-17◀, Fig. 19-8◀, Eq. 19-17◀

very distant objects, up to a few hundred thousand kilometers away. The Ground-based Electro-Optical Deep Space Surveillance (GEODSS) one-meter telescopes use low-light-level TV cameras to track basketball-sized deep-space objects from about 5,000 km out to GEO altitudes. There are 4 GEODSS sites located in the vicinity of Socorro, New Mexico; Maui, Hawaii; Diego Garcia, British Indian Ocean Territories; and Morón, Spain.

The SSN includes some space-based assets, which will be described later in Sec. 19.2.2.2.

Deep Space Network (DSN). NASA operates an international network of antennas to support interplanetary spacecraft missions. DSN has provided navigation data derived from spacecraft radio signals since the early 1960s for lunar and interplanetary space exploration. With its current sensors, DSN can provide a geocentric angular accuracy on the order of 1 nrad which corresponds to a position accuracy of 150 m at a distance of one astronomical unit (AU).

The DSN currently consists of 3 deep-space communications facilities placed approximately 120 degrees apart in longitude around the world, located in the vicinity of Los Angeles, California (known as the Goldstone site), Madrid, Spain, and Canberra, Australia. This geographic distribution permits constant observation of spacecraft as the Earth rotates, and helps to make the DSN the largest and most sensitive scientific telecommunications system in the world. Each complex consists of 4 or more deep space stations equipped with ultrasensitive receiving systems and large parabolic dish antennas. All DSN antennas are steerable, high-gain, parabolic reflector antennas. The smallest are 26 m diameter antennas used for tracking LEO spacecraft. The 70 m diameter antenna is the largest and most sensitive DSN antenna, and is capable of tracking spacecraft traveling more than 16 billion kilometers from Earth.

Russian Space Surveillance System. This system is operated by a space-surveillance division of the Russian military [Podwig, 2010]. To monitor objects in low Earth orbits the system uses the Russian early-warning radar network which currently includes 9 stations. The space surveillance network includes the Krona system at Zelenchukskaya in the North Caucasus which has dedicated X-band space surveillance radars. Another system of this type is being deployed near Nakhodka in the far-eastern part of Russia. To monitor objects in high-altitude orbits, the space surveillance system uses optical sensors. The main optical observation station, Okno, located in Nurek, Tajikistan, began operations in 1999. Its telescopes allow detection of objects in all orbital regimes up to altitudes of about 40,000 km. However, this optical sensor network lacks coverage outside the borders of the former Soviet Union and thus does not have global radar coverage of the GEO belt.

International Scientific Optical Network (ISON). This system is a network of optical telescopes around the world, managed by the Russian Academy of Sciences. ISON is a partnership of many academic and scientific institutions, currently including 30 telescopes in 20 observatories in 10 countries [Weeden et al., 2010]. ISON is a heterogeneous mix of telescopes of various sizes and capabilities, but as a network it can track a wider range of object sizes throughout deep space (GEO belt) and provide a significant number of observations.

European Space Surveillance and Tracking System. The role of this planned system is to detect, catalog, and predict the movement of objects orbiting the Earth. The Space Surveillance Test and Validation Centre is located in Spain. A precursor system utilizes existing radar and optical sensors; dedicated sensors are planned to developed and deployed after 2011.

There are some individual radar sites currently operational in Europe [Weeden et al., 2010]. They include the French GRAVES bi-static radar fence, the German tracking and imaging radar (TIRA) system that tracks and images LEO objects, and the Norwegian GLOBUS II tracking radar that tracks and images GEO objects.

Indian Deep Space Network (IDSN). The IDSN complex located about 100 km from Bangalore has 3 antennas. The technical facilities in IDSN include a 32 m Deep Space Antenna, an 18 m Antenna Terminal, an 11 m Antenna Terminal, the Indian Space Science Data Centre, and a technical services complex. Operational since 2008, the ISDN was used during the Chandrayaan-1 lunar mission for tracking, orbit control, and housekeeping operations.

Chinese Space Surveillance System. China is believed to have a network of phased array radars at various locations within its borders, each likely to have 3,000 km range and 120 degree of azimuth coverage [Weeden et al., 2010]. It is also believed to have long-range precision mechanical tracking radar. China's main optical sensor capabilities are coordinated by the Purple Mountain Observatory, which operates multiple telescopes in 4 separate locations that can track satellites throughout all orbital regimes. However, China lacks coverage outside of its borders and thus does not have global coverage of the GEO belt.

19.2.2.2 Space-based Navigation Systems

Space-based navigation systems have sensors on one or more spacecraft orbiting the Earth; the overall system may include ground stations to receive/process data from the satellites. Space-based sensors can detect debris, spacecraft or other distant space objects without interference from weather, atmosphere, or lighting conditions. The principal systems are described below.

Tracking Data Relay Satellite System (TDRSS). NASA established this system in the early 1980s to replace its worldwide network of ground tracking stations. The TDRSS space segment currently has 8 operational satellites (including spares) located in geosynchronous orbit. This constellation provides 100% visibility to user spacecraft operating at altitudes from 10,000 km down to 1,200 km, decreasing to 85% visibility at 300 km altitude [Teles et al., 1995] due to line-of-sight limitations. The TDRSS ground segment is located

near Las Cruces, New Mexico, and consists of two functionally identical ground terminals collectively known as the White Sands Complex. An additional ground terminal complex in Guam communicates with the satellites not visible from New Mexico.

The user spacecraft needs an onboard TDRSS transponder. High-precision range and range-rate tracking of the user spacecraft relative to the tracking data relay satellites is performed according to a resource schedule. Highly accurate estimates of the tracking data relay satellites' position and velocity are also available, enabling the TDRSS ground segment to determine the user spacecraft's position with a 3-sigma accuracy of about 50 m. This navigation data is relayed from the TDRSS ground segment to the user's ground station.

Space-Based Surveillance System. The US military's Midcourse Space Experiment (MSX) missile defense test satellite, launched in 1996, carried the first dedicated space-based optical device (a visible-band electro-optical camera) for space situational awareness. Until its end-of-life in 2008, the MSX satellite used its space-based visible (SBV) sensor to augment the US Space Surveillance Network's ground-based sensors by expanding surveillance and tracking of objects in the GEO belt. MSX is being replaced by a more advanced constellation of orbiting sensors (and supporting ground infrastructure) known as the Space-Based Surveillance System (SBSS). The first "pathfinder" satellite of this system was successfully placed into a 630 km Sun-synchronous orbit in 2010. It has a 30 cm telescope on a two axis gimbal with a 2.4 megapixel image sensor. Today, many GEO objects go days without being tracked. The SBSS will provide the ability for uninterrupted scanning of the entire GEO belt every 24 hours.

19.2.2.3 Autonomous Navigation Systems

Autonomous navigation systems utilize sensors onboard the spacecraft to determine navigation states in real time. These instruments may need signals transmitted from other satellites and/or ground stations (semi-autonomous), or not need any transmitted signals at all (fully autonomous).

Global Navigation Satellite Systems (GNSS). These satellite constellations provide geo-spatial positioning data with global coverage. An overview of operational and some planned GNSS constellations is provided below. Position (e.g., latitude, longitude, altitude) of receivers are determined using time signals transmitted along a line-of-sight by radio from the satellites. Signals from 4 or more satellites are used to compute the receiver's horizontal and vertical position vector components. Receivers using civilian signals typically provide position data with an accuracy on the order of 10 m [Hofmann-Wellenhof et al., 2008].

GNSS satellites have an orbital altitude of about 20,000 km (exact value depends on the constellation), and their antennas are designed to provide signals only in a cone covering the Earth's surface. Hence global coverage is not assured at altitudes well above LEO, but it is possible to use the signals for navigation at altitudes up to and beyond GEO by using the spillover of the beam beyond the edge of the Earth's disk.

US Navstar Global Positioning System. Commonly known as GPS, this military-operated system has been globally available since 1994. There are 24 satellites (plus spares) in 6 orbital planes with a period of about 12 hours (2 revolutions per day) at about 20,200 km altitude. The orbital planes are evenly spaced by 60 deg in right ascension, with an inclination of 55 deg [Langley, 1992]. There should always be 6 satellites within line-of-sight from almost any point on Earth; this requires uneven spacing in mean anomaly along an orbit. Each satellite repeats its ground track on a 1-day cycle and hence will pass the same Earth location every day. Commercial GPS receivers are available for spacecraft, and they are commonly used in LEO. For example, the EO-1 mission used GPS for orbit determination.

Russian GLONASS. This military-operated system is nearly complete and is planned to become globally available in 2011. The full constellation provides 24 satellites in 3 orbital planes with a period of 11.3 hours at 19,100 km altitude. The orbital planes are evenly spaced by 120 deg in right ascension, with an inclination of 65 deg. There is even spacing in mean anomaly within each orbital plane, and an argument-of-latitude displacement of 15 deg between successive orbital planes. Each satellite repeats its ground track on an 8-day cycle; since there are 8 satellites in each orbit, a satellite will pass the same Earth location every day.

European Galileo. This civilian-operated global navigation system is under development by the European Commission and the European Space Agency. Initial operational capability is anticipated around 2015, with full capability expected sometime after 2020. The full constellation design calls for 27 satellites (plus 3 spares) in 3 orbital planes with a period of 14 hours at 23,222 km altitude. The satellites will be spread evenly around each orbital plane with an inclination of 56 deg.

Other Systems. The Chinese Beidou-1 regional navigation system is operational. There are 3 satellites (plus 1 backup) in GEO providing coverage over a substantial portion of eastern and southern Asia. Beidou-2, commonly known as Compass, is a system of 35 satellites that is planned to provide global coverage by 2020. Eight Beidou-2 satellites had been deployed by April 2011.

The Indian Regional Navigational Satellite System (IRNSS) is currently in development. It consists of 3 geostationary satellites and 4 geosynchronous satellites at 29 deg inclination that are anticipated to provide coverage over the Indian sub-continent by 2014.

TDRSS On-board Navigation System (TONS). This NASA system, developed in the early 1990s, utilizes the space segment of the TDRSS described earlier in Sec. 19.2.2.2. TONS works by forwarding the ground communication signal from the TDRSS to the user satellite, measuring the Doppler effect on that signal, and processing it using an onboard extended Kalman filter to accurately estimate the user satellite's state vector.

Table 19-17◀ Fig. 19-8◀ Eq. 19-17◀

TONS provides position data with a 1-sigma accuracy of 10 m [Gramling and Long, 1995]. Navigation using TONS requires the user satellite to carry more onboard equipment than is required for navigation using the basic TDRSS, but the advantages are autonomy and increased accuracy of navigation data.

Doppler Orbitography and Radiopositioning Integrated by Satellite (DORIS). Operated by the French space agency, DORIS is a world-wide network of about 60 ground beacons that emit radio signals. The signals' frequency shift caused by the movement of the user satellite (Doppler effect) is measured by a receiving antenna on board the satellite to generate its own navigation data. DORIS has been used by several satellites, including TOPEX/Poseidon and Envisat, for precise orbit determination on the order of a few centimeters.

Optical Stellar Navigation. Imaging devices are used to observe a local celestial body against a known background star field. Navigation states are calculated using optical measurements: the angle formed by a star and the horizon of a celestial body, and the apparent angular diameter of the celestial body. In some cases, the observation tasking request is planned by the ground station and uplinked to the spacecraft well in advance; the images acquired by the spacecraft are then downlinked to the ground station for processing to determine the navigation states. In other cases, onboard systems command image acquisition and perform the calculations to determine the navigation states.

Throughout the primary mission of Deep Space 1, about once per week, its automatic navigation system (AutoNav) was invoked by the operating sequence to acquire optical navigation images. It turned the spacecraft and the integrated camera and imaging spectrometer to record images of asteroids and stars, and then used on-board image analysis for navigation. The apparent position of an asteroid relative to the much more distant stars enabled AutoNav to calculate the location of the spacecraft in the solar system.

Optical Landmark Navigation. Image-based on board navigation is accomplished by tracking landmarks such as craters. Tracking individual craters enables orbit determination accuracies on the order of the camera resolution, e.g., several meters for small bodies such as asteroids.

Navigation relative to the celestial body is accomplished by comparing the current image against a reference map stored on the spacecraft; spacecraft attitude information is also required. Generating the reference map in advance of onboard use requires a substantial effort at the ground station. For example, the Near Earth Asteroid Rendezvous (NEAR) mission used an on-board landmark-tracking navigation system during its rendezvous with the asteroid Eros in 2000–01. In preparation for this event, ground crew manually extracted and identified craters from thousands of optical images—about 10,000 craters were identified on Eros' 32-kilometer long and 16-kilometer wide body. Feature recognition software could be used to assist this process in future missions.

Table 19-17◀, Fig. 19-8◀, Eq. 19-17◀

In 2005, the Hayabusa spacecraft mission to the asteroid Ikotawa used optical navigation (with some ground assistance) for proximity operations within a few kilometers of the asteroid's surface. Additionally, a target marker was released from the spacecraft to provide a visual target for the landing site. Hayabusa's navigation sensors included a lidar and laser range finder.

Horizon Scanner Navigation. These systems feature spinning sensors that move along the surface of a cone. Each scan generates horizon entry/exit points on the edge of the celestial body disc seen by the satellite. These points are used to determine the spacecraft's distance and direction relative to the celestial body. For example, the Microcosm Autonomous Navigation System (MANS) system uses two spinning sensors that provide Earth/Sun/Moon position measurements to determine satellite position and attitude. For most low-Earth orbit conditions, the position accuracy is 500 m with one sensor and 100 m with two sensors. Other horizon scanner systems include Honeywell's Earth Reference Attitude Determination System (ERADS) and the Compact Autonomous Navigation System (CANS) developed by the Chinese.

19.2.3 Trajectory Control

Trajectory control utilizes the spacecraft's propulsion system to alter the magnitude and/or direction of the vehicle's velocity vector. This requires a thruster burn of a specific duration, and often requires a prior attitude change maneuver to properly orient the thruster(s) used for trajectory control.

Most small spacecraft do not have onboard propulsive systems; this has the advantage of reducing spacecraft complexity, weight, and cost. However, no trajectory control is possible after the spacecraft has been placed into orbit, and hence the orbital parameters will drift significantly over time due to the cumulative effect of secular gravitational-field perturbations. LEO satellites will also experience a continuous decay of orbital altitude due to atmospheric drag. An uncontrolled trajectory may be acceptable for some satellites with a mission life of 1 to 3 years. Other spacecraft will generally require some type of trajectory control.

In some cases, the propulsive force for trajectory change may be provided by another spacecraft. The Space Shuttle has boosted the Hubble Space Telescope's orbit during servicing missions. The International Space Station (ISS) periodically has its orbit raised by a few kilometers by visiting spacecraft such as Progress supply ships, Space Shuttles, and Jules Verne automated transfer vehicles. In an emergency, the ISS can use the thrusters on its own Zvezda service module for orbital maneuvering.

This section provides basic information about various types of trajectory changes from a controls perspective. Some typical values for the frequency and magnitude (ΔV) of the required control are also presented. Section 9.6 covers orbital maneuvers from an astrodynamics perspective.

19.2.3.1 Orbit Maintenance

After a spacecraft has been placed into the desired orbit, minor adjustments are generally required to compensate for disturbances so that the orbital parameters remain within mission tolerances. The disturbances can arise from a non-uniform gravity field of the central body, atmospheric drag (at lower altitudes), solar radiation pressure, and gravity effects of celestial bodies other than the central body. Orbit maintenance maneuvers typically require a small ΔV, but are performed numerous (~100) times over the mission life of the spacecraft.

For Earth orbits, the equatorial bulge creates significant secular perturbations as well as long-period (order of days or weeks) and short-period (order of minutes or hours) cyclic perturbations. Solar radiation pressure produces a small force that can affect the orbital trajectory over long periods of time; this force depends in part on the spacecraft structure and its orientation with respect to the Sun. High altitude orbits are affected by the gravitational forces of the Moon and Sun. Orbits below 1,000 km altitude are significantly affected by atmospheric drag that varies over time according to the solar cycle which has a period of about 11 years.

A commonly used maintenance maneuver for LEO satellites is orbit raising to compensate for altitude decay due to atmospheric drag. The frequency of performing this maneuver depends on the nominal orbital altitude (more frequent at lower altitudes) and the tolerance within which the altitude must be maintained (depends on mission performance requirements). A spacecraft adjusts altitude by firing its thrusters to provide an in-plane ΔV at the appropriate point(s) along the orbit. The ΔV required for altitude maintenance is quite small; for example a 1 km adjustment at 1,000 km altitude requires a ΔV of 0.5 m/s. Orbit raising maneuvers for LEO satellites are typically performed at intervals of a few weeks. For example, Envisat maintains its orbital altitude within tens of meters on a cycle of roughly 15 days to maintain its nominal altitude of 800 km.

Satellites in Sun-synchronous orbit (SSO) need to perform periodic corrections in inclination to maintain the SSO precession rate of the orbital plane, and in longitude of ascending node to maintain the desired mean solar time at the ascending/descending node. This is accomplished by firing thrusters to provide an out-of-plane ΔV at the appropriate point(s) along the orbit. Satellites with repeating ground track orbits (SSO or otherwise) also need to perform maintenance maneuvers to adjust their orbital parameters. For example Envisat maintains its ground track to within 1 km, its mean solar time to within 5 minutes, and makes small corrections to its inclination every few months.

GEO satellites are assigned slots to avoid possible radio interference and to maintain adequate physical separation from neighboring satellites. These slots are distributed along the Equator approximately 2 deg apart in longitude (about 220 km distance), and the associated station-keeping box is ±75 km in each of the three spatial dimensions. GEO satellites experience a resonance effect due to the Earth's non-uniform gravity field, which causes perturbations along the orbital path, resulting in East-West drift. Hence they need to perform phasing adjustments to remain within the along-track boundaries of their station-keeping box. GEO satellites also experience out-of-plane perturbations due to Sun-Moon effects, which cause North-South drift. Hence they need to perform inclination adjustments to remain within the cross-track boundaries of their station-keeping box. See Sec. 9.5.1 for coverage of orbital perturbations experienced by GEO satellites.

19.2.3.2 Moderate Orbit Changes

Moderate orbit changes are sometimes required in response to planned or opportunistic modifications of mission objectives. They typically require a moderate ΔV, and are performed only a few times over the mission life of the spacecraft. Some applications of moderate orbit changes are described below.

A change in mission objectives of a SSO satellite may require a modification of the mean solar time, which in turn requires a change in longitude of ascending node. This can be done by making a small change in altitude which changes the nodal rotation rate caused by Earth oblateness; see Eq. (9-54). After the ascending node has reached the desired longitude, the altitude is changed back to the original value. Another approach is to first determine the two points at which the initial and final orbits intersect. A single burn that applies a ΔV normal to the orbital plane at either of these intersection points provides a change in longitude of ascending node.

GEO satellites are moved into their (original or new) longitude slot with a phasing maneuver. This is typically done by placing the spacecraft in a drift orbit by making a small change in altitude which modifies the along-track velocity. After the spacecraft has drifted to the desired position along the orbital track, the altitude is changed back to the original value. For example, the process of moving a GEO satellite by 1 deg of equatorial longitude (about 110 km distance) over 1 day requires a total ΔV of 5.7 m/s; see Eq. (9-81).

All GEO satellites are required to raise their altitude by ~300 km for end-of-life disposal to a graveyard orbit; this requires a ΔV burn of ~11 m/s which corresponds to roughly 3 months of station-keeping propellant. Some LEO spacecraft are deorbited according to the 25-year disposal rule [NASA, 2009]. The spacecraft performs a deorbit burn that results in an initially slow altitude-decay rate that subsequently becomes faster as atmospheric density increases at lower altitudes. The intent is to complete the deorbit within 25 years after end-of-life, based on a conservative estimate of the solar cycle. From Table 9-21, a spacecraft in a 600 km circular orbit needs to perform a ΔV burn of about 107 m/s to lower its perigee to 216 km so that it will complete its deorbit in approximately 25 years. For more information on spacecraft disposal, see Sec. 9.6.4 and Chap. 30.

Table 19-17◀, Fig. 19-8◀, Eq. 19-17◀

19.2.3.3 Major Orbit Changes

Major orbit changes correspond to substantial changes in one or more orbital parameters. They typically require a large ΔV, but are performed very few times (usually just once) over the mission life of the spacecraft.

A common type of major orbital maneuver is a large change in semi-major axis. A Hohmann trajectory provides the minimum-energy transfer between two coplanar circular orbits; see Sec. 9.6.1 for more information. For example, a Hohmann transfer from a 191 km parking orbit to geosynchronous altitude (35,781 km) requires a ΔV of 3.9 km/s with a flight time of 5.26 hours [Vallado, 2007]. In practice, most orbit transfers are relatively close to a Hohmann transfer, although high-energy transfers are sometimes used when transfer time is critical. If the spacecraft design trade-offs do not permit the use of large rocket engines to provide the impulsive (short-duration) burns required for a Hohmann transfer, then low-thrust chemical or electric propulsion systems can be used to achieve orbit transfers over much larger periods of time than a Hohmann transfer. For orbit transfer maneuvers, the propulsive force is applied in the orbital plane. Hohmann transfers require that the thrust be tangential to the orbital path.

Another common type of major orbital maneuver involves a substantial change in orbital plane inclination, e.g., a satellite launched from a high-latitude site that needs to be placed into a low-inclination orbit. A spacecraft changes the inclination of its orbit by firing its thrusters to provide a propulsive force normal to the orbital plane at the ascending or descending node; Eq. (9-78) provides the ΔV relationship for circular orbits. For example, a 10 deg change in inclination requires a ΔV equal to almost 18% of the orbital velocity; at 1,000 km altitude this translates to a ΔV of 1,283 m/s. For an elliptical orbit, the thruster burn should be performed at the node where the satellite has the smaller velocity because this requires a smaller ΔV.

Some maneuvers require a major change in both semi-major axis and inclination, e.g., transfer from an inclined low-altitude parking orbit to GEO. In such cases, performing a combined orbit-raising and inclination-change maneuver requires less ΔV than performing two separate maneuvers. The thrust direction for such combination maneuvers has both in-plane and out-of-plane components. The first burn of the combination maneuver typically provides only a small fraction of the inclination change; most of the velocity vector change is devoted to orbit raising. This burn puts the spacecraft on a highly elliptical geosynchronous transfer orbit (GTO) with inclination slightly less than that of the parking orbit. The GTO perigee is equal to the parking orbit altitude, and the apogee is equal to GEO altitude. The second burn of the combination maneuver occurs at GTO apogee; it provides most of the inclination change and also circularizes the orbit by raising the GTO perigee to GEO altitude.

Occasionally, a major orbit change is made in response to large-scale revisions of mission objectives. For example, after the International Sun/Earth Explorer 3 (ISEE-3) completed its original mission at the Sun-Earth L1 Lagrange point, it was repurposed and renamed the International Cometary Explorer (ICE). Three years later, ICE passed through the tail of Comet Giacobini-Zinner.

19.2.3.4 General Trajectory Changes

The above categories of trajectory change maneuvers involved closed orbital paths. Some other types of trajectories are covered below.

Interplanetary Flight. During long-duration interplanetary missions, some trajectory errors build up over time. To correct such errors in the spacecraft's solar or planetary orbital parameters, a Trajectory Correction Maneuver (TCM) is planned and executed. This adjustment involves computing the direction and magnitude of the velocity vector required to get back on the desired trajectory; this is known as guidance. An appropriate time is determined for making the change. The spacecraft is commanded to rotate to the appropriate attitude and its thrusters are fired for a pre-calculated time interval. TCMs generally involve a ΔV on the order of 1–10 m/s, using control thrusters that provide a thrust force on the order of 10 N. The velocity change magnitude is necessarily small due to the limited amount of propellant mass available on interplanetary spacecraft. Information on interplanetary orbits can be found in Sec. 10.7.

The Cassini spacecraft provides an example of the accuracy achieved in TCM burns. Burn durations were executed within about 0.1% of the planned value, and the pre-burn pointing maneuvers were executed within about 0.4 deg. Over the course of 7 years from launch to arrival at Saturn, the Cassini spacecraft executed only 17 such planned, small-velocity adjustments.

Flybys or gravity-assist trajectories can save much energy (ΔV) in interplanetary flight, albeit at the expense of increased transit time. For example, the MESSENGER spacecraft arrived at Mercury after performing flybys past Earth and Venus. Between flybys, trajectory control was accomplished using 4 monopropellant thrusters (22 N each) and "solar sailing" by rotating the spacecraft and tilting its solar panels to use the very small solar radiation pressure to gradually alter the spacecraft's trajectory. For more information on planetary assist trajectories, see Sec. 9.6.3.

Atmospheric Entry and Descent. This phase of flight begins when the spacecraft enters the atmosphere shortly after a deorbit burn. The subsequent trajectory of the spacecraft is governed by gravitational and aerodynamic forces—there are no propulsive forces. The aerodynamic force acting on the spacecraft can be adjusted by reorienting the vehicle in the atmosphere. Capsule spacecraft (e.g., Apollo and Soyuz) use reaction control system (RCS) jets to perform these attitude-change maneuvers; winged spacecraft (e.g., Space Shuttle) transition from RCS jets to aerodynamic control surfaces at

Table 19-17◀, Fig. 19-8◀, Eq. 19-17◀

higher dynamic pressures. In both capsule and winged spacecraft, trajectory control is accomplished by changing the vehicle bank angle (orientation of the lift vector relative to the horizontal) to steer the vehicle toward the desired landing/splashdown site.

Using aerodynamic control surfaces on a winged spacecraft provides very precise trajectory control; the Space Shuttle has made over 130 successful landings on different runways. Capsule spacecraft lack the high-precision trajectory control available to winged spacecraft at low altitudes, but they are generally within 10 kilometers horizontal distance from their touch/splash down target at parachute deployment.

Powered Descent and Landing. This is a good option if the atmosphere of the planet/moon is too thin for aerobraking and parachute landing. Trajectory control in the vertical axis can be accomplished by a throttled descent engine that provides the desired descent-rate profile. Trajectory control in the horizontal axes is accomplished by tilting the thrust vector of the descent engine, with an appropriate throttle increase to compensate for the reduction in vertical component of thrust. The thrust vector can be tilted by rolling/pitching the vehicle to the appropriate attitude using RCS jets; this approach was used by the Apollo lunar landers. The Phoenix Mars Lander used a system of multiple descent thrusters for a powered descent from 570 m altitude to touchdown.

19.2.4 Design Considerations

This section presents some general guidelines for the selection of navigation and trajectory control systems during the design cycle of a spacecraft.

A key decision is the choice of autonomous vs. traditional navigation and trajectory control. Another high-level decision is the choice of combined vs. separate systems for attitude and trajectory control. There are pros and cons for each of these options and the final choice depends on the interplay between mission requirements such as performance, reliability, and cost. These choices in turn affect the options available for subsystem selection and design.

19.2.4.1 Navigation Systems

Ground-station tracking performed by the spacecraft operator requires transponder and telemetry equipment on board the spacecraft. Note that this type of navigation data is available only when the spacecraft is visible to the operator's ground tracking station. Even if the tracking stations are geographically distributed, like the SSN or DSN, resource sharing/allocation constraints will limit the update rate of navigation data. It is possible for a satellite operator to acquire navigation data without any onboard equipage by accessing publicly available tracking data from the US Space Surveillance Network. Two-line element sets are available to users at no cost, but they are updated only twice a day and the accuracy of the data will likely be less than that achievable by transponder signals. However, this may be acceptable for some low-cost nanosatellites.

A space-based tracking system such as TDRSS can provide high-accuracy navigation data to the spacecraft operator; this requires a specific type of transponder on board the user spacecraft. With some additional on-board equipage, the TONS system can use TRDSS signals to provide autonomous navigation capability. Another autonomous navigation option for spacecraft in LEO is to install commercially available GPS receivers on board the spacecraft. For interplanetary flight, autonomous navigation is available with optical stellar navigation. Autonomous navigation relative to a celestial body such as an asteroid can be performed using optical landmark navigation. Space-based and autonomous systems can generally provide navigation data at any desired update rate, whereas ground-based systems can provide navigation data only at times when the spacecraft is visible to the ground tracking station(s).

Note that all navigation systems requiring on-board equipment have varying amounts of direct (acquisition) costs as well as indirect costs associated with mass and power usage.



Table 19web-1 provides some data on mass, power, and accuracy for flight-qualified navigation systems.

19.2.4.2 Control Systems

Propulsion options for trajectory control include cold gas, liquid (monopropellant, bipropellant, dual mode, hybrid), and electric; see Chap. 18 for details. Cold gas propulsion systems are inexpensive, low performance systems that are rarely used unless there is an overriding requirement to avoid the hot gases and safety concerns of liquid propulsion systems. A widely used type of propulsion for spacecraft control is monopropellant hydrazine which decomposes explosively when brought into contact with an electrically heated metallic catalyst in the decomposition chamber of a control thruster. Although its efficiency is lower than that of many other options, propulsion using monopropellant hydrazine is popular because of its simplicity and reliability. Some spacecraft produce control thrust using bipropellants: a fuel such as hydrazine and an oxidizer such as nitrogen peroxide. This bipropellant combination is hypergolic and ignites spontaneously upon being mixed in the combustion chamber of a control thruster. The Hayabusa spacecraft used/tested ion propulsion engines during its deep-space mission to the asteroid Ikotawa. Deep Space 1 also demonstrated ion propulsion during its mission.

The size and number of thrusters depend on the ΔV requirements of the various maneuvers that need to be performed over the life of the spacecraft. The total number of control thrusters on a spacecraft can range from a few to as many as 30. Note that in some cases, the same thrusters may be used for attitude (rotational) control as well as trajectory (translational) control. In general, long steady burns are used for trajectory control while short-pulse (~10 ms) firings are used for attitude control.

The spacecraft's flight control computer selects the appropriate control thruster(s) and commands a burn

Table 19-17◀, Fig. 19-8◀, Eq. 19-17◀

duration by utilizing on-board control laws (autonomous control system) or simply executing commands uplinked from mission control. The thruster burn profile varies widely, depending on the specific parameters of the desired trajectory change maneuver. The flight computer controls thruster burns by sending commands to open/close the appropriate valves that regulate the flow of propellant to the combustion chamber.

Several types of thrusters are available for trajectory control. For example, a small thruster (mass of 0.1 kg) can provide a propulsive force of 0.1 N, while a large thruster (mass of 10 kg) can provide a propulsive force of 3,000 N. Note that limits on payload acceleration may set an upper limit on propulsive forces. The specific impulse of control thrusters typically lies in the range of 200–300 sec.

As an example, orbit control (and backup momentum unloading) on Landsat 7 was provided through a blow-down monopropellant hydrazine control system with 12 jets that each provided 4.4 N of thrust. A single tank contained 122.5 kg of hydrazine, corresponding to roughly 6% of the total spacecraft mass.

Table 18-6 lists flight-qualified thrusters available for trajectory control. Performance, mass, and power data can be found in Table 18-5 (cold gas thrusters) and Table 18-8 (bipropellant thrusters).

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20 Spacecraft Subsystems III—On-Board Processing

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20.1 Computer System Baseline
Requirements Definition; Requirements Flow-Down; Architecture Evaluation
20.2 Preliminary Design
Defining Processing Tasks; Computer Selection Guidelines; Software Development; Integration and Test
20.3 FireSat II Example
20.4 Modular Approaches to Processing—*L. Jane Hansen and Paul Graven*

Mission-supporting computer systems include the computers on board the spacecraft, as well as those on the ground, as illustrated in Fig. 20-1. On board the spacecraft, both large computing systems and microcontrollers have become an integral part of the overall system, as well as being part of most spacecraft subsystems. A key feature of on-board flight software (FSW) is the hard, real-time and deterministic nature of the execution. Ground based computer systems are often used to test the on-board systems, prior to launch, and then again, after launch they are used to support daily operations. The post-launch systems are often derived from systems originally used for developing and testing space-based elements.

Computer systems often cross traditional subsystem and organizational boundaries. In previous chapters, we have described some of the spacecraft subsystems that are implemented by specific disciplines, such as sensors and actuators, for *attitude determination and control* (ADCS). While the sensors and actuators, as well as the control law algorithms, are specified by an ADCS specialist, we often include the processing associated with each as part of the on-board processing. This means that in most cases, the task of defining computer system requirements and associated costs takes on a scope that includes many, potentially integrated disciplines. When subsystem functionality is migrated to a centralized processor, the computer system resource estimation process takes on a larger scope than when each functional subsystem contains its own processing element. This is primarily due to the fact that domain experts from different disciplines, with different implementation styles and processes, will be required to integrate their parts of the system into a single processing product. When we create this single product, we must attempt to integrate the development approach and styles of the different organizations by creating a programming standard and style guide. We can achieve some synergy in the final product by developing a standard set of "utilities" such as mathematical functions that can be used by all discipline specific software. In smaller spacecraft, many of the functional software components are co-resident on a limited number of processing elements with limited resources which may require additional management to assure there are no duplications and utilities are properly defined. However, when the management overhead is traded against the potential for each discipline carrying their own computing resources, the weight and power associated with processing is often prohibitive. Thus, when designing spacecraft software it is important that we include both flight software experts and the subsystem domain experts who are familiar with the various disciplines.

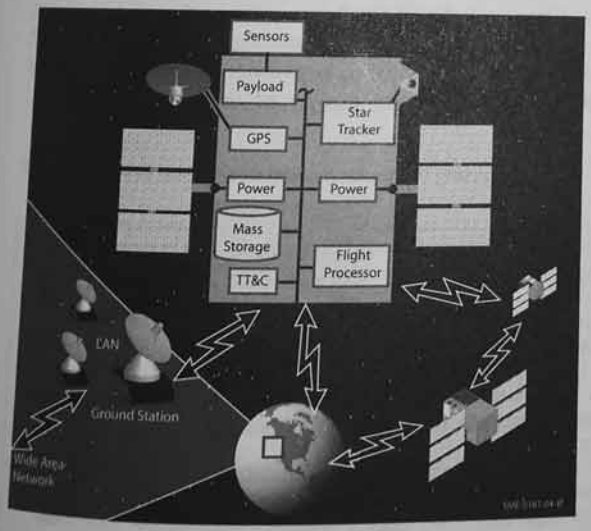


Fig. 20-1. Mission Computer Systems. Notice that there are many interfaces and managing their compatibility is critical to reducing cost and risk. Also, notice that the development tools and environment required to build, integrate, and test the computer hardware and software are included as part of the mission computer system

In this chapter we discuss how to refine the computer system requirements, generate computer system resource estimates, evaluate the implementation effort, and define the tasks associated with developing computer systems on board a spacecraft. Additionally, we will briefly examine the interactions between the on-board computer systems and the ground-based computer

Table 20-1. Computer Systems Development Process. This iterative process defines top-level requirements during a program's concept through development phases.

Define Requirements
• Evaluate Mission Objectives for direct implications to computer systems
• Evaluate derived requirements from all disciplines to find synergy and define utilities
• Develop Top-Level Computer Requirements
Allocate Top-Level Computer Requirements
• Evaluate Candidate Architectures and Interfaces
• Perform Functional Partitioning
• Evaluate Reliability Requirements
• Establish System Baseline
Define Computer System Requirements
• Define Processing Tasks
• Establish Software Size and Timing Requirements
• Establish Computer Size, Weight, Power and Throughput Requirements
• Identify Target Hardware
Define Development and Deployment Strategy
• Identify Operating System and Development Environment
• Identify Required Support Tools
• Detail a Configuration Control Process
• Document Test and Integration Approach
• System Deployment and Long-term Support

systems, with an emphasis on deployment through on-orbit operations. In most cases, given a reasonable set of requirements, we can create multiple approaches for implementation that will all meet the mission needs. It is important to understand the prioritization of the mission requirements and the specifics of the flow-down requirements, as well as the often unspoken implementation requirements such as keeping employees involved on the projects or using specific subsystem elements. In this chapter we will discuss how these overlapping requirements influence the baseline computer system design. Finally, we will examine various approaches to modular processing that may be incorporated to facilitate rapid assembly, integration, and test (AI&T) and reduce the timeline associated with early on-orbit checkout.

As outlined in Table 20-1, we discuss the iterative process used to estimate computer resources, based on mission requirements and operational needs, flowed into processing requirements and software development approaches, continuing with deployment and operational constraints. We will also explore alternative approaches and constraints being put on on-board computer systems such as *reusable flight software* (FSW) modules, the use of *plug-and-play* (PnP) technologies, and other modular implementation concepts. We will accomplish this by first discussing the computer system elements and the process of specifying the requirements, we will move on to the task of creating a baseline computer system architecture from top level requirements. Next, we will evaluate the resources required to achieve the baseline system. This includes hardware and software, in addition to development and on-orbit support. We will use the FireSat II example to clarify some of the key components and concepts of the estimation process. In the final section we will address modular approaches to processing which can be used for rapid integration and rapid response missions, to reduce inventory on a constellation or fractionated space system, or as a means of standardizing interfaces to increase the potential market for subsystems functionality. Table 20-2 provides definitions for terms frequently used in estimating computer system resource requirements.

20.1 Computer System Baseline

Chapter 6 discusses the requirements definition process while Chap. 14 discusses the top-level spacecraft design process, identifies key design drivers, and looks at the evolution of the design of the mission lifecycle.

Table 20-2. Definitions Associated with Computer Systems. Often when discussing computer system design and development we use terms which have a specific meaning to those involved in the discipline.

Embedded Systems	A built-in processor or microprocessor, providing real-time control as a component of a larger system, often with no direct user interface.
Real-Time Processing	Handling or processing information at the time events occur or when the information is first created. Typically, embedded or on board processing is real-time.
Hard Real-Time	Requiring precise timing to achieve their results, where missing the time boundary has severe consequences. Examples include attitude control software and telemetry downlink. [Stankovic and Ramamritham, 1988].
Soft Real-Time	Requiring only that the tasks are performed in a timely manner, the consequences of missing a time boundary are often degraded, but continuous, performance. Examples include orbit control software and general status or housekeeping.
Operating System Software	Manages the computer's resources such as input/output devices, memory, and scheduling of application software.
Application Flight Software (FSW)	Mission specific software which does work required by the user or the mission rather than in support of the computer.

Through the design process, we determine system requirements and allocate them to subsystems, identify operational modes for the spacecraft bus and payload, allocate top-level requirements to the computer system (among other spacecraft elements), define the subsystem interfaces in a broad sense, and identify peripheral system requirements. Defining requirements for the computer system begins with these results. Figure 20-2 illustrates that the computer system baseline must include hardware and software, as well as the associated documentation. Documentation is critical to the success of any program, whether it is small or large, short timeline or long, inexpensive or very costly. Documentation often starts at the system level and flows down to the components. We often use different documents for processing hardware and software. Because software is often hard to manage, we create additional management documents, such as the software development plan (SDP) for software configuration items (CI). A list of necessary documentation, by type, is shown in Table 20-3. Not all of the documents listed in Table 20-3 are created for every program.

For a software product the four most important documents, in order of importance are the Software Requirements Document (SRS), the Software Design Document (SDD), the Software Test Plan (STP), and the Software Test Descriptions (STD). In addition, a Software Development Plan (SDP), a Software Standards and Practices (SSP), and an Algorithm Design Document (ADD) are often needed. There are guidelines for developing software documentation in terms of military and NASA standards, data item descriptions (DIDs), as well as templates that are provided by IEEE, AIAA, SAE, and other professional organizations. For smaller, shorter programs we can tailor the documentation to meet the needs of the program. However, when tailoring occurs, it is imperative that documents of different types are not merged together. For example, a requirements document will outline the functional capabilities of the system, without regard to the implementation. A design document will delineate the implementation, providing traceability back to the requirements that generated the need. A design document, such as an Algorithm Design Document (ADD), cannot be included in a requirements specification because this mixes two different types of documentation that have different purposes in the development life-cycle.

For computer hardware, documentation is developed to describe the actual hardware design, including the instruction set architecture (ISA), form factor (size), weight and power. Additionally, because we use the processing hardware to execute the software, we require documentation that describes how the software developer will use the hardware, often defined as a hardware-software Interface Control Document (ICD). This includes information about the number of bits per word and the speed, as well as memory type, access speed, and scrubbing mechanisms (discussed in more detail later in this chapter). Other hardware design features that we need to have documented include the use of specialized processing chips such as digital signal processors (DSPs), the implementation of mathematical functions — those that are supported in hardware versus software (a hardware implementation is faster than those implemented using software), fixed versus floating point processing availability, the number and types of input/output (I/O) lines—digital and analog, power requirements and interface, and available operating system support packages. For computer systems that meet military standards, we consult documentation that describes testing requirements such as MIL-STD-883H

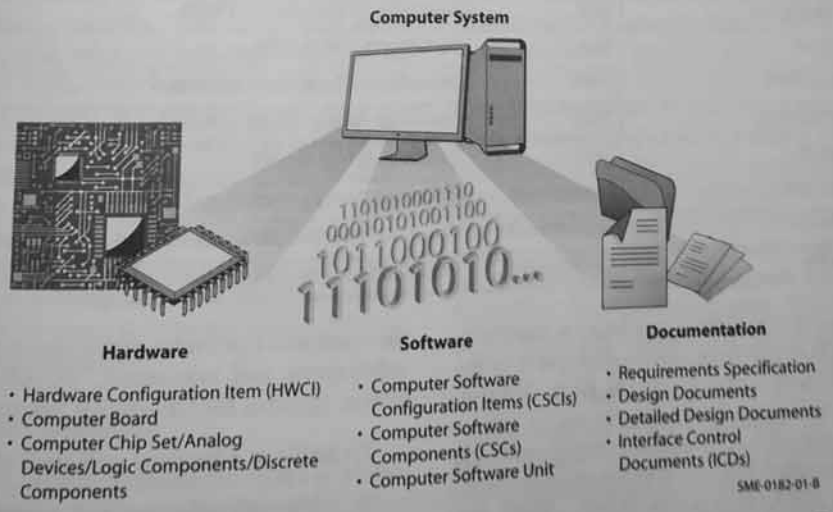


Fig. 20-2. Hierarchy of Elements in a Computer System. Computer systems consist of hardware, software, and their interface definitions. Hardware and software components are in a hierarchy, building to the final configuration item. Documentation provides a mechanism to transfer pertinent information about each element and how they interact. Documentation also has a hierarchy, but it starts with top-level requirements and leads to increased implementation detail.

Table 20-2◀, Fig. 20-2, Eq. 20-0

Table 20-3. Documentation for Computer Systems. Computer systems documentation address all aspects of the software development life-cycle. Documentation may be tailored for specific programs but it is important that tailoring does not combine documentation from different types.

Document	Type	Description
System Segment Specification (SSS)	Requirements	Defines the top level requirements derived from the mission objectives for the spacecraft bus components
Configuration Item (CI) Functional Product Specification (FPS)	Hardware Requirements	Defines the requirements, performance, design, test, manufacture, and acceptance test requirements for hardware elements
Hardware Interface Control Document (ICD)	Requirements	Defines the physical interface requirements for the component
Hardware-Software ICD	Requirements	Defines how the processing hardware is used to implement software
Software Requirements Specification (SRS)	Software Requirements	Defines the requirements for the software flowed down from the SSS and how to meet requirements
Interface Requirements Specification (IRS)	Requirements	Defines the internal and external data interface requirements for the software system
Critical Item Product Fabrication Specification	Program Management	Establish requirements for manufacturing and acceptance of critical end items
Software Development Plan (SDP)	Program Management	Outlines software development policies, standards, and procedures
Software Resource Document (SRD)	Program Management	Describes necessary resources for software development
Software Standards and Practice (SSP)	Program Management	Identifies policy and styles for standardized software development
Hardware Detailed Design Document	Design	Defines hardware design
PCB Layout / Schematic	Design	Details printed circuit board (PCB) layout and design via schematics
Enclosure / Power / Connectors	Design	Defines hardware enclosure, power supply, connectors, and other physical device requirements
Software Design Document (SDD)	Design	Defines software design
Interface Design Document (IDD)	Design	Details Software internal and external interfaces
Algorithms Design Document (ADD)	Design	Provides mathematical description of algorithms to be implemented
Engineering Support Log	Implementation	Documents all aspects of use during hardware development and subsequent laboratory utilization
Unit Development File (UDF)	Implementation	Documents software development process and decisions
Hardware Test Plan	Test	Describes hardware test approach
Hardware Test Procedures	Test	Describes how to perform tests
Hardware Test Report	Test	Documents test results
Software Test Plan (STP)	Test	Describes software test approach
Software Test Procedures	Test	Describes how to perform tests
Software Test Report	Test	Documents test results
Hardware User's Guide	Deployment	Describes how to operate and utilize hardware
Software User's Guide	Deployment	Describes how to install, initialize, and operate software
System User's Guide	Deployment	Describes how to communicate with and operate processing system



[DoD, 2010a] test method standard for microcircuits, or MIL-STD-202G [DoD, 2003] test methods for electronic and electrical parts. Finally, when we develop computer hardware for use in spacecraft applications we also review and comply with the applicable documentation that outlines the required environmental testing (MIL-STD-810G [DoD, 2008] test methods for determining the environmental effects on equipment) and power distribution and loading requirements (MIL-STD-461F [DoD, 2007] requirements for the control of electromagnetic interface characteristics of subsystems and equipment, [AIAA, 2007], Electrical Power Systems for Unmanned Spacecraft Standard).

To arrive at a baseline computer system, we:

1. Allocate mission and system requirements to functional on-board processing components
2. Partition and allocate the computational requirements to space or ground, payload or

spacecraft, individual subsystems, and to hardware or software

3. Define the computer system requirements, detailing the type of processing (e.g., mathematical, database) and associated data storage requirements
4. Develop computer system operational modes and states, based on the computer system requirements
5. Evaluate the internal and external interfaces of the processing system (analyze data flow and throughput requirements), while establishing the candidate architectures and topologies iteratively
6. Select a baseline architecture
7. Form a baseline computer system specification from the architecture, modes and states, and system level requirements and create the appropriate documentation
8. Establish an implementation approach with associated processes
9. Determine a product deployment and operational use strategy

The first six steps are associated with requirements definition and typically occur before the System Requirements Review (SRR). These general activities are discussed in more detail in Sec. 20.1. We usually complete steps 7 through 9 by the Preliminary Design Review (PDR) although iterations on the products from the first six steps are shown again as part of the PDR. These preliminary design activities are discussed in more detail in Sec. 20.2. Revisiting and iterating between steps occurs frequently because requirements are often contradictory or not complete, especially in the early design stages. Requirements can also be unreasonable or too narrow in scope. For example, if we determine that the star tracker must have a specified level of accuracy on its own to meet pointing or mapping requirements, without synergy that might come from other available attitude determination sensors such as Earth sensor or gyro; we might over-specify the star tracker. By using an iterative process, we can refine contradictory computer requirements and question the validity of others. An assumption made for one subsystem to reduce its complexity may increase the complexity and cost for another subsystem dramatically. Often, a compromise solution is needed and can be achieved by using an iterative design process between the various disciplines.

20.1.1 Requirements Definition

When designing computer systems for space applications, we want to optimize the availability, capability, flexibility, and reliability of the system while minimizing cost and risk, yet meeting size, weight, and power constraints. Additionally, the reduction of schedule time, from mission need to space vehicle implementation, may also be important. This leads to the use of the phrase "faster, better, cheaper" and current experience shows that achieving two out of the three will result in the optimal solution. Often achieving one is a lofty goal. Our objective is to meet the system and mission requirements, whether the resulting system is implemented on the ground, in space, or distributed between the two. With the onset of more powerful space qualified computer systems, we can implement more of the requirements in a space-based system which makes the spacecraft more autonomous and often reduces the volume of data that is transmitted to the ground station. Because there is always a desire for more data to be downloaded, we use data compression techniques to reduce the size of the downloaded data. However, the compressed data must be "re-created" on the ground so we need to assure that time critical functions can be completed with minimal delay whether on the spacecraft or on the ground.

As mission objectives evolve, we blend complex hardware and software to meet them and we continue to evaluate the location of their implementation. As we increase complexity in the systems, particularly the space based systems, there is an exponential increase in the associated testing that is required because when we implement requirements in space based assets we must be certain that they have been implemented accurately. In particular, the portion of the on-board system that is responsible for preservation of the spacecraft must be guaranteed to work. Thus, we make the critical implementation as simple as possible and assure that if a failure occurs, the system will fail safe. In general, we strive to keep the on-board processing systems simple, implementing requirements at the lowest level, then building up the capabilities needed to meet evolving mission requirements. Additionally, the size, weight, and power consumption of the on-board assets is often closely scrutinized and thus we must continue to make trade-offs between space based implementations and ground based implementation. It is important that we maintain margin in space-based assets so that we can use the resources to enhance system capability post launch or fix problems that occur on orbit. We often can mitigate problems such as subsystem failures or malfunctions by changing their usage in the flight software implementation. This has some inherent risks and so we must make these enhancements carefully.

The primary design drivers used to measure our success in optimizing the computer system design are shown in Table 20-4. Mission requirements, shown on the left, typically dictate the system-level drivers, shown in the next column. These flow down to the subsystems where we establish driving requirements for computation assets. Finally, logistics support personnel set down the requirements which we feed back against the computer and system level drivers, helping to manage the overall design process. We weigh each of the design drivers based on mission objectives and constraints. Again, this iterative process requires multi-discipline participation and often crosses traditional subsystem and organizational boundaries.

Table 20-3◀, Fig. 20-2◀, Eq. 20-0

Table 20-4. Design Drivers for Computer Systems. These are factors that we evaluate throughout the design process. When flowing down mission requirements, including system level processing requirements, we must be careful to design hardware and software with the "ilities" in the fourth column in mind.

Mission Requirements	System Level Processing Requirements	Computer Level Requirements	Additional Requirements "ilities"
• Customer Needs • Expected Availability – Weeks – Months – Year or More • Number of Satellites • Number and Location of Ground Stations • Level of Autonomy • Security Requirements • Programmatic Issues – Cost – Schedule – Risk	• Functional Capabilities • Processing Partitioning – Payload vs. Spacecraft – Onboard vs. Ground • Physical Characteristics – Size – Weight – Power – Radiation • Communication Protocol – Commercial Digital Standards – Commercial Analog Standards – Protection / Encryption	• Throughput • Memory • Radiation Hardness • Development Tools • COTS Software availability • Emulator / Engineering Model availability	• Testability • Feasibility • Usability • Reusability • Reliability • Flexibility • Maintainability • Interchangeability • Replaceability

As with all systems and subsystems, poor computer system requirements definition results in an inferior product and erroneous requirements are very expensive to correct. Requirements volatility is often the primary cause of program development failures. Thus, requirements have high leverage, meaning that a small improvement early in the process helps to avoid many problems later. Next, in the process of defining top level requirements, we overlay the mission timeline and mission modes onto the mission needs and requirements. For example, we need to evaluate when the mission must be executed, at what frequency or interval the spacecraft functions are supposed to occur, where, when, and how the mission data will be utilized, and how the system will be deployed. Some of these issues are discussed in Chaps. 3, 4, and 5. It is with all of this information that a mission timeline is developed as described in Sec. 4.2.3 and simplified for illustration in Fig. 20-3. For spacecraft, one key element of the timeline will be the availability of a launch system. Once a potential launch system is identified, we add the launch system requirements (environmental, location, time frame, and data interfaces) into the mix of mission requirements and needs. Flowing mission needs into system and subsystem requirements is difficult, subjective, and time consuming. One approach to doing this is to study a set of questions, such as those shown in Table 20-5, which will motivate needed trade studies and identify areas of negotiation for the overall requirement development process.

20.1.2 Requirements Flow-Down

The requirements that are documented in the System Segment Specification are split between the hardware and software implementation. We document the requirements for software, in terms of processing tasks and system interfaces, in Software Requirements Specifications (SRS) and

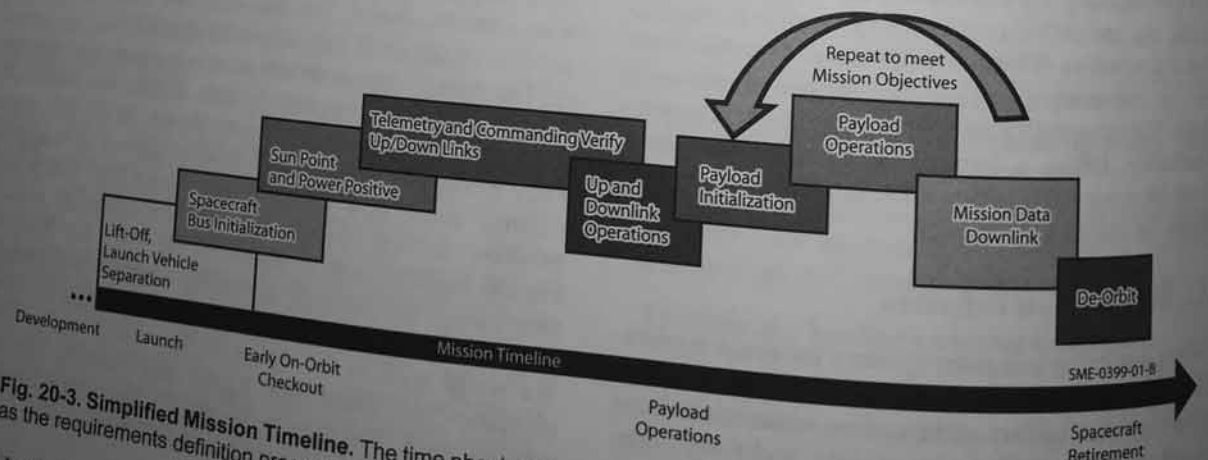


Fig. 20-3. Simplified Mission Timeline. The time phasing of a mission helps to identify top level mission modes which evolve as the requirements definition process matures and all of the on-board processing requirements are established.

Table 20-4, Fig. 20-3, Eq. 20-0

Table 20-5. An Approach to System Requirements Definition. General questions which we ask in all aspects of life can be directly applied to computer system requirements derivation by evaluating the specific parameters listed below. (R. Holmes, S. Jacobs, and R. Lane of TRW.)

Questions to Ask	Parameters to Review
What Must The System Do?	Evaluate and establish basic functional requirements.
Why Must It Be Done?	Establish traceability from functions to mission objectives. Be sure to challenge the requirements and assess their validity.
How Can We Achieve It and What are The Alternatives?	Evaluate candidate architectures including network topology, data flow, control flow, and UML diagrams to understand the implications of various approaches on the final design. Investigate emerging technologies and perform trade studies as needed to assure a robust solution. A combination of traditional architectures is an acceptable "hybrid" solution.
What Functions Can We Allocate to Which Parts of the System?	Perform functional partitioning to development block diagrams of ground station and vehicle, as well as spacecraft and payload.
Are all Functions Technically Feasible?	Determine if the value of state-of-the-art technology outweighs the risk. Evaluate the potential use of off-the-shelf hardware and software. Look for data flow bottlenecks and reallocate functions to evenly distribute the data flow. Review baseline block diagrams for potential holes.
Is the System Testable?	Develop nonintrusive test plan and procedures that ensure the system will meet mission objectives. Are test points available outside the system for easy "black-box" testing.

Interface Requirements Specifications (IRS). We define the Computer Software Components (CSCs) and the lower level Computer Software Units (CSUs) which are integrated together to create the Computer Software Configuration Items (CSCIs). We document the hardware requirements in the Hardware Requirements Specification (HRS) and establish the necessary Interface Control Document (ICD) based on the system topology and network specification. The detailed design for hardware includes Printed Circuit Board (PCB) layout, the chips and components that are used to populate the PCB, the power subsystem, I/O transceivers, and the enclosure interconnect structure and form factor. We use traceability analysis to make sure the requirements are complete as depicted in Figure 20-5. We must trace the computer system's requirements to parent requirements, which come from various sources. For example, we may derive computer system requirements from top-level requirements, operational concepts, or launch-vehicle interfaces. We must also trace the flow of requirements to the components to reduce excessive capabilities. Whenever a top-level requirement changes, good traceability allows us to examine the effect of this change on lower-level requirements and how we will meet the new requirements. Often, we re-evaluate a top level change in requirements when we see its effects at the lower level implementation. In any case, traceability allows us to identify all areas where we must evaluate the design to incorporate changes. If we flow the requirements properly during conceptual design, we can accurately run tests at each level during development.

We use the baseline requirements that are generated at the top level, to flow down requirements to specific subsystems and implementations. To define requirements for a computer system, it is convenient to develop a system *state diagram*. The state diagram shows valid states of the system (such as "off" or "initialized") and the conditions required to achieve them, based on mission requirements as they overlay on mission modes via the mission timeline. For each mission mode, such as sun point, safe mode, or navigate, we might want to construct different state diagrams. The computer system states and state transitions must be consistent with the allocated requirements and the mission concept of operation (See Chap. 4). Figure 20-4 is an example of a simple state diagram for a computer system, showing the general states and the source of their transitions. On and off are the obvious first choices for system states. Even when a system must be on at all times, we should consider having an off state to allow graceful degradation if the system were to fail or shut down for some reason. This is where we consider the fail-safe concept discussed earlier. Other states relate to what the system must do and include related state transitions or causes for change of state. For a specific mission and mission requirements we might have several substates in place of one state shown in Fig. 20-5. Or we might not have a state shown in the figure if it's not applicable to our mission. For example, several fail-safe conditions will be associated with the error contingency state shown. However, we may implement the contingency concepts differently for each specific spacecraft based on mission requirements and mission phase.

When developing the state diagram for a spacecraft computer system, we must keep in mind implications for the ground system. Complex state transitions influence ground station software that deals with the spacecraft's limitations and constraints. Other domain experts that help define the spacecraft and the engineers that specify the ground station are also needed to review the spacecraft computer system state diagram to assure that the complexity and risk of the resulting implementation is balanced between the various subsystems or domains, and whether the requirements are implemented on the spacecraft or as part of a ground-based system.

Functional partitioning is a structured methodology which begins with decomposing requirements into their

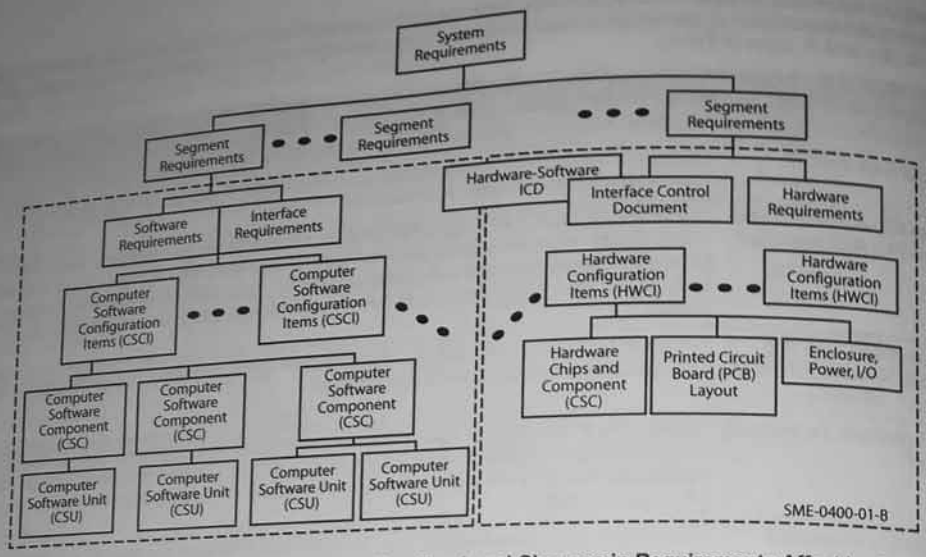


Fig. 20-4. Traceability Analysis Helps Understand How Top Level Changes in Requirements Affect Implementation.

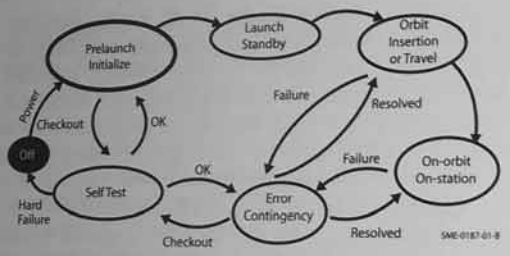


Fig. 20-5. Typical State Transition Diagram for an On-board Computer System. The state diagram shows the valid states of the system and the conditions needed to achieve each state.

lowest functional component and ends in the creation of multiple candidate architectures. This method allows us to group similar functions in subsystem definitions without unnecessary influence from traditional subsystem or organizational boundaries. The processing for a spacecraft system is usually partitioned between various processors in the vehicle and on the ground. The top-level considerations determine where the processing will be performed to meet the system performance requirements (both technical and programmatic) Table 20-6. This allocation of processing or functional partitioning is performed after the major processes have been determined and estimates of the processing timelines, or at least the time dependencies, are available.

If the initial analyses call for on-board processing, we further partition functions between self-contained components, a silicon based implementation, firmware, or software. Self-contained components perform a complete function with the implementation being transparent to the end user. Silicon based implementation includes field programmable gate arrays (FPGAs) where specific functional elements are committed to hardware via a single approach. *Firmware* is the software which resides permanently in non-volatile memory but is not intrinsically part of the computer itself. Silicon implementations and firmware both reduce the susceptibility to upset, but we cannot modify the implementation after launch. Certain elements of the system are clearly hardware: space-qualified computers and processors, the data bus, and so on. Software is often used for state transitions via evaluation of state data, and processing complex mathematical algorithms for attitude and orbit determination and control. Software typically executes out of random access memory (RAM) where read/write functions can occur nearly an infinite number of times. We store data elements that are generated as part of the processing in either Dynamic RAM (DRAM) or Static RAM (SRAM) depending on whether the data is changing frequently or relatively static. DRAM requires periodic refreshing due to the implementation approach while SRAM uses bistable latching circuitry to assure the data is electrically maintained. However, all RAM is only maintained when power is applied, and because RAM is susceptible to single-event upsets (Chap. 7), firmware is often the answer for critical processes such as initialization or contingency operations. Firmware often executes out of read only memory (ROM) or programmable ROM (PROM) where we can write a finite number of times. Direct memory access (DMA) allows access of the memory without utilizing the central processing unit (CPU) which makes reading and writing to memory faster and does not impact other ongoing processing tasks. This is particularly important for command and data handling (C&DH) functions or payloads that produce large quantities of data that need to be stored for upcoming downlink opportunities.

We next evaluate the *Instruction Set Architecture* (ISA). This is the machine code format used by a specific processor, such as the Intel X86 family of processors, 68040, RAD6000, RAD7000, RH3000, and MIL-STD-1750A [DoD, 2010b]. The ISA defines

Table 20-5◀, Fig. 20-5, Eq. 20-0

Table 20-6. Functional Partitioning Requirements for Spacecraft Computer Processing. We partition functions in a general sense using mission timelines as the starting point. We must partition functions to various processors and subsystems so that each element of the system maintains an acceptable level of complexity. A large increase in the complexity of any element will greatly impact our test requirements.

Perform Processing in Space	Perform Processing on Ground
• When processing delays would be intolerable • When data can be fused or reduced to provide a more coherent solution to the ground when downlink bandwidth is limited	• When human interaction with processing is necessary • When the downlink bandwidth from vehicle to ground is satisfactory
Perform Processing in Hardware	**Perform Processing in Software**
• When very high performance is needed • When processing is highly mathematical • When well-defined, inexpensive hardware for the process is available • When a function will be required for multiple vehicles and a cost-effective hardware component can be created	• When processing complexity exceeds that available in hardware • When changes in processing need to be made after hardware is acquired • When expensive, custom hardware can be replaced by software • When there is considerable unused computer capacity
Allocate Processing Between Spacecraft Bus and Payloads	**Do Not Allocate Processing Between Spacecraft Bus and Payloads**
• When payload processing is distinctly different from spacecraft bus processing or developed by completely different organizations • When payload performance accountability is critical	• When payload processing is minimal • When synergy exists between standard bus process and payload processing
Allocate Processing Along Organizational Lines	**Do Not Allocate Processing Along Organizational Lines**
• When there are geographical, contractual, or other impediments to effective inter-organizational communication • When there are standard subsystems and accurate interface control documents which are typically managed within a defined organization	• When the project is small enough that there is a single organization with strong top-down authority • When subsystems are so complex that specific disciplines and experienced organization personnel are required

the software developer's interface to the processor at the lowest level. To evaluate hardware architectures, we examine instruction sets, recognizing advantages and disadvantages of the two basic types: *general-purpose* and *custom ISA*. General purpose ISA's support all kinds of processing but with only moderate performance. They allow us to modify algorithms more easily, but they slow down processing because they are not designed for a specific algorithm. Custom ISAs, such as *Digital Signal Processors* (DSPs) or *Application Specific Integrated Circuits* (ASICs) support specific algorithms or classes of functions very well but often require additional engineering support to optimize the performance of the software functions in the hardware implementation. We can then define our computer systems as a "black box" that we load flight software into, or we can build up our processing capability using single board computers (SBCs) or microcontrollers that are resident in a single enclosure or distributed throughout the spacecraft. For larger spacecraft, with extensive processing requirements and long on-orbit life time requirements, we typically use an integrated flight computer, while smaller spacecraft, such as CubeSats, as well as those with limited lifetime requirements of 5 or less years, we often use stand alone SBCs and microcontrollers.

Additionally, we look at computer system attributes such as error detection and correction (EDAC) mechanisms, network protocol, data up and down link capabilities, to determine if there is a viable hardware implementation or if the feature must be implemented in software. We typically prefer hardware implementations of these lower level capabilities as they operate more efficiently. The term EDAC can include errors in memory due to environmental issues (i.e., Single Event Upsets, or SEUs) or errors in processing due primarily to undiscovered software features. Memory errors often occur at the bit level due to cosmic rays or alpha particle emissions. It is important that we check each memory location periodically for errors which we call *memory scrubbing*. Memory scrubbing is often performed as a background function but should not require more than 5–10% of the overall processing capability at any time. We can find errors in processing using a software error handling utility that understands typical processing errors, such as divide by zero, and performs a work-around to assure that the error does not halt the processing unnecessarily. Finally, we can implement a *watch-dog timer*, in either hardware of software, which causes a hard reset (re-boot) of the computer or other corrective action, if it is not serviced at a predetermined frequency. Thus, if there is an error in processing that causes the software to hang, the watch-dog is not serviced and the system will reset.

20.1.3 Architecture Evaluation

An *architecture* is a framework for developing a system that will guide the development of that system, but at a gross level. We mold the architecture to meet mission requirements and operational needs, creating a baseline system. The architecture shows us the system's parts and how they interact, often through a block diagram. The computer system architecture shows the system structure (topology), the data network and system inter-

actions (data architecture and protocol), the hardware, and software architectures. Architecture studies for computer systems must also address the data, hardware, and software architectures. *Topology* is the form taken by interconnecting the various subsystems and components. *Data architecture* addresses the physical structure of the data network or bus, as well as the protocol or logical interaction across the bus. A *protocol* is a set of rules for sending data between computers, or between computers and peripherals. The *hardware architecture* defines the *instruction set architecture* (ISA) and the functional elements that are available in hardware. Finally, the *software architecture* defines how the processing instructions execute. Software processing can function as a single thread, executing from top to bottom, repetitively, or as scheduled modules, where processing order is based on major and minor frames organized by system states. Alternatively, software processing can be event driven, where interrupt service routines preempt normal execution in a deterministic way when hardware interrupts occur, or as multi-thread where varying functions execute concurrently within a processor.

After partitioning functions, performing trades, creating state diagrams with state transitions, evaluating the data architectures, and analyzing data flow, we can develop a block diagram for our specific computer system implementation. The system *block diagram* illustrates how we implement an architecture, showing types and numbers of processors and networks, including topology and protocol when reasonable. It provides a point of departure for developing more detailed software, hardware, and interface requirements. An initial block diagram, such as the one shown in Fig. 20-6, shows the basic topology of the system indicating the type of subsystems that we might need and the fact that each will interact with the central processor. Figure 20-6 implies a point-to-point connection between each of the subsystems and the central processor.

We next analyze the flow of data to determine how to manage interfaces between components. We want clean, simple interfaces—a data path is inefficient and slows down the flow if it calls for data to pass through a component without being examined or used. We might have the data flow in a circular path, known as a ring configuration, which is shown in Fig. 20-7. We might incorporate a multi-drop data bus, such as the ones shown in Fig. 20-8, with either a dedicated processing resource or distributed processing resources. This data bus topology allows point to point data transmissions across a common network with interconnections to the various components integrated into the bus structure.

Once we have considered various topologies and their applicability to our system requirements, we create a baseline system architecture. The architecture will highlight the topology and the necessary subsystems that we need to meet the system objectives. This block diagram drawing, in conjunction with the state and state transition diagram will provide the foundation for the baseline architecture. As a baseline we typically select a hybrid

Table 20-6◀, Fig. 20-7, Eq. 20-0

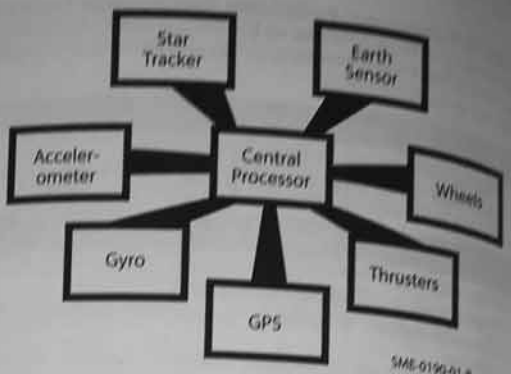


Fig. 20-6. Initial Block Diagram Showing a Centralized Topology. This drawing depicts a *Centralized Topology* which has point-to-point interfaces between processing units and a single management computer, or central node, or hub. This is a highly reliable topology that works best with a few, well-defined subsystems. The negative attributes of the topology include large and distributed wiring harnesses as well as the need for new wiring and interface software if a new component is desired.

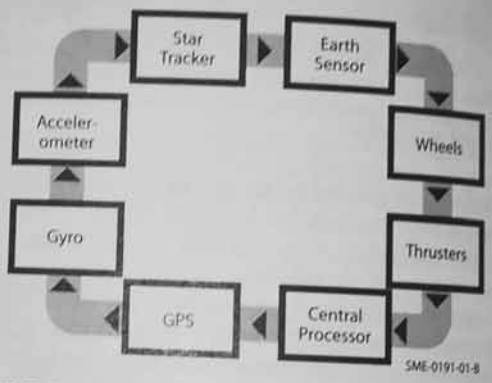
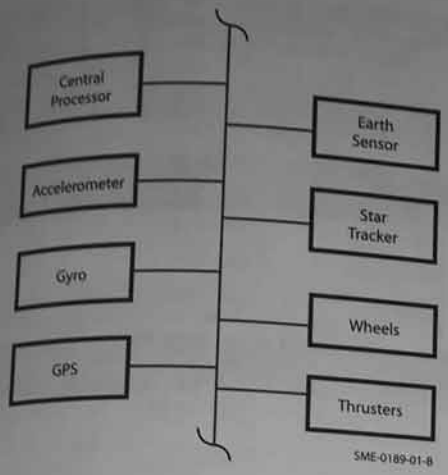


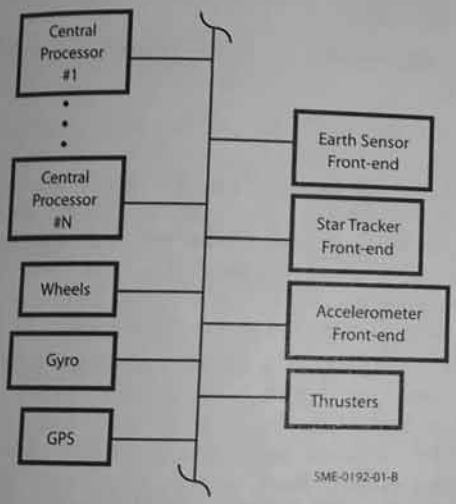
Fig. 20-7. Block Diagram Showing Ring Topology. This drawing depicts a *Ring Topology* which allows data to traverse the ring, providing identical information to multiple nodes. In the configuration the wiring harnesses are relatively small and can be distributed about the spacecraft. However, the negative attributes of the ring topology include the fact that each data transmission is dependent on the previous node to reach its destination.

system architecture, where we combine the positive attributes of several of the topologies while eliminating or reducing the risk associated with the negative attributes. Iteratively, we now evaluate the impact of mission requirements on the baseline computer system. Information regarding the selected orbit, expected period of operation, and any high performance requirements such as large field of view with continuous coverage or tight pointing and mapping criteria will affect the fidelity of the hardware and software needed for the specific mission.

Next we look at specific hardware implementation for the processing resources identified in the system block diagram. Several mission parameters drive the hardware selection. For example, the orbit we select will define the radiation environment. When we increase the required level of fidelity or include a requirement for autonomous operations, we often require a more capable computer



Federated Bus Block Diagram



Distributed Bus Block Diagram

Fig. 20-8. Block Diagram Showing Bus Network Topology. A *federated bus architecture* uses a common data bus with all processors sharing the bus. This encourages the use of standard protocols and communication schemes for all nodes and results in deterministic message transfer across the network. A *distributed architecture* uses multiple "like" processors to execute all software on an as-needed basis. This results in a highly reliable system that can implement fault tolerance and redundancy with limited impact to physical constraints such as size, weight, and power.

system with additional redundancy. Either of these conditions can develop if we have to perform rapid transitions between differing orbits, or travel to distant targets. These mission requirements may impact software development. More complex requirements lead to more complex software implementation, which requires a more robust design and more test cases to accomplish a desired level of preflight validation. Inadequate requirements definition may cause cost and schedule risk for both hardware and software, as modifications and last minute changes may be required.

When the mission environment is not as demanding or multiple copies of a satellite will perform the same tasks, we can select computer hardware for the mission based on less stringent environmental consideration. Often we can use commercial rather than space-qualified parts. Additionally, as the possibility for achieving unknown states goes down, we may reduce the software complexity and the level of pre-flight testing. As we increase the expected mission life or mission criticality, we should specify the use of more robust hardware, specifically space-qualified components and systems. We also increase the risk that software will operate in a manner that was not predetermined, and thus we should do more thorough pre-flight testing and system validation.

When we impose high performance requirements on the spacecraft computer systems we also increase the performance requirements of both the hardware and the software. High-data rate payloads such as imaging devices or communication subsystems impose the need for higher bandwidth data busses and often increase the processing performance requirements. When we require tight attitude knowledge and control, we not only impose a requirement for increased accuracy on the sensors, we also increase the computational complexity of the software, which in turn affects the processing performance requirements.

20.2 Preliminary Design

During the preliminary design phase of the software development process, we add detail to the baseline architecture by developing the SRS, IRS, STP, and SDP. Using these requirements specifications we estimate the size of the software to be developed as well the processing throughput. From these estimates and an understanding of the overall project, we can select a target computer system for implementation of the embedded flight software. We also establish a software development approach, documented in the SDP, that includes the level of automation and reuse that will be used in the development, the software development environment including cross compiler if necessary, other types of tools that will be utilized, and the build process that will be executed, as well as the product deployment and *concept of operations* (ConOps) that will be employed. The general test philosophy that will be implemented and the type of testing that will be performed are documented in the STP.

20.2.1 Defining Processing Tasks

While establishing requirements for the spacecraft, we define processing tasks by classifying what the spacecraft must do. It is important that we identify all the potential processing tasks early, including adequate margin, to eliminate or accommodate surprises later in the development cycle. Software for on-board processing falls into five principal classes.

Operating system (OS) software directly manages computer resources and controls their allocation to spacecraft and mission tasks. This includes basic

executive functions such as scheduling tasks for execution, time management, interrupt handlers, input/output device handlers and managing other peripheral drivers, carrying out diagnostics and built-in tests, and memory fault management. All computer systems must manage these processes. We often consider software for the operating system as overhead to application software.

System management software includes such items as fault detection and correction, long duration event schedulers (such as reconfiguring the power system during eclipses), and payload system management. Software in this class manages control flow and is therefore logic intensive. Simple instruction sets are sufficient for this class because it requires few floating-point computations.

Control system software, such as attitude or orbit determination and control, requires an input stimulus and responds by changing the state of the system. This software is often mathematically intensive—requiring high accuracy and strict timeliness.

Command and data handling (C&DH) software typically collects state of health (SoH) data for downlink to the ground station, is also manipulates and compacts large quantities of data as they are collected and readied for transmission. This function often demands special computer architectures, such as signal processors, as well as large storage capacity for collected data.

Payload management software includes functions such as monitoring the payload status, managing multiple payloads, and interfacing with the payload to mode it and to assure the spacecraft is providing adequate support (i.e., pointing), as well as collecting data, preparing it for downlink to the ground station, and potentially compacting large quantities of data as they are collected. This function may be integrated with the System management software and the C&DH software.

As we refine the list of processing tasks to be developed, we need to decide if we can use existing software components or must develop new ones. When using existing software components it is important to understand its pedigree and have continued contact with the original developer as well as any other organizations or projects that are reusing the component. In general, we want to use established, proven algorithms because new ones involve technical risk, added costs, and a longer implementation schedule. However, if much tailoring is needed to make the algorithm fit to need, it may be better to start fresh so that risk associated with unknown attributes can be avoided. Early emphasis on risk assessment and reduction is part of defining processing tasks.

There are many different types of operating systems that can be implemented for a spacecraft computer system. They are often characterized by proprietary versus non-proprietary, free versus cost, as well as footprint and overhead when compared with the application software modules. When looking for a spacecraft implementation, the most likely candidates are those that have a moderate cost and are highly reliable with a moderate footprint and many associated tools or those that are free, with a small footprint and open source, shareware type tools. These include the operating systems listed in Table 20-7, along with many others that are included in the Wikipedia listing.

Table 20-7. Operating Systems for Spacecraft Computer Processing. When selecting an operating system for a spacecraft application it is important to research the reliability of the product, the footprint or size consumed in memory, the throughput or overhead added to the processing load, and if the product is free or requires licensing as well as the development tools that are associated with the operating system.

For Cost	Supplier	Operating System
	Wind River	VxWorks
	Green Hills	Integrity
	QNX	QNX Posix microkernel OS; usually a real time embedded OS
For Free	**Supplier**	**Operating System**
	GNU (free software foundation)	UNIX based
	Linux	UNIX based
	BSD (Berkley Software Distribution)	UNIX based
	OAR Corporation	RTEMS (real-time executive for Multi-processor systems)

Run-time kernel software normally supports higher-order languages. For example, it may represent, store, optimize, and pack data; drive input or output; handle exceptions or errors; and interact or interface with other programs, other devices, or even other mixed-language programs. The *I/O handler* controls data movement to and from the processor, as well as packing data for any specific interface. Likewise, the *device handler* or device driver software manages interfaces and data between the processor and any peripheral devices.

Utilities are software routines which several functions use. For example, different components of application software in a single processor might access a set of mathematical operations called *math utilities*. Their size and complexity vary directly with the application and its mathematical requirements. If the processor provides such utilities implemented in hardware, they are referred to as *built-in functions* (BIFs), and the hardware specification will often define these functional elements.

Built-in test software provides initial, periodic, or continuous testing for computer elements under the control of software or firmware. Diagnostic software not only identifies faults or failures but also isolates them. We can make it sophisticated enough to recover from some of them. For built-in testing and diagnostics, we can write the software as firmware. The computer vendor

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3. Define the real-ti basic elements.
4. Estimate the sou needed for each A. **Similarity:** Fin characteristics the known fun size directly. A development l order languag used.
B. **Bottoms-Up:** the lowest lev one time only executable ar Determine th
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Table 20-8. Software Estimation Process. The *estimation-by-similarity* technique uses existing, well characterized functions and their relationship to functions under development to estimate processor memory and throughput needs for the new functions. The *bottoms-up estimation* process forces the estimator to break the functions into the smallest components, which are then evaluated based on experience.

Step	Notes
1. List all application functions allocated to the given computer.	Document any assumptions.
2. Break down the functions from step 1 into basic elements.	This often requires several iterative steps. We should continue to break down the functions until we have reached the lowest level we can identify. Group functions back into utilities if they are used by multiple states or modes and unique software elements if they are only used in one instance.
3. Define the real-time execution frequency for each of the basic elements.	We only need to perform this step for those functions which are time critical. Execution frequency is not required for utility or background functions.
4. Estimate the source lines of code (SLOC) and memory needed for each function by: A. **Similarity:** Find a function with similar processing characteristics and known size. Compare the complexity of the known function with the new one and adjust the code size directly. Adjust the code estimate for differences in development language, such as assembly versus higher order language. Determine the SLOC from the memory used.	**Similarity**—Requires knowledge of both the existing functions and the new functions in terms of specific requirements, complexity, and general implementation. Some rules of thumb are: • A 25% increase in complexity implies a 25% increase in code • If the known function is in assembly code, increase the code size by 25% for a higher order language. • If the known function is in a higher order language, decrease the code size by 20% for assembly code.
B. **Bottoms-Up:** Identify the SLOC for executable elements of the lowest level functions as well as the data structures and one time only initialization software. Sum the SLOC for all executable and nonexecutable functions separately. Determine the memory requirements, based on SLOC.	**Bottoms-Up**—Requires knowledge of general elements of each function and how to implement the capability.
5. Estimate throughput requirements based on: A. **Similarity:** Find a function with similar processing characteristics and known throughput requirements. Compare the complexity of the known function with the new one and adjust the throughput value for differences in complexity directly. Based on the frequency of execution from step 3 above, compute the total throughput requirements for the new function.	Throughput should be expressed as instructions per second.
B. **Bottoms-Up:** Based on an average number of computer instructions executed per SLOC for a specific compiler and processor, multiply the number of instructions by the execution frequency. Define a "loop factor" for executable SLOC to represent internal loops within the function.	The internal loop factor for an ACS function associated with a three axis stabilized spacecraft will be 3—each function will be repeated for each axis. In order to estimate throughput from SLOC, the lines of code related to initialization and data must be separated from the executable lines of code. The loop factor is applied only to the executable SLOC.
6. Determine the operating system and overhead requirements by similarity to other implementations. Compare the complexity of the known operating system and overhead functions to the new and adjust accordingly.	You must identify all application code prior to this step. Complexity for operating system functions is based on number of tasks scheduled per second, the number of interrupts to be handled, and the amount of I/O data.
7. Determine the margins for growth and on-orbit spare based on where you are in the development cycle and Fig. 20–10 that outlines the margin at each milestone.	Growth and spare requirements are important and should be strictly calculated.

may even supply them. If not, we would have to decide whether we need this added reliability, despite the processing and cost overhead.

We measure software size by words of memory, and processing time by *throughput*, usually expressed in thousands or millions of instructions per second (*KIPS* or *MIPS*, respectively). We estimate the size and throughput of on-board software for several reasons. When we begin defining a computer system, we use the software estimates in conjunction with requirements for spare processing to determine how much computing power we need to perform the mission. During system development, we revise the estimates to make sure hardware capacity is not exceeded. We can also use software size estimates to estimate cost.

Processor throughput is a function of the instruction set and the clock speed. With only one instruction, a computer's throughput is proportional to clock speed. If it has two instructions, one (A) requiring two clock cycles and the other (B) requiring seven cycles, the computer's throughput also depends on the instruction mix. The *instruction mix* is the proportion in which the software uses the instructions. For example, if the software is 60% type A instructions and 40% type B, the throughput available with a 10 MHz clock is $(10/[(0.6 \times 2) + (0.4 \times 7)]) = 2.5$ MIPS. If the mix is reversed, the throughput avail-

able is 2.0 MIPS. Table 20-8 shows the basic steps used to estimate the software processing throughput. Both an estimation based on similarity and a bottoms-up evaluation of each element will yield a potential throughput estimate. These two approaches are often used as they bound the problem, with the similarity technique leading to a smaller throughput estimate than the bottoms-up approach.

System Requirements Review is the milestone when we formally identify computer resource requirements. A good rule of thumb is to set the amount of computer memory and throughput at the System Requirements Review at four times the estimate of what has been estimated to date for software size and throughput. Empirically, initial software size and throughput estimates double from this review to launch because early requirements are uncertain, and changes in software are easier to make than changes in hardware during late stages of spacecraft development. We also want spare memory and throughput at launch to correct anomalies or to increase performance after system calibration. Thus, we need to establish reserve capacity when initially defining requirements. A reasonable value for post-launch reserve is 100% spare (equivalent to 50% of machine capacity).

Most on-board flight software is executed under a *real-time operating system (RTOS)* which provides the timing and system services for the processing resource. The size and functionality of the RTOS should be tailored for each mission, based on mission needs. The RTOS software is typically quite small (<1K SLOCs) and is not included in the estimate of source lines of code (SLOCs) for costing purposes. This is primarily because we typically purchase the RTOS and then tailor the run-time version of the software based on the system services used by the specific software implementation. RTOS procurement usually includes the tools needed to create the embedded flight software and there is often a small fee ($100–$1,000) for each time the RTOS is embedded in a processor and flown. Table 20-9 lists general categories of application software and estimates for size and execution frequency. In some cases, the actual code size may not be significant, however, the size of the storage required may be more extensive. For example, with command processing, each of the potential ground commands must be stored with the associated command sequence for execution. In general, typical spacecraft of moderate size (~450 kg) will contain less than 100K SLOC in total. Not only do the number of SLOCs drive the cost but the amount and type of testing that we require is based on the SLOCs as well. The more complex we make the vehicle processing, the more we must test to be sure all possible paths within the software have been tested and the software not only does what we require, but also does not do anything that is undesirable.

An *Executive* is often used to manage states and state transitions, which are developed based on the mission timeline and the state diagram and make the embedded flight software custom for the specific mission. This software is usually quite concise and often tailored from a previous version of state machine software. This element

Table 20-9, Fig. 20-8◀, Eq. 20-0

Table 20-9. Estimating Source Lines of Code (SLOC) for Typical Spacecraft Functions. These are SLOCs for typical spacecraft functions with nominal performance. As performance requirements increase, the complexity of processing often increases as well which ultimately leads to an increase in the SLOC estimates. NtH= Nice to Have

Computer Software Component (CSC)	Source Lines of Code (SLOC)	FireSat II
Executive	1,000	Req.
Communication		
Command Processing	1,000	Req.
Telemetry Processing (standard)	1,000	Req.
Telemetry Processing (with compression)	3,500	
Attitude / Orbit Sensor Processing		
Rate Gyro	800	Req.
Sun Sensor	500	NtH
Earth Sensor	1,200	
Magnetometer	200	Req.
Star Tracker (without star id)	400	
Star Tracker (output quaternion)	2,000	Req.
GPS (output position/velocity)	300	Req.
GPS (output with pseudo ranges)	2,500	
Attitude Determination and Control		
Kinematic Integration	2,000	Req.
Kalman Filter	6,000–10,000	
Error Determination	800	
Precession Control	3,500	
Ephemeris Propagation	2,000	Req.
Complex ephemeris	4,000	
Orbit Propagation (linear models)	3,000	
Orbit Propagation (complex environment)	8,500	Req.
Attitude Actuator Processing		
Magnetic Control (torque)	600	Req.
Thruster Control	600–1,200 (catalyst bed)	Req.
Reaction Wheel Control	1,200	Req.
CMG Control	1,500	
Fault Detection		
Monitor	4,000	Req.
Fault Identification	2,500	Req.
Fault Correction	5,000	
Utilities		
Basic mathematics	800	Req.
Transcendental mathematics	1,500	Req.
Matrix mathematics	1,200–2,300	Req.
Time Management and conversion	700	Req.
Coordinate conversion	1,500–3,500	Req.
Other Functions		
Momentum Management	3,000	Req.
Power Management	1,200	Req.
Thermal Control	800	Req.

of the FSW will not be a cost driver for the software development effort, but will add the structure needed to make testing more straight forward.

Communications software includes processing external commands and collecting internal data for transmission to an external source. The information in Table 20-9 assumes a modest number of ground commands (~100) and collection of data for telemetry to the ground. It does not include inter-processor commanding, but we can allow for this commanding by similarity of function. The actual code used to de-commutate commands is relatively straight forward but as the number of commands increases, the storage space required to maintain the command lists and associated command sequences will increase. Similarly, to pack information into digital data packets for distribution in the telemetry stream is a relatively small function. However, if extensive compression is used or if decisions are being made on-board the vehicle based on the data that is collected, then the processing complexity will increase and thus the SLOCs and the associated cost will increase as well.

Attitude-sensor software handles data from various sensors, compensates for sensor misalignments and biases, and transforms data from sensor to spacecraft body coordinates. Processing for gyros, accelerometers, Sun and Earth sensors, and magnetometers involves decoding and calibrating sensed data. Processing for star trackers involves identifying stars against a star catalog, which can require extensive data and memory resources. It is also possible that the star tracker will provide an attitude quaternion as the output which greatly reduces the I/O handling and associated processing but makes fault tolerance when the star tracker fails potentially more cumbersome.

The *Attitude Determination and Control* category covers various control methods that make use of the data processed by the attitude sensor software. In *kinematic integration*, we estimate current attitude by integrating sensed body rates using gyros. Using *error determination*, we find how far the spacecraft's orientation is from that desired. For spin-stabilized spacecraft, we maintain attitude control using precession control. The precession-control size and throughput numbers reflect those of a thruster-based system and account for the processing that determines how much torque is needed on the vehicle to achieve the desired attitude. The thruster control function listed in the table is for a 3-axis stabilized control system using thrusters. This element of the software is responsible for assigning the specific thrust to each thruster, based on the mounting location of the thruster and the total vehicle torque that is desired. Table 20-9 also lists control algorithms using reaction wheels, control moment gyros, and magnetic torquers. Also in this category are object ephemerides, which we can maintain using crude table look-ups and curve fits, or very complex algorithms with increasingly complex environmental models. The orbit propagator integrates the spacecraft's position and velocity information.

Fault detection is closely tied with autonomy. Monitors exist to identify failures or adverse conditions in on-board equipment and we typically down-link these failures to the ground for diagnosis and commanding. Size and throughput vary widely depending on the system. Processing for corrective actions usually depends on tables of pre-stored procedures and, therefore, requires considerable data but not many actual lines of code. The more autonomous the fault detection becomes, the more complex and thus larger the embedded flight software becomes.

Momentum management monitors the reaction wheels looking for saturation and then unloads or shed momentum using magnetic torquers or other non-saturating mechanisms for attitude control. *Power management* and *thermal control* are support functions that often reside in on-board computers. Through power management, the computer controls battery charge and discharge and monitors the power bus. Active thermal control involves monitoring and controlling temperatures throughout the spacecraft.

We should not attempt to use all of the available memory or throughput. Asynchronous processing, such as interrupt handlers, introduces a level of uncertainty in throughput. Costs also rise dramatically as we shoe-horn the software into existing memory [Boehm, 1981]. As a rule of thumb, we should use 70% or less of available throughput. After System Requirements Review, we continuously update estimates for software size and throughput as requirements solidify. We plot them as *reaction curves* to ensure we can detect whether the software is growing too much. Figure 20-9 shows a typical reaction curve for software development, measured as a percentage of maximum use. As long as the estimates fall below the curve, no extraordinary action is required. When they exceed it, we must pare down the requirements, relax the restriction on reserve capacity, or increase the resources available.

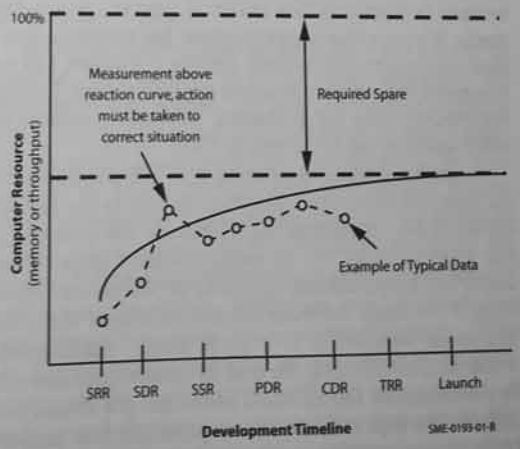


Fig. 20-9. Reaction Curve for Using Computer Resources. Reaction curves mitigate risk. They discipline our management of onboard computer resources. Whenever an estimate exceeds the reaction curve, we correct the situation.

20.2.2 Computer Selection Guidelines

After examining what the computer system must do, based on the software requirements, we assess the size and throughput requirements for the software. Based on this initial sizing estimate, we can determine our hardware resource requirements. We must find a computer system which meets all of our basic needs, as well as the spare allocation, and has the required support environment. Each computer considered must have suitable system software (operating system or kernel and built-in functions such as mathematics). Additionally, we must determine if radiation susceptibility is an issue for our mission. This is based on the mission life-time and the orbit selected for the mission. In almost all cases, processors should be purchased rather than developed to avoid paying the nonrecurring development costs, and incurring schedule risks. The alternative to purchasing a black box processor includes the integration of a single board computer (SBC) with other computer boards to create a custom processor box, or developing a custom microcontroller from Commercial off-the-shelf (COTS) parts such as an FPGA and various soft core ISA implementations.

Hardware processing architectures for spacecraft on-board processing have followed the commercial market, albeit with slower general acceptance due to the space-based environment. As commercial processing resources improve exponentially, so do space-based processing resources. While resources continue to increase, the basic functional elements of a typical spacecraft have remained relatively stagnant. Thus, we find that on-board processing resources are usually more than sufficient for baseline processing tasks. However, as the processing resources become more capable, we find ways to continue to enhance our processing performance by making the error checking more intricate or incorporating higher fidelity models on-board the vehicle. When pairing on-board processing resources with on-board processing requirements, it is important to have a quantitative mechanism for evaluating performance. We evaluate hardware architectures to determine their ability to meet our processing needs. We use a *benchmark program* that contains a specified instruction mix so that various computers can be compared to a standard measure of performance. Typical benchmarks used for evaluating computer resources in space applications are shown in Table 20-10.

When we select a benchmark it is important that we use one that has a similar instruction mix to the one we expect in our operational flight software. For example, if our flight software will be mathematically intensive we could use Khornerstone, Linpack, or Whetstone. However, if our flight software focuses on integer math, the Dhrystone benchmark is a good match. Additionally, when we are evaluating the type of mathematics that will be performed by the on-board processor, we should also consider the type of functions the processor is to support. If the processor will be performing significant commutation or multiplexing at the bit level (I/O handling and manipulation) we should find a suitable hardware ISA that supports this type of processing. If the primary purpose of the on-board processor will be to perform command and data handling (C&DH), we should consider a processor that supports bulk data moves such as *Direct Memory Access (DMA)* command or a block move command. This allows the hardware to traverse the data, moving larger portions of data without software intervention.

Table 20-10. Benchmark Programs Used to Evaluate Computer Performance. Listed below is a set of traditional benchmark programs highlighting their strength in measuring the performance of a computer system.

Benchmark	Measures
Dhrystone	A test designed primarily to measure a CPU's integer performance. An outcome of the Dhrystone test is the MIPS rating.
Khornerstone	A suite of 21 tests developed in 1987 by SRS/Workstation Laboratories to measure the overall performance of a computer's CPU, floating-point capabilities, and disk I/O.
Linpack	This test measures a CPU's floating-point performance. Results reported in mflops (millions of floating point ops per sec).
MIPS	Millions of instructions per second. Refers to a CPU's ability to process integer operations.
Whetstone	This test, written in Fortran in the late 1960s at the National Physical Lab in Whetstone, England, measures floating-point performance. An optimized compiler can improve Whetstone performance, thus making the test somewhat misleading.

In many cases, the processor performance for any candidate hardware component will far exceed the requirements for traditional spacecraft processing needs or the processing required for a single processor is so diverse that focusing on point solutions such as DMA is not suitable. Often in the case of smaller spacecraft, we even consider incorporating payload management as well as payload data processing, refinement, and/or compression functions into a single on-board processor. In this case, the trade off is to reduce size, weight and power (SWaP) by using a single, multi-purpose processor, rather than optimizing multiple processors' performance for various aspects of the spacecraft functionality.

If we determine that several hardware solutions can meet the performance needs we outlined in the software resource estimation process, a computer's heritage can be a major selection criterion. We often select a computer previously used in space by NASA, ESA, the DoD, or a major commercial space venture if we are supporting a long-life, high radiation environment, or national security mission. By starting with computers which have prior use in space, the major development, qualification testing, and documentation risks and costs will have been borne by the prior programs. However, while older space computers are often highly reliable, they are typically more expensive and less capable. Thus, if the mission has a reduced life-time (less than 3 to 5 years), will be in a low-earth orbit (LEO), and can be quickly and easily replaced if failed, COTS processors and SBC integrated boxes are often the better solution.

Table 20-11. COTS Processors Used with Either ADCS and C&DH Functional Focus. Listed below are several COTS processors and their associated mission, along with the focus of the embedded flight software and rate of execution for the ADCS which is typically the driving factor in terms of execution speed.

Year	Mission	Processor	ADCS Rate	% ADCS	% C&DH
1995	SWAS/ WIRE	386/387 @16Mhz	10 Hz	23%	55%
2000	Triana	Rad6000 @20 Mhz	10 Hz	28%	63%
2007	LRO	Rad750 @132 Mhz	5 Hz	1.7%	48%

Building an operational system depends on the development philosophy and environment we select. Procuring COTS hardware and software is the easiest way we can build a capability. Table 20-11 shows several COTS processors that were used in real missions along with the focus of the embedded flight software, either ADCS or C&DH. However, off-the-shelf items often don't meet our needs exactly. Thus, we must tailor the COTS products or adjust the need to match what is available. The break-even point is different for each program depending on the number and types of requirements, individual skills, our knowledge of the product, and whether the COTS products are maintainable and of high quality. If we opt for custom development, we must also consider the development and test tools we will need to create the operational unit. We may need special hardware, software, or integrated systems. For single-unit or unique systems, developing support tools can cost as much as developing the operational units. We must evaluate these support tools during conceptual design, and include their cost in the overall development cost. As with the cost of other developmental tools, we may be able to amortize these costs over multiple products or projects.

20.2.3 Software Development

Many issues associated with the development cycle affect conceptual design and long-term life-cycle cost. For example, we select hardware and software design concepts during requirements definition but we draw on implementation and development experience to do so. Because we often have to cost activities for developing a computer system before a complete design is available, we must take into account the many aspects of software and hardware development, testing, and integration. Software-based tools and standards such as MIL-STDs, IEEE and SAE Specification Guide, and ANSI or ISO standards help us structure our methods and give us more management and technical controls. The Software Engineering Institute (SEI) as well as the International Standards organization (ISO) have established a rating system for the software development process. Companies can apply to SEI for increasing levels of performance ratings and can become certified with ISO as being ISO compliant. The SEI also supports Capability Maturity Model Integration (CMMI) as process improvement approach that increases an organizations performance. According to the Software Engineering Institute, CMMI helps "integrate traditionally separate organizational functions, set process improvement goals and priorities, provide guidance for quality processes, and provide a point reference for appraising current processes."

In addition to the development process, long-term issues, such as life-cycle costs, require attention during conceptual design. The complete life-cycle costs include conceptual design, detailed design, implementation, system integration, test, and on orbit maintenance as well as the tools associated with each life-cycle phase. There are many software development life-cycle models we can use. However, the key for developing costs estimates for software development is to ensure that life-cycle issues such as launch and early check-out, on-orbit maintenance, and potential development, test, and uploading of new software components are taken into account.

Additionally, we must consider the availability of tools, their maturity and availability, based on the processing and processors selected. This includes such tools as cross-compilers, linkers, assemblers, and the host processors on which they reside. A *host machine* is the computer where the development activity is to take place. The *target machine* is the embedded microprocessor or ground-based computer where the code will perform throughout its lifetime. A *cross-compiler* executes on the host, compiling software for the target processor. For both assembly language and higher order languages, the loaders and debuggers allow us to store software in the target processor and evaluate its performance based on either symbolic or physical information. Sometimes the host computer and the target computer have the same ISA. In this case the compiler is not the issue but library and other functions' availability on the target should be explored. Table 20-12 summarizes the various life-cycle issues which we must address during the early phases of program development.

Modeling tools are another set of tools that need to be evaluated and considered as we create spacecraft onboard processing system. The *unified modeling language* (UML) for software intensive programs is used to specify, visualize, modify, construct and document the artifacts of an object-oriented software-intensive system under development while the system modeling language (SysML) is a general-purpose modeling language for systems engineering applications. It supports the specification, analysis, design, verification and validation of a broad range of systems and systems-of-systems. UML and SysML are implemented in many tools to combine techniques for architecting software and systems in terms of stakeholder use cases and requirements to diagram functional, performance, and interface requirements. Additionally, the tools provide a standardized mechanism for us to visualize the system implementation and maintain traceability to the requirements.

Table 20-12. Summary of Life-Cycle Cost Issues. Through the use of development and protocol standards, auto-code generation, and re-use of common software and hardware modules, as well as increasingly capable development tools we create a means of controlling and streamlining the costs associated with computer system development [Boehm, 1989].

	Standards	Automation and Re-Use	Development Tools
Example	• DoD Standards • IEEE Standards • ANSI Standards • AIAA Standards • SAE Standards • ISO Standards • MIL-Specification • Data Item Description (DID) • Structured Development • Object Oriented Development	• Commercial Off-the Shelf (COTS) Software • Computer Aided Software Engineering (CASE) • Automated Code Generation	• Compiler, Assembler, and Linker • Cross Compiler • Loader and Debugger • Code Analyzers and Optimizers • Test Case Generator • Code Management Software • Simulators and Emulators
Benefits	• Adds structure to development activities • Standardizes documentation between and among programs	• Reduces development time which implies lower costs	• Allows for more generic software to be developed and then compiled specifically for target hardware • Aids in configuration management • Simulation and emulation increase testing efficiency effectiveness
Notes	• Tailoring is critical • Must balance required amount of documentation with associated costs	• Documentation must accurately reflect the implementation • MUST continue to test at the same level for all sources of software	• Be sure that a development environment is available for the hardware selected for the project

Many of the major aerospace organizations participated in developing the SysML standard including Northrop Grumman Corporation, Lockheed Martin Corporation, The Boeing Company, NASA, BAE Systems, Thales, EADS Astrium GmbH, and the Israel Aircraft Industries. Additionally, component manufacturers such as Raytheon Corporation, IBM, and Motorola, software development support suppliers such as Gentleware AG, Pivot Point Technology Company, Telelogic AB, Artisan Software Tools, Ceira Technologies, Deere & Company, EmbeddedPlus Engineering, Eurostep Group AB, I-Logix Inc, and oose.de Dienstleistungen für innovative Informatik GmbH participated in the standardization of SysML. Finally, organizations such as the Georgia Institute of Technology and INCOSE (International Council on Systems Engineering) participated in the standards process. Several tools are COTS products that support the UML/SysML visualization for the requirements and development process. These are shown in Table 20-13.

It is important that we manage the software development activities throughout the program life cycle. There are many tools that help in this effort. The UML and SysML tools described above, as well as compilers and debuggers associated with a particular software development environment can be licensed for the equivalent of several full-time equivalent (FTE) software engineers. However, there are also tools available for free that provide some aspects of the more expensive licensed tools. An important part of planning for a software development project is determining which tools will be needed to improve development efficiency and provide continued traceability between requirements and implementation.

Table 20-13, Fig. 20-9◀, Eq. 20-0

Table 20-13. UML Based Software Development Tools. UML based design and CASE (computer aided software engineering) tools are designed to aid in the software development process by reducing the complexity of dynamic systems through the use of visualization and modeling schemas.

Supplier	Tool Name
Atego (Merger of Artisan Software and Aonix)	Artisan Studio
Visual Paradigm	Simulacian
Telelogic (owned by IBM)	TAU Generation 2 Rhapsody (rational rose, doors)
Embedded Plus	Sys. Modeling and Sim Toolkit
Microsoft	Visio
Sparx	Enterprise Architect

Finally, the use of configuration management tools allows us to maintain control of code revisions, providing traceability between major and minor software releases, as well as assuring that various software releases can be recreated. Again, we can license configuration management tools, or there are tools available that provide the basic capabilities for free. A critical aspect of software development is that we maintain management control throughout the life-cycle of the product.

20.2.4 Integration and Test

As Fig. 20-10 shows, testing usually begins at the lowest level and builds incrementally. By building our test scenarios from the bottom up, we can reduce the complexity and thus the risk. Testing must be rigorous at

all stages from the unit level to the system level [Kaner et al., 1999]. Software and hardware testing follow the same general path, with the subsystem resulting when we integrate the hardware and software. At this level, we test the entire computer system. We perform path testing, functional testing, and performance testing. We must document each test thoroughly using test procedures that trace back to the original test plan, and thus the requirements, and outputting a test report that notes the details of the test execution. The test report can become crucial to identifying anomalous behavior prior to launch or on-orbit. It is more important that we document behavior, even if off-nominal, than that we justify why test outcomes are acceptable.

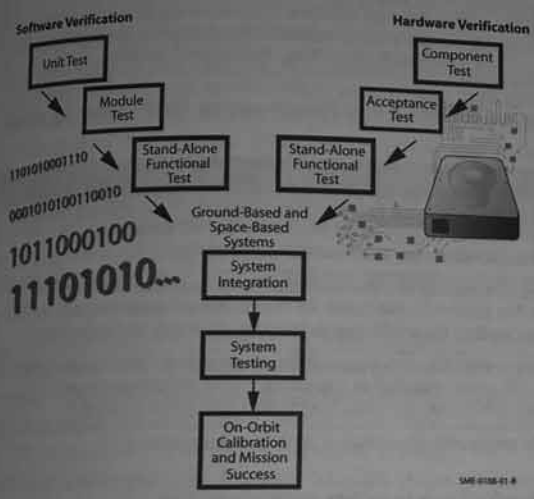


Fig. 20-10. Levels of Testing. Testing builds incrementally as the product develops from component to system. It begins at the lowest level building up into system and mission requirements verification as the elements are integrated.

Finally, we test the whole spacecraft, both in terms of functional performance through "day-in-the-life" test scenarios, and through environmental testing such as thermal, vacuum, and vibration testing. Unfortunately, we cannot completely determine whether the processing system performs properly until it is in orbit. Once operational, the system needs general testing to ensure it continues to perform as required following launch and environmental exposure. Just as acceptance test procedures or inspections check systems for damage on delivery, retesting on-orbit checks for damage during launch. This testing is often referred to as *on-orbit check-out and calibration*.

In general, integration and test pulls disciplines and subsystems into a configuration that meets top-level, system requirements. In this sense, integration and test is much the same as systems engineering. System testing includes all activities that increase confidence in the system's performance. It ensures that we have met requirements and that anything happening beyond these requirements does no harm to the system, while preserving specified functions. Testing for these "extras" is the most difficult because we do not always know what we are looking for. Testing, especially for the software-intensive computer resources associated with space systems, is a complex undertaking. It can consume up to half of the development cost and a significant percentage of support costs over the life-cycle.

20.3 FireSat II Example

Using the FireSat II example from the previous chapters, we will examine the actual flow down process from mission requirements to derived requirements for on-board processing. We will also develop a baseline architecture for a possible on-board processing or avionics implementation to meet the requirements and walk through the software sizing and processor selection process. Subsystem requirements that will directly impact the on-board processing as well as those that will be affected by the on-board processing implementation will be discussed. It is important to note that with the traditional subsystems so highly integrated in this digital processing implementation, we must coordinate and iterate on-board processing requirements with those generated by other subsystems (such as attitude determination and control systems (ADCS), telemetry, tracking, and command (TT&C), and power/thermal management) to assure that all derived requirements are met at each level of design.

We will use Table 20-1 Computer Systems Development Process as the roadmap for performing the FireSat II example. First, we define requirements. These requirements are based on mission objectives and their direct implication to computer systems. We evaluate derived requirements from other system areas to determine if they will impact the on-board processing requirements. Specifically, when we determine processing specific requirements from mission level objectives, we partition the information we have into traditional subsystem requirements as outlined in Table 20-14.

For the FireSat II example, we begin with Table 3-5, finding all of the requirements that may affect on-board processing. Table 20-15 shows the requirements that we identified as discussed in other chapters, but, where the outcome of decisions made will affect the on-board processing; however, as was stated before, it is imperative that we work other disciplines.

Next, we define the top-level computer or on-board processing requirements. All launch and early operations will be handled by the launch vehicle provider. As the spacecraft on-board processing component, including the telemetry tracking and command (TT&C) we will provide a fixed set of go/no-go functional tests and results to be used on the launch pad and at power up following launch vehicle separation.

This step also includes functional partitioning of requirements to implementation and the development of a candidate architecture and the associated interfaces. Because of the number of requirements that can be

Table 20-14. Flow Down of Mission Objectives into Potential On-board Processing Requirements.

System Attribute	Typical Trade Space	Baseline System Implications
System Timing	Event or data driven system versus a systematic scheduled data delivery system.	Usually payload centric processing is event or data driven and maintenance of the spacecraft functions are then secondary.
Asset Preservation	Hard Real-time data delivery and network determination versus Dependable Performance in terms of fault detection and system safe mode protection.	A hard real-time system typically focuses on performance of the ADCS to achieve performance while an expensive asset that must be maintained might allow for data to be less precise.
Communications	Type of ground contact, often based on orbit (i.e., LEO implies infrequent contact while GEO implies continuous contact) or amount of data to be stored, compressed, parsed, and forwarded.	These elements will drive the on-board storage and antenna design. It is often very difficult to download a lot of raw data in LEO with limited ground contact.
Data Storage	Acceptable data reduction or maintenance of complete raw data sets.	Extensive data storage often requires high radiation tolerance which is often expensive.
Growth	Is the mission partially defined and might there be on-orbit enhancements or mission growth?	This affects the system design in terms of maintainability and ease of enhancement, which directly affects processing margins, need for software uploading, and ground testing.

Table 20-13. Derived On-board Processing Requirements Based on Mission Level Requirements Established by Other Disciplines. Requirements from Table 3-5. (See Sec. 3.5)

Requirement	Description	Implication to On-Board Processing
Performance	Identify emerging fire within 8 hours of start and with less than 10% false positive	• LEO (~450 km) provides for recurrence within 2.5 hours/less radiation than high elevations/orbit inclination may still be an issue • Payload responsible for false positives reliability
Interpretation	Interpreted data to end user within 5 minutes	• On-board processing of information/autonomous evaluation of potential fire (autonomous rules for fire id. must be well-wrung out) • Additional testing for additional autonomy (funding and schedule)
Timeliness	Commandable within 3 minutes of events, stored coverage areas executed on-board	• Must have continuous up-link capability, potentially via multiple ground stations OR when location is identified and notification occurs, immediate response of stored command sequence (storage)
Mission Design Life	8 years	• Long-Life parts with reasonable radiation tolerance
System Availability	95% excluding weather and 24 hour maximum down time	• Potential redundancy to assure limited down time
Data Distribution	Up to 500 fire-monitoring offices and 2,000 ranges worldwide (maximum of 100 simultaneous users)	• Bandwidth and frequencies covered by communications discipline, on-board processing must be able to handle up to 500 event driven sequences with different downlink data files (Storage)
Data Content, Form, and Format	Location and extent in lat/long for local plotting, avg. temp. for each 40m^2 grid	• High fidelity mapping / knowledge to identify ground location. • Temperature from what sensor and correlated to ground how
Cost	Non-Recurring <$50M Recurring <$10M /year	• Processing components and RTOS, with Software Development costs limited to 7.5% ($3.75M) • Schedule Requirement—increase costs by <10% ($4.2M)
Schedule	Operational within 3 years	• Increase cost for rapid test and integration, quicker delivery of subsystem components
Risk	Probability of success >90%	• Autonomy must be tested, upload capability for enhancements and potential feature fixes
Regulations	Orbital debris and civil regulations	• Must have end-of-life retirement—deorbit
Interfaces	Interoperable through NOAA ground stations	• Antenna issue but orbit and pointing must be maintained for NOAA locations

directly traced to the potential payload and payload management, we will determine that payload processing will be separate from the spacecraft general processing and it will be the responsibility of the payload supplier. Thus, the autonomy functions needed for identification (within 10%) will be implemented and tested by the payload supplier. Additionally, fault monitoring and identification will be part of the baseline on-board processing functions but the ability to work-around a fault or provide graceful degradation will be done on the ground and uploaded since there is ample time within the 24 hour window for multiple passes.

The standard "rules" used to determine stored command sequences based on fire identification will be pro-

Table 20-15, Fig. 20-10◀, Eq. 20-0

vided by the payload supplier but implemented within the C&DH functional elements of the spacecraft bus processing. Storage required for this function based on the potential identification of up to 500 fires at one time will be provided by on-board processing. Additionally, storage for the potential up-linked stored sequences based on the 3 minute commandability to the vehicle following a fire identification will also be provided by on-board processing. Control will require thrusters that must have enough propellant to perform de-orbit functions following the 8 year nominal life of the vehicle. Fine attitude control will be provided via reaction wheels and the associated momentum management will be provided by torquers and an associated magnetometer for direction identification. A sun sensor has been identified as nice-to-have for maintaining power, and since the associated processing is not significant, this will be part of the system baseline. A star tracker with quaternion output, a gyro with delta-V and acceleration, as well as a high resolution GPS receiver will be part of the baseline system configuration for attitude (star tracker/gyro) and orbit (GPS) determination. On-board orbit control will assure that a recurring path is created during each orbital pass but service will only be provided to those stations that are in the up/down-link performance of the system. Other data distribution can be handled by ground-station to ground-station data transfers.

Interoperability with NOAA ground stations will drive antenna size and frequency allocation which we will assign to the communication discipline but antenna pointing will impact on-board processing. To achieve this level of interoperability with NOAA ground stations, we will focus on coarse pointing resolution with a larger antenna. We make this selection from the on-board processing perspective because the knowledge requirement for the fire identification is higher than the control requirement and we don't want to implement a costly function in the main processing, if we don't need to. However, on-board storage in terms of C&DH and stored command sequences will be supplied by the on-board processing. Basic utility functions, math and coordinate transformations, as well as power and thermal control will be part of the baseline on-board processing. It is assumed that the thermal control will be passive and the power will maintain a 20% margin throughout operations.

The total SLOC count for the baseline spacecraft processing associated with the FireSat II mission is 38,900 SLOC as detailed in Table 20-16. This value, with 100% margin since we are well before SRR, will be used to determine the size and throughput requirements for the on-board processing and a mass storage device will be added to the system to assure that the data and stored sequences/commands can be accommodated. The ADCS is approximately 40% of the available processing (15,700/38,900) and the C&DH is only 5% (2,000/38,900) so the best processor, with past mission history shown in Table 20-11 is the RAD600 or the RAD750 whichever is available in the most radiation tolerant configuration for a reasonable amount of money.

Table 20-16. Estimating Source Lines of Code (SLOC) for FireSat II Spacecraft Functions. Items selected as Required and Nice-to-Have are shown in Table 20-9. Since the Nice-to-Have items are relatively small, we will include them in the baseline. Payload functions (including pointing) will be done within a payload processor that is outside the responsibility of the on-board processing organization.

Computer Software Component (CSC)	Source Lines of Code (SLOC)
Executive	1,000
Communication	2,000
Attitude/Orbit Sensor Processing	3,800
Attitude Determination and Control	12,500
Attitude Actuator Processing	2,400
Fault Detection	6,500
Utilities	5,700
Other Functions	5,000
Total	**38,900**

The Wind River VxWorks RTOS will be used as the baseline operating system and associated development environment because the board support package (BSP) for necessary RTOS functions is quite mature for the RAD600 /750 series of processors. A baseline system architecture is shown in Fig. 20-11 based on the need for a reasonable amount of throughput on the spacecraft bus as well as the payload side and the decision to reduce risk associated with on-board processing, by partitioning the payload functioning and processing from the on-board spacecraft bus functions.

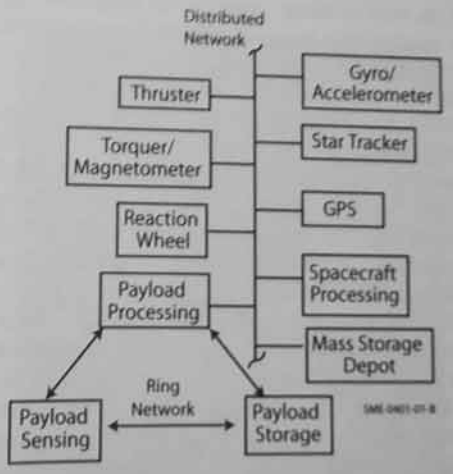


Fig. 20-11. FireSat II Spacecraft Baseline On-board Processing Architecture.

This architecture takes into consideration a distributed network for the primary spacecraft bus functions. The network protocol is a nominal command response system ([DoD, 1996]/ARINC 667 part 4) and provides deterministic timing with standardized interfaces. The payload on-board processing system has been baselined

as a ring architecture since there are a limited number of items that need to interact with each other. The ring provides a certain level of redundancy and fault tolerance. This baseline will be iterated with the various other subsystem domain experts until a final configuration is selected to best meet the needs of all of the stakeholders.

Next, we select a tailored set of documentation to be provided as part of the FireSat II effort. This will include an SRS/IRS integrated document and a hardware PFS with integrated ICD. Test plans and procedures will be developed for the flight software and then for the spacecraft bus avionics as a stand-alone system. The User's Guide will be developed at the system level implying that some of the development staff may be required for long-term support. With the high reliability requirement and 24 hour down minimum, this is probably a good trade (personnel for documentation) in the long run. Finally, we evaluate reliability based on the mission objectives and data that might be provided by the supplier of various components. Again, the baseline system design is iterated based on the reliability, availability, and cost/schedule requirements and the functional performance needs.

20.4 Modular Approaches to Processing

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Spacecraft development has become excessively and increasingly expensive in terms of cost and time, which has led to intolerance for failure, further driving higher costs and longer schedules driving ever more intolerance for failure. This spiral has led to very complex processing systems, one's that are highly reliable and custom built, as well as being expensive to create and replace. With an emphasis on processing systems and FSW, we will examine an alternative approach where we use modularity and software reuse to create components that are essentially "off-the-shelf". These components can be rapidly assembled to create a spacecraft or we can create a number of less expensive spacecraft that can be quickly and inexpensively replaced. We will first examine an approach to create modular processing and potentially less expensive spacecraft and later we will address issues associated with software reuse.

"Modularity is a general systems concept: it is a continuum describing the degree to which a system's components can be separated and recombined, and it refers to both the tightness of coupling between components and the degree to which the "rules" of the system architecture enable (or prohibit) the mixing and matching of components."[Schilling, 2000]. In this context we see that space systems offer numerous potential opportunities for application of modularity. However, the conventional wisdom is that space systems are so unique, and typically so customized, that there would be little benefit from the introduction of modular architectures. Historically, we have viewed spacecraft as tightly integrated, non-modular, custom designs that are driven by the desire for maximizing the performance in a specific mission scenario. Until a decade or so ago we believe that this was essential to meet many mission needs. Currently, we see that component, computer, and digital network performance capabilities have dramatically increased such that they are more than adequate to support modular approaches without sacrificing mission capabilities. Certainly there is overhead in terms of mass, processing and time that can be reduced by tight integration, but in most cases the increased costs will not be offset by improved mission performance.

The inclusion of modular designs and modular components has become prevalent across numerous high-technology industries and markets, and the reasons are fundamentally economic. We know that modular architectures reduce the cost of offering variety and of the introduction of new technologies into products. We have seen that innovations and improvements can be rapidly introduced at the module level without the need for broader design changes. Modularity can also enable supplier competition, which drives cost reductions, by allowing easy substitution of alternative components based on performance, price, or availability. These benefits have been clearly demonstrated in the computer industry.

The first significant foray into modularity in the computer industry was the development of the IBM System/360 which was announced in 1964 and reviewed by Baldwin and Clark [1997]. Prior to the development of the System/360 architecture, offerings from every computer manufacturer and even new designs from a given producer were wholly incompatible requiring custom hardware and software and "upgrades" required costly and risky migration efforts. Development of a robust modular architecture turned out to be more challenging than anticipated, however, the rewards were remarkable. By 1969, IBM stock accounted for more than 70% of the total market value of the computer industry.

One attempt at change in the spacecraft industry is the incarnation of Operationally Responsive Space (ORS). ORS is rooted in the concepts that if spacecraft were available off-the-shelf (OTS) or if many smaller and cheaper, potentially expendable spacecraft, could be assembled quickly, with system performance that is "good enough," rather than optimal, then cost and schedule would become more manageable. In the ORS paradigm we have a quick turn-around from identifying a mission need, to designing and implementing a solution, then deploying that solution for a reasonable, if not limited, on-orbit life-time. The *CubeSat* (Sec. 25-4) is a standardized 10 × 10 × 10 cm (1U or 1 unit) spacecraft that performs a very specific function over a limited lifetime. The CubeSat originated at Stanford University and the early specifications and standards for CubeSats were developed at Cal Poly, San Luis Obispo. CubeSats evolution was originally due to university experimentation and the 1U CubeSat was scaled to include multiple units (e.g., "2U" is 20 × 10 × 10 cm, a "3U" CubeSat 30 × 10 × 10 cm). As we see these more complex CubeSats evolve and become operational, we also see their application expanding beyond universities, including commercial, military and ORS applications.

Table 20-16◀, Fig. 20-11◀, Eq. 20-0

We see another mechanism for meeting the ORS concept of rapid response in the development of spacecraft busses that are built to inventory so that we can quickly pair the bus with a COTS payload to meet a range of ORS mission requirements with "good enough" functionality, rather than optimal performance. Alternatively, spacecraft can be built quickly either with "existing," fully tested, building blocks or modular components that are capable of flexible configuration to meet a diverse range of mission requirements. We see that there are differing opinions as to the granularity of the built-to-inventory components:

1. Complete, ready-to-fly spacecraft
2. Spacecraft busses and payloads held separately in inventory
3. Functional subsystem elements combined to create functional services, then stocked on shelves and snapped together to create a complete spacecraft, or
4. Lower-level components being rapidly assembled with the aid of a configuration wizard that determines the parts that are needed to create a spacecraft that will meet specific mission requirements.

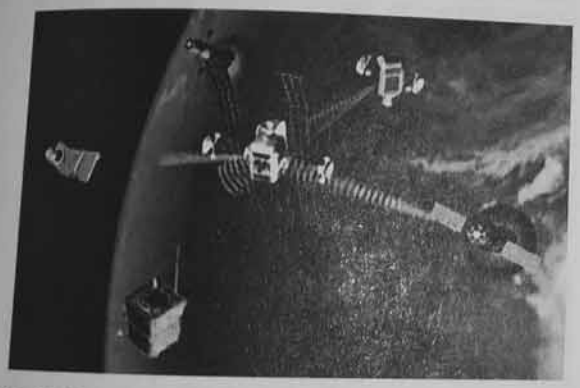


Fig. 20-12. A Fractionated Spacecraft Each of a Subset of the Overall Functionality that is Desired. (Courtesy of DARPA)

We have also seen that there are several implementation concepts or approaches that are associated with creating modularity for spacecraft and spacecraft processing architectures. First, there is *fractionated spacecraft* where we create several individual spacecraft, each with a subset of the desired functional capabilities as shown in Fig. 20-12. Figure 20-13 shows how the fractionated spacecraft operate together as a cluster or constellation to have all the functions needed to meet the desired mission objectives [Brown and Eremenko, 2006]. In a second approach, we create a few types of *modular buses* that meet different mission performance parameters yet maintain a standard interface to the launch vehicle and to the payload. This concept is shown is shown in Fig. 20-14. In this implementation we might have a

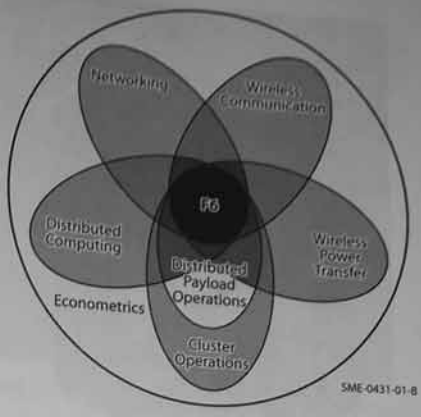


Fig. 20-13. Sample Functional Elements of the Fractionated Spacecraft.

spacecraft bus that has a high-performance ADACS (attitude determination and control system) but moderate radiation tolerance (suitable for LEO imaging spacecraft), and another with moderate ADACS capabilities but high radiation tolerance (suitable for MEO communication spacecraft). In a third approach to modularity, we develop *Plug-and-Play components*, (PnP) at the subsystem and down to the individual device, so that a specific mission can be met by rapidly assembling the necessary components or devices to an equipment manifest that is generated by a design tool, given the specific mission requirements and objectives. Figure 20-15 shows a car-



Fig. 20-14. In the NRL/ORS ISET Approach Several Modular Spacecraft Busses are Created to Mate with Standard, Off-the-shelf Payloads to Meet Specific Mission Needs. (Courtesy of NRL)

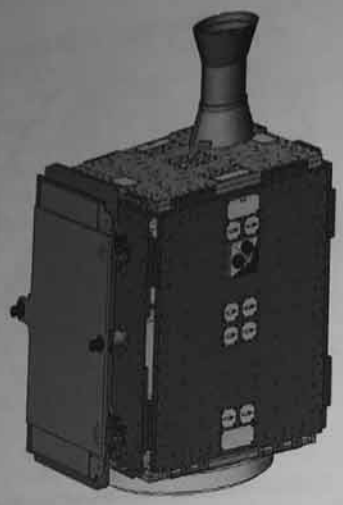


Fig. 20-15. Cartoon of the Plug-and-Play (PnP) Satellite (PnPSat) with Modular Panels and Embedded Wiring Harnesses.

toon of the AFRL PnPSat. An interesting attribute we find with the AFRL PnP vehicle is that modular panels have been created with standard grid pattern for easy assembly of components onto the bus structure. Additionally, within each modular panel is embedded wiring and routers creating the PnP data network.

The fractionated spacecraft was introduced by DARPA in the early 2000s. The basic concept is that we would implement different functional elements on each of a small number of spacecraft that we operate as a complete functional system. For example, downlink capabilities on one spacecraft, high bandwidth processing on another, and imaging on yet another. If we were to implement the fractionated spacecraft, we would create a cluster of fairly specific functional elements that are straightforward to replace individually if failure occurs. However, we must assure that all the elements of the cluster can communicate with each other (perhaps through an intermediary) and we must maintain precision control of both the attitude and orbit of each of the spacecraft and the integrity of the formation. (Sec. 23.2)

The concept of modular spacecraft has been explored by numerous organizations and companies over the last 25 years. Recently, the Integrated Systems Engineering Team (ISET) project, established by the Naval Research Laboratory (NRL) and the Office of the Secretary of Defense/Office of Force Transformation (OSD/OFT), developed specific requirements for "what" a spacecraft bus must do to meet the needs of various mission categories (i.e., imaging, communications), including a description of the various elements that are required to create a "system". For each type of spacecraft bus we would implement a standardized interface from the bus to the launch vehicle and from the bus to the payload making them virtually interchangeable. The general concept is that once we identify the mission requirements, we select the appropriate spacecraft bus from a list of available units and then integrate the payload. After a few instances of integration with this standardized interface, we hope that the integration time and effort is reduced, thus reducing the overall cost of the spacecraft. Also, because the spacecraft was ready to go, we have significantly reduced the time to launch. (Sec. 23.6).

Another mechanism for reducing the time to launch is by creating PnP spacecraft components that can be rapidly assembled into a spacecraft. In thinking about plug-and-play (PnP) for spacecraft we borrow terminology and even concepts from the commercial implementation of PnP. Specifically, we look at the PC-based concept of PnP, where the user plugs a mouse into a PC resident USB socket and invokes the operating system to find the correct driver, configures the system parameters, and then seamlessly allows the user to begin operations. In this example, and in general, PnP attempts to hide interface complexity from the user by establishing standard interfaces. The issue here is that the standards must be agreed to by all parties that would use or develop the components, and there must be a reasonable return on investment (ROI) for the spacecraft community to adhere to them.

ESA has also published a draft standard for a PnP SpaceWire Protocol Definition (SpW-PnP-PD). This draft standard was created by the University of Dundee and focuses on the data transfer protocol, the plug aspect of PnP, without specifically addressing the play element. With help from the AIAA, a group of people from the spacecraft community are working to create a complete set of standards for a Spacecraft PnP Architecture (SPA). The standards focus on the interface definition, based on the Open System Interconnection Reference Model (OSI Reference Model or OSI Model) which includes seven layers. The OSI seven layer model begins at the lowest level with the physical medium, moves to the data-link, network and transport layer, then adds the session, presentation and application layers. It is the middle layers, the data-link, network, and transportation layers that provide the functional and procedural means to effectively transfer data from a source to a destination. We can define the data source as individual devices, subassemblies, subsystems, or the spacecraft bus itself. When we are discussing FSW, we can consider a data source as functional services, such as navigation and attitude determination services.

For PnP, as defined by SPA, we would create individual spacecraft components that are PnP compatible which would include implementing a standard physical interface as well as a data protocol that allows the component to describe its function, command and control, and data output using a machine negotiated mechanism. The Transducer Electronic Data Sheets (TEDS) used by the automotive industry (IEEE 1451) has been augmented for use with spacecraft components creating an extended TEDS or xTEDS. This machine negotiated interface is based on extensible mark-up language (XML) and includes information about the component that a subscriber or controller would need to

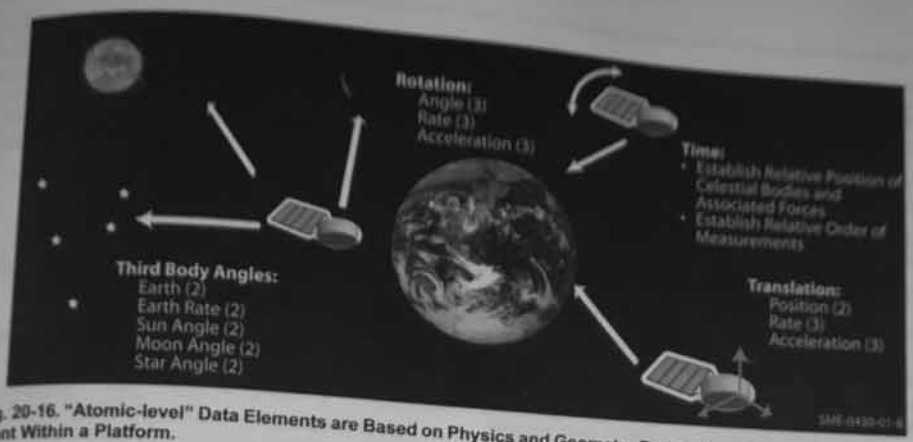


Fig. 20-16. "Atomic-level" Data Elements are Based on Physics and Geometry Rather than Specific Devices/Equipment Within a Platform.

know to make full use of that component. Specifically, the xTEDS provide a "service description" for components, such that functions not detailed in the xTEDS are effectively invisible to SPA. In terms of PnP hardware, the AFRL PnPSat team is making great strides toward acceptance of traditional commercial PnP practices as viable in the spacecraft industry. Through the implementation of appliqué sensor interface modules (ASIMs) we can augment existing, legacy hardware to meet the SPA standards. AFRL and the ORS Office have supported several activities to develop ASIM technology and PnP enabled components.

To fully achieve the concept of PnP spacecraft, we find that innovation in development of FSW is required. This innovation can range from process oriented enhancements to actual software implementations that facilitate new missions, including operationally responsive space. Specifically, the reuse of off-the-shelf software modules is highly desirable. As discussed earlier in this chapter, software costs are traditionally based on the number of source lines of code we develop for each specific discipline. If we can reuse software modules, much in the same way that we might make use of an off-the-shelf modular spacecraft bus, the only recurring cost is the cost of integration and system level test.

If we are to support software reuse we must develop an architecture that is data-centric such that the resulting FSW implementation can be implemented, tested, and ready off-the-shelf. To achieve this, we need to partition software and data at the lowest levels, using terminology that originates in the physics of the device rather than the manufacturer. Much like atomic elements in chemistry, we call those low level data elements *atomic data elements* to imply that they are the smallest elements that can be observed. Additionally, to achieve software reuse the core algorithms must be designed to be generic and the software must be developed and tested, before the specifics of the mission may have been identified. Thus, we will define specific mission parameters associated with the vehicle configuration and characteristics in a way that will facilitate the reuse of the generic core algorithms we have developed and tested. To create reusable FSW we must define data elements at the *atomic data level*. Figure 20-17 shows atomic data for an attitude determination and control system (ADCS) application. Here we define data as they are measured or sensed such that there is a direct tie to physics and geometry (e.g., spacecraft motion, rates, expressed in body coordinates), rather than subsystem or component implementation specific parameters. By having all of the components that supply spacecraft rotation present their data interface using atomic data definitions, sensors can be interchanged based on availability without impacting the primary algorithms.

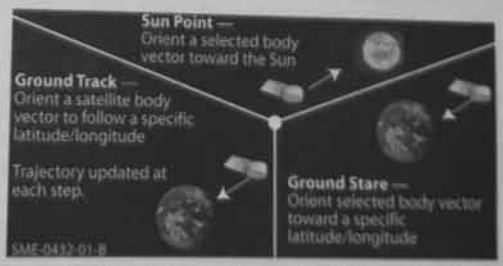


Fig. 20-17. Atomic Level Moding Facilitates One Control Law Designed to Reduce the Error Between Sensed and Desired State.

The control algorithms remain abstracted from the data sets and data is resolved to the atomic level. Additionally, we can create the command and control aspect of a PnP ADCS system with its elements reduced to the atomic level. By using atomic level command we allow a user to build-up any capability that is required, without necessarily being aware of the capability when the FSW is created. Traditional modes such as sun point, ground track, and ground stare, are all achieved using atomic level mode commands. These modes include the following:

- SLEW—Slew to a vector (time, position, velocity)
- TRACK—Maintain pointing to a vector (time, position, velocity)
- RATE—set body rates to a vector (de-tumble, spin-up)
- STANDBY—transitional or cruise (minimum power consumption)
- MOMENTUM-DUMP—reduce stored momentum

By using this atomic level moding, there may be varying modes of operation that will be performed with respect to varying objects. As shown in Fig. 20-16, the objects or pointing directions are defined to include celestial objects (e.g. Sun, Moon), nadir, and zenith. Additionally, the ability to point to a vector (position) or target (state vector including rates) can be accommodated. This leads to an additional concept for the reuse of the PnP ADCS FSW. Since the inputs, processing, and outputs are the same for any of the traditional modes, there can also be a single control law. While this may not allow for optimal gains in each of the ADCS modes, we have created responsive FSW that is modular and OTS.

If we standardized the input and output (I/O) mechanisms of all components, for any vehicle whether spacecraft, Unmanned Aerial Vehicle (UAV), Autonomous land vehicle or sea vehicle, the hardware components and even the application FSW can be similar which provides many benefits to the all autonomous vehicle industries. First, this level of interface transparency facilitates reuse at the algorithm, component, and subsystem level. As electronics become smaller, and micro-electro-mechanical systems (MEMS) are used to implement GN&C sensing, sensors and potentially actuation/propulsion devices, could be shared between the various vehicle types. The environmental concerns will differ, but as the electronics become more and more capable, it is possible that other methods of accommodating the environment, such as radiation hardening (rad-hard) by design, may help to make devices reusable for different vehicle types.

If more advanced PnP techniques are applied to the standardization process, we can create a resulting system that will be able to support self-discovery and configuration. This will further facilitate rapid assembly and test, while providing for a robust, fault-tolerant performance with graceful degradation in the event of device or subsystem failures. Moreover, this approach will greatly improve the interoperability of the various types of vehicles, helping with economies of scale for the smaller market vehicles such as spacecraft and submarines by leveraging the larger market vehicle products such as those developed in support of aircraft and UAVs. And finally, with the development of vehicle agnostic components with PnP interfaces, we will create a completely modular spacecraft, as well as other autonomous vehicles, which provides a natural transition into an integrated system of systems from underwater to land and air, and into space.

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21 Spacecraft Subsystems IV—Communications and Power

21.1 Telemetry, Tracking, and Command (TT&C)
Preliminary Design Process; TT&C Requirements; Frequency Bands of Operation; Preliminary Link Analysis and Modulation/Coding Tradeoffs; Hardware Design and Key Technologies; Subsystem Design Examples; Special Topics; Future Trends

21.2 Power
Mission and Design Requirements; Power Generation/Sources; Energy Storage; Power Regulation and Control; Power Distribution; Future Power Systems

21.1 Telemetry, Tracking, and Command (TT&C) Subsystem

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The *telemetry, tracking, and command (TT&C) subsystem* provides for communications and radiometric tracking between the spacecraft and ground stations. These functions are accomplished through the transmission of electromagnetic signals, typically at radio frequencies (RF). The TT&C subsystem provides the spacecraft with an uplink (or forward link) for commands, a downlink (or return link) for mission data, and radiometric links (i.e., Doppler shift and ranging signals) that enable the velocity and distance to the spacecraft relative to the ground station to be measured for the purpose of determining the spacecraft orbit. The mission data includes both the spacecraft engineering data (sometimes referred to as *housekeeping data*) and the sensor/instrument data generated by the spacecraft payload. The latter data may be integrated with the engineering data on the same link or returned to Earth via a separate wideband downlink.

This section will provide us with a solid understanding of the typical requirements on the TT&C subsystem and how these requirements can be implemented in a basic design. We will learn to perform a link analysis and use the results to make top-level system decisions, including subsystem transmitter power and antenna size requirements. We will learn the architecture of a TT&C subsystem and be able to create a block diagram design and perform tradeoffs that lead to sizing of the elements in the subsystem based on spacecraft mass and power requirements. Finally, we will gain familiarity with the various technologies currently available to implement the subsystem and the emerging technologies that hold promise for the future. An emphasis will be placed on high bit rate communications, and the challenges associated with bandwidth.

21.1.1 Preliminary Design Process

The TT&C preliminary design process begins with the definition of top-level requirements and constraints as flowed down from the mission-level requirements. A preliminary RF link analysis is typically performed early-on to "size" the subsystem, particularly the spacecraft power amplifier and high gain antenna. A hardware block diagram is then developed, followed by various tradeoff studies to refine the equipment selection. A point design can then be generated to determine bus power, mass, data return, and link margins. The frequency assignment process can then be initiated through the National Telecommunications and Information Administration (NTIA). Table 21-1 presents an outline of the preliminary design process for a typical TT&C subsystem.

21.1.2 TT&C Requirements

Many requirements levied onto the TT&C subsystem are determined by various aspects of the mission itself. For instance, the range between the spacecraft and the ground station is set by the nature of the mission. The spacecraft attitude control system (spinning, 3-axis, or pointing control accuracy) is typically determined by bus power and concept of operations (CONOPs) considerations. The ground station antenna sizes and capabilities are often dictated by mission type (near-Earth, NASA, military, deep-space, or commercial). There may be limits on the size, placement, and number of spacecraft antennas. While the TT&C system engineer can of course influence these requirements and is an important player in the overall spacecraft system design, these types of requirements are often considered drivers for the TT&C design. The TT&C system engineer typically has more flexibility in defining downlink data rate, an important, first-order requirement in sizing the TT&C system that also determines (or is determined by) data volume and number of passes. The downlink data rate is set to balance the achievable RF output power with antenna size and placement constraints for the mission. This calculation can be relatively straightforward for some missions, yet require substantial analysis for other missions. For example, consider a mission in low-Earth orbit with a daily science return requirement of 10 Gbits. If the satellite is in view of the ground station four times per day with 10-minute pass durations, then the required downlink data rate is calculated rather easily to be 4.2 Mbps. In

Table 21-1. TT&C Subsystem Preliminary Design Process.

Step	Details	Where Discussed
Define Top-Level Requirements and Constraints	• Redundancy • Ground network/ground station compatibilities • Distance to ground station (orbit geometry) for normal, mission-critical, and emergency operations • Data rate/data volume and associated data quality (bit error rate, frame error rate) • Frequency band and emission constraints • Radiometric tracking accuracy (Doppler, ranging) • Antenna coverage • Standards (CCSDS, SGLS, other) • Power, mass constraints • High-level electrical interfaces with other subsystems such as MIL-STD-1553, RS-422, and low voltage differential signaling (LVDS). • Environmental (especially temperature and radiation)	Sec. 21.1.2, Sec. 21.1.3
Perform Preliminary Link Analysis to Size the Subsystem	• Determine antenna characteristics (gain, beamwidth, size) and RF output power based on (1) mission data return and (2) emergency mode communications. • Determine driving requirements for fixed-pattern (low and medium gain) antenna coverage: LEOP, emergency mode, maneuvers, Doppler tracking	Sec. 21.1.4
Develop Subsystem Block Diagram	• Major hardware components including transponders, power amplifiers, diplexers, RF switching assemblies, and antennas • Top-level subsystem electrical interfaces • Preliminary cost estimate is often generated at this stage.	Sec. 21.1.5
Perform Tradeoff Studies to Refine Equipment Selection	• Transponder vs transceiver technologies • Antenna technologies (reflector, waveguide, wire, patches, others) • Solid-state vs traveling wave tube power amplifier technologies • Passive network technologies (waveguide vs coaxial diplexers, RF switches, transmission lines)	Sec. 21.1.5
Develop Point Design Based on Tradeoff Selections	• Balance performance vs cost and power/mass constraints. Outputs of this step are: – Detailed subsystem block diagram – Power, mass, and volume table – Refined link analysis, link margins – Data return projections – Refined cost estimate	Sec. 21.1.6
Initiate Frequency Assignment Process	• Application via government sponsor to the NTIA for specific frequency of operation assignments	Sec. 21.1.3
Documentation	• Internal memoranda, design review presentations, requirements document.	—

contrast, a mission in a highly elliptical Earth orbit experiences substantial variations in ground station distance and viewing time. A fixed downlink data rate would be inefficient in this case. Calculation of a variable downlink data rate schedule and associated average daily science return requires a combination of link analysis, orbital analysis, and on-board recorder utilization analysis.

Table 21-2 presents a summary of the typical TT&C subsystem requirements for near-Earth and deep-space missions.

21.1.3 Frequency Bands of Operation

The TT&C subsystem operating frequency band is selected based on several considerations. The nature of the mission (science, military, commercial) may dictate the use of specific ground stations with dedicated bands of operation. In selecting from the available bands, the primary consideration is meeting downlink data rate requirements (the downlink is usually the driver because of the limited transmit power on the spacecraft, compared to the transmit power available at a ground station). A close review of the Friis link, Eq. (21-1) through Eq. (21-3) reveals that the operating frequency of a communications link can be optimized to a first order by considering the nature of the antennas used in the link. Three basic cases exist:

1. **Low Gain/Low Gain:** Links with low gain antennas on both ends (driven by pattern coverage requirements) favor operation at lower RF frequencies such as VHF and UHF (ex: proximity operations).

Table 21-2. Typical TT&C Subsystem-Level Requirements.

Parameter	Typical Requirement	
	Near-Earth Mission	Deep-Space Mission
Ground Network Compatibility *(see also Chap. 28)*	**Near Earth Network** (NEN) for NASA Category A missions (within 2 × 10^6 km from Earth) **Space Network** (SN) for near-Earth missions that have a requirement for nearly continuous contact with Earth. Also known as the Tracking and Data Relay Satellite System (TDRSS). **Air Force Satellite Control Network** (AFSCN) for military missions **European Space Agency (ESA)** **Indigenous National Networks** (Japanese Aerospace Exploration Agency JAXA, Indian Space Research Organization ISRO) **Independent Ground Station Assets** (Universal Space Network, or dedicated mission antennas)	**Deep Space Network** (DSN) for NASA Category B missions (beyond 2 × 10^6 km from Earth) **ESA** Deep Space Stations
Earth-Spacecraft Distance	Worst-case links are typically calculated at the maximum slant range distance (lowest ground antenna elevation angle)	Worst-case links are typically calculated at the maximum distance from Earth in the spacecraft trajectory
Data Rates	**Uplink:** 125 to 1 kbps emergency mode 1 to 2 kbps normal mode 1 Mbps memory reload **Downlink:** Up to 10 kbps engineering data & emergency mode Up to ~5 Mbps mission data on same link with engineering data Greater than 5 Mbps mission data on separate wideband downlink May be variable depending on mission geometry and dynamics.	**Uplink:** 7.8 bps emergency mode 15.6 to 2000 kbps other modes 1 Mbps memory reload (future) **Downlink:** 5 to 10 bps emergency mode Up to 10 kbps engineering data Up to 10's Mbps science data (100's of Mbps future) Usually variable due to widely varying geometry of a deep space mission.
Frequency Band (B_N= necessary bandwidth as defined by the NTIA; see Table 21-3)	S-band for uplink S-band for downlink B_N up to 6 MHz X-band for downlink B_N up to 400 MHz K_a-band for downlink B_N > 400 MHz	X-band for uplink X-band for downlink up to several Mbps K_a-band for downlink > several Mbps
Radiometric Tracking Accuracy	**Doppler:** < 5 cm/sec over 1 to 10 sec (1σ) **Ranging:** < 10 m over 1 to 10 sec (1σ)	**Doppler:** 0.1 mm/s over 60 sec (1σ) **Ranging:** 1–10 m over 60 to 600 sec (1σ)
Antenna Coverage	Narrow or moderate beam coverage for mission data return is dependent on attitude control requirements and CONOPs Omni-directional pattern usually required for launch and early operations (LEOPs) and emergency mode	Narrow beam (high gain) antenna usually required for science return Moderate beam (medium gain) antenna may be needed depending on CONOPs Omni-directional pattern usually required for LEOPs and emergency mode
Standards	CCSDS (NASA and military) SGLS (military)	CCSDS
Data Quality	**Uplink:** Bit error rate ≤ 10^{-5} to 10^{-6} (typically uncoded data) **Downlink:** Frame error rate ≤ 10^{-4} (typically w/ forward error correction coding)	
Redundancy	Highly dependent on high-level mission cost/mass/reliability tradeoffs. Usually flowed down directly from mission requirements.	
Power and Mass	Highly mission dependent and managed through an iterative process between the spacecraft systems engineer and TT&C system engineer.	
Electrical Interfaces	RS-422 (complementary digital) (simple, low-power) Low voltage differential signaling (LVDS) (simple, very low-power) Universal asynchronous receiver-transmitter (UART) (simple, low-power) MIL-1553 bus (for inter-box interfaces) I^2C bus (for telemetry status signals)	
Special Environmental Conditions	Radiation, temperature, shock	

2. **High Gain/Low Gain:** Links with a high gain antenna on the ground and a low gain antenna (driven by pattern coverage requirements) on the spacecraft are RF frequency independent to a first order. Constraints such as component availability, cost, size, and passive losses often drive the designer to choose a lower microwave frequency such as S-band. Many near Earth links fall in this category.

3. **High Gain/High Gain:** Links with high gain antennas on both ends (driven by maximum antenna size constraints) favor operation at higher RF frequencies such as X, K_u, and K_a-band. X-band is commonly used for space-to-Earth links for near-Earth and deep space missions. K_u-band is currently used for spacecraft-to-spacecraft links via TDRSS but is being phased out. K_a-band is rapidly gaining prominence for spacecraft-to-Earth and spacecraft-to-spacecraft links for near Earth and deep space. Pointing requirements, atmospheric loss, and atmospheric noise must be carefully accounted for at K_a-band.

Communications links operating at frequencies 10 GHz and above can experience severe degradation due to rain attenuation (fades of 10 dB or more). Link parameters such as transmit power and antenna gain are sized to overcome a selected amount of rain loss; this determines the expected availability of the system and drives cost. Below 10 GHz, and especially at S-band and below, rain loss is a relatively minor consideration.

The use of lower frequencies also results in less expensive hardware on the spacecraft. There are multiple commercial providers of L-band and S-band TT&C communications equipment, with generally increasing cost as one moves to X-band and above. However, spectral crowding in the heavily-used S-band from terrestrial cellular communications is creating pressure to move all satellite links to higher frequency bands. Additionally, limitations are set on the *necessary bandwidth*, B_N, of the transmitted signal in certain bands. These limitations could drive a system to a higher frequency band if a wide bandwidth is required. The National Telecommunications and Information Administration (NTIA) defines necessary bandwidth as the minimum emission bandwidth necessary that allows an acceptable quality of performance, and provides guidance on calculating B_N for various emission classes. Refer to Annex J of the NTIA Redbook [NTIA, 2010]. Table 21-3 provides guidance on frequency band selection and the necessary bandwidth recommendations set forth by the Space Frequency Coordination Group (SFCG) [SFCG, 2011].

Once the operational frequency band and bandwidths are selected, the frequency approval process can begin. For US government NASA or military missions, this is a 4-stage process with the NTIA that results in authority to operate in the desired bands. This process is also used to coordinate with other agencies around the world via the International Telecommunications Union (ITU) to ensure that the mission will operate without interference with the spacecraft and ground systems of other countries and vice-versa. Depending on the type of mission, not all stages need to be completed. For an operational space mission, only stages 2 (Experimental, or Preliminary) and 4 (Operational) are typically required (see Fig. 21-1). Stages 1 (Conceptual) and 3 (Developmental) may be skipped as less-evolved versions of the Stages 2 and 4 submittals, respectively.

Commercial missions typically provide their TT&C links within their payload service allocation, often at the band edges. Licenses are sought from the FCC through a similar process to the NTIA, and the FCC also coordinates submittals to the ITU for international assignment. Frequency approval within other countries follows their domestic approval authority process.

21.1.4 Preliminary Link Analysis and Modulation/Coding Tradeoffs

Initial sizing of the TT&C subsystem is done by calculating the required combination of RF transmit power and downlink antenna size. Both the RF transmit power and antenna aperture area are proportionally related to the downlink bit rate, which becomes the driver for this analysis. We will review the basic link equations needed to perform the sizing and discuss the various tradeoffs involved. The link equations can also be applied to the uplink and emergency downlink scenarios. In some cases, the emergency mode downlink will drive the transmit power requirement because a broader beam (lower gain) antenna is used for emergencies. This section will present an overview of the link analysis process and provide examples that will allow a systems engineer to make basic tradeoffs in sizing the TT&C subsystem and verifying the robustness of the links. A wealth of detailed information on RF link analysis and design can be found in the CCSDS standards documents, particularly CCSDS 401.0-B-20 [CCSDS, 2011], and the DSN Telecommunications Link Design Handbook [DSN, 2011].



A communications link with a transmitter system located a distance d from the receiver system can be described mathematically by a relatively simple equation. Assuming *free space propagation* where the transmitted power P_t spreads isotropically (uniformly) in space as it propagates but does not suffer other propagation effects such as weather, diffraction, or multipath loss, the power density at the receiver system can be described as $P_t/(4\pi d^2)$ where $4\pi d^2$ is the surface area of a sphere of radius d. The transmit antenna will often be designed to focus the radiated power density in the direction of the receiver system. This focusing effect is quantified by the *gain* of the antenna, described mathematically as the power density radiated in a given direction relative to the power density that would have been radiated in that direction by an isotropic radiator. If the transmitter system incorporates an antenna with gain G_t, then the power

Table 21-2◀, Fig. 21-0, Eq. 21-0

Table 21-3. RF Channel Allocations for Space Missions. Note: necessary bandwidth (B_N) limitations are given as guidance where applicable

Frequency Band	Channel Allocations (GHz)		Examples
	Near Earth	Deep Space	
VHF/UHF	0.144 to 0.146 (amateur) 0.435 to 0.438 (amateur)	0.435 to 0.450 (forward) 0.390 to 0.405 (return)	Microsats, amateur radio satellites, Mars/Electra relay link
S-Band Military	1.7 to 1.9 (up) 2.2 to 2.3 (down) $B_N \leq 6$ *MHz*	N/A	Many missions
S-Band NASA	2.025 to 2.12 (up) 2.2 to 2.3 (down) $B_N \leq 6$ *MHz*	2.0 to 2.1 (up) 2.2 to 2.3 (down)	Many near-Earth and deep-space TT&C links and TDRSS multiple access links. Note: S-band not supported for future deep space missions
C-Band Commercial	3.7 to 4.2 (down) 5.925 to 6.425 (up)	N/A	Communications satellites
X-Band Earth Science	8.0 to 8.4 (down) $B_N \leq 400$ *MHz*	N/A	Landsat, RADARSAT
X-Band Space Science	8.4 to 8.45 (down) $B_N \leq 10$ MHz	8.45 to 8.5 (down)	Many near Earth and deep space missions.
K_u-Band Commercial	~11 (down) ~14 (up)	N/A	Direct broadcast. Specific frequencies vary by region.
K_u-Band TDRSS	13.9 (forward) 15.0 (return)	N/A	Shuttle and Space Station links to TDRSS. Note: frequency band being phased out for space missions.
K_a-Band Commercial	17.7 to 20.2 (down) 27.5 to 30.0 (up)	N/A	Direct broadcast. Fixed Satellite Service.
K_a-Band TDRSS and Lunar	22.6 to 23.6 (forward) 25.2 to 27.2 (return) 40 to 40.5 (forward) 37 to 38 (return)	N/A	High data rate links through TDRSS. Lunar trunk link.
K_a-Band Deep Space	N/A	34.2 to 34.7 (up) 31.8 to 32.3 (down) 37 to 38 (return) 40 to 40.5 (forward)	Mars Reconnaissance Orbiter (MRO) high data rate planetary science link. Deep space trunk link.

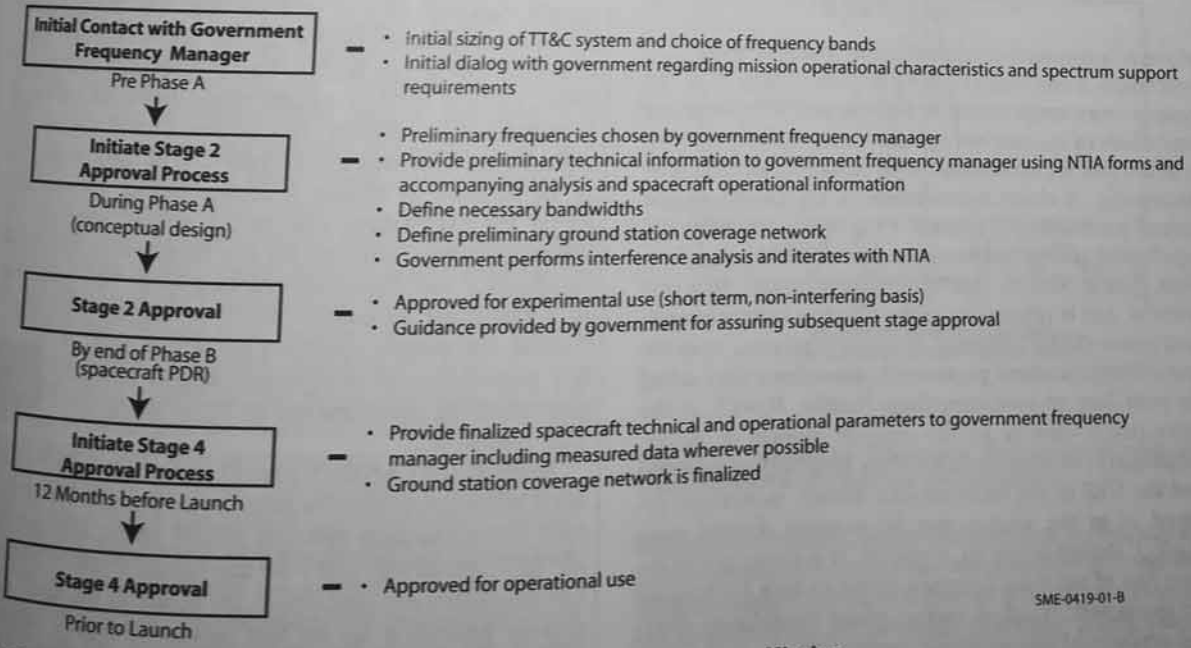


Fig. 21-1. Example Flow of NTIA Approval Process for an Operational Space Mission.

density at the receiver system becomes $P_tG_t/(4\pi d^2)$. The receiver system will have an antenna with an *effective aperture area* A_{er} that intercepts the power density at that location and converts it into a received power level at the output port of the antenna. The received power, P_r is:

$$P_r = \frac{P_t G_t}{4\pi d^2} A_{er} \qquad (21\text{-}1)$$

This equation is intuitive for large aperture antennas where one can envision the physical aperture of the antenna to be intercepting the power density present at its surface (in practice the effective aperture area is less than the physical aperture area by a factor of η which represents the overall efficiency of the antenna). Antenna theory can be used to relate the effective aperture area of any antenna to its gain, resulting in the expression $A_e=(G\lambda^2/4\pi)$ where G is the gain of the antenna (either transmit or receive) and λ is the wavelength of the RF signal. Using this expression, the received power of the communications link can now be expressed as either:

$$P_r = \frac{P_t A_{et} A_{er}}{d^2 \lambda^2} \qquad (21\text{-}2)$$

$$P_r = \frac{P_t G_t G_r}{(4\pi d/\lambda)^2} \qquad (21\text{-}3)$$

where A_{et} is the transmit antenna effective area and G_r is the receive antenna gain. Equations (21-1) through Eq. (21-3) are three forms of the Friis link equation. The equations can be modified to include the effect of thermal noise in the receiver system by dividing the total received power by the total system noise power. Using Eq. (21-3), the received signal-to-noise ratio (SNR) is:

$$SNR = \frac{P_r}{N} = \frac{P_t G_t G_r}{(4\pi d/\lambda)^2 k T_s B} \qquad (21\text{-}4)$$

where N is the receiver system noise power in Watts, k is Boltzmann's constant (1.38×10^{-23} W/K·Hz), T_s is the system noise temperature in Kelvin, and B is the noise bandwidth of the receiver system in Hz. Additional losses in the link are accounted for by reducing the SNR proportionally. A closer examination of Eq. (21-4) reveals several parameters of interest. First, P_tG_t is numerically equal to the power that would be transmitted by an equivalent system with an isotropic radiator instead of a real antenna, and is referred to as the *effective isotropic radiated power* (EIRP). Second, $(4\pi d/\lambda)^2$ is referred to as the *space loss* (a unitless parameter), sometimes also called the *path loss* or *spreading loss*. Finally, $N_o=kT_s$ is the *noise power density* of the receiver system in W/Hz. Equation (21-4) is general and may be modified to represent the *SNR* of the received data stream, commonly referred to as the *energy per bit-to-noise density ratio* (E_b/N_o). Before doing so, however, it is important to review two of the most common forms of data modulation for spacecraft: residual carrier phase modulation (PM) and suppressed carrier phase modulation.

Table 21-3◀, Fig. 21-1◀, Eq. 21-4

Modulation Techniques

Residual Carrier Phase Modulation (PM) is the most commonly used TT&C modulation technique for both military and civilian space applications. It apportions part of the transmitted power into the data and part into a discrete carrier component to enable precise phase tracking of the carrier by the ground station for demodulation and Doppler tracking purposes. In the absence of ranging and assuming rectangular (unfiltered) data bits, the proportions of transmitted power in the data and carrier are $P_t\sin^2(\beta)$ and $P_t\cos^2(\beta)$, respectively, where β is the *phase modulation index*. Each data bit (after any forward error correction coding) is typically split in time with the polarity reversed for the second half of the bit time to produce a spectrum that lacks energy around 0 Hz to minimize interference with phase tracking of the carrier. This type of data formatting is often referred to as bi-phase or Manchester line coding. Figure 21-2a illustrates the resulting modulated RF spectrum of residual carrier PM. Note that the "main lobe" of the spectrum (defined as the portion between first nulls on each side of the carrier) is 4 times the coded bit rate, resulting in a relatively inefficient transmitted spectrum. Residual carrier PM is often used for uplinks and for downlinks with low-to-moderate bit rates where the transmitted bandwidth is not a driver. It can also be used in the absence of Manchester line coding to conserve spectrum width if suitable measures are taken to ensure that long strings of 1s or 0s in the data are broken up. Pseudo-randomization of the data before modulation is a common way to accomplish that objective. The transmitted spectrum will then appear as shown in Fig. 21-2b except with the residual carrier component at center frequency still present. A derandomizer must be used in the ground receiver system to recover the original data pattern.

The phase modulation index, β is set by the designer usually between 45 deg and 90 deg. The modulation technique approaches the theoretical best efficiency possible as β approaches 90 deg. Values of 60 deg and higher are typically used for normal-mode downlink bit rates to maximize the received data power. Lower values are typically used for very low bit rate (10s of bps) emergency mode links to ensure adequate carrier component power for maintaining lock in the ground receiver. A subcarrier is typically used for low bit rate data to reduce interference with the carrier tracking loop. The presence of a sinusoidal subcarrier and/or a ranging waveform changes the simple equations presented here. A full explanation of modulation modes and the corresponding equations can be found in the CCSDS standards.

Suppressed Carrier Phase Modulation, which places all of the power into the data, is also popular for spacecraft communication systems. *Binary phase-shift keying (BPSK)* corresponds to $\beta = 90$ deg (phase symbols spaced 180 deg apart) and maximizes the power in the data. It is popular because it has the best power efficiency theoretically possible and is relatively simple to implement on

the spacecraft. Each data bit (after any forward error correction coding) is modulated directly onto the carrier resulting in the spectrum illustrated in Fig. 21-2b. The main lobe width is 2× the coded bit rate, or half the width of the residual carrier PM spectrum because the data bits are modulated directly (not split in polarity) onto the carrier. BPSK is considered for higher bit rate uplinks (i.e., memory uploads) and moderate bit rate downlinks. *Quadrature phase-shift keying (QPSK)* encodes pairs of coded bits into phase symbols spaced 90° apart to reduce the transmitted main lobe width by a factor of two compared to BPSK (Fig. 21-2c). It is popular for high data rate communications because its power efficiency is also the best theoretically possible (same as BPSK) but in half the transmitted bandwidth. A filter is often included in the transmitter to ensure that the QPSK spectrum rolls off at the NTIA-required rate of –12 dB per octave beyond the necessary bandwidth points. Care must be taken in designing the filter to prevent substantial spectrum "re-growth" once the signal passes through a non-linear power amplifier.

Recovery of either BPSK or QPSK must be done in the ground receiver using a nonlinear operation to regenerate a coherent carrier reference signal for demodulation. This operation introduces ambiguities in the absolute phase of the coherent carrier reference signal. Compensation for this ambiguity must be incorporated into the spacecraft design and might be accomplished through differentially encoded data, transmission of a known pattern, or error correction coding. Other downsides of using BPSK or QPSK are their sensitivity to phase noise and additional "squaring" or "quadrupling" noise generated in the nonlinear carrier recovery tracking loop. Nevertheless, these modulation techniques are very popular due to their relatively simple implementation on the spacecraft, high power efficiency, and, in the case of QPSK, relatively narrow main lobe width.

Further reduction of the transmitted spectrum width and sidelobe levels can be achieved through more advanced modulation techniques. Techniques that encode more bits into each transmitted symbol, such as 8-PSK and 16-PSK, can be used to reduce the width of the main lobe of the spectrum at the expense of substantially reduced power efficiency relative to QPSK. *Gaussian Minimum-Shift Keying (GMSK)* applies special filtering and encoding of the data to generate symbols with gradual phase transitions between them, thereby substantially reducing the sidelobe levels at the expense of a slightly wider main lobe bandwidth compared to QPSK. GMSK has been used widely for cell phone applications and is gaining popularity in the space community for high data rate transmissions. The downside is its reduced power efficiency relative to QPSK and increased complexity on both the spacecraft and ground ends of the link. Generally speaking, bandwidth-efficient modulation techniques, whether seeking to reduce the width of the main lobe or reduce the level of the sidelobes, will increase the complexity of the implementation and reduce the power efficiency of the link relative to basic modulation techniques. Nevertheless, bandwidth-efficient techniques are gaining popularity as data rate requirements increase and implementation becomes easier as a result of advanced digital processing technologies [Deutsch, 2008], [Wesdock, 2008].

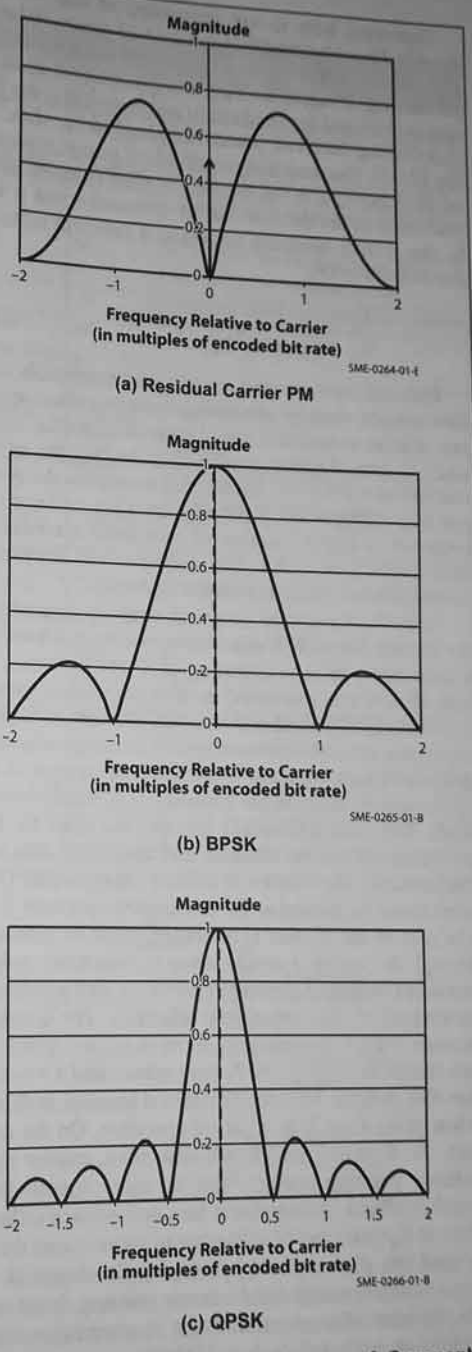


Fig. 21-2. Normalized Magnitude Spectra of Commonly Used Spacecraft Modulation Techniques.

Returning now to our discussion of link analysis, Eq. (21-3) can be modified for residual carrier PM modulation by recognizing that the energy in each transmitted bit (E_b) is equal to $(P_r/R)\sin^2(\beta)$ where R is the bit rate (substituted for bandwidth) and $P_r\sin^2(\beta)$ is the fraction of total received power contained in the data. See Eq. (21-5). The remaining received RF power, represented by $P_r\cos^2(\beta)$, is the fraction of total received power contained in the discrete carrier component and is used by the ground receiving system as a coherent reference for demodulation.

$$\frac{E_b}{N_o} = \frac{P_t G_t G_r}{(4\pi d/\lambda)^2 k T_s R}\sin^2(\beta) \qquad (21\text{-}5)$$

Forward error correction coding adds symbols to the data stream, thereby increasing the transmitted symbol rate relative to uncoded data. Common examples include rate-1/2 convolutional coding that doubles the symbol rate and rate-1/6 Turbo coding that multiplies the symbol rate by a factor of six. Generally speaking, codes that incorporate a higher number of additional symbols perform better in noise but increase the transmitted bandwidth, resulting in a classic tradeoff.

Table 21-4 provides downlink analysis examples for the FireSat II and SCS missions operating at S-band, and a deep space mission operating at X and K_a-bands. The unit of power is expressed in dBm (decibels relative to 1 milliwatt). The FireSat II and SCS missions incorporate a low gain antenna that must provide coverage over an entire hemisphere, resulting in a worst-case gain of –4 dBic (dB relative to an isotropic antenna that is circularly polarized). The main differences between the links for these two missions are the distance and associated data rates. Additionally, the FireSat II mission incorporates QPSK modulation to minimize the transmitted spectrum width. The cost of the FireSat II hardware might be minimized through the use of a single-string architecture, radiation resistant (instead of hardened) hardware, and an appropriate tradeoff of risk versus parts reliability. The deep space mission is fairly aggressively sized to achieve a high data rate return, including 100 W transmitters and a 3 m diameter dish antenna. Note the substantial increase in data rate when going from X to K_a-band operation. On the downside, the K_a-band system will cost more, require tighter antenna pointing control, and be more susceptible to weather effects. Atmospheric loss and other weather effects at K_a-band can be mitigated to some extent through ground site diversity and by varying the downlink data rate as a function of ground antenna pointing elevation angle. Weather effects would not be an issue for crosslinks with other spacecraft such as TDRSS.

Downlink margin requirements are typically determined by uncertainties in the link analysis, sponsor requirements, and company policy. Typical values for planning purposes might be 6 dB for a near Earth mission (where more uncertainty in the spacecraft antenna patterns and ground station performance exist) and 3 dB for a deep space link (where the ground station performance is highly characterized). Lower operational link margins can be used, especially when the data rate is adjustable or retransmission of erroneous frames is automatically done. In the latter case the use of the CCSDS File Transfer Protocol (CFTP) provides the capability of automatically retransmitting files, thereby allowing for lower operational link margin and a higher frame error rate. Uplink margins are treated in the same manner but are often set to be substantially higher than the downlink margins to ensure mission robustness in the event of unforeseen attitude control problems. Generally speaking, higher link margins on both the downlink and uplink, especially for emergency modes, result in additional mission robustness but can be expensive and hard to justify because of the difficulty in quantifying unforeseen problems at the design phase stage.

Table 21-3◀, Fig. 21-2◀, Eq. 21-5

The Decibel

A bel is simply the base-10 logarithm of a unitless power ratio: 2 converts to 0.3 bel, 100 to 2.0 bel. It is more convenient to work with tenths of bel's, so the decibel (abbreviated dB) is used in practice (2 converts to 3 dB, 100 converts to 20 dB). The satellite communications link equation is multiplicative, with more than a dozen terms depending on the complexity of the link. Some terms differ by 30 orders of magnitude (Boltzmann's constant is on the order of 10^{-23} W/Hz-K, and satellite distances may be millions of km, or 10^9 m). Converting all terms in the link equation to decibel form allows for easier manipulation, because multiplying two numbers is equivalent to adding their logarithms. Many elements in a link equation have units related to absolute power. These elements can be converted to dB-form by comparing them to a reference value and appending a suitable identifier. For example, transmit powers are frequently listed in units of dBW (power relative to 1 W) or dBm (power relative to 1 mW). In this way, a 2 kW transmit power can be represented as 33 dBW (= $10\log_{10}(2000)$). Other factors like bandwidth and noise temperature are proportional to noise power, and the same conversion applies. The term dB-Hz refers to a bandwidth relative to 1 Hz, and dB-K to an absolute temperature relative to 1 K. Antenna gain is commonly denoted in dBic (gain relative to an isotropic, circularly polarized antenna).

21.1.5 Hardware Design and Key Technologies

Figure 21-3 shows a representative block diagram of a redundant TT&C subsystem. The subsystem typically consists of five basic hardware elements: transponder, power amplifier, diplexer, RF switching/combining network, and antennas. A wideband transmitter might be in-

Table 21-4. Downlink Analysis Examples.

Parameter	Sym	Near Earth Links		Deep Space Links		Notes
		FireSat II	SCS	Mars	Mars	
RF Frequency (GHz)	f_c	2.2	2.2	8.4	32	
Distance to Ground Station	d	2,560 km	26,000 km	2.5 AU	2.5 AU	Antenna ground elevation angle =5 deg
Information Bit Rate	R	5 Mbps	50 kbps	500 kbps	2.1 Mbps	Before any error correction coding
Phase Modulation Index (rad pk)	β	QPSK	1.2	1.2	1.2	Typical for non-emergency data rates
Transmit Power (dBm)	P_t	36	36	50	50	4 W SSPA for FireSat II and SCS; 100 W TWTA for deep space
Transmit Passive Loss (dB)	L_p	–2	–2	–2	–2	Between the transmitter and antenna
Transmit Antenna Gain (dBic)	G_t	–4 (Hemi)	–4 (Hemi)	46 (3-m dish)	58 (3-m dish)	Includes pointing loss
EIRP (dBm)	$P_tG_tL_p$	30	30	94	106	
Path Loss (dB)	$(4\pi d/\lambda)^2$	–167.5	–187.6	–282.4	–294.0	
Atmospheric Loss		~0	~0	–0.3	–1.0	No rain
Ground Antenna Gain (dBic)	G_r	45.0	45.0	68.2	79.0	10-m dish for FireSat II and SCS; 34-m dish for deep space
Total Received Power (dBm)	P_r	–92.4	–112.6	–120.5	–110.0	
Data-to-Total Power (dB)	$\sin^2(\beta)$	0	–0.6	–0.6	–0.6	
System Noise Density (dBm/Hz)	$N_o = kT_s$	–174.6 (T_s=250 K)	–174.6 (T_s=250 K)	–183.8 (T_s=30 K)	–179.6 (T_s= 80 K)	
Received E_b/N_o (dB)	$\frac{P_r \sin^2(\beta)}{N_o R}$	15.2	14.5	5.7	5.7	
Required E_b/N_o (dB)		5.5	5.5	1	1	Rate ½ convolutional coding for FireSat II and SCS; Rate 1/6 Turbo coding for deep space
Receiver System Loss (dB)		–3	–3	–1	–1	Relative to theory
Link Margin (dB)		6.7	6.0	3.7	3.7	

Note: losses are represented by negative decibels. Other small losses not included in example.

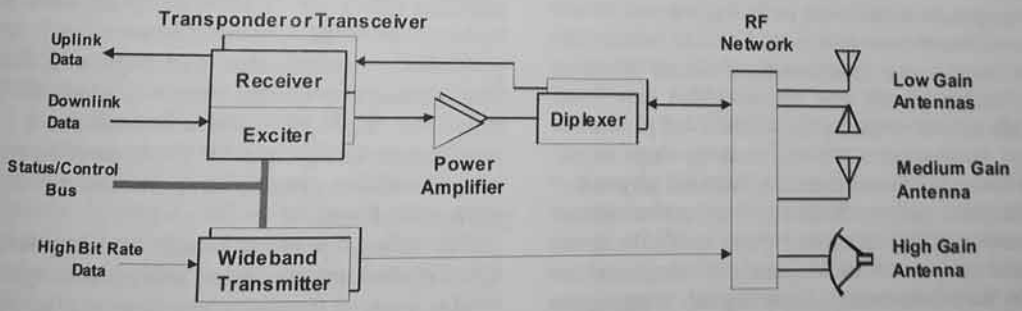


Fig. 21-3. TT&C Subsystem Block Diagram.

cluded depending on the data return requirements of the mission. The power amplifier is often built into the transponder and wideband transmitter when the RF output power is relatively low ($\leq$ 5W) but may be a separate standalone component for higher power levels. Cross-strapping between redundant systems is not shown but can be added at the cost of greater complexity.

Transponders/Transceivers: the transponder or transceiver is the heart of the subsystem because it provides for modulation and demodulation and controls the modes of operation of the subsystem. The transponder incorporates receive, transmit, two-way Doppler tracking, and ranging functions. It generates the downlink signal by phase-locking to the uplink signal and deriving the downlink signal in phase coherence with the uplink signal. This process results in a fixed "turnaround ratio" for the transponder that enables accurate orbit determination from observation of the two-way Doppler shift as a function of time. Standard downlink/uplink turnaround ratios are 256/205 for SGLS, 240/221 for NASA S-band and 880/749 for NASA deep space X-band transponders, although other turnaround ratios are possible for special applications. A basic transpon-

Table 21-4, Fig. 21-3, Eq. 21-5◀

der will incorporate serial data and clock signals for the downlink telemetry and uplink command data. Transfer of the downlink data can also be done via spacecraft bus (i.e., MIL-STD-1553) [DoD, 1978] at the expense of additional power and cost. A trend for the future is to incorporate a packetized data interface for the downlink telemetry to enable the use of commercial standards (i.e., USB) and increase the flexibility of the transponder in selecting bit rates and coding modes. Mode controls and status telemetry for the transponder can be implemented in several ways including discrete signal lines, universal asynchronous receiver/transmitters (UARTs), or the spacecraft bus (i.e., MIL-STD-1553). The transponder often includes a "pass-through" channel that enables uplink ranging modulation to be turned around and modulated onto the downlink signal, thereby enabling accurate range measurements to be made by the ground station for orbit determination. Most ranging systems today are simply "elbows" or "bent-pipes", where the uplink modulation and associated noise are passed through to the downlink. A trend for the future is to incorporate regenerative ranging where the transponder locks to the uplink ranging signal and regenerates it for modulation onto the downlink, as in a digital repeater. This technique results in less power wasted in the downlink on turnaround ranging noise and more power available for data return and ranging.

A simpler and lower-cost alternative to the transponder is a *transceiver system* that provides separate receive and transmit functions without phase coherence between the signals. The downlink signal frequency will be dependent on the stability of the on-board oscillator, which will limit the accuracy of Doppler shift measurements. The transceiver alternative is viable for missions that rely on alternate means for orbit determination such as an on-board GPS receiver or radar skin tracking.

The transponder/transceiver technology area is poised for substantial improvements in the future as modern analog and digital signal processing technology becomes available for spacecraft use. Great strides have been made in the past 20 years on the reliability of plastic encapsulated microcircuits (PEMs), making them candidates for use in spacecraft instead of hermetically sealed parts. This trend enables the designer to take advantage of low-power, highly integrated parts available in the commercial sector such as frequency synthesizers, upconverters/downconverters, direct digital synthesizers (DDS), and vector modulators. Digital signal processing technology is already incorporated into most modern transponders, and the potential for greatly improved performance in the near future exists. Current designs make use of application-specific integrated circuits (ASICs) to achieve a combination of small size, low power, and radiation hardness. The ASIC design cycle, however, can be relatively expensive resulting in a tendency to resist changes after the circuits are fabricated. Future transponder designs will make increasing use of radiation-resistant field programmable gate arrays (FPGAs) to provide the designer with more flexibility for incorporating custom signal processing functions and making late design changes. These FPGAs are currently available with millions of gates. Open-source processors are available for downloading into the FPGAs and custom digital signal processing functions can be designed into them. Software design tools are relatively mature at this point. As the digital technology continues to improve, software-defined radios (SDRs) will become increasingly viable for spacecraft use. These radios have great functionality and flexibility but typically have high power and mass relative to conventional TT&C transponders. Improvements in technology, particularly the availability of low-power general-purpose processors, will make SDRs a more attractive choice for some missions in the future. Improvements in on-board digital processing will naturally enable more functionality and reconfigurability of the TT&C transponder; for example, increasing use of uplink forward error correction coding, reconfigurable receiver loop bandwidths, downlink low density parity-check coding (LDPC), FFT-based uplink signal acquisition, and bandwidth-efficient modulation techniques such as Gaussian minimum-shift keying (GMSK).

Power Amplifiers: power amplification of the downlink signal is accomplished either with a solid-state power amplifier (SSPA) or traveling wave tube amplifier (TWTA). The SSPA is relatively low mass and size compared to the TWTA, simple to implement, and power efficient at RF power levels of 15 W and less operating at RF frequencies of X-band and below. The TWTA becomes an attractive choice at RF power levels of approximately 15 W and above because of its superb power efficiency at those levels. Table 21-5 illustrates typical mass and efficiency performance achieved with current spaceborne power amplifiers. The overall efficiency is defined as the RF output power divided by the spacecraft bus input power. A substantial drop in solid-state power amplifier efficiency occurs when going from X-band to K_a-band operation. This performance may represent a problem for some missions that require K_a-band operation but have very limited bus power available. Improvements in K_a-band solid-state efficiency and the development of K_a-band TWTAs operating at lower RF power levels are potential areas for technology development in the future.

The choice of power amplifier technology is often a fairly straightforward one. Near-Earth missions operating at S-band or lower in frequency benefit most often by the small size and relative low cost of a solid state amplifier built into the transponder or transceiver. Missions that require high EIRP due to a high data rate requirement or large Earth distance may benefit from the high RF power and efficiency of a TWTA that is purchased separately from the transponder or transceiver. Some special requirements, such as high shock, rapid power cycling, or the need for an active phased array antenna, will drive a mission to solid-state technology at any of the frequency bands.

Diplexers: the diplexer is a filtering component that couples the receiver and transmitter signals together onto a single RF cable or waveguide. It provides filtering for the receiver input to prevent interference and overload

Table 21-4◀, Fig. 21-3◀, Eq. 21-5◀

Table 21-5. Power Efficiency and Mass for Spaceborne Power Amplifiers.

Frequency Band	SSPA		TWTA	
	Overall Efficiency	Mass	Overall Efficiency	Mass
S-band	30–40%	0.5 to 1.5 kg, depending on RF output power, including power converter	60%	~2.5 kg including power converter
X-band	20–28%		50–60%	
K_a-band	11–17%		50–60%	

from the transmit signal and other signals including those on the spacecraft and those present during launch and flight operations. It provides filtering at the transmitter output to reduce output noise at the receive frequency and reduce harmonics. Because of its highly resonant nature, the diplexer must be carefully specified and designed to prevent a high vacuum breakdown phenomenon called *multipactor discharge*. This phenomenon can occur at RF power levels as low as a few watts at S-band. Many missions have suffered schedule delays late in the development cycle because of a multipactor discharge problem. Multipactor discharge is of particular concern in filters but can occur anywhere that high RF power levels exist including the power amplifier, RF cables, the RF switching/combining network, and antennas.

RF Network: the RF network design can vary greatly depending on the redundancy and operational requirements on the TT&C subsystem. In its simplest form, the network is simply an RF power divider that splits the RF signal between two hemispherical coverage antennas. This type of network is often used on missions that require omni-directional antenna coverage. The actual coverage pattern is relatively constant over the main lobe area of each antenna but becomes highly variable where the two antenna patterns meet because of interferometry nulls. Alternatively, the network could consist of an RF switch that selects between two hemispherical coverage antennas. This arrangement creates higher overall antenna gain (the signal is not split) than the omni-directional pattern but the spacecraft must select which antenna to use. Some missions incorporate a combination of low gain antennas, a medium gain antenna, and a high gain antenna, thereby driving the RF network to a more complex combination of switches. Care must be taken to mitigate potential single point failures when designing the RF network.

Antennas: the design of the spacecraft antenna system is tailored to accommodate the mission coverage requirements, orbital parameters, spacecraft attitude control characteristics, and bit rate requirements. Antennas are usually custom-designed, although some might be purchased as "off-the-shelf" items. The antenna directs the radiated power into the appropriate regions of space to maximize the efficiency of the RF transmission through free space. A key metric for an antenna is its *gain*, defined as the power density radiated in a given direction relative to that of an isotropic (uniform) radiator. The antenna gain includes inefficiencies in the design such as spillover and resistive losses and is usually expressed in dBi for a linearly polarized antenna and dBic for a circularly polarized antenna. Most civilian and military space missions incorporate circularly polarized antennas to prevent mismatches in physical orientation between the spacecraft and ground station antennas as a function of orbit location. In contrast, some commercial geosynchronous satellites with a fixed orientation will incorporate linearly polarized antennas.

Key tradeoffs when specifying antennas are between beamwidth, gain, and effective aperture area (size). The beamwidth is inversely related to both the gain and effective aperture area. A narrow-beam antenna will have relatively high gain and large size because the radiated power is focused in a given direction. It is typically used for high rate mission payload data return. Conversely, a broad-beam antenna will have relatively low gain and small size because the radiated power is spread over many directions. It is often used as the primary antenna for near-Earth missions with modest downlink bit rate requirements. It is also used whenever the spacecraft attitude is not precisely controlled or known, for example, launch and early operations, emergency mode recovery, thrusting maneuvers, and Doppler tracking during science acquisition periods. A conical or fan beam antenna, often termed "medium gain", is useful on some missions with special requirements. For example, they can be used on deep space missions with limited RF transmit power that cannot establish an emergency downlink using a low gain antenna. For initial design purposes, one can use the information and rules-of-thumb in Table 21-6 to determine the characteristics and applications of various antenna types.

Shaping the beam pattern of an antenna can be used to provide advantages to missions. For example, in the common LEO case of a nadir-pointed broad-beam antenna, the edges of the beam pattern can be increased in gain compared to the boresight to account for the extra path loss to the ground station at these look angles. This technique provides a more constant receive power at the ground station, and increases the link data rates that are typically sized for the worst-case path loss/look angle. The fan beam antenna has a narrower beamwidth in one axis than the conical-beam antenna. If the spacecraft can rotate in one dimension, then the fan beam can be used to apply more gain to TT&C links, particularly useful in deep space missions.

Antenna placement can be challenging, and should be addressed early in the design phase. Integrating broad-beam, omni-directional antennas onto a spacecraft with large extended structure or solar arrays can substantially

Table 21-5, Fig. 21-3◀, Eq. 21-5◀

perturb the beam pattern. The beam pattern will be affected by any protrusion into or near its field of view, or by reflective surfaces in the back-lobe of the antenna. Achieving a perfectly clear field of view is not always possible. Gain measurements should be taken with the antenna on a representative satellite mock-up at an antenna range to check for destructive interference in the antenna gain pattern.

Spin-stabilized missions present a particularly challenging problem to the antenna designer. The antenna pattern must often be co-aligned with the axis of rotation, resulting in the need to keep the spin axis pointed at Earth or to use a relatively broad antenna pattern to cover the Earth direction as the geometry varies. A "pancake" or "donut" antenna pattern with constant gain about the equator of the spacecraft can be used on missions that keep the spin axis aligned in a direction perpendicular to the Earth orbit plane. Electronically despun antennas, which switch between radiating elements as the spacecraft rotates, can be used to keep a relatively narrow beam pointed in the Earth direction. These antennas incorporate complex feed and switching networks and depend on accurate knowledge of the Earth direction as the spacecraft rotates.

A few words should be said about electronically steered phased-array antenna technology. This technology can simplify spacecraft packaging and operations by eliminating the need for a mechanical gimbal to point the antenna. Its use in spacecraft designs to date, however, has been relatively limited because of the high cost and mass of the antenna, especially when two-dimensional scanning is required. Two notable accomplishments are the X-band one-dimensional scanned antenna flown on the MESSENGER mission to Mercury and the X-band two-dimensional scanned antenna flown on the Earth Orbiting-1 (EO-1) mission. Both of these antennas operated

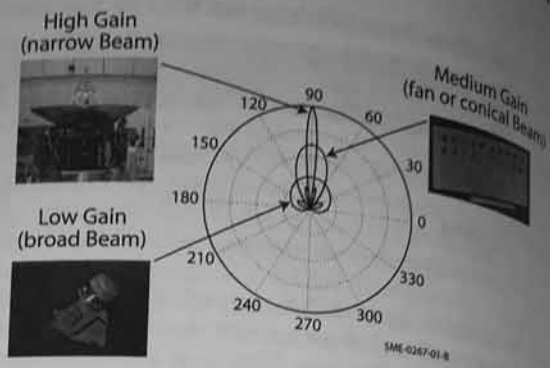


Fig. 21-4. Representative Antenna Technologies and Associated Beam Patterns.

in transmit-only mode and had distributed solid-state power amplifiers to maximize transmit efficiency. Hopefully, further applications for electronic scanning will drive the technology developments in packaging and electronics necessary to enable low-cost, low-mass electronically steered antennas in the future.

Wideband Transmitters: the wideband transmitter is an optional element depending on the data return requirements of the mission. Moderate or low data rate requirements (up to, say, 6 Mbps) can often be handled by the TT&C transponders assuming that the transmitted spectrum is consistent with NTIA limits. Additional wideband transmitters, typically operating at X-, K_u-, or K_a-band, can be added to the subsystem for higher data rate operations. These transmitters may have an integral power amplifier depending on RF output power requirements and are typically connected directly to the high gain antenna to minimize loss.

Table 21-6. Antenna Characteristics and Applications.

Antenna Type	Full 3 dB Beamwidth	Gain	Applications	Notes
High Gain Parabolic Reflector Antenna	$BW_{3db} = 66\lambda/D$ λ = wavelength D = diameter The beamwidth is typically in the single-digit degrees.	$G = 10\log(4\pi A_e/\lambda^2)$ $= 10\log(\pi^2 D^2/\lambda^2)\ \eta$ A_e = effective aperture area λ =wavelength η= overall efficiency	High data rate downlink/return link.	Effective aperture area is equal to the physical aperture area times the overall efficiency, typically between 55 and 65%.
High Gain Phased Array Antenna	Dependent on element and array pattern design. Basic tradeoffs between beamwidth, gain, and size still apply.		Flat packaging requirement. Electronic beam steering requirement.	Electronically steering antenna can be used to eliminate mechanical gimbals.
Medium Gain Antenna	Typically between 10 deg and 90 deg Can be conical beam or fan beam.	6 to 25 dBic	Deep space missions for emergency mode or to fill coverage gap at large Earth distances.	Might be used to reduce transmitter power required to support emergency mode communications. Examples include a small patch antenna array, waveguide horn, small dish reflector.
Low Gain Antenna	90 deg to 180 deg	0 to 6 dBic	Normal mode communications for Earth orbiters. LEOP coverage. Emergency mode coverage. Extended Doppler tracking for deep space missions.	Examples include single patches, quadrifilar helix, open-ended waveguide.

Table 21-6, Fig. 21-4, Eq. 21-5◀

21.1.6 Subsystem Design Examples

Tables 21-7 and 21-8 present TT&C subsystem design examples for representative near-Earth and deep-space missions, respectively. Included are component quantities, bus power consumption, and masses for representative redundant systems. These examples correspond to the traditional "box and harness" approach to spacecraft design where the major hardware functions (transponder, power amplifier, diplexer) are implemented as standalone electronics boxes that are integrated together at the subsystem and spacecraft levels. This approach provides for a structured, systematic approach to the spacecraft bus design. However, the box and harness approach may result in excessive mass and power consumption for extremely small spacecraft designs such as microsats and nanosats. In these cases, the TT&C hardware could be implemented as board-level products that are carefully integrated with the other subsystems within the spacecraft structure. Such a highly integrated approach dictates the need for a small team environment and careful attention to interface and electromagnetic compatibility details. The approach can dramatically reduce total TT&C mass to the 500 g region. Single-board UHF radios are available with receive-only DC power consumption in the 100 to 200 mW range, and RF transmit powers around 5 W (40% overall efficiency).

Table 21-7. Representative Power and Mass Summary for a Near-Earth TT&C System.

Unit	Qty	DC Bus Power (W)	Total Power (W)	Unit Mass (kg)	Total Mass (kg)	Notes
S-Band Transponder	2	4 (Rx) 40 (Rx & Tx)	8 (2Rx) 44 (2Rx & Tx)	3.5	7	Built-in 5 W SSPA
S-Band Diplexer	2	0	0	0.2	0.4	
S-Band Cables	All	0	0	3	3	Approximately 10 RF cables
S-Band Antennas (low gain)	2	0	0	0.4	0.8	Forward and aft combined to form omni pattern
X-Band Wideband Transmitter	2	60	60	3.5	7	For wideband payload data return. Built-in 5 W SSPA.
X-Band Cables	All	0	0	0.5	1	2 RF cables, typically
X-Band Antenna (high gain)	1	0	0	1.5	1.5	0.5 m diameter parabolic reflector. Gimbal is separate.
TOTAL:			44 W (S only) 104 W (S and X)		20.7	

Table 21-8. Representative Power and Mass Summary for a Deep Space TT&C System.

Unit	Qty	DC Bus Power (W)	Total Power (W)	Unit Mass (kg)	Total Mass (kg)	Notes
X-Band Transponder	2	12 (Rx only) 14 (Rx and Tx)	24 (2Rx) 26 (2Rx & Tx)	3	6	Low-power exciter output (no built-in SSPA)
X-Band TWTA	2	50	50	2.5	5	RF output power= 25 W
X-Band Diplexer	2	0	0	0.6	1.2	
X-Band Switching Network	2	0	0	0.5 to 1.0	1 to 2	Dependent on number of switches. Includes interface plate.
X-Band Cables	All	0	0	5	5	18 RF cables (with SMA connectors)
X-Band Low Gain Antennas	2	0	0	0.7	1.4	Switched waveguide horns with 70 deg. Beamwidth (3 dB)
X-Band Medium Gain Antenna	1	0	0	1.5	1.5	0.3 m waveguide horn with 5 deg beamwidth (3 dB)
X-Band High Gain Antenna	1	0	0	6	6	1.5 m diameter parabolic reflector
K_a-Band Exciter (Modulator)	2	3	3	0.3	0.6	incremental increases to X-band transponder
K_a-Band TWTA	2	81	81	2.8	5.6	RF output power= 40 W
K_a-Band Waveguide	All			3	3	
K_a-Band Antenna (High Gain)	1			2.5	2.5	Additional feed and structure added to existing 1.5 m diameter parabolic reflector. Gimbal is separate.
Ultrastable oscillator	2	2.5	5	1.3	2.6	
TOTAL:			81 W (X Tx/Rx) 165 W (X Tx/Rx & K_a Tx)		41.4 to 42.4	

21.1.7 Special Topics

Very High Bit Rate Communications

For almost all missions, significant data return is critical to mission success. Data rates have increased with technology, and the restricted bandwidths in the S- and X-bands are pushing missions to utilize K_a-band frequencies for data rates in excess of 10's of Mbps.

The move to higher frequencies (even to optical bands) comes with substantial challenges that require significant investments in advanced technology. Atmospheric losses and outages at K_a-band and optical frequencies are much higher, especially losses due to rain, which may require spatial diversity on the ground and the use of transmission protocols to overcome. Higher operating frequencies demand a tighter surface accuracy specification on the antennas and more accurate pointing from the attitude control system. Though the TT&C functions can be included in the high bit rate links, concerns regarding communications during spacecraft emergencies, such as attitude control anomalies, require that a lower rate, lower frequency capability is retained for robustness.

Emergency Mode Recovery

The design of the TT&C subsystem will frequently be driven by how the system must support the spacecraft during emergencies. The spacecraft may experience problems that cause it to lose knowledge or control of its attitude. With unknown or anomalous attitude, maintaining communications is critically important in reestablishing control before batteries drain too low or faults progress further, ending in mission failure. It is the unfortunate lot of the TT&C subsystem engineer that many emergencies on the spacecraft manifest themselves first as a loss in communications.

The brute force approach to providing emergency mode TT&C capability is to design the antenna system to provide continuous uplink and downlink coverage over the entire sphere about the spacecraft. Typically two low gain antennas are oriented towards opposite hemispheres on the spacecraft, and a power splitter connects them to the rest of the RF system. Commands can be transmitted successfully to the spacecraft and telemetry received no matter what attitude the satellite exhibits. The drawback to this approach lies in the use of the splitter. Where the two hemispherical antenna patterns meet, the resultant sum pattern will experience significant nulls when the patterns of the two LGAs combine destructively. Additionally, half the RF power is transmitted through the opposite antenna. This scheme works best in LEO missions with robust link margins. Alternatively, a switch may be used instead of a power splitter, which removes the nulling effect and focuses all of the RF power into one hemisphere but requires the satellite to autonomously switch hemispheres over time to guarantee that a link can be established.

For missions at longer ranges, a hemispherical coverage antenna may not provide sufficient gain to establish a low rate emergency link, and a higher gain TT&C antenna must be used. Most spacecraft are able to robustly point in the direction of the Sun, even in emergency situations. If the maximum angular separation of the Earth from the Sun is known, a medium gain antenna might be used to provide coverage. One solution is to point a conical-beam antenna at the Sun and wobble the rotation axis of the spacecraft to cause the antenna beam to sweep out a wider volume of space than its actual beam pattern. Another solution is to use a fanbeam antenna with a rotating spacecraft. The spacecraft rotates slowly about the spacecraft-Sun line, and the RF power received from the spacecraft is monitored on the ground. When the fanbeam antenna sweeps through the Earth direction, the increase in received RF power is seen on the ground. The spacecraft can then be commanded to stop rotating at precisely the correct time to leave the fanbeam antenna pointed directly at Earth, reestablishing communications. This approach was used successfully on the NEAR and MESSENGER deep-space missions.

Ultrastable Oscillators

Ultrastable oscillators (USOs) are precision, temperature-compensated, oven-controlled crystal oscillators that exhibit extraordinary short-term frequency stability, with Allan deviations* of approximately $\Delta F/F = 3\times10^{-13}$ measured in time intervals of 1 to 100 seconds and excellent phase noise. This performance is achieved for relatively low mass and power compared to an atomic clock which provides better long term stability. USOs consume approximately 2.5 W or less with about 1 kg mass, and are trending lower in both categories.

USOs can be used to the advantage of the TT&C subsystem and satellite design in several ways. They permit precise timekeeping on the satellite over large time intervals. One-way Doppler tracking becomes feasible using a USO, permitting autonomous satellite navigation for landing assists. Precision ranging between tandem spacecraft is enabled for gravity field mapping. As a reference source for the TT&C radio's downlink signal, USOs can provide such a stable signal that the received signal on the ground can be used to deduce properties of planetary atmospheres as the planet occults the satellite when viewed from Earth.

21.1.8 Future Trends

Spaceflight TT&C technology tends to lag commercial, terrestrial RF communications technology due to the difficulties and expense involved in qualifying parts for space, particularly due to high-reliability, packaging, and radiation constraints. The gap is closing, as indicated earlier, with the increased use of qualified PEMs and highly integrated RF parts. With advances in space-qualified digital technology, the conversion from RF (analog) to digital is migrating closer to the antenna. Similarly,

* Allan deviation is a measure of the fractional change in oscillator frequency due to noise processes. It excludes systematic changes such as frequency drift and temperature effects.

Table 21-8◀, Fig. 21-4◀, Eq. 21-5◀

software-defined radios (SDRs) are becoming more prevalent with improvements in digital processors and FPGAs. SDRs promise increased in-flight flexibility, reduced hardware assembly costs for subsequent missions (since potentially only the software settings need to be changed), and ease of technology insertion (via software or firmware, without changing the hardware). Mission data rate requirements generally increase with time, requiring wider bandwidths, more power, or more efficient transmitters or coding. The drive to higher data rates and the squeeze at lower frequency bands due to competition from cellular systems and wireless applications (especially at S-band) serve to impel TT&C frequencies to higher bands such as K_a-band. The ultimate limiting factor for the use of higher frequencies is weather effects, which drives complexity in operating the link efficiently.

21.2 Power

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This section describes the *Electrical Power System* (EPS), which provides, stores, regulates, and distributes electrical power to payloads/instruments, and other flight subsystems (i.e., thermal, communication, guidance, and navigation). Nothing on the spacecraft can function without electrical power. The need for more power is growing as power begets powerful machines and enables missions that will change the world. The challenges for space power systems focus around maximizing efficiency, safety, reliability, and radiation harness; while minimizing mass, volume, thermal requirements, and costs (development, procurement, integration, and operational).

The power systems defined in this section are mostly applicable for today's LEO, GEO, planetary, and deep space spacecraft missions and emerging high altitude air ships. In this section we focus on the fundamentals of space electrical power systems, including their mission and requirement drivers, with an emphasis on the processes and methodologies for determination of the power system architectures. The 4 basic components of an EPS, shown in Fig. 21-5, are independently addressed in Secs. 21.2.2 through 21.2.5, along with a mission (FireSat II) sizing example, to provide a basic understanding of these subsystems and the processes used to size and optimize the EPS.

A GEO communications spacecraft power system mass comprises ~30% of the spacecraft dry mass, with percentages to the 4 basic EPS components as shown in Fig. 21-6. Figure 21-6 also depicts the typical cost distribution for the 4 main EPS components. The EPS mass and cost distributions may vary from mission to mission, although one can typically expect the energy source and energy storage systems to be the main players for both

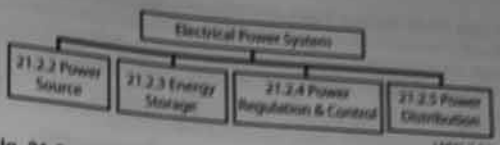


Fig. 21-5. Functional Breakdown for the Spacecraft's Power Subsystem. We start with these 4 basic functions and must determine requirements for the hardware, software, and interfaces for each.

EPS mass and cost. The large mass and cost allocations for the power source and energy storage components dictate early trade studies, usually focused around selection and optimization of the solar arrays and batteries. Sections 21.2.2 and 21.2.3 go into detail for the power source (solar array) and energy storage (battery) systems to help identify the key early EPS trade studies.

Section 21.2.6 summarizes where investments and research are focused for future power systems that may offer new and improved power system architectures and topologies to further support and enable future missions. Based upon the power source and energy storage mass and cost allocations shown in Fig. 21-6, advancements in power generation technologies and energy storage systems are the focus of Sec. 21.2.6.

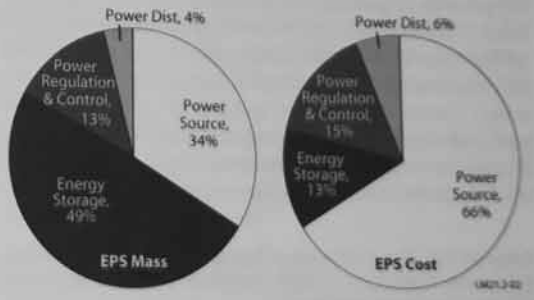


Fig. 21-6. Mass and Cost Breakdown for the Spacecraft's Power Subsystem. The large mass and cost allocations for the energy source (solar array) and the energy storage system (batteries) result in power system trade studies focused around these EPS components prior to design and manufacture.

21.2.1 Mission and Design Requirements

Systems engineering and power system design must begin with an understanding and definition of requirements. Table 21-9 lists typical functions performed by the EPS. The 3 most important EPS sizing requirements are the demands for average and peak electrical power; the orbital profile (inclination and altitude); and mission life. We must identify the average and peak electrical power loads for mission operations during the transfer orbit where there may be limited power generated from the solar array; for operations at beginning-of-life (BOL); and operations at end-of-life (EOL). For many missions, the EOL power demands must be reduced to compensate for power source (e.g., solar array) perfor-

Table 21-8◀, Fig. 21-6, Eq. 21-5◀

Table 21-9. Typical Top-Level Power Subsystem Functions. Each of these functions consists of sub-functions with myriad design characteristics which we must develop to define and meet mission requirements.

- Supply a continuous source of electrical power to spacecraft/vehicle loads during the mission life.
- Control and distribute electrical power to the spacecraft / vehicle.
- Support power requirements for average and peak electrical loads.
- Provide converters for AC and regulated DC power buses, if required.
- Provide command and telemetry capability for EPS health and status, as well as control by ground station or an autonomous system.
- Protect the spacecraft payload against failures within the EPS.
- Suppress transient bus voltages and protect against bus faults.
- Provide ability to fire ordnance, if required.

mance degradation. The average electrical power needed at EOL determines the size of the power source. For any mission, the identification and tracking of spacecraft component average and peak power loads as a function of mission phase are critical to proper requirements definition. Section 14.5.3 shows a sample power budget with typical input data that we may use to begin the sizing process. If peak power requirements are not fully defined, a first-estimate is to multiply average power by 2 or 3 to obtain peak power requirements for attitude control, payload, thermal, and EPS (when charging the batteries). Fortunately, all the systems do not require peak power at the same time during the mission. Table 21-10 summarizes the power subsystem design process, which we discuss further in the following subsections.

Table 21-11 shows the principal effects of mission requirements on the power system design. The definition and tracking of power system requirements are critical for any mission. Requirements will change as the program matures from concept to flight, and we must fully understand requirement changes to properly assess spacecraft/vehicle impacts. The power budget table in Sec. 14.5.3 provides an example template for tracking subsystem electrical power requirements. The key to ensuring a spacecraft power system is properly sized is in the detail and tracking of a power system budget via a mission-specific power budget profile.

With proper requirements definition, we are now ready to examine and select the best power system architecture (Sec. 21.2.2), and then we can work through the design process beginning with the power source.

21.2.2 Power Generation/Sources

A *photovoltaic cell* or *solar cell* is a device that converts sunlight into electricity. Solar cells are connected in series to form a *string*; the number of cells in a string determines the string voltage. A *solar array* is made up of one or more strings in parallel. Earth-orbiting spacecraft at low-Earth to geosynchronous orbits have usually employed photovoltaics as their power source. Often, photovoltaics were the only real candidate for these low-power missions (less than 15 kW) because solar cells were well-known and reliable. Photovoltaic sources are not attractive for interplanetary missions to the outer planets because solar radiation decreases with the square of the distance, thus rapidly reducing the available energy from a solar array. To configure and size a solar array, we must understand cell types and characteristics; solar-

Table 21-10. The Preliminary Design Process for the Power Subsystem. All of these design steps must link back to mission requirements to satisfy the owner and users. Note that derived requirements may impact previous design decisions and force designers to iterate the design process.

Step	Information Required	Derived Requirements	References
1. Identify Requirements	Top-level requirements, mission type (LEO, GEO), spacecraft configuration, mission life, payload definition	Design requirements, spacecraft electrical power profile (average and peak)	Secs. 10.1, 10.2
2. Select and Size Power Source	Mission type, spacecraft configuration, average load requirements for electrical power	EOL power requirement, type of solar cell, mass and area of solar array, solar array configuration (2-axis tracking panel, body-mounted)	Secs. 10.1, 10.2, 14.3.3, 21.2.2, Table 14-20
3. Select and Size Energy Storage	Mission orbital parameters, average and peak load requirements for electrical power	Eclipse and load-leveling energy storage requirement (battery capacity requirement), battery mass and volume, battery type	Sec. 21.2.3
4. Identify Power Regulation and Control and Distribution	Power-source selection, mission life, requirements for regulating mission load and thermal-control requirements	Peak-power tracker or direct-energy-transfer system, thermal-control requirements, bus-voltage quality, power control algorithms	Secs. 21.2.4, 21.2.5
5. Estimate the Mass, Average Power and Peak Power Requirements	Power profile budget from systems engineering that combines all payload and spacecraft housekeeping loads. Mass largely driven by power generation and energy storage.	Average and peak power (W)	Secs. 21.2.1, 21.2.3

Table 21-11. Effects of System-Level Parameters on the Power Subsystem. Most aspects of the mission affect the power subsystem because so many other subsystems require specific power attributes.

Parameter	Effects on Design
Average Electrical Power Requirement	Sizes the power-generation system (e.g., number of solar cells, primary battery size) and possibly the energy-storage system given the eclipse period and depth of discharge
Peak Electrical Power Required	Sizes the energy-storage system (e.g., number of batteries, capacitor bank size) and the power-processing and distribution equipment
Mission Life	Longer mission life (> 7 yr) implies extra redundancy design, independent battery charging, larger capacity batteries, and larger arrays
Orbital Parameters	Defines incident Solar energy, eclipse/Sun periods, and radiation environment
Spacecraft Configuration	Spinner typically implies body-mounted solar cells; 3-axis stabilized typically implies body-fixed and deployable solar panels

array design issues, sizing calculations, configurations, regulation; and radiation and thermal environments. Key design issues for solar arrays include spacecraft configuration, required power level (peak and average), operating temperatures, shadowing, radiation environment, illumination or orientation, mission life, mass and area, cost, and risk. Table 21-12 shows the solar array design process.

Step 1. Mission life and the average power requirement are the two key design considerations in sizing the solar array for most spacecraft. We size a photovoltaic system to meet power requirements at EOL, with the resulting solar array often oversized for power requirements at BOL. This excess power at BOL requires coordinated systems engineering to avoid thermal problems. The longer the mission life, the larger is the difference between power requirements at EOL and BOL. We usually consider photovoltaics a poor power source for missions lasting more than 10 years because of natural degradation in the solar array. Section 21.2.4 (Power Regulation and Control), discusses how we manage excess power from the solar array. The average power requirement can be obtained from Sec. 10.1 and 10.2.

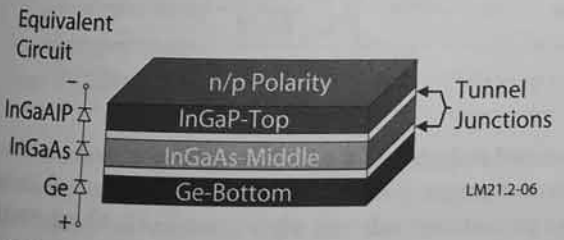


Fig. 21-7. Typical Triple Junction Cell Schematic. Each junction collects energy over a different range of the Solar spectrum.

In designing a solar array, we trade off mass, area, cost, and risk. Silicon cells cost the least but with lower efficiency, so when the total array cost, area, mass and environmental effects are considered, it may not be a better choice than the more costly gallium-arsenide (GaAs) cells. Triple junction GaAs cells have become the standard for most space applications, due to their higher collection efficiency and suitability for use in the higher radiation environment of most space missions. Due to their extensive use, many thousands of hours of on-orbit data is available to predict performance, reducing program risk. In the US, these cells are produced by Emcore and Spectrolab. Figure 21-7 shows a typical triple junction solar cell structure.

A solar array's illumination intensity depends on orbital parameters such as the Sun incidence angles, eclipse periods, solar distance, and concentration of solar energy. Tracking and pointing mechanisms on the solar array often adjust for these influences. If we mount the cells on the body of the spacecraft, we must orient them so they will generate adequate power throughout the mission.

Step 2. To estimate the solar array area required for a spacecraft, we first determine how much power, P_{sa}, the solar array must provide during daylight to power the spacecraft for the entire orbit

$$P_{sa} = \frac{\left(\frac{P_e T_e}{X_e} + \frac{P_d T_d}{X_d}\right)}{T_d} \qquad (21\text{-}6)$$

where P_e and P_d are the spacecraft's power requirements (excluding regulation and battery charging losses) during eclipse and daylight, respectively, and T_e and T_d are the lengths of these periods per orbit. The terms X_e and X_d represent the efficiency of the paths from the solar arrays through the batteries to the individual loads and the path directly from the arrays to the loads, respectively. The efficiency values for eclipse and daylight depend on the type of power regulation: direct energy transfer or peak-power tracking. (A description of these methods follows in Sec. 21.2.4. For direct energy transfer, the efficiencies are about $X_e = 0.65$ and $X_d = 0.85$; for peak-power tracking they are $X_e = 0.60$ and $X_d = 0.80$. The efficiencies of the former are about 5% to 7% greater than the latter because peak-power tracking requires a power converter between the arrays and the loads.

Step 3. Table 21-13 shows the efficiencies and radiation-degradation sensitivities of typical solar cells. Triple

Table 21-12. Solar Array Design Process. In the FireSat II example column, I_d represents inherent degradation, θ is the Sun incidence angle, L_d is life degradation, and X_e and X_d represent the efficiencies of the power distribution paths. The material following the table further explains these quantities

Step	Reference	FireSat II Example
1. *Determine requirements & constraints for power subsystem solar array design*		
• *Average power required during daylight and eclipse*	Input parameter, Secs. 3.5, 10.1, 14.1, 14.3	110 W during daylight and eclipse
• *Orbit altitude & eclipse duration*	Input parameter	700 km 35.3 min
• *Design lifetime*	Chaps. 2, 3	5 yr
2. *Calculate amount of power that must be produced by the solar arrays, P_{sa}*	Step 1 Eq. 5-5 (Orbit period – T_e) Eq. (21-6)	$P_e = P_d = 110$ W $T_e = 35.3$ min $T_d = 63.5$ min Assume a peak power tracking regulation scheme with $X_e = 0.6$ and $X_d = 0.8$ $P_{sa} = 239.4$ W
3. *Select type of solar cell and estimate power output, P_o, with the Sun normal to the surface of the cells*	*Si: $P_o = 0.148 \times 1{,}368$ W/m^2 = 202 W/m^2 *GaAs: $P_o = 0.185 \times 1{,}368$ W/m^2 = 253 W/m^2 *Multijunction: $P_o = 0.28 \times 1{,}368$ W/m^2 = 383 W/m^2	Si solar cells $P_o = 202$ W/m^2
4. *Determine the beginning-of-life (BOL) power production capability, P_{BOL}, per unit area of the array*	Tables 21-13, 21-15 Eq. (5-7) Eq. (21-7)	$I_d = 0.77$ $\theta = 23.5$ deg (worst case) $P_{BOL} = 143$ W/m^2
5. *Determine the end-of-life (EOL) power production capability, P_{EOL}, for the solar array*	Performance degradation Si: 3.75% per yr, GaAs: 2.75% per yr, Triple Junction: 0.5% per yr Eq. (21-8) Eq. (21-9)	Performance degradation is 3.75% per year $L_d = 0.826$ for 5 yr mission $P_{EOL} = 118.1$ W/m^2
6. *Estimate the solar array area, A_{sa}, required to produce the necessary power, P_{sa}, based on P_{EOL} an alternate approach*	Eq. (21-10)	$A_{sa} = 2.0$ m^2 $A_{sa} = 2.5$ m^2
7. *Estimate the mass of the solar array*		$M_a = 9.6$ kg
8. *Document assumptions*		

junction GaAs cells have become the standard for most space applications due to their high efficiency and radiation resistance, but at a high cost. Silicon solar cell technology is mature and can be lower cost for some applications where radiation is not a concern. Single junction GaAs and InP cells are shown here for reference, but are not widely available commercially.

To complete this step, we identify the type of solar cells and how their performance will degrade during the mission. The *energy conversion efficiency* of a solar cell is defined as the power output divided by the power input. The power input value for a planar solar array is the 1,368 W/m^2 or the *solar constant* (the amount of energy received at the top of the Earth's atmosphere on a surface oriented perpendicular to the Sun's rays). Ideally, silicon and gallium arsenide and triple junction GaAs solar cells have efficiencies of about 24.7%, 21.8%, and 33.8%, respectively. These solar cell efficiencies give us ideal solar cell output performance per unit area, P_o, of 338 W/m^2, 298 W/m^2 and, 462 W/m^2. We also need to note that these efficiency values often refer to laboratory cells and not production cells, which have lower average efficiencies. In the next step, we will identify losses inherent to panel assembly (diodes, interconnect cabling, transmission losses) to size the array adequately.

Table 21-12, Fig. 21-7◀, Eq. 21-6◀

Table 21-13. Performance Comparison for Photovoltaic Solar Cells. Note that the stated efficiencies are for single solar cells, not solar arrays.

Cell Type	Silicon	Thin Film Amorphous Si	Gallium Arsenide	Indium Phosphide	Triple Junction GaAs
Planar Cell Theoretical Efficiency	29%	12.0%	23.5%	22.8%	40+%
Achieved Efficiency:					
Production	22%	8.0%	18.5%	18%	30.0%
Best Laboratory	24.7%	11%	21.8%	19.9%	33.8%
Equivalent Time in Geosynchronous Orbit for 15% Degradation					
– 1 MeV electrons	10 yr	10 yr	33 yr	155 yr	33 yr
– 10 MeV protons	4 yr	4 yr	6 yr	89 yr	6 yr

Step 4. Next, we must determine the realistic power production capability of the manufactured solar array. As shown in Table 21-14, an assembled solar array is less efficient than single cells due to design inefficiencies, shadowing and temperature variations, collectively referred to as *inherent degradation*, I_d. Solar cells are applied to a substrate, usually honeycomb aluminum, and interconnected, resulting in losses of 10% of the solar array's substrate area. This accounts for the design and assembly losses. If we configure the spacecraft well, its appendages will shadow few cells, and shadowing losses should be slight. The temperature of a typical flat solar panel receiving normal incident radiation ranges from about 67 C in LEO to 53 C in GEO. The reference temperature for silicon solar cells is 28 C, with performance falling off 0.5% per degree above 28 C. Body-mounted arrays on nonspinning spacecraft are typically about 5 C warmer than deployed solar arrays because they can't radiate heat into deep space as efficiently.

Table 21-14. Elements of Inherent Solar Array Degradation. Although individual solar cells may have adequate efficiency, after we manufacture the solar array, these elements cause some degradation in the cumulative efficiency by the amounts indicated.

Elements of Inherent Degradation	Nominal	Range
Design and Assembly	0.85	0.77–0.90
Temperature of Array	0.85	0.80–0.98
Shadowing of Cells	1.00	0.80–1.00
Total Inherent Degradation, I_d	0.72	0.49–0.88

We commonly refer to the current-voltage characteristics of a solar cell as the I-V curves. Figure 21-8 depicts a first-quadrant I-V curve for a planar array in LEO. This curve characterizes BOL and EOL performance. As the figure illustrates, the three significant points for solar-array design are:

- *Short-circuit current*, I_{sc}, where voltage = 0
- *Peak-power point*, where voltage times current is maximized
- *Open-circuit voltage*, V_{oc}, where current = 0

We must also consider how temperature affects the I-V characteristics. While a LEO spacecraft is in eclipse, the solar-array temperature can get as low as –80 C. The highest operating temperature for an LEO spacecraft solar array is 100 C, occurring near the end of a full Sun period during an orbit.

The operating temperature of the array is a key issue because the solar cell's performance depends on temperature. A *current-voltage*, or *I-V plot*, illustrates the performance of a solar array (see Fig. 21-8). A change in the operating temperature of the solar cell or array causes 3 changes in the I-V curve:

- A scaling of the I-V curve along the current axis
- A translation or shifting of the I-V curve along the voltage axis
- A change in the I-V curve shape affecting the roundness of the knee region

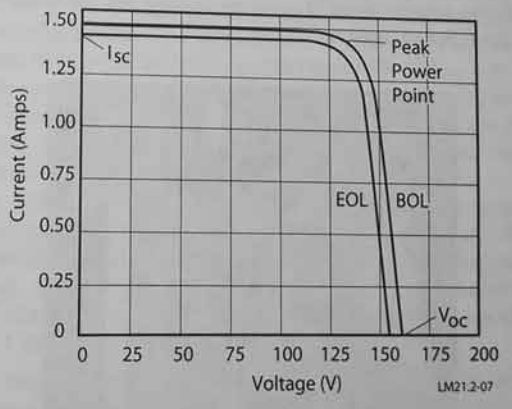


Fig. 21-8. I-V Plot for a Planar Array. The power available is simply the area under the curve.

The *temperature coefficient* or percent degradation of performance with increasing temperature, for any solar cell, depends on factors such as the type of cell and its output-power characteristics, actual operating temperature, and radiation environment. Gallium arsenide and indium phosphide have lower temperature coefficients, but higher temperature still means reduced performance. Solar arrays using gallium arsenide and indium phos-

Table 21-14, Fig. 21-8, Eq. 21-6◀

phide also resist radiation better than silicon and provide greater EOL power for a given area. We must establish a profile for operating temperatures during a mission, so the photovoltaic system can generate adequate power throughout.

The peak-power point depends on the array's operating temperature at BOL and EOL. Thus, an array often provides maximum power exiting an eclipse period because it is at its coldest operating temperature. Transient voltage excursions often occur when leaving eclipse, so we may need to clamp voltages to protect spacecraft loads. By understanding how the array's performance relates to these variables, we can get the highest output power from the array.

Usually, solar cells in series-parallel combinations make up a *solar array*. The number of series-connected solar cells in one string establishes the bus voltage available at EOL at the operating temperature; the number of parallel strings depends on the required current output. Isolation (bypass) diodes mounted within the solar array typically minimize the effects of shadowing and reversed-biased solar cells.

Solar-array configurations are either planar or concentrator, and either type can be body- or panel-mounted. Most photovoltaic applications to date have employed a *planar array* in which solar cells are mounted onto a surface (typically a composite panel with either insulated aluminum or carbon facesheets) with an adhesive. A Kapton®, Kevlar®, or fiberglass sheet usually insulates the solar cell from the support structure. Figure 21-9 shows a planar array on the Genesis program, while Fig. 21-10 illustrates an A2100 rigid panel. Figure 21-11 shows a large flexible solar array wing for the International Space Station.

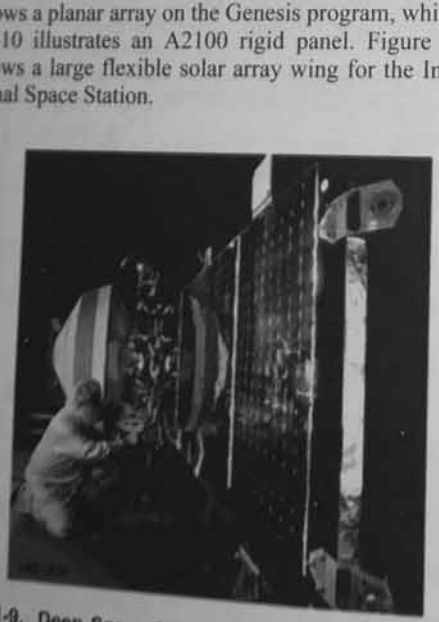


Fig. 21-9. Deep Space Planar Solar Array During Testing with the Satellite.

Concentrator solar arrays increase the solar cell's output by using mirrors or lenses to focus more solar radiation on the cells but can add significant complexity to the array design. For example, depending on the design

Table 21-14◀, Fig. 21-11, Eq. 21-6◀



Fig. 21-10. Geostationary Spacecraft Rigid Panel Solar Array. Each panel is approximately 7.1 m^3 and produces 1.4 kW.

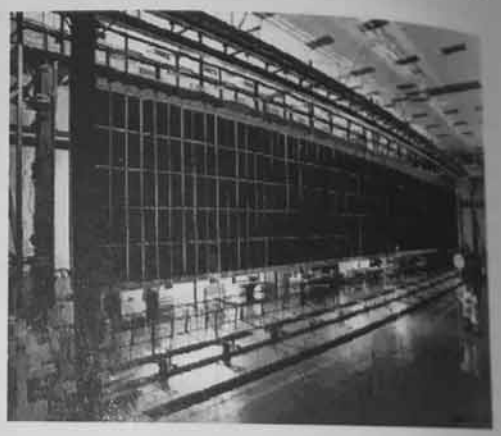


Fig. 21-11. Space Station Flexible Solar Array. The Space Station array is made up of 8 wings, each wing produces 32 kW, for a total of 256 kW.

of the concentrator, collection may occur over a narrower angle of offset from the Sun, meaning that more accurate pointing of the solar array would be required to take advantage of the concentrating affect. Also, the thermal design is an important consideration in concentrating systems, to ensure that no components will exceed acceptable operating temperatures.

Panel-mounted solar arrays usually apply only to 3-axis stabilized spacecraft. The panel-mounted approach tracks and points the solar array to get the best Sun incidence angle. The body-mounted approach reduces the requirements for tracking and pointing on any spacecraft (spinning or stabilized). But the less effective Sun incidence angle and increased array temperature of body-mounted cells produce a lower efficiency in orbit. Panel-mounted solar arrays are usually mounted on a boom. Deployable panel arrays are either flexible or rigid, according to the type of substrate material employed for mounting. For most spacecraft, we try to place the solar array away from the payload and other spacecraft subsystems because of the variable and often high temperature of the solar cells.

Body-mounted planar cells are typical on spinning spacecraft, which tend to even out the environmental heating of the spacecraft by exposing all areas to Sun and space as the craft rotates. In general, this raises the operating temperature of the body mounted solar cells, but helps to avoid hot spots on the spacecraft. Since body mounted solar cells tend to operate at a higher temperature than deployed solar cells, they operate less efficiently. However, the fact that not all the cells are collecting solar energy at the same time creates a larger impact on the power output of body mounted solar arrays. The decrease in power depends on the spacecraft's configuration and the drive mechanisms of the solar array (if any). For example, a stabilized array using Sun-tracking and pointing on two axes would fully use the solar array's surface area. But the array's reduction in output power per total surface area would be approximately π for body-mounted cells on a cylindrical, spinning spacecraft and 4 for body-mounted cells on a cubic-shaped spacecraft that does not employ active tracking. The output power decreases because not all cells are illuminated. The solar array impacts must be considered as part of a system level trade study, which includes the cost and design for the solar array's total surface area against the cost and complexity of stabilizing the spacecraft and using a drive system for the solar array. If the spacecraft is planned to spin for other reasons not related to the solar array, then it generally makes sense to use body mounted cells, avoiding the complexity of a deployed solar array. However, if the spacecraft will be 3-axis stabilized, the additional energy collection per square meter of solar cells provided by a deployed array that orients to the Sun may well be worth the additional complexity and expense to the overall system.

Shadowing considerations are important because a solar cell will go into open circuit (become high resistance) when not illuminated. In a simple series-connected string of solar cells, the shadowing of one cell results in the loss of the entire string. Shadowing may be caused by spacecraft components such as transmitting or receiving antennas, deployment mechanisms, or structures such as the solar array. We can reduce shadowing effects by actively pointing and tracking solar arrays on 3-axis stabilized spacecraft, using diodes, or designing series-parallel arrays. On spinning spacecraft, we must lay out solar cells so all solar cells within a string are illuminated. Diodes, which bypass groups or single solar cells in a string, help prevent damage to reduce the effects of shadowed solar cells.

We can improve solar cell performance with coverslides, coatings, and back-surface reflectors. Coverslides provide a hermetic seal yet allow the cell to receive sunlight and reject heat. They are textured or smooth. A textured coverslide is used for body-mounted solar cells that do not actively point toward the Sun. It reflects incident solar energy back onto the solar cell, improving the overall efficiency. Smooth coverslides are used for spacecraft whose arrays actively track and point. By decreasing reflective losses on solar cells, coatings allow cells to use more of the incident energy. Back-surface reflectors direct incident solar radiation that passes through the solar cell back through the cell again to improve overall efficiency. By reducing solar absorptance, they help the solar array manage thermal energy. Solar-cell vendors are continually improving the mechanical and thermal characteristics of coverslides, coatings, and back-surface reflectors. Thus, we must coordinate mechanical and thermal characteristics of these cells with the vendors.

At beginning-of-life, the array's power per unit area is Eq. (21-7)

$$P_{BOL} = P_o I_d \cos\theta \qquad (21\text{-}7)$$

where P_o was calculated in step 3 is I_d inherent degradation from step 4, and $\cos\theta$ is referred to as the cosine loss. We measure the Sun incidence angle, θ between the vector normal to the surface of the array and the Sun line. So if the Sun's rays are perpendicular to the solar array's surface, we get maximum power. Obviously, the geometry between the array and the Sun changes throughout the mission and different solar array panels will have different geometry. We configure the solar array to minimize this cosine loss. For example, in GEO with a flat, silicon solar array and a worst-case Sun angle of $\theta = 23.5$ deg angle between equatorial and ecliptic planes and the nominal value of I_d, the power output at beginning-of-life is 143 W/m^2.

Step 5. Radiation damage severely reduces a solar array's output voltage and current. At geosynchronous altitude, we must guard against solar-flare protons on-station, trapped electrons on-station, and trapped electrons and protons during transfer orbits. (Chapter 7 explains these terms.) Electrons and protons trapped in the Earth's magnetic field cause most degradation of solar cells. Silicon solar cells protected by coverslides lose 15% of their voltage and current (shielding assumed) when exposed to a total fluence of 10^{16} of 1-MeV equivalent electrons (4 to 5 years for a LEO spacecraft). As mission planners, we should coordinate degradation characteristics with the solar-cell manufacturer, based on the radiation environment the spacecraft will encounter. Degradation of a solar cell also depends on its design. More modern technologies, such as triple junction GaAs cells are more radiation hardened. They lose 11% of their power when exposed to 1-MeV, 5×10^{14} electrons/cm^2 fluence.

Next, we must consider the factors that degrade the solar array's performance during the mission. *Life degradation*, L_d, occurs because of thermal cycling in and out of eclipses, micrometeoroid strikes, plume impingement from thrusters, and material outgassing for the duration of the mission as well as material property degradation, such as darkening of solar cell cover glass. In general, for a silicon solar array in LEO, power production can decrease by as much as 3.75% per year, of which up to 2.5% per year is due to radiation. For gallium-arsenide cells in LEO, the degradation is about 2.75% per year, of which radiation causes 1.5% per year. The actual lifetime degradation can be estimated using Eq. (21-8).

Table 21-14◀, Fig. 21-11◀, Eq. 21-7

Table 21-15. Typical Solar Array Design Parameters. Most commonly used cell technologies. Note: Power values assume a 0° (zero) inclined GEO orbit.

Technology	Typical BOL Efficiency	Typical Size (cm)	Cell Watts/sq meter SS BOL (approx)	Array Watts/sq meter SS BOL/15 Yr EOL (approx)	kg/sq meter BOL SS (approx)	$/W at BOL SS (Approx)
Si	14%	4 × 6	191	116/92.6	2.3	378
GaAs (S.J)	18.5%	4 × 6	252	158/132	2.7	902
GaAs (M.J)	22.6%	4 × 6	308	194/163	2.8	696
GaAs (improved M.J)	26%	4 × 6	354	202/188	2.8	617
GaAs (ultra M.J)	28	32.3 cm²	382	245/210	2.8	617
SPL: Spectrolab, SS: Summer Solstice	—	—	—	—	—	—

$$L_d = (1 - D)^L \tag{21-8}$$

where D is the degradation per year and L is the satellite lifetime in years.

Step 6 The array's performance per unit area at end-of-life is Eq. (21-9).

$$P_{EOL} = P_{BOL}\, L_d \tag{21-9}$$

Using the FireSat II example array in Table 21-12 for a 5-year mission, L_d is 82.6%, resulting in a P_{EOL} of 118.1 W/m². The solar-array area, A_{sa}, required to support the spacecraft's power requirement, P_{sa}, is

$$A_{sa} = P_{sa} / P_{EOL} \tag{21-10}$$

The resulting solar-array area for the example spacecraft is about 2.0 m². If we had used a perfectly pointed array, the BOL power would have been 155 W/m², resulting in an EOL power of 128 W/m² and an array area of 1.9 m². So, having to account for the cosine loss costs us 0.1 m² in array size and the equivalent mass.

Solar-array sizing is more difficult than it appears from the above discussion. Typically, we must consider several arrays with varying geometry. Also, the angle of incidence on the array surface is constantly changing. We must predict that angle continuously or at least determine the worst-case angle to develop an estimate of P_{EOL}. Table 21-15 shows a comparison of solar performance at the cell level, as well as at the array level for BOL and EOL.

The power source generates electrical power within the spacecraft. Launch vehicles such as Atlas or Delta use primary batteries (discussed in Sec. 21.2.3) as the power source for electrical loads because the batteries usually need to last less than an hour. But batteries alone are too massive for missions that last from weeks to years. These missions need a source that can generate power over many orbital cycles to support electrical loads and recharge the batteries.

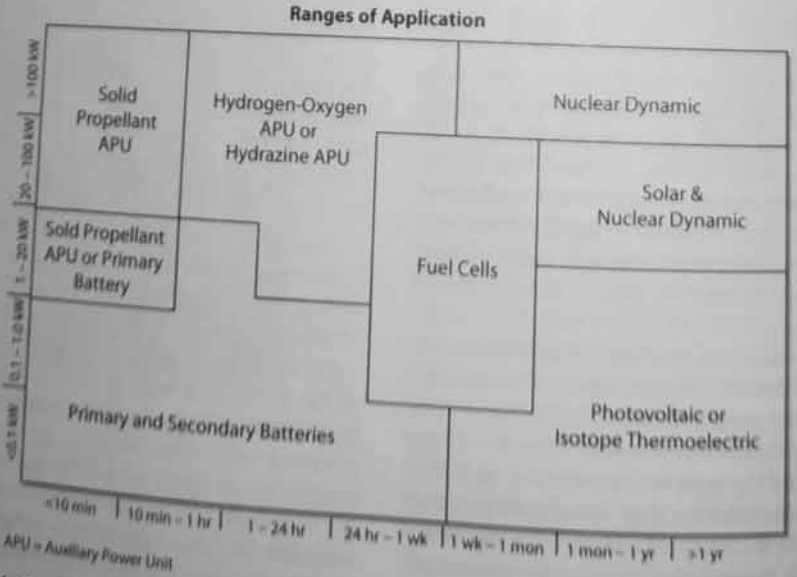


Fig. 21-12. Ranges of Application for Various Power Generation Technologies.

Table 21-15, Fig. 21-12, Eq. 21-10

Typically, we use 4 types of power sources for spacecraft as shown in Fig. 21-12. Photovoltaic solar cells, the most common power source for Earth-orbiting spacecraft, convert incident solar radiation directly to electrical energy. Static power sources use a heat source—typically plutonium-238 or uranium-235 (nuclear reactor), for direct thermal-to-electric conversion. Dynamic power sources also use a heat source—typically concentrated solar radiation, plutonium-238, or enriched uranium—to produce electrical power using the Brayton, Stirling, or Rankine cycles. The fourth power source is fuel cells, used on manned space missions such as Gemini, Apollo, SkyLab, and the Space Shuttle. These will be covered in more detail in Sec. 21.2.6.

Advanced Technologies

There are several promising areas of new technology, or new applications of technology. For example, NASA's Phoenix Lander, shown in Fig. 21-13 used circular fold up solar arrays to successfully operate on Mars.

The large circular objects on either side of the spacecraft are solar array wings, showing that solar arrays do not have to be square or rectangular. This design uses typical space cells, but mounts them on flexible gores that make up fan-like structures that fold up for launch, and unfold after the satellite has been deployed. This is an example of a creative way to maximize use of the volume and mass available for the spacecraft within the launch vehicle fairing.



Fig. 21-13. Phoenix Lander Photo Applied to Picture of Mars Estimating Actual Landing Conditions. The round solar array wings fold up like a fan for storage in the launch configuration, then deploy for operations.

21.2.3 Energy Storage

Energy storage is an integral part of the spacecraft's electrical-power subsystem providing all the power for short missions (<1 week) or back-up power for longer missions (>1 week). Any spacecraft that uses photovoltaics or solar thermal dynamics as a power source requires a system to store energy for eclipse periods and peak-power demands. High energy density storage technologies include batteries, super-capacitors, fuel cells, and fly wheels. This section will focus on batteries for energy storage, recognizing that research and development of super-capacitors, flywheels and fuel cells may result in promising energy storage systems for future missions. Figure 21-14 lists the main battery chemistries/types that are available today along with expected performance for both *primary* (non-rechargeable) and *secondary* (rechargeable) energy storage applications. The secondary batteries shown in Fig. 21-14 are capable of thousands of charge/discharge cycles for most space applications. The silver zinc (AgO/Zn) battery shown in Fig. 21-14 is typically a primary battery, although it may be applicable for recharge for applications that require limited (<100 charge/discharge cycles) recharge capability.

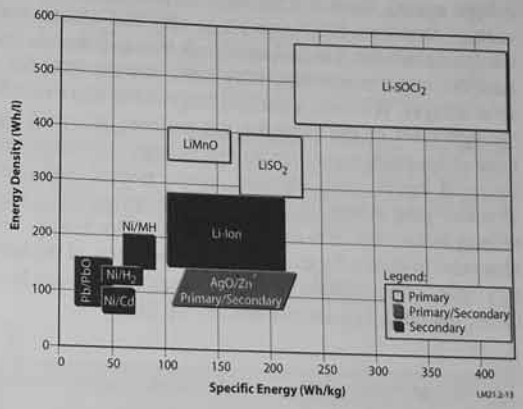


Fig. 21-14. Aerospace Battery Types. Energy density and the specific energy characteristics are the key parameters for battery selection to support energy storage for a mission. Note the performance characteristics shown above are at the cell level, and the packaging associated with a battery can reduce these values by ~10–25%. Technology advances for the shown battery types, along with chemistries under development, should be monitored to properly assess trade studies for battery type selection..

A *battery* consists of individual cells connected in series. The number of cells required is determined by the bus voltage requirements. The amount of energy stored within the battery is the *ampere-hour capacity* or watt-hour (ampere-hour, Ah, at times operating voltage) capacity. Batteries can be connected in series to increase the voltage or in parallel to increase the current output, with the net result being an increase in watt-hour capacity.

Table 21-16 lists battery issues to consider early in the conceptual phase of any program. Most of all, we try to provide a stable voltage for all operating conditions during the mission life because load users prefer a semi-regulated bus voltage. The difference in energy storage voltage between end-of-charge and end-of-discharge often determines the range of this bus voltage.

Table 21-15◀, Fig. 21-14, Eq. 21-10◀

Table 21-16. Issues in Designing the Energy Storage Capability. Energy storage usually means large batteries and we must consider all their characteristics when designing this subsystem.

Physical	Size, weight, configuration, operating position, static and dynamic environments
Electrical	Voltage, current loading, duty cycles, number of duty cycles, activation time and storage time, limits on depth-of-discharge, and short-circuit (fault) recovery
Programmatic	Cost, shelf and cycle life, mission, reliability, maintainability, produceability and safety

Figure 21-15 highlights the charge-discharge characteristics of a typical battery system. At the left edge, the voltage is low because the spacecraft just came out of eclipse where it used battery power. During the charge phase, there is positive current from the power regulator, so the battery voltage rises. In the discharge phase (in eclipse again), there is a negative current, so the battery voltage decreases. Discharge phase may also occur when the spacecraft is not in eclipse if peak power demands exceed the output power capability of the power source (i.e., solar arrays). We want a flat discharge curve that extends through most of the capacity, yet we need to avoid the knee of the discharge curve for bus voltage regulation and to avoid battery cell damage due to reverse voltages. Overcharging quickly degrades most batteries, and a key to long battery life is to minimize battery overcharge. We also need to match the electrical characteristics of the battery cells as charge imbalances stress and degrade the batteries, resulting in a shorter life for the EPS.

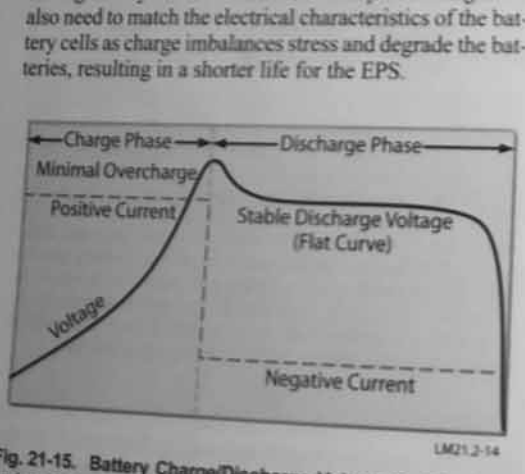


Fig. 21-15. Battery Charge/Discharge Voltage Profile. Secondary batteries may cycle through this type of profile hundreds or thousands of times during their mission life. At the left edge, the voltage is low because the spacecraft just came out of eclipse where it used battery power. Battery charge power is available when the solar array output is greater than all mission critical power loads, which will increase the battery voltage. In the discharge phase (eclipse again), the battery is supplying spacecraft power (negative current), resulting in decreasing battery voltage.

All battery cells are either primary or secondary. *Primary battery cells* convert chemical energy into electrical energy but cannot reverse this conversion, so they cannot be recharged. Primary batteries typically apply to short missions (less than one day) or to long-term tasks such as memory backup, which use very little power. The most common primary batteries are highlighted in Table 21-17, and one can see the wide ranges in each couple's specific-energy density. We cannot specify a value for specific energy density because cells vary in design and depend on mission requirements. We must coordinate mission requirements with the battery manufacturer to specify battery performance. The typical application comments for Table 21-17 are important factors for launch vehicles, the most predominant user for space qualified primary batteries.

Table 21-17. Characteristics of Selected Primary Batteries. Primary battery manufacturers can meet power requirements within these ranges of specific energy density. We must tradeoff cost and mass with capacity while ensuring mission accomplishment.

Primary Battery Types	Specific Energy (WHr/kg)	Typical Application
Silver Zinc	60–130	High rate, short life (minutes)
Lithium Thionyl Chloride	175–440	Medium rate, moderate life (<4 hours)
Lithium Sulfur Dioxide	130–350	Low/medium rate, long life (days)
Lithium Carbon Monoflouride	500–800	Low rate, long life (months)
Thermal	30–60	High rate, very short life (minutes)

A *secondary battery cell* can convert chemical energy into electrical energy during discharge and electrical energy into chemical energy during charge. It can repeat this process for thousands of cycles. A secondary battery provides power during eclipse periods on spacecraft that employ photovoltaics and can also level loads. Secondary batteries recharge in sunlight and discharge during eclipse. The spacecraft's orbital parameters, especially altitude, determine the number of charge/discharge cycles the batteries have to support during the mission life. A geosynchronous satellite needs to store energy for two 45-day eclipse periods per year with eclipses lasting no more than 72 min each day. The geosynchronous orbit demands few charge/discharge cycles during eclipse periods, thus allowing a fairly high (> 60%) depth-of-discharge. On the other hand, LEO spacecraft encounter at most one eclipse period each orbit or about 15 eclipse periods per day, with maximum shadowing of approximately 36 min. Therefore, the batteries must charge and discharge about 5,000 times each year, so the average depth-of-discharge is limited to ~30%, much lower than for geosynchronous spacecraft. Table 21-18 highlights performance characteristics and system impacts for today's most common secondary aerospace batteries.

Depth-of-Discharge (DOD) is simply the percent of total battery capacity removed during a discharge period. *State-of-Charge (SOC)* is the amount of total battery capacity at any given time. For example, if a 10 Ah battery has 1 Ah of capacity removed, the DOD was 10%

Table 21-17, Fig. 21-15, Eq. 21-10◀

Table 21-18. Char
energy densities tha
spacecraft.

Performa Characteris Rechargeable
Energy Density
Energy Efficienc
Thermal Power (scale 1–10)
Self Discharge (% per day)
Temperature R
Memory Effect
Energy Gauge
Trickle Charge
Modularity
Heritage

(1 Ah/10 Ah), whi
ing/10 Ah total). F
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life for Nickel Cad
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Fig. 21-16 is base
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Table 21-18. Characteristics of Selected Secondary Batteries. Though secondary batteries have much lower specific energy densities than primary batteries, their ability to be recharged makes them ideal for backup power on solar powered spacecraft.

Performance Characteristics for Rechargeable Batteries	Ni-Cd	Ni-H_2	Li-Ion	System Impact
Energy Density (W-hr/kg)	30	60	125	Mass savings and vehicle center of gravity
Energy Efficiency (%)	72	70	98	Reduction of charge power reduces solar panel mass & size
Thermal Power (scale 1–10)	8	10	1	Reduction of radiators, heat pipes sizes
Self Discharge (% per day)	1	10	0.3	Simple management at launch pad & more margin during transfer
Temperature Range (°C)	0 to 40	–20 to 30	10 to 25	Management at ambient and thermal control reqs
Memory Effect	Yes	Yes	No	No reconditioning management
Energy Gauge	No	Pressure	Voltage	Easier state of charge assessment
Trickle Charge	Yes	Yes	No	Balancing need prior to eclipse. Li-Ion typically requires cell equalization circuitry.
Modularity	No	No	Yes	One cell design
Heritage	Yes	Yes	Yes	Risk Assessment. Continued Li-Ion LEO testing and missions establishing heritage.

(1 Ah/10 Ah), while the SOC is now 90% (9 Ah remaining/10 Ah total). Figure 21-16 illustrates that the higher the depth-of-discharge (DOD) then the shorter the cycle life for Nickel Cadmium (Ni-Cd), Nickel Hydrogen (Ni-H_2), and Lithium Ion (Li-Ion) secondary batteries. The cycle life versus DOD characteristics depicted in Fig. 21-16 is based upon extensive flight and test data for both Ni-Cd and Ni-H_2. Li-Ion data is just becoming available that will mature and further qualify its cycle life characteristics as a function of DOD. Once we work with the cell/battery supplier to fully understand the batteries' cycle life versus DOD characteristics, we can determine the total capacity of the batteries based upon the mission power budget profile.

The Nickel-Cadmium (Ni-Cd) battery was the most common space battery type prior to the 1980s. A 28 V_{dc} aerospace Ni-Cd battery usually consists of 22–23 series-connected cells. Ni-Cd battery cells for aerospace missions had nominal capacities of 5 to 100 Ah. Today, Ni-Cd batteries are selected for spacecraft based upon an extensive, mission-specific Ni-Cd data base that results in very low mission risk relative for the energy storage system.

The Nickel-Hydrogen (Ni-H_2) battery became the common energy storage system of choice for aerospace applications during the 1980s and 1990s. The three space-qualified design configurations for Ni-H_2 are individual pressure vessel, common pressure vessel, and single pressure vessel. The individual pressure vessel was the first Ni-H_2 technology used for aerospace application. Here, a single electrochemical cell is contained within a single pressure vessel. It has a working terminal voltage of 1.2 to 1.5 V_{dc} depending upon discharge loads. The typical individual pressure vessel battery design consists of multiple cells/pressure vessels connected in series to obtain the desired battery voltage. Cell diameters are typically 9 to 12 cm, with capacity ranges from 20 to over 300 Ah. Figure 21-17 shows a Ni-H_2 battery module for a geostationary spacecraft.

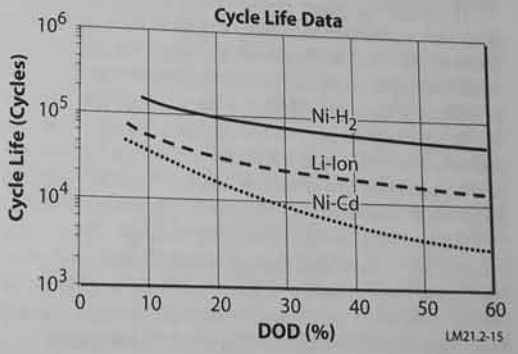


Fig. 21-16. Depth-of-Discharge vs. Cycle Life for Secondary Batteries. Increased cycle life reduces the amount of energy available from the batteries during each cycle — DOD decreases with cycle life.

The common pressure vessel Ni-H_2 battery is very similar to individual pressure vessel technology, with the primary difference in the wiring connection of the internal electrode stacks. In the individual pressure vessel, the electrode stacks are all connected in parallel. In a common pressure vessel, there are two sets of electrode stacks within the pressure vessel that are series connected, yielding a working minimal terminal voltage of 2.4 to 3.0 V_{dc} depending upon discharge loads. This design has a higher specific energy at the battery level since there are half as many pressure vessels and a significant reduc-

Table 21-18, Fig. 21-16, Eq. 21-10◀

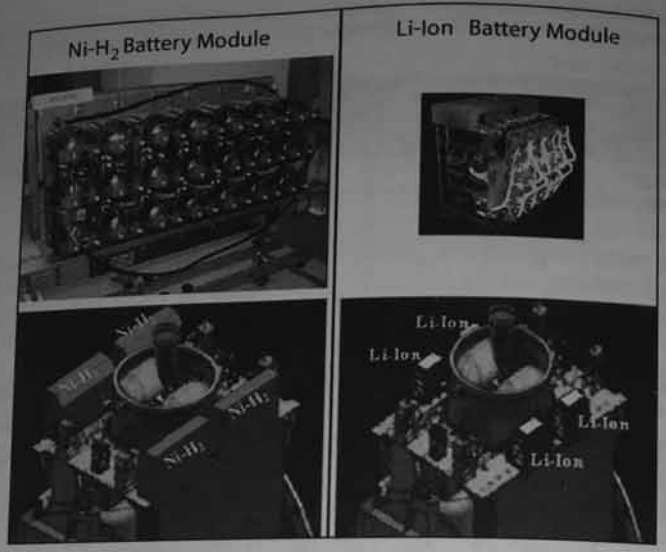


Fig. 21-17. Ni-H_2 vs Li-Ion Battery Comparison. The above geostationary spacecraft utilizes 4 batteries for energy storage. The relative volume required for these 4 batteries is shown in the lower schematics. The Li-Ion battery has obvious volumetric advantages and offers ~82 kg mass savings along with significant cost savings. However, its main disadvantage is a higher risk assessment due to a maturing data base and a tighter temperature operating range.

tion in cell piece-parts. Common pressure vessel Ni-H_2 technology has been space qualified in the 6 cm and 9 cm cell diameter configuration for capacities in the 12 to 20 Ah ranges.

The single pressure vessel Ni-H_2 battery is designed such that a common hydrogen supply is used by three or more series connected cells with a single pressure vessel. Each cell stack contains its own electrolyte supply which is isolated within individual cell stack containers. The key operating characteristic of this design is to allow the free movement of hydrogen within the cell stacks while maintaining cell stack electrolyte isolation. These batteries are presently available in a 12.5 cm or 25 cm diameter design.

Lithium Ion (Li-Ion) battery technology offers a significant volumetric and energy density advantages over Ni-Cd and Ni-H_2 battery types. In addition, the higher recharge efficiency of Li-Ion require a less complex and less costly thermal control system and Li-Ion have a very low self discharge rate on the order of 5% per year. The nominal operating voltage for a Li-Ion cell is 3.6 to 3.9 V_{dc}, which allows us to reduce the number of cells by approximately two-thirds when compared to Ni-Cd or Ni-H_2 cells resulting in significant battery volume and mass reductions. The Li-Ion secondary battery system offers a 65% volume advantage and a 50% mass advantage for most present day aerospace battery applications. Figure 21-17 shows a Li-Ion battery module designed and under qualification for the same A2100 spacecraft as the Ni-H_2 battery module. The advantages of a Li-Ion battery are easily seen for these two battery types on the same spacecraft. Li-Ion batteries have been qualified for numerous GEO and LEO missions. Aerospace Lithium batteries typically require cell equalization/conditioning and bypass circuitry to meet mission life requirements, and Range Safety requires individual cell voltage monitors for Lithium batteries prior to launch. Thermal control requirements for a Li-Ion battery are tighter due primarily to the fact that the design and emerging life cycle database for Li-Ion batteries has been focused around operation and performance in the 10–25 C temperature range. As the test database expands over the next several years resulting in lower overall mission risk, Li-Ion will likely become the preferred rechargeable battery for most, if not all, spacecraft missions.

To size a secondary battery, we must identify the parameters and apply the equation in Table 21-19. The parameter values used in the equation can vary significantly with battery type. The ideal battery capacity is the average eclipse load, P_e, times the eclipse duration, T_e. This ideal capacity must be increased to include the battery-to-load transmission efficiency, n, and the depth-of-discharge constraints. For LEO, we limit the battery's DOD to be 40–60% for Ni-H_2, 20–40% for Li-Ion, and 10–20% for Ni-Cd batteries. We base these expectations on the average DOD over 24 hours and assume the batteries are fully recharged at least once during this period. The number of batteries, N, may be equal to one for this calculation if you simply require battery capacity. Two to five batteries are typical for most spacecraft. We must have at least two (unless the battery uses redundant cells) because the spacecraft needs redundant operation with one unit failed. But more than five batteries require complex components for recharging. Battery capacity may also be required to help meet peak power loads during full Sun conditions. For some missions, the peak power loads may drive the required battery capacity rather than the average load resulting from an eclipse period.

Table 21-18◀, Fig. 21-17, Eq. 21-10◀

Table 21-19. Steps in the Energy Storage Subsystem Design. To obtain the required battery capacity in Amp-hr, divide by the required satellite bus voltage.

Step	Consider	FireSat II Example
1. ***Determine the Energy Storage Requirements***	• Mission length • Primary or secondary power storage • Orbital parameters – Eclipse frequency – Eclipse length • Power use profile – Voltage and current – Depth of discharge – Duty cycles • Battery charge/discharge cycle limits	• 5 yrs • Secondary power storage • 16 eclipses per day • 35.3 min per eclipse (T_e) • Eclipse load 110 W (P_e) – 26.4 V, 4.2 Amp (max) • 20% (upper limit)
2. ***Select the Type of Secondary Batteries***	• Ni-Cd (space qualified) • NiH_2 (space qualified) • Li-Ion (space qualified)	• Ni-Cd, Ni-H_2 or Li-Ion. Selection based upon trade studies and program risk assessment
3. ***Determine the Size of the Batteries (Battery Capacity)***	• Number of batteries • Transmission efficiency between the battery and the load	• N = 3 batteries (non redundant) • n = 0.90 • Cr = 119 W-hr • Cr = 4.5 Ah (26.4 V bus)
Battery Capacity: $C = (P_e T_e/(\text{DOD})Nn)$ W-hr (for battery capacity in Ah, divide by bus voltage)		

21.2.4 Power Regulation and Control

Many loads within a spacecraft require a voltage at a different level than that provided by the voltage source. When this is the case, DC-DC voltage converters are generally used. A DC-DC converter can also keep the load voltage within a range specified by the load. To know what voltage levels are acceptable to specific loads we start with the load requirements. Input power requirements we look for in a load are: voltage range, current, acceptable normal/abnormal transient time duration, and voltage ripple. Table 21-20 shows an example of a load's input power requirements. If a voltage regulator can provide the load with the specified input power then the supplier of the load is guaranteeing their product will operate as specified.

Table 21-20. Example of a Load's Input Power Requirements.

Condition	Range
Normal Steady State Voltage Range	25 to 31 V DC
Normal Transients	20 to 36 V DC*
Abnormal Transients	0 to 40 V DC*
Nominal Ripple	1 V peak to peak
Max Current	15 Amp
Inrush Current	25 Amp*

* Note: Include time duration and wave shape of transients as applicable.

Many techniques are used to regulate and control power going into a load. When charging, a battery acts as a load on the spacecraft's power bus. Because the energy source output voltage does not always meet the requirements specified by the battery, we must, at a minimum, control the source's output power so that the battery input power requirements are met. The energy source determines how we regulate a spacecraft's power. Since most aerospace applications use solar photovoltaics, we will examine power regulation emphasizing that viewpoint. Power regulation at the main power bus divides into 3 main categories: controlling the solar array, regulating bus voltage, and charging the battery.

We must control electrical power generated at the array to prevent battery overcharging and undesired spacecraft heating. The two main power bus control techniques, illustrated in Fig. 21-18 are a *peak-power tracker (PPT)* and a *direct-energy-transfer (DET)* subsystem. The main difference between these two approaches is how they deal with the input power from the solar array. A DET system directly transfers the power from the solar array to the power bus. This implies that the solar array must operate at the same voltage as the rest of the power bus (ignoring line loss). With a PPT, the solar array and bus can operate at different voltages, and the peak power tracker is functionally between the solar array and the bus, processing all the solar array power.

When a power bus is dominated by the battery voltage, such as in a DET system, the solar array also operates at that voltage and cannot deliver the full power it is capable of at all times. Since the solar array is designed to never exceed a voltage past the peak power point seen in Fig. 21-8 it will reach its full power producing capability only when the battery is at its highest voltage which occurs when the battery is fully charged. When the battery is at its minimum voltage, at the beginning of its charge cycle, the solar array operates well below the peak power point.

A solution to this issue is the PPT which is a DC-DC converter operating in series with the solar array. The objective of the DC-DC converter is to keep the voltage of

Table 21-20, Fig. 21-17◀, Eq. 21-10◀

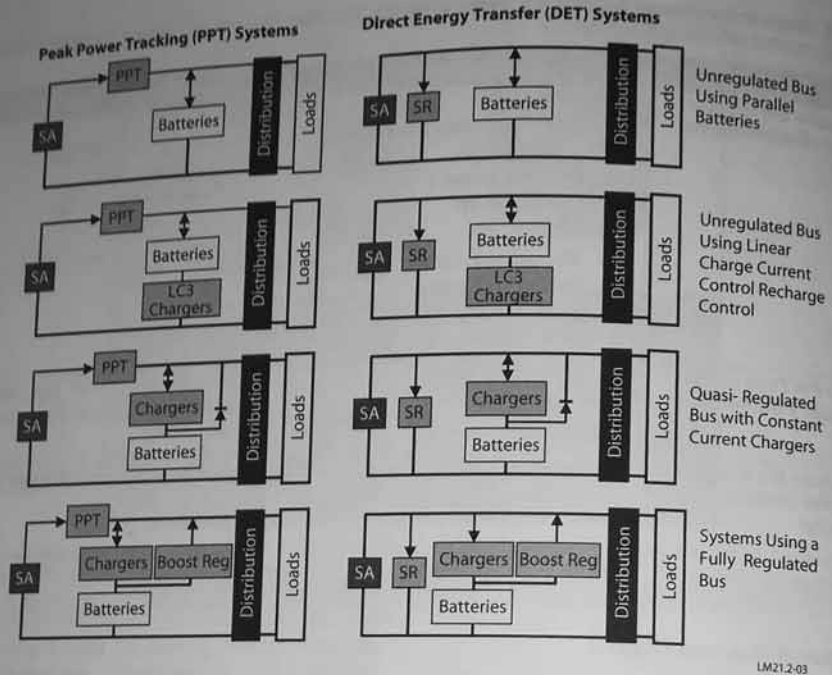


Fig. 21-18. Techniques for Power Regulation. The basic approaches are *Peak Power Tracking (PPT)*, which places a regulator in series with the solar arrays and the load, and *Direct Energy Transfer (DET)*, which uses a regulator in parallel with the solar arrays and load.

the solar array at a value slightly below, for stability reasons, the maximum power point. When the solar array is at this value it is utilizing its full power producing capability. In order for the DC-DC converter to achieve its goal, it uses a buck or boost converter to track the battery voltage then will adjust the duty cycle of the converter to bring the solar array voltage to the desired max power point. The buck and boost converter will be discussed later in this section. A disadvantage of the PPT is the additional weight, complexity and cost it adds to the system. PPTs also have a 3–8% power loss to contend with and are a significant noise source to the rest of the system.

For direct energy transfer systems, a *shunt regulator* operates in parallel to the array and shunts the array current away from the subsystem when the loads or battery charging do not need power. Power subsystems with shunt regulation are extremely efficient. A shunt-regulated subsystem has the advantages of fewer parts, lower mass, and higher total efficiency at EOL. The main disadvantage of the shunt-regulated subsystem is that the full power potential of the solar array cannot be utilized.

Batteries can be charged individually or in parallel. A parallel charging system is simpler and lower cost, but does not allow flexibility in vehicle integration. It can also stress batteries so they degrade faster. When batteries are charged in parallel, the voltage is the same but the current and temperature are not. Because current is not rigidly controlled, one battery could receive all the available charge current, and a thermal runaway condition could result if the bus voltage of the hottest battery is not controlled. Parallel batteries eventually end up balancing out, so we could use them for missions under five years. Individual charging optimizes the battery use by charging all the batteries to their own unique limits. It also forgives battery deviations in systems with several batteries. Unfortunately, individual chargers add impedance, electronic piece parts, and thermal dissipation not present in a parallel system.

When charging the battery a control loop must be implemented to prevent overcharging. There are several different methodologies used to determine when a battery is at its maximum capacity; voltage, current and pressure are three of the parameters commonly used as a capacity gauge. When one of these parameters reaches a predetermined level a control algorithm will usually go into a trickle charge mode or stop charging the battery all together. The fully charged mode depends on the battery chemistry used. A configuration comparison of the EPS topologies shown in Fig. 21-18 is provided in Table 21-21.

Power Regulators

A power regulator is an electrical device used to bring an unregulated bus to within the specifications required by the various loads throughout the system. Here we will consider two regulator topologies; linear regulators and switch-mode regulators. (See Tables 21-23 and 21-24.)

Linear regulators were once widely used in all spacecraft power subsystems for both small and large loads but have lost favor for utilization on large loads due to the development of the more efficient switch-mode reg-

Table 21-20◀, Fig. 21-18, Eq. 21-10◀

Table 21-21. Configuration Comparison.

System	Pros	Cons	Best Application
Unregulated Direct-Energy Transfer (U-DET)	High solar array and battery to load efficiency Fewer EPS components	More complex load converters Potential "lockup" problems and large solar array	Small load variations Small power source variation for most of sunlight period
Regulated Direct-Energy Transfer (R-DET)	Tightly regulated input to spacecraft component Simpler, lighter, more efficient load converters	Additional EPS converters Series power loss between battery and load	Heritage spacecraft loads requiring tight regulation Large solar array power/voltage variations
Maximum Power Tracking	Shunt regulator or battery charge regulator not required Can make maximum use of solar array energy in some cases	Less efficient than DET at EOL in many cases More dissipation inside spacecraft	Large variations in solar array input throughout mission

ulators in the 1970s. Some newer devices such as the low drop out regulator (LDO) have improved the efficiency of linear regulators and are used for local voltage regulation and to provide low frequency noise rejection. Increased efficiency of the linear regulator is achieved by reducing the input-to-output voltage differential.

A series regulator is seen in Fig. 21-19. The basic operating principle of the series regulator is to measure the voltage at the load then make adjustments on the series resistor to maintain a desired voltage at the load. The series resistor is the main cause of inefficiency in this system. As can be seen in Table 21-22 efficiency of the series regulator is dependent on both the line (input) voltage and the voltage at the load. As the line voltage further deviates from the load voltage the efficiency decreases.

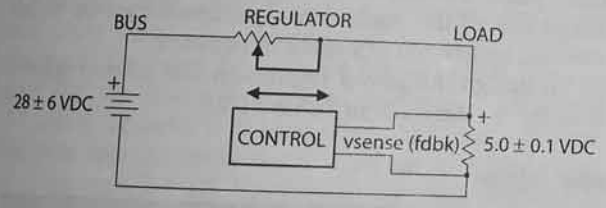


Fig. 21-19. Series Regulator. Constant voltage load efficiencies are shown in Table 21-22.

In response to the poor efficiencies achieved by most linear regulators, switch-mode regulators were developed. Today these are the most commonly used regulator topologies. They regulate power using active electronics, switching semiconductors, transformers, inductors and capacitors which conserve energy rather than simply bleeding it off the way a linear regulator would. The advantage of switch-mode regulators is high efficiency which means low power losses and low heat dissipation. Figure 21-20 shows a block diagram of the basic operating principal of switch-mode regulators.

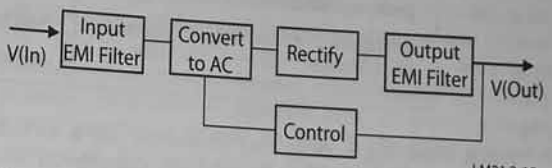


Fig. 21-20. Basic Operating Principle of Switch-Mode Regulators.

The most common regulator topologies are the Buck and Boost regulators which either reduce (buck) or increase (boost) the voltage at the output of the regulator. A buck-boost regulator can increase or decrease the input voltage. By adding more components, and thus complexity, to these basic topologies, greater efficiencies can be achieved. Common buck or boost derived regulators, from least complex to most complex, are the flyback, single switch forward, two switch forward, push-pull, and full bridge regulators. As complexity increases in the above list, so does each regulator's ability to process more power, ranging from 20 W to greater than 600 W. Further increases in efficiency can be achieved by implementing resonant mode switching within one of the above regulators. This technique saves power by turning on solid-state switches when the sine-wave voltage across the transistor is at zero, thus eliminating switching losses; of course further complexity is added when implementing resonant mode switching.

Table 21-22. Series Regulator Efficiency. Efficiency of the series regulator is dependent on the line voltage and load voltage.

V (BUS) =	Load Power =	R (Reg) =	Reg. Loss	Total Source	Efficiency
22 V	5 W	17 Ω	17 W	22 W	22.7%
28 V	5 W	23 Ω	23 W	28 W	17.9%
34 V	5 W	29 Ω	29 W	34 W	14.7%

Table 21-23. Linear Regulator Advantages/Disadvantages.

Advantages	Disadvantages
• Wide gain-bandwidth and excellent transient response • Excellent ripple rejection below 1 MHz • No spectral line noise added to the output • No-load operation possible • Simple single pole transfer function	• Efficiency is dependent on input to output voltage differential and output current • High dissipation concentrated in single device, cooling may be a problem

Table 21-24. Switch-Mode Regulators Advantages/Disadvantages.

Advantages	Disadvantages
• Most commonly used topology • Usable at switching frequencies from 20 KHz to 2 MHz • Mature topology, developed in 1970's, supported and predictable reliability, parts available • Efficiencies greater than 95 percent are possible	• Ripple and Noise generated at the switching frequency and integral modules. • Transient requirements forces most buck PWM circuits to be operated at low duty cycle and with high voltage and current stresses • More complex than linear regulators • Harder to analyze and simulate than linear regulator • Stability issues can occur if not thoroughly analyzed

When choosing a switching frequency for a switch mode regulator several factors to be considered are packaging, efficiency, performance, and cost. As the frequency increases, the magnetic components and filter capacitors reduce in size. However, the switching losses are higher and gate drive circuitry consume more power as frequency increases. To compensate for lower efficiencies at higher frequencies a soft-switching topology can be considered to eliminate switching losses. Correct shielding and grounding becomes more critical as frequency and noise emissions increase.

When constant (non-varying) power is required by a load, the converter supplying the power can be modeled by negative impedance, meaning that as voltage decreases at the input, the current into the converter increases in order to maintain constant power. Negative impedance can cause stability issues when thorough analysis has not taken place. To maintain stability, it is necessary that the magnitude of the equivalent load impedance be higher than the magnitude of source impedance.

To design the power regulation and control subsystem, follow the steps in Table 21-25.

Table 21-25. Steps in the Power Regulation and Control Subsystem Design.

Step	Consider	Possibilities
1. Determine Load Requirements	Voltage Range Current Transients Ripple	
2. Determine the Power Source	All spacecraft loads, their duty cycles, and special operating modes	Primary batteries Photovoltaic Static power Dynamic power
3. Design the Electrical Control Subsystem	Power source Battery charging Spacecraft heating	Peak-power tracker Direct-energy transfer
4. Develop the Electrical Bus Voltage Control	How much control does each load require? Battery voltage variation from charge to discharge Battery recharge subsystem Battery cycle life Total system mass	Unregulated Quasi-regulated Fully regulated Parallel or individual charging < 5 yrs—parallel charge > 5 yrs—independent charge

Table 21-25, Fig. 21-20◀, Eq. 21-10◀

21.2.5 Power Distribution

A spacecraft's power distribution system consists of cabling, fault protection, and switching gear to turn power on and off to the spacecraft loads. It also includes command decoders to command specific load switches on or off. The power distribution system is a unique feature of the electrical power subsystem and often reflects individual spacecraft loads and power-switching requirements. Power distribution designs for various power systems depend on source characteristics, load requirements, and subsystem functions. In selecting a type of power distribution, we focus on keeping power losses and mass at a minimum while attending to survivability, cost, reliability, and power quality. An equivalent circuit of a distribution system is shown in Fig. 21-21. The DC source is represented by a battery and a series resistance. Elements for consideration in the distribution line are lumped inductance, noise pickup, capacitance from parallel lines and resistance from line copper, fuse and switching elements. Solid state relays have surpassed mechanical relays for power switching because of increased flight history, reliability, and low power dissipation. It is usually necessary to add a decoupling capacitor at the load to minimize the effects of transient caused by varying loads. The capacitor is represented by a capacitor in series with an inductor and a resistor. Since distribution systems are made up of long wires, they act as an antenna to pickup noise and to radiate out noise to the rest of the system. Cable length is normally dictated by mechanical considerations and is therefore beyond the control of the electrical designer. Wire inductance is approximately 0.36 μH per foot, don't forget to count both the power and return wires when determining the cable's total inductance. For example: A 10-foot power cable, run as a pair, will be found to have an inductance of around 7.2 μH. If the cable is twisted, the inductance will be reduced to approximately one third of the untwisted value, or in this case, 2.4 μH. Even relatively small inductances can become a source of noise.

The *load profile* of a spacecraft is a key determining factor in the design specifications of a power distribution subsystem. Predominant spacecraft loads (radar, communications, motors, computers) may require low to high-voltage anywhere from 1–120 V_{dc} converted from the 28-V_{dc} power bus. Spacecraft power loads often turn on or off or otherwise vary their power consumption. When a load's current varies, voltage spikes are created due to the inductance in the distribution lines. This transient behavior may produce noise that the distribution system translates to other loads, potentially harming working components. With ever-decreasing voltage levels for spacecraft loads, (currently as low as one volt for computer processors) and rising current needs, proper shielding, twisting of pairs and grounding must be considered at the beginning of the power distribution design phase.

High current cables are frequently improperly designed. The distribution cable voltage drop, in the supply and return path, must meet the load voltage regulation requirements. In addition, the cable harness often lacks adequate de-rating for the worst case environment, temperature, and the effects of vacuum. Copper I^2R power loss is the most basic of considerations in power design. High voltage distribution helps reduce the current and thus the I^2R loss. At higher current, the greatest concern is cable power dissipation and insulation damage due to heating. This is especially a problem in cables with large numbers of conductors because, when bundled together, it is harder for a cable to dissipate heat. At lower voltages, loss of regulation due to IR drop is the most prevalent problem as discussed earlier. The trend of digital circuits to progressively lower logic voltages and increase current transients has aggravated the problem. The remote sense capability of power converters may be used to reduce the voltage variation associated with cable resistive voltage drop. The remote sense feature moves the converter voltage feedback sense point from the output terminals to the load. The converter then adjusts the duty and output voltage to maintain the required voltage at the load. The remote sense is capable of compensating for a cable drop of around 105–110% of the nominal output. If the load varies over a large range and there is great difficulty associated with additional copper, then remote sense is a good option. If the remote sense feature is not available, additional copper is the simplest and most efficient option to meeting the load regulation requirements.

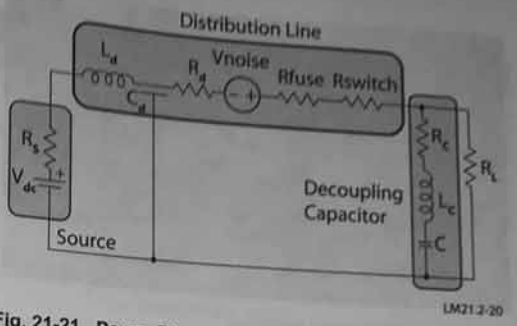


Fig. 21-21. Power Distribution System Equivalent Circuit.

The harness or cabling that interconnects the spacecraft's subsystems is a large part (10–25%) of the electrical-power subsystem's mass. As seen in Fig. 21-22, the harness connected to the power distribution unit (labeled PDDU in this example) takes up a considerable amount of volume inside the spacecraft, and therefore, routing must be considered early in the design phase. To avoid strain on the wiring inside the harness a minimum bend radius of 10 times the diameter of the harness should be used when routing. We must keep harnesses as short as possible to reduce voltage drops, noise and to regulate the bus voltage. A maximum voltage drop across the distribution line of 2% of the minimum source voltage is an industry standard. For example, lets say we have a heater which has a maximum current draw, I, of 10 A and 26 V_{dc} is our minimum bus voltage, we must then design the cabling so that the load will see a voltage, V_{min}, no less than:

Table 21-25◀, Fig. 21-21, Eq. 21-10◀

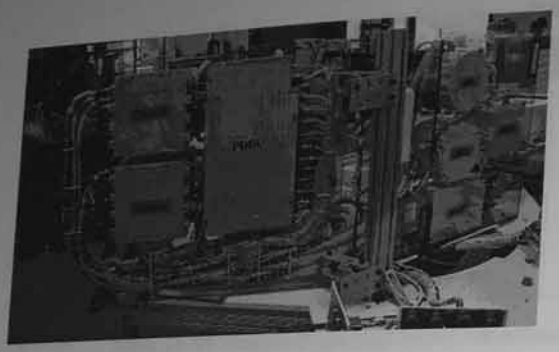


Fig. 21-22. JUNO Flight Harness.

$$V_{min} = 26\ V_{dc} \times 0.98 = 25.48\ V_{dc} \qquad (21\text{-}11)$$

Our voltage drop must be less than 0.52 V_{dc}. This means our wire resistance, R_w, needs to be less than

$$R_w = V/I = 0.52\ V_{dc} / 10\ \text{A} = 0.052\ \Omega \qquad (21\text{-}12)$$

If the heater is a long way from the source, a large gauge wire would be used or multiple smaller gauge wires in parallel would be used to decrease the wire resistance. Remember to include the return wires when calculating the total length of your wire, so if the heater is six meters from the source the total wire length will be twelve meters.

Figure 21-23 depicts the relationship between current and cable mass.

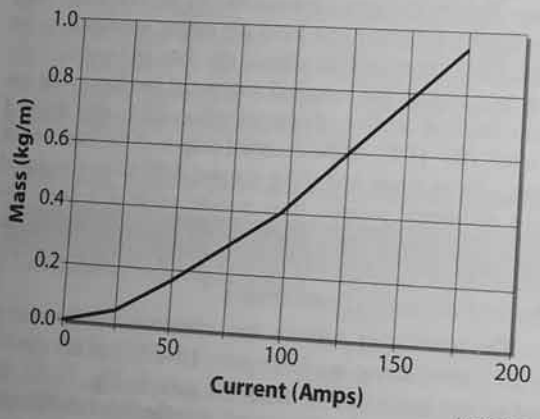


Fig. 21-23. Cable Mass vs. Current. We must account for the cable and harness mass when designing the power subsystem. Operating low current (less than 30 A) devices helps keep this mass low.

Systems for distributing power on spacecraft have been predominantly DC because solar arrays and batteries output a direct current power. Conversion to AC would require more electronics, which would add mass to the EPS.

Power distribution systems are either centralized or decentralized, depending on the location of the converters. The decentralized approach places the converters at each load separately, these are called *point of load (POL)* converters, whereas the centralized approach regulates power to all spacecraft loads within the main bus. The decentralized approach implies an unregulated bus because distributed converters regulate power. A regulated power bus typically has some power converters at the load interface because electronics may require different voltages (+5, ±12 V_{dc}). An advantage of the centralized system is that we do not have to tailor-design the EPS for different applications. Larger spacecraft with high power levels use the decentralized distribution systems, with an unregulated bus, usually.

Fault protection within the EPS focuses on detection, isolation, and correction of faults. Its main purpose is to isolate an electrical short that could eventually degrade or cause loss of the mission. A short circuit will draw excessive current, and if this condition continues, the short may stress cables and drain the energy storage reserve. Typically, we would isolate these faults from the EPS bus with fuses (sometimes resettable). *Fuses* are pieces of wire specially designed to pass a reasonably well defined current, and because current causes the fuse material to heat up, a current higher than that of what the fuse is designed to pass will melt and burn open. Fuses are normally specified to blow in some defined period (e.g., 5 sec) at currents of say 200%. The higher the current through the fuse the faster the fuse will blow (timing is not very reproducible). When sizing a fuse, be conservative. When a fuse in orbit is blown, the function powered off that circuit is lost. A rule of thumb for currents greater than 1 A is to size the fuse to twice the maximum current expected on the circuit when operating normally. At lower currents greater margin is needed. Most spacecraft power loads have some sort of fuse in series with the power bus to isolate faults. If the mission requires us to know where load faults occur, fault-detection circuits can be added.

To design the Power Distribution subsystem, follow the steps in Table 21-26.

21.2.6 Future Power Systems

Future power systems will focus around more power at a lower cost to enable new missions. The enabling factor for future power systems will focus around the power source and the energy storage subsystems. The most common spacecraft power sources with EPS design parameters are shown in Table 21-27. Solar photovoltaics remain the optimum power source selection for most spacecraft missions today. Solar thermal dynamic, RTG, and fuel cell power sources are preferred when solar photovoltaics is no longer a viable candidate. Nuclear reactors will likely be required for manned Lunar or Mars based missions due to power requirements > 100's kW. However, the costs associated with a reactor are not within the projected budget for the majority of US space missions planned for the next decade.

Planetary missions beyond Mars and Jupiter, and Deep Space missions, have an inherent limitation of solar power (solar energy flux) as shown in Fig. 21-24.

Table 21-25◀, Fig. 21-23, Eq. 21-12

Table 21-26. Steps in the Power Distribution Subsystem Design.

Step	Consider	Possibilities
1. *Determine the Electrical Load Profile*	• All spacecraft loads, their duty cycles, and special operating modes • Inverters for AC requirements • Transient behavior within each load • Load-failure isolation	• Low-voltage DC: 5 V • High-voltage DC: 270 V • High-voltage 1-phase AC: 115 Vrms, 60 Hz • High-voltage, 3-phase AC: 120/440 Vrms, 400 Hz
2. *Decide on Centralized or Decentralized Control*	• Individual load requirements • Total system mass	• Converters at each load—for a few special loads • Centralized converters control voltage from the main bus (no specialized power requirements)
3. *Determine the Fault Protection Subsystem*	• Detection (active or passive) • Isolation • Correction (change devices, reset fuses, work around lost subsystem)	• Cable size (length and diameter) and excess current-carrying ability • Size of power storage in case of a short circuit • Location of fuses and their type

Table 21-27. Matrix for Comparing Most Common Spacecraft Power Sources. We may use different factors to select the correct power source but specific power and specific cost are used extensively. Almost all Earth orbiting missions use solar photovoltaic and nearly all missions beyond Mars use radioisotope thermoelectric generator (RTG) based power systems. Solar photovoltaic is the most common power source for spacecraft missions today.

EPS Design Parameters	Solar Photovoltaic	Solar Thermal Dynamic	RTG	Nuclear Reactor	Fuel Cell
Power Range (kW)	0.2–300	5–300	0.2–10	5–300	0.2–50
Specific Power (W/kg)	25–200	9–15	5–20	2–40	275
Specific Cost ($/W)	800–3,000	1,000–2,000	16K–200K	400K–700K	50K–100K
Hardness *– Natural Radiation* *– Nuclear Threat* *– Laser Threat* *– Pellets*	 Low–Medium Medium Medium Low	 High High High Medium	 Very high Very high Very high Very high	 Very high Very high Very high Very high	 High High High Medium
Stability & Maneuverability	Low	Medium	High	High	High
Low-orbit Drag	High	High	Low	Medium (due to radiator)	Low
Degradation Over Life	Medium	Medium	Low	Low	Low
Storage Required for Solar Eclipse	Yes	Yes	No	No	No
Sensitivity to Sun Angle	Medium	High	None	None	None
Sensitivity to Spacecraft Shadowing	Low (with bypass diodes)	High	None	None	None
Obstruction of Spacecraft Viewing	High	High	Low	Medium (due to radiator)	None
Fuel Availability	Unlimited	Unlimited	Very low	Very low	Medium
Safety Analysis Reporting	Minimal	Minimal	Routine	Extensive	Routine
IR Signature	Low	Medium	Medium	High	Medium
Principal Applications	Earth-orbiting spacecraft	Interplanetary, Earth-orbiting spacecraft	Interplanetary	Interplanetary	Interplanetary

Solar photovoltaics have supported most Earth orbiting missions and planetary missions out to Mars. The large solar arrays on Juno (~60 m^2) combined with low overall power requirements (~ 200 W average) enabled NASA's New Frontiers Juno mission.

Sections 21.2.1 through 21.2.5 primarily focused on power systems based upon solar photovoltaics as the power source with batteries for energy storage for LEO and GEO missions. Research and development continues on photovoltaic power source systems, and Fig. 21-25 shows historical advancements along with the anticipated potential for multi-junction cell efficiencies. Even with the solar cell efficiency advancements shown in Fig. 21-25, there are missions where the "Sun does not Shine." For these missions, one must consider the other power sources shown in Table 21-27.

Table 21-27, Fig. 21-23◀, Eq. 21-12◀

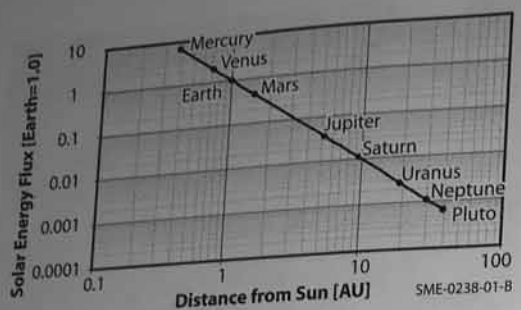


Fig. 21-24. Solar Energy vs. Distance from the Sun. The quick reduction of solar flux past Mars requires very large solar arrays, as shown for NASA's Juno spacecraft and its mission to Jupiter. Missions past Jupiter require a non-solar power source, such as a Radioisotope Thermoelectric Generator (RTG) based power source.

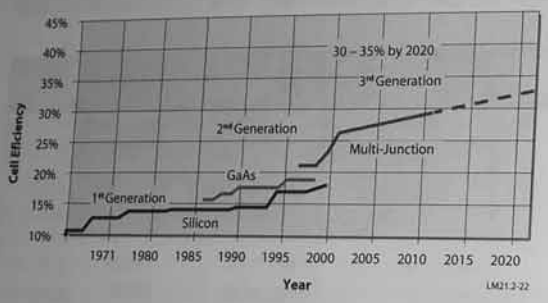


Fig. 21-25. Solar Cell Efficiency Advancements. Multi-Junction solar cells described in Sec. 21.2.3 offer the potential for reduced power generation mass and costs for future missions. Additional work continues on solar cell technologies, especially in areas that promise to reduce the mass or volume of the completed solar array.

For missions that require hundreds of Watts for power where solar energy is no longer viable, the most common power system utilizes heat transport from a reactor (typically fueled with plutonium-238) to drive a power converter based upon the power conversion technologies shown in Fig. 21-26. Static power conversion technologies have no moving parts, while dynamic power conversion technologies utilize a working engine. The most common static power conversion technologies are Thermoelectric and Thermionic. The most common static power source for spacecraft is the thermoelectric couple. This basic converter uses the temperature gradient between the p-n junction of individual thermoelectric cells connected in a series-parallel arrangement to provide the desired DC electrical output from each converter. This temperature gradient comes from slow decay of the radioactive source. The thermal-to-electric conversion efficiency for a thermoelectric source is typically 5–8%.

Thermionic energy conversion produces electricity through a hot electrode (emitter) facing a cooler electrode (collector) inside a sealed enclosure that typically contains an ionized gas. Electrons emitted from the hot emitter flow across the inter-electrode gap to the cooler

Table 21-27◀, Fig. 21-26, Eq. 21-12◀

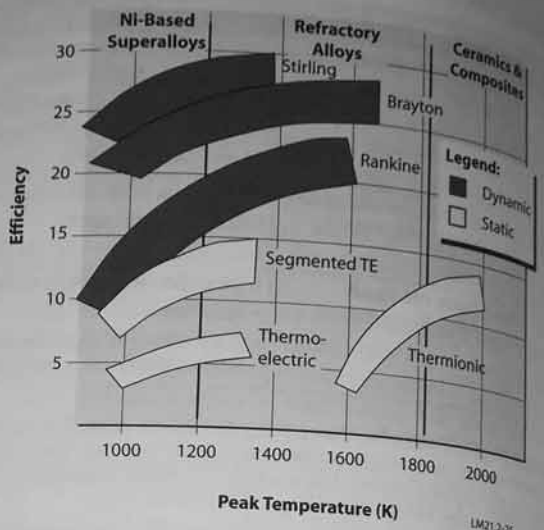


Fig. 21-26. Heat from Reactor to Electric Power Conversion Technologies. Static power conversions technologies have lower efficiencies but high reliability, while dynamic power conversion technologies have high efficiencies with less demonstrated life, thus higher mission risk.

collector. There they condense and return to the emitter through the electrical load connected externally between the collector and the emitter. We choose the collector and emitter temperatures for best overall system performance. In choosing the collector temperature, we try to decrease the weight and size of thermal radiators, and we choose materials based on mission life requirements. Thermionic power sources usually rely on a reactor heat source because of the high temperature required for efficient thermionic conversion. Power efficiencies for a thermionic power conversion are typically 10–20%.

In contrast to static sources, dynamic power sources use a heat source and a heat exchanger to drive an engine in a thermodynamic power cycle. The heat source can be concentrated solar energy, radioisotopes, or a controlled nuclear-fission reaction. Heat from the source transfers to a working fluid, which drives an energy-conversion heat engine. For a dynamic solar-power source, the balance of energy remains as latent and sensible heat in a heat exchanger (molten eutectic salt), which provides continuous-energy to the thermodynamic cycle during eclipse periods. A dynamic power source using a nuclear reactor or plutonium-238 decay does not require thermal-energy storage because the source provides continuous heat. The most common and practical power dynamic technologies are Stirling, Rankine and Brayton. Stirling-cycle engines use a single-phase working fluid as the working medium. The thermodynamic cycle consists of two isothermal processes (compression and expansion) and two constant-volume processes (heating and cooling). Power-conversion efficiencies for Stirling engines are 25–30%. Rankine-cycle engines are dynamic devices that use a 2-phase fluid system employing a boil-

er, turbine, alternator, condenser, and pump. This power-conversion cycle is essentially the same as that used to generate electricity from fossil and nuclear energy on Earth. Power-conversion efficiencies for Rankine-cycle engines are 15–20%. Brayton-cycle engines are dynamic devices that use a single, compressible working fluid as the working medium. The thermodynamic cycle consists of adiabatic compression and expansion stages separated and coupled by stages that add or reject heat at constant pressure. Placed after the turbine, a recuperator-heat exchanger improves the cycle's efficiency. Power conversion efficiencies for the Brayton cycle are 20–35%. Significant mission risk issues for radioisotope based power sources include the availability of the radioactive element, the costs for the fuel, launch safety requirements, and the generally negative public perception towards nuclear power systems.

Fuel cells are a common power source selection for short-duration missions, such as the Space Shuttle. Fuel cells convert the chemical energy of an oxidation reaction to electricity. They are self-contained generators that operate continuously without sunlight, but must carry their own reactant supply. The hydrogen-oxygen (referred to as "alkaline" because of the potassium hydroxide (KOH) electrolyte) fuel cell is the most popular type today because of its relatively high specific power (275 W/kg on the Space Shuttle), low reactant mass (hydrogen and oxygen), and useful by-product (water). A typical single cell produces a voltage of 0.8 V_{dc}. In combination, a fuel cell unit can create many kilowatts of power (each Shuttle fuel cell produces 16 kW peak or 12 kW continuous). The energy conversion efficiency can run as high as 80% for low current draws, but as current increases, the efficiency drops to 50–60%, due to activation over-potential and electrical resistance in the electrolyte solution between electrodes. However, compared with other power sources shown in Table 21-27, fuel cell efficiencies are high. The three Space Shuttle fuel cells are state-of-the-art power generators that produce all of the Shuttle electricity for the 28 V_{dc} bus. Their high efficiency (70%), low weight (260 kg), and excellent reliability (> 99% available) attest to their quality. Other important factors are their 15-minute start-up time, instantaneous shutdown, and long lifetime (2,400 hours before refurbishment). Besides electricity, these fuel cells produce crew drinking water, at a rate of 0.36 kg/kWh, or about 104 kg a day.

Research is underway to solve the short-mission limit with fuel cells, caused by carrying large reactant masses. Because the fuel cell reaction is reversible, we can use electrolysis to create more reactants from the water by-product. To optimize each process, however, we have to use separate units for generating electricity and separating the water. Any long-duration mission could use this regenerative system if it had some input electricity from solar cells, nuclear generators, or other power system during periods of low electrical load. Figure 21-27 shows an example of a regenerative system utilizing electricity from a solar array to drive the electrolyzer unit. The solar array provides power to the electrolyzer and the rest of the electrical bus during the day. The electrolyzer cracks water from the fuel cell into hydrogen and oxygen which are stored in tanks to be used by the fuel cell at night. The fuel cell powers the electrical bus at night by creating electricity from the hydrogen and oxygen as well as creating water that will used by the Electrolyzer.

Space battery improvements over the next 10 years will be focused around the battery chemistries and types defined in Sec. 21.2.3, with focus around Lithium batteries for primary (non rechargeable) applications; and Lithium batteries and regenerative fuel cells for secondary (rechargeable) applications. Fuel cells should be closely monitored as a viable and enabling energy storage technology for future space missions.

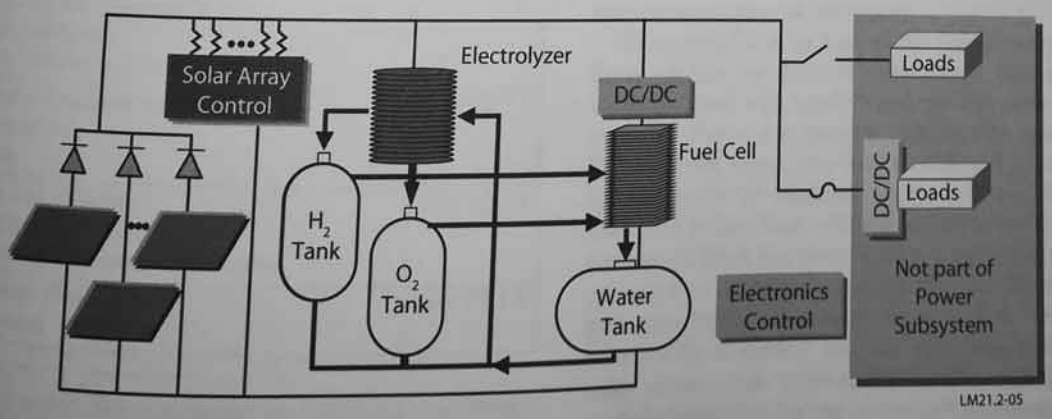


Fig. 21-27. Regenerative Power System. A solar array to power the loads electrolyze water during the day, and a fuel cell to produce electricity and water at night.

Table 21-27◀, Fig. 21-27, Eq. 21-12◀

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22 Spacecraft Subsystems V—Structures and Thermal

22.1 Spacecraft Structures and Mechanisms
Structural Requirements and Constraints; Design Concepts; Structural Analysis and Sizing; Structural Design Verification Philosophy and Criteria; Mechanisms and Deployables; Typical Space Mechanisms

22.2 Spacecraft Thermal Control
Requirements; Radiant Energy Heat Balance in Space; Sources of Radiant Energy in Space; Applied Radiant Heat Transfer and Temperature Control in Space; Thermal Model for Analysis and Design; FireSat II Thermal Analysis and Design; Thermal Hardware, Tools, and Techniques; Thermal Control Components

22.1 Spacecraft Structures and Mechanisms*

Michael Pryzby, *ATK Space Systems*

Having defined the initial spacecraft bus design, it is now time to develop a more detailed spacecraft bus design and arrive at a bottom up mass estimate. At this point in the process, there should be considerable understanding and documentation of the mission and spacecraft systems level requirements. These spacecraft systems requirements will flow down to the mechanical subsystem and allow the mechanical team to arrive at a detailed concept. Following preliminary analysis and sizing, a bottom up mass estimate can be derived for the subsystem. This section will focus on the critical topics that affect the conceptual layout and sizing of the mechanical subsystem. It is not intended to be an analytical tutorial on how to size the system.

Table 22-1. Spacecraft Structure Design Steps.

Step	Reference
Overview Layout of Section	Sec. 22.1
Define Objectives and Constraints Define Structural Requirements and Constraints	Sec. 22.1.1
Define Concepts / Designs Configuring the Spacecraft Design Modeling Design Options and Trade Studies	Sec. 22.1.2
Preliminary Analysis and Sizing Requirements Analysis	Sec. 22.1.3
Evaluate and Validate Verification Philosophy Mechanical Verification and Testing	Sec. 22.1.4
Mechanisms Overview Mechanisms	Sec. 22.1.5

Chapter 14 details the process for establishing a preliminary concept for the spacecraft. That chapter should be re-read or scanned prior to reading this chapter. The mechanical structure is the backbone for the spacecraft, and touches every other subsystem. One should have a good understanding of how all the subsystems interrelate prior to delving into the mechanical layout.

Table 22-1 describes the basic process used to design a preliminary structure. Before you start to layout the structure, you have to understand your design space. The mission level requirements and performance requirements provide the outline of that design space. Several key documents provide the key information to begin to shape the design space. The *Mission Requirements* document can provide substantive details on the mission design life, orbit parameters, launch vehicle class, and payloads specifications. They can also provide constraints, like schedule and cost. *The Payload* or *Science Specification* defines how well the flight segment must perform. From here, you can gather constraints like pointing accuracy and knowledge. Payload performance specifications also provide the interface details for the payloads. A preliminary *Concept of Operations* will partition the mission into numerous phases. For each phase, you can derive the mechanical functions required to meet mission goals. You will also get an understanding of the different spacecraft states and configurations for each phase. From all of these documents, you can then generate a top level mechanical subsystem requirements document.

Once you have an idea of the requirements and functions of the mechanical subsystem, you can start to partition them into smaller and smaller segments or subsystems. For a mechanical system, the system usually breaks down into a couple of modules. Typically, the following types of modules apply: primary bus (primary load path to interface with the launch vehicle), propulsion module (if there is a prop system, it consists of tanks, propellant, thrusters, associated control electronics, and support structure), instrument module (can be an optical bench with or without payload electronics: stability typically drives what is mounted on the module), avionics module (structure for the spacecraft bus electronics, faraday cage, radiation shielding) and, if needed, deployables (solar arrays, antenna systems, covers, and/or shields).

* The author would like to recognize the following contributors to Sec. 22.1: Ed Devine–ATK Space, Mike McEachen–ATK Space-Goleta, Roger Kraus–ATK Space and Michael Golob–Vantage Systems, Inc. (photo contributions).

As you partition the mechanical subsystem, you can also can start a preliminary concept of the spacecraft. As the concept is fleshed out, the requirements and constraints will become more obvious. The iterative process of collecting requirements and constraints will continue indefinitely or until time or money runs out. One of the great unwritten benefits of major mission reviews is that it forces engineers to have to stop designing, document their work, and move on.

At the end of this process detail, a section was included describing the selection and sizing of mechanisms. Mechanisms can be complex little subsystems on their own. They are often organized in the project structure to be part of the mechanical team or subsystem. However, due to their complexity, spaceflight mechanisms can create major havoc with project costs and schedules and are best elevated on par with other subsystems, not just a subset to the mechanical team.

22.1.1 Structural Requirements and Constraints

Figure 22-1 shows a typical process for technical requirement definition per the NASA Systems Engineering Handbook. As mentioned earlier, the inputs to the process (right hand side) are the top level requirements and a baselined concept of operations. In a typical flow down of requirements, the level 2 documents would flow down a mechanical subsystem (level 3) document.

The mechanical systems lead should work closely with the Mission System team during creation of level 2 documents. During that process, the top level requirements for the subsystem can be initiated and traced to its higher level requirements. The process requires numerous discussions with the key stakeholders for the mission. Five of the key personnel for the mechanical team to work with are Instruments/Payloads, Thermal, GNC, I&T, and Launch Vehicle personnel. Keeping the systems team in the loop will only help minimize finding unknowns later. The system team will help find any other critical drivers that may have been overlooked. This will save you from more design changes and iterations later.

The instrument/payload team is the primary reason the mission is flying. Their needs should be the primary focus of all subsystems. I have yet to meet a *principal investigator* (PI) who did not want more capability from the spacecraft than what was originally proposed. PIs will always be willing to negotiate—typically for more. If you recognize that any procurement or selection process is not perfect, you know that PIs cannot always propose their exact needs or wants. There are many trades the PI must go through to be competitive in the selection process. They have to make assumptions about where to best compromise to win the competition. Therefore, some flexibility by the spacecraft leads early in the design process can yield better mission success and results. Pointing is almost always an area where increased performance is welcome. For example, if a PI proposes extremely tight pointing requirements from a bus, their proposal may be judged as too expensive to accommodate. To meet the extreme pointing they desire, the PI may be attempting herculean efforts on their side of the interface or else they may be required to perform a lot of data post-processing to meet mission (level 1) requirements. In discussing needs, you may find a simple option on the subsystem side that can make a huge difference on the instrument side. Quite possibly, the bus structure might require a composite frame to meet weight and stiffness requirements. The low coefficient of thermal expansion (CTE) or stiffness of the composite could increase stability by an order of magnitude and reduce risk on the PI's side of the mission. These key discoveries

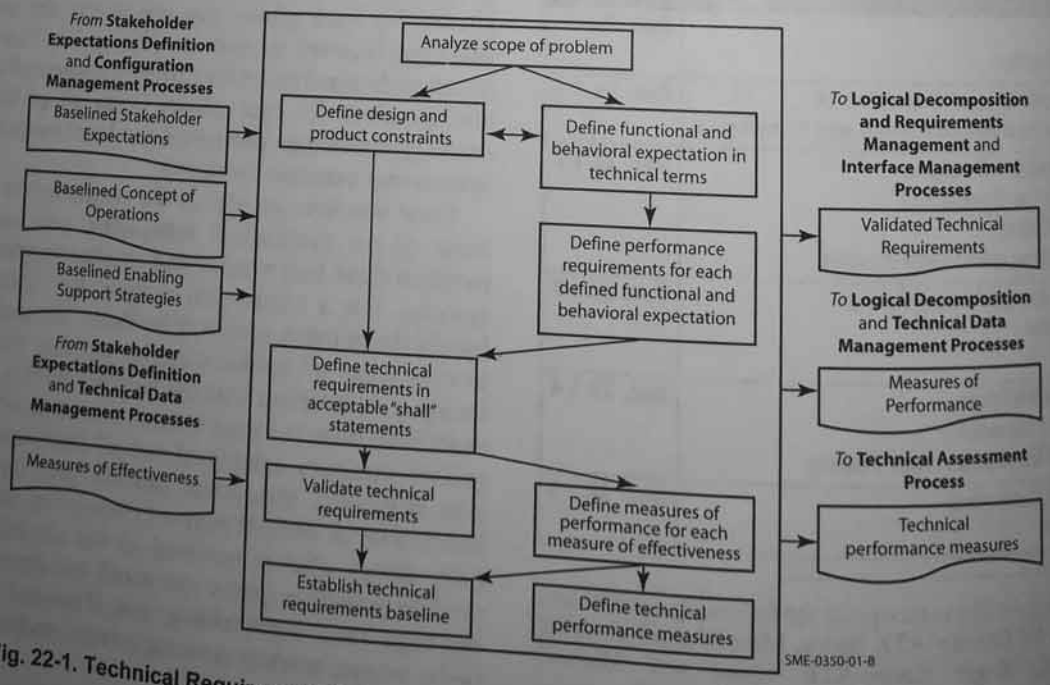


Fig. 22-1. Technical Requirement Definition Process.

Table 22-1◀, Fig. 22-1, Eq. 22-0

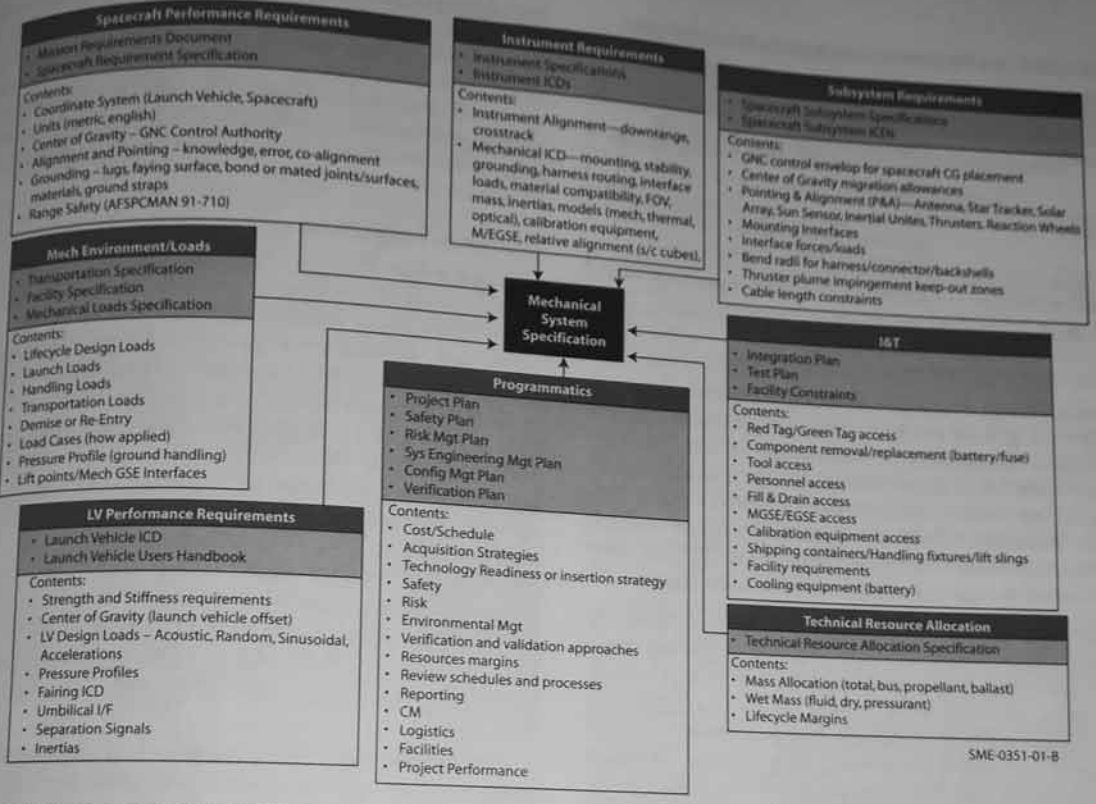


Fig. 22-2. Top Level Document Flowdown to Mechanical Systems.

will be what makes a mission successful. Finding more elegant solutions reduce risk, costs, and schedules by simplifying mission designs, saving resources, and reducing iterations. Unfortunately, for many engineers, the only way to make this happen is to get out and interact with people outside your normal sphere of peers.

> An excerpt from **A Project Manager's Lessons Learned,** [Madden, 2005]
>
> ***Item 75.*** A scientific proposal takes about 9 months to put together. It takes NASA HQ about 9 months to a year to select the winning proposals. Then, it takes 3 to 4 years to sell the program. This means 5 to 6 years after the initial thoughts, the real work starts. Managers, for some strange reason, do not understand why a scientist wants to build something different than proposed. Managers are strange people.

Figure 22-2 lists typical level 1 and 2 documents that will flow requirements down to the mechanical subsystem. Depending upon where you are in the design phase, the state of these documents and requirements can vary quite a bit. The more discussions you can have about these range of requirements and the more questions you can answer, the less chance you will have costly design iterations later.

The following subsections give a quick overview on the types of requirements or constraints the lead mechanical engineer can use to derive the mechanical system specification. This section will list the areas requirements are derived from and a short blurb regarding how the requirement might affect the design.

Performance/Mission

Performance and Mission requirements will provide details and constraints regarding top level items affecting the mechanical subsystem. This will include items like mission life (key for mechanisms to determine number of cycles of the life of the item), pointing and stability requirements, grounding, range identification, launch vehicle class, and orbit parameters.

Environment

Given the mission orbit parameters, numerous environmental requirements can be derived from the influences or concerns highlighted below. See Table 22-2.

Thermal: The thermal environment and mission orbit dictate the size and placement (field of view) of radiators. It also dictates range of temperatures used to analyze thermally induced stresses on the structure or mechanism, including the survival and operational ranges of the hardware. It determines the range of temperatures used to calculate temporal structural stability (how much an optical bench will flex over temperatures, or

Table 22-2. Environmental Concerns by Mission Phase.

Orbit Class or Mission Phase	Environmental Concerns or Influences
Launch	Venting, pressure profile, atomic oxygen, aerodynamic heating, launch loads
LEO	Atomic oxygen, thermal radiation (solar, planetary/albedo, reflected), vacuum, micrometeoroids, magnetic fields
MEO or GEO	Additional cosmic radiation, solar wind
De-Commissioning	Re-entry (loads, thermal) insertion orbit loads

even pointing). It can also dictate the amount of thermal shift from ground test to on-orbit temperatures (key for cryogenic operations). The environment can also constrain component placement options for thermal shielding (mount components internally to the bus (relatively warm/thermally stable) or external to bus (more variability/easier to radiate high power). See Chap. 22.2 for more on thermal design and effects on other subsystems.

Radiation: The radiation environment dictates the amount of radiation shielding required to protect sensitive electronics (more shielding can be provided by adding more metal/material around sensitive components via thicker walls in electronic box packaging or by increasing bus panel thickness), faraday cage shielding (RF shielded/sealed enclosures) grounding paths (faying surface specification, grounding lugs, surface/joint mating materials/designs, materials and coatings).

Launch: The launch vehicle typically drives the design loads (accelerations, acoustic, shock/separation). It provides the spacecraft bus volume constraints (the spacecraft must be packaged within launch vehicle fairing volume), pressure profiles (venting paths of structure and electronics), center of gravity (CG) location (offset from thrust line), umbilical access, propulsion fill and drain access, strength and stiffness requirements of the spacecraft bus, and moment of inertias (affects component placement and packaging). See Chap. 26 for more information on launch vehicle considerations and their affects on other subsystems.

Interfaces/Subsystems

Alignments: There are often requirements that relate the alignment of one item to another. Examples of these types can include Instrument to Instrument (including cross-track, downrange alignments), Instrument to Sensor (Star Tracker, IRU, positional and pointing components relative to an instrument), Antenna to Sensor (ST, IRU to high gain antennas). The requirements can cover one time shifts or deformation usually from a result of launch loads or a system deployment. They can be for on-orbit deflections like from orbit period thermal changes or jitter from an operating mechanism (moving arrays, antenna, optical steering mirrors). There are also longer duration effects, like temporal thermal affects as the spacecraft processes in an orbit over the seasons. See Fig. 22-3 for a sample of a typical alignment budget allocation and flow down. Also see Sec. 6.2 and Sec. 14.5.4.

Mechanical ICD: Many requirements come from just mounting a component. Details like component mounting (area, location, fastener types/size), stability, grounding, interface loads (load specifications can go both ways including high frequency response/jitter, deployments forces, mechanisms forces), material compatibility, field of view (glint, stray light, un-impeded), mass, moment of inertia, models (mechanical, thermal, optical), calibra-

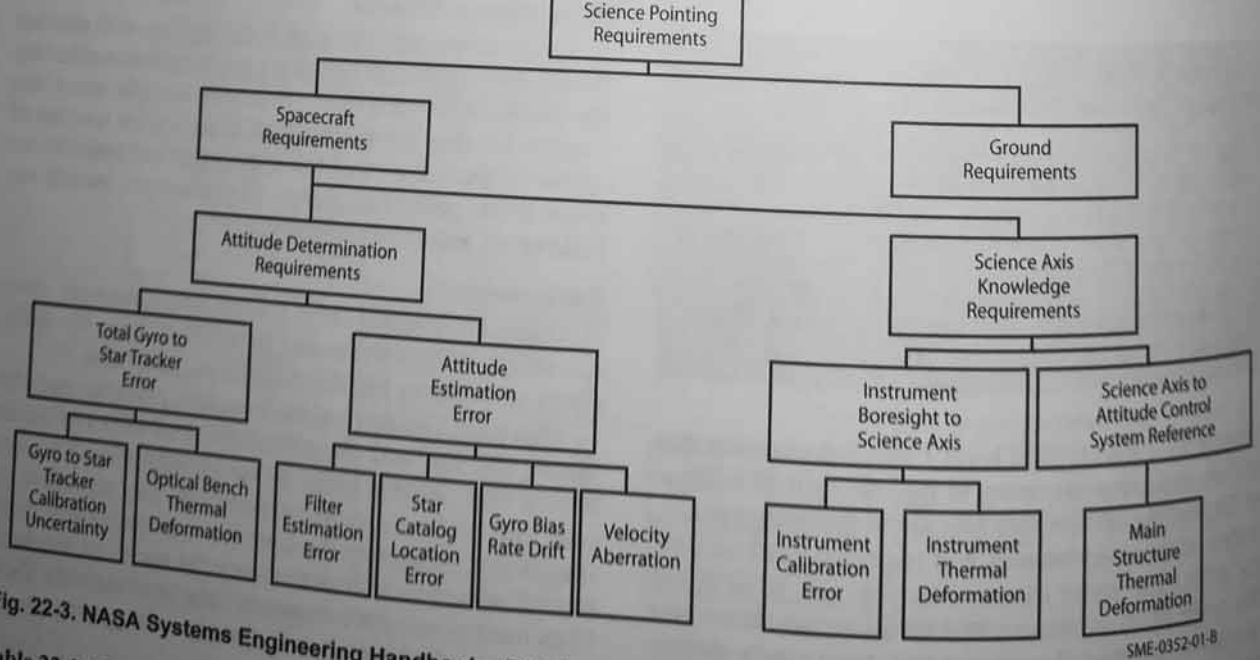


Fig. 22-3. NASA Systems Engineering Handbook—Pointing Requirement Allocation and Flowdown.

Table 22-2, Fig. 22-3, Eq. 22-0

tion equipment access, mechanical/electrical GSE access and interfaces, alignment cube positioning and access for ground alignment, harness routing and strain relief for electrical connector backshells.

Center of Gravity: The launch vehicle and the spacecraft ACS will also provide requirements for the spacecraft CG. The net thrust of a system should travel through the system's CG for better control and more efficient use of propellant. The farther the CG is off the thrust line, the more control pulsing the system will need to do to maintain direction. Care should also be taken to account for CG migration controls (depleting propulsion fuel on-orbit, deployed or jettisoned components/systems).

Programmatic

Maybe surprisingly to many, programmatics can affect many areas of the design. These needs should be addressed early. The effects can be the source of numerous iterations.

Programmatics will define how requirements are levied on the project and how the requirements will be managed. They will address cost and schedule constraints. They can determine acquisition strategies (make or buy decision criteria), technology readiness thresholds or insertion strategy (what level of readiness must all technology be to be considered for the mission and how else do you increase the technology readiness level), safety, risk, environmental management (allowed materials, processes and waste), verification and validation approaches, resource margins, review schedules and processes, reporting, configuration management (CM), logistics, facilities, and project performance. All of these items can affect execution of the mission and mechanical subsystem. There should be time dedicated early in the mission to understand these requirements and make sure they are properly flowed down into the mechanical subsystem specification.

22.1.2 Design Concepts

David Watson, ATK Space

Configuring the Spacecraft

Significant program cost and schedule can be saved by efficiently configuring the spacecraft early in the design process. Decisions made on how the spacecraft looks, where both major and minor components are placed, how easily the structure is to design and analyze, and how efficient the load-paths are through the structure, all greatly affect the success or failure over the life of the program. Therefore, considerable time, resources, and experience should be applied during this phase of the program so that early and accurate assessments of mass, strength, and stiffness can be achieved.

An efficient and robust bus module that is able to support all of the payloads, the equipment boxes, the solar arrays, and the propulsion module should be the goal when configuring the spacecraft. Decisions on placement of components should be made with these thoughts in mind:

1. Effect on center of gravity
2. Length of electrical harness required for components
3. Electrical grounding needed for components
4. Amount of support structure required
5. Stiffness required

An example of this packaging effort would be placing as many of the large mass components and subsystems (e.g., main payloads, propulsion module) along the centerline of the Spacecraft. This is important because the launch vehicle interface is typically the center of the structure where all of the load must be reacted. If the major mass items are centered over the launch vehicle interface, then the eccentric loading, and resulting necessary structure to support it, will be minimized. Also, most launch vehicles require a lateral center of gravity to be within 12.5 cm of the centerline. This requirement becomes difficult to achieve if major mass items are offset from the center.

An efficient structure can also be achieved by placing as many large mass items as low as possible on the bus structure. This packaging will reduce the overall bending moment at the launch vehicle interface and, in the process, increase the overall spacecraft launch frequency.

Once the large mass items are conceptually identified, the engineer can estimate the vertical and lateral center of gravity using a simple sketch and spreadsheet. Calculating the $\Sigma(Md)$ divided by the ΣM, where M is the mass of each major component and d is the distance of the same component from each axis, vertical or lateral as applicable, will produce an approximate center of gravity (CG). This calculation can be done with a limited number of major mass components or subsystems, typically the payloads, the propulsion system/tanks, solar arrays, the bus structure (assume 20% of wet mass), batteries, and any other unique heavy spacecraft component. Initially, I like to use a rule of thumb of any component used in this preliminary calculation should be at least 5% of the total mass of the system. (One exception will be for post launch deployed systems. The distance of a deployed system can be large enough to shift the CG, especially if any fuel or propellant has been expended, even for smaller masses.) The center of gravity sensitivity can then be monitored by varying the location of these items within the spreadsheet.

Design simplicity should always be maintained throughout the configuration process. If the primary structure of the spacecraft is difficult and complex to design, then it will be equally hard to analyze and build. If the structure is not robust then the risk of failures during component, subsystem, and system-level testing is increased. Increases in complexity and risk invariably lead to significant cost and schedule pressures on the program.

Design Modeling

All of the spacecraft design concepts and configurations that are generated in the conceptual and preliminary

Table 22-2◀, Fig. 22-3◀, Eq. 22-0

design phases of the program are usually documented and visualized using a *Computer Aided Design* (CAD) tool. The most popular CAD tools in the spacecraft industry are Pro-E, NX (formerly IDEAS and Unigraphics), and Catia. These CAD tools are 3-dimentional modeling suites that spacecraft designers and engineers use from conceptual design through detailed modeling and then ultimately to production models and drawings that will be used to manufacture all of the structural components. ISO 10303, STEP AP-203 provides a standard format for exchanging data between CAD tools.

In the conceptual design phase, it is important to model the major bus, propulsion, and payload modules to determine overall system envelopes and interfaces. This CAD modeling effort includes, but is not limited to, the following tasks:

1. Assembly layout of all major components of the spacecraft to determine required interfaces and envelopes. These mechanical interfaces will need to be managed throughout the program with Mechanical Interface Control Drawings and Documents (MICDs).
2. Placement of assembled model in the launch vehicle fairing to verify static and dynamic clearances or interferences. This launch vehicle check will also be used to design the interface between the bus module and the launch vehicle adapter.
3. Modeling of all component and instrument Fields of View (FOV) to ensure full performance within requirement specifications. Any interferences must be negotiated at a system level early in the program to make the necessary mechanical or performance changes.
4. Modeling of all major bus and payload equipment to ensure adequate volume and access as well as assist in determining mass and center of gravity (CG) limits.
5. Transfer of CAD model to other spacecraft subsystem engineers for concurrent modeling. The structural analysis and thermal analysis engineers will use the CAD model to translate geometry into a system level finite element model and thermal model, respectively. The power engineers will use the CAD model to verify expected solar array area.

When done properly, the CAD model can also be used to help the engineer and program estimate the total mass of the spacecraft. In general, the structural mass of the spacecraft will range between 15% and 20% of the overall wet mass of the spacecraft. This rule of thumb can be used as an early program guideline from which to confirm the validity of the conceptual design. The CAD model and the finite element model can be used to capture the mass of the primary structural elements but the design team should carefully add other items that are often overlooked:

1. Secondary structure such as brackets for antennas, star trackers, and reaction wheels
2. Fasteners and inserts
3. Adhesives
4. Honeycomb panel inserts and fittings
5. Paints and thermal coatings

The program assumed Weight Growth Allowances (WGAs), not intended to account for these excluded items and often are not enough to cover the increases in mass that are produced when all of these are calculated.

Design Options and Trade Studies

Performing trade studies is an important part of the design process. Trade studies are needed early in the program to ensure that the most informed decision can be made by the program with the most available information at that time. The goal of a trade study is to choose the most efficient and cost-effective technical solution that meets all of the necessary requirements. Risk, reliability, producibility, cost, and schedule should all be included and evaluated when performing a trade study and making a decision. Engineers have a tendency to want to make a design optimal, instead of acceptable, for a given set of requirements.

When performing a trade study, be sure to understand customer priorities. For example, over-optimization of mass may lead to expensive materials or complicated structures. Often, cost and schedule benefits can be realized by sacrificing mass.

Simplicity should also be considered when performing a trade study. The simplest design may not be the lightest but it is often the most cost-effective solution. A simple design usually has fewer parts to design, analyze, manufacture, and assemble which results in cost and schedule savings at every stage of the program.

Structural trade studies often involve the general construction of the bus, the component materials, and the method of attachment of the bus structure. Typically, the primary structure is composed of a combination of machined metallic components, composite beams and struts, and honeycomb panels (two facesheets bonded to a core structure). An attempt should be made in the conceptual design phase to decide between a truss structure versus honeycomb panels, primary composite versus metallic components, and whether to bolt the components together or bond them. Each of these decisions will allow the rest of the assembly to be packaged and configured.

A truss design is often a lightweight solution of beams (composite or metallic) that efficiently transfer spacecraft load from one area to another. A truss works well when a large expanse of space needs to be covered without many secondary features needed to be attached in

that area. A truss structure beam can be of various cross-sections; rectangular, circular, and I-beams are the most typical. These truss shapes connect at "nodes" that are typically metallic fittings with arms that attach to each truss member. A honeycomb panel will often be a better solution over the truss if structure is required to be available to mount or attach equipment boxes, thermal hardware, or other spacecraft components. The honeycomb panel may weigh more than the truss beam, but often less secondary structure is required. All of the spacecraft components can attach to the honeycomb panels using standard structural inserts.

The material trade is also important. Choosing a composite material or a metallic material affects all aspects of the structure from early design ultimately to assembly and testing. Since composite materials typically require significant development, qualification, and testing, they should only be used when necessary. Composite materials are often required when instrument pointing accuracy is required. Composites typically vary minimally when exposed to temperature changes and, therefore, will remain very stable in many launch and on-orbit spacecraft environments. This thermal stability is important for a structure that is supporting a very accurate and sensitive instrument. When designed properly, composite materials can also show a mass benefit over traditional metallic structures. This mass benefit must be weighed against the potential increased cost and schedule impact. Composites cost more because the raw materials are expensive and the development costs of the tooling and building-block testing are non-trivial. These development costs can often be offset if multiple units are being made or if tooling can be reused.

Even in the early design stages, the method of structural attachment should be considered and baselined. Most structure is either bolted or bonded together. A bolted connection is typically made with a high strength steel bolt and a corresponding nut or other retention device. A bonded connection is made with an adhesive that is applied between the interfaces and cured at the appropriate temperature. In general, the bonded attachment is used with composite connections but the bolted attachment can be achieved with either metallics or composites. The decision to go with bolts or bonds should primarily be made based on the assembly plan. Bonded joints are permanent while bolted joints can be removed. Bonded joints also often require a non-destructive inspection to confirm the structural integrity which can be time-consuming. Bolting and bonding the same components is unnecessary and should be avoided.

22.1.3 Structural Analysis and Sizing

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Requirements

Structural analysis is often required to perform preliminary sizing of the primary structure. This allows a more thorough resource estimate to be made. It also allows an assessment of the preliminary design against the mission requirements. These mission requirements are generally defined by the following:

- Loads & Environments
- Stiffness
- Design & Strength Criteria
- Performance (i.e., pointing)
- Accommodations (e.g., Interfaces, Layout)
- Mass Properties (e.g., DNE Mass, CG Location)

The loads and environments are often considered the largest design driver for a structure. These loads and environments come from manufacturing, ground operations, ground testing, launch (landing), and on-orbit.

The launch loads and environments are often the driving case for most spacecraft structures. These loads are often launch vehicle and mission dependent. The landing cases are often only considered for STS payloads, sounding rocket recovered payloads, and interplanetary structures. Preliminary loads and environments are defined in launch vehicle mission planner's guides which are often available to the public. See Chap. 26.

Manufacturing stresses can be significant for brittle parts such as Beryllium and for parts that require high temperature curing such as graphite fiber reinforce plastic (GFRP) parts. Procedures and inspections should be well-planned to mitigate the risk of failure due to manufacturing.

Ground operations such as crane lifts and transportation should also be considered. Lifts load the structure in configurations that are not seen in flight. These lifts often drive the design of large structures. Transportation loads from an air-ride truck or aircraft can also be significant for large spacecraft as well. A shock isolation system is often considered to minimize the exposure of sensitive hardware.

Ground testing such as thermal vacuum testing in 1 g and workmanship screening of avionics boards can drive structural designs as well.

On-orbit environments are usually driven by the thermal design. The thermal environments should be estimated during ascent and operations. The structure should be designed to survive exposure to these environments as well.

Structural stiffness is often driven by the launch vehicle loads and environments. The stiffness is often measured by the natural frequency of the structure. The first natural (fundamental) frequency of the spacecraft is usually required to be greater than a value to prevent dynamic coupling with the launch vehicle fundamental frequency. These values are also defined in launch vehicle mission planner's guides.

The design and strength criteria are often defined at the organizational level. Standards such as NASA-STD-5001 [2008], AIAA-S-110-2005 [AIAA, 2005], and MIL-STD-1540B [DoD, 1989] have



Table 22-2◀, Fig. 22-3◀, Eq. 22-0

been developed for this purpose. Other, more detailed standards have been created at lower organizational levels such as the General Environmental Verification Standard GEVS (GSFC-STD-7000) developed at NASA Goddard Space Flight Center [2005]. These standards define the minimum strength margin required for a structural design. Most implement this through the use of design factors of safety that depend on the material, failure mode, and verification philosophy.

The structural design and analysis is also often driven by mission performance requirements. These requirements such as pointing (bias and stability) require significant engineering and analysis of structure material and interfaces. Interfaces are designed using a kinematic (flexure) mounting scheme to allow the structure to expand and contract during operation thermal excursions. Interfaces are also designed with press-fit shear pins to prevent shifting during launch and operation.

The physical layout of the structural design is most driven by the accommodation of the launch vehicle interface, payloads, and other spacecraft subsystems. The launch vehicles often use standard interfaces (such as marmon bands of pre-defined diameter) that should be used for the design. Interfaces to the launch vehicle and heavier items, such as propellant tanks, are important to define early in development because the primary structural load path is defined using their geometry. Payloads generally have requirements for field of view (regard) that must be met.

Structural Analysis

Analysis of the structure can begin once most of the requirements and preliminary design have been defined. The first step in the analysis procedure is to create a structural model of the preliminary design. Following the creation of a structural model, the model is analyzed to demonstrate compliance of the most driving requirements. This analysis is often divided into the areas of loads and dynamics, and stress analysis.

Finite Element Model

Preliminary sizing analysis typically utilizes a finite element model (FEM) of the primary structure and significant mass items. The spacecraft geometry can be imported into finite element pre-processors in STEP and IGES formats. The primary structure is modeled using a mesh of two-dimensional (e.g., plate, laminate) elements to represent rings and panels. Truss and frame structures are represented using beam and bar elements. Typical spacecraft mesh sizes can easily maintain a node-spacing of approximately 1 inch or less using today's computers. Instruments and components are readily represented with lumped mass elements at the center of mass locations and rigid-body elements (RBEs) connecting the lumped mass node to the interface locations. Rigid-body interpolation elements (such as RBE3s) that provide no artificial stiffness are recommended at this stage to ensure that all structure is sized adequately. Large and flexible subsystems such as solar arrays should be modeled similar to the primary structure if a design has been established. These items often drive the fundamental frequency and the size of the structural members. Instruments and other significant mass items that require special (semi-kinematic) mounting should be modeled due to their effect on the stiffness and strength of the structure. Physical models of the mounts should be included if possible or appropriate interface degrees of freedom should be released at a minimum. Nonstructural mass should be added to equipment panels and other structure to account for the mass allocated to other nonstructural items. Finally, the mass of the model should be scaled to maximum mission capability at this early development phase. This should be defined by the mission systems engineer and is usually limited by the launch vehicle or spacecraft propulsion capabilities.

The model is constrained at the launch vehicle interface which is usually the center of the separation plane. The constraints depend on the type of separation system used for the mission. For a standard marmon band, the model is constrained in all degrees of freedom at the nodes around the circumference of the launch vehicle interface ring.

Finally, the model should be verified using the appropriate validity checks before beginning any analysis. The free-free modes check, unit gravity check, and the unit displacement check should be performed at a minimum. These checks demonstrate that there are no numerical or modeling issues and that the model is free of unintended constraints. A thermal stress check should also be performed on models used for any thermal stress and distortion analysis.

Loads and Environments

The first step in the analysis procedure begins with an assessment of the mission loads and environments. The results of the assessment should reveal which loads and environments drive the design. These results will allow the analysis effort to take focus.

The primary structure launch limit loads (load factors) are the best place to start for preliminary sizing of the spacecraft structure. These load factors are usually computed for each significant launch event, such as liftoff and maximum dynamic pressure (max Q). The load cases are typically provided as combination of axial and lateral loading (relative to the launch vehicle coordinate system. See Sec. 26.5.1 for typical loads. It should be noted that smaller launch vehicles often have higher load factors. It is important to note that these preliminary loads are typically only valid for spacecraft of typical mass and stiffness. In other words, spacecraft must meet the launch vehicle mass properties and frequency requirements to ensure that the preliminary loads are adequate and stable. In all cases, these load factors should be verified using mission-specific launch vehicle coupled loads analysis (CLA) as described in Sec. 26. The first CLA cycle is typically performed approximately around the preliminary design review phase of project development.

Sine vibration environments are derived from analysis of the launch vehicle dynamic environments. The sine vibration environments are estimated by computing the shock response spectra of the launch vehicle transient events and sustained sinusoidal environments (such as POGO). See Sec. 26.5.2. These environments are used for analyzing the dynamic response of the spacecraft, estimating the loads at subsystems and components, and testing the spacecraft for sensitivity to the low-frequency dynamic environment. The sine vibration environments are derived at the spacecraft interface for both axial and lateral directions and are typically limited to frequencies between 5 and 100 Hz. The maximum expected flight levels are typically on the order of 1G peak amplitude. Limiting or notching is often required due to differences between the launch vehicle and test/analysis configurations. Limiting the sine vibration responses to not exceed the CLA levels is typical. Therefore, the sine vibration environment does not typically drive the design of the primary structure.

The vibroacoustic environment is induced by high-intensity acoustic noise created during the launch phase. See Sec. 26.5.3 for more details. The term is used because the environment manifests itself throughout the payload in the form of structure-borne random vibration and as transmitted acoustic excitation. Therefore, the common practice is to specifically consider both the acoustic and random vibration environments and test levels [NASA-STD-7001 1996]. This vibroacoustic environment primarily ranges from 20 Hz to 10,000 Hz. Most will find that the spacecraft level random vibration environments are negligible for larger launch vehicles and spacecraft. The acoustic environments often drive the design of large, lightweight panels with area-to-weight ratios greater than 0.15 m^2/kg. Analysis and/or testing of the spacecraft should be used to create vibroacoustic test specifications for components and instruments. This should be done early in the mission development because the random vibration environment is typically the driving case for the design of small (<20 kg) components and instruments. However, for smaller spacecraft, random vibration testing is sometimes performed instead of acoustic testing. Again, test levels from NASA-STD-7001 or random environments specific to the particular launch vehicle will be applied to the base of the spacecraft.

The shock environment is induced by actuation of shock producing devices such as pyros. Section 26.5.2 provides further explanation of this environment. These devices typically produce large high-frequency accelerations that damage small, brittle hardware. The environment at shock sensitive hardware is most often driven by the launch vehicle separation system. The peak values often range from 0.5 kg to 2 kg. Due to difficulty of analysis and screening of workmanship issues, these shock-sensitive items are often verified by testing.

The thermal environment is derived using thermal analysis of the spacecraft in significant ascent trajectories, orbits, and operational modes. Typical spacecraft temperatures range from 200 K to 330 K. A more extreme thermal environment can significantly drive the design to minimize thermal stresses and distortion. It may also require the selection of more exotic materials to maintain adequate material properties, maximize thermal isolation, and minimize outgassing.

Once assessment of the loads and environments is complete, an analysis of the structure under the most stringent loading conditions can begin.

Dynamics

Preliminary sizing of the spacecraft structure for dynamics begins with analysis of the spacecraft stiffness and mass properties. It is important to understand these properties because they largely determine the stresses experienced by the spacecraft during launch. The stiffness and mass properties are best characterized by performing a finite element analysis, (FEA) and normal modes analysis. The normal modes analysis finds the eigensolution to the undamped, harmonic, multiple degree of freedom system equations of motion. The equations of motion include the stiffness and mass matrices created from the FEM described above. The results of the normal modes analysis are both eigenvalues and eigenvectors which correspond to the square natural frequencies and natural mode shape, respectively. Spacecraft fundamental frequencies in launch configuration are typically between 8 Hz and 50 Hz.

The normal modes analysis mode shapes and element strain energy should be reported as well. These plots can be used to better understand the dynamics and troubleshoot any modeling issues. The element strain energy is an important design tool that measures the strain energy in each element for each mode. The natural frequency can be most easily modified by changing the stiffness of the elements with the highest strain energy.

Finally, modal effective mass fractions should be reported for each mode. The modal effective mass fraction measures what fraction of the overall mass is participating in each direction of a mode. The modal effective mass fraction in each direction (degree of freedom) sums to 1.0 over all system modes. This important parameter indicates which modes may be a significant contributor to the loads of the primary structure. Typically, modes with modal effective mass fractions greater than 0.05 (5% of the total mass) are considered significant. The significant mode with the lowest natural frequency (fundamental mode) should be evaluated against the launch vehicle and/or spacecraft frequency requirements described above. Typically, the results should show a margin of 15% over the frequency requirement during preliminary sizing analysis.

Another aspect of dynamics that should be considered is that of appendage deployments. Deployment hinges, particularly the root hinge, as well as hinge interfaces must be evaluated against the potential sudden start/stop loads of deployments, especially at the end of travel. Excessive loads/stresses generated at deployment hinges

Table 22-2◀, Fig. 22-3◀, Eq. 22-0

may lead to the necessity of dampers or other devices to control the deployment motion.

Spacecraft Fundamental Frequency

We can quickly estimate the fundamental frequency of a spacecraft as a sanity check on the configuration. Most spacecraft have a fundamental mode shape similar to beam bending mode when constrained at the launch vehicle interface. Therefore, we can use a cantilever beam assumption and estimate the fundamental frequency using the following equation [Young and Budynas, 2001].

$$f_n = \frac{1}{2\pi}\sqrt{\frac{3EIg}{WL^3}} = \frac{1}{2\pi}\sqrt{\frac{3EI}{ML^3}} \qquad (22\text{-}1)$$

where f_n is the fundamental frequency (Hz), E is the modulus of elasticity, I is the area moment of inertia, g is the acceleration due to gravity, W is the weight of the beam, M is the mass of the beam (W/g), and L is the length from beam root to center of mass.

We can also use empirical data to estimate the fundamental frequency of the spacecraft. The plot below shows historical data which plots the spacecraft (Observatory) fundamental frequency versus the launch mass of several missions. The data generally shows that the fundamental frequency decreases with launch mass. However, the data and trending analysis also shows that the fundamental frequency can vary significantly for spacecraft of similar mass. These frequency variations are due to spacecraft design differences such as structural materials, payloads, propellant mass, separation systems, launch vehicle interface, and spacecraft layout. Therefore, care must be taken when applying these results to a particular mission. Alternatively, this data also shows that the systems and structural engineers can have a significant influence on the spacecraft stiffness.

Example: FireSat II

We can apply the above equation to our FireSat II example mission. We first need to address some key assumptions based on our initial understanding of the mission.

Key Assumptions:

- 2 spacecraft stacked in launch configuration
- Octagonal spacecraft 1.2 m across the octagon and 1.1 m tall
- Separation systems have negligible effect on stiffness
- 260 kg total mass
- Center of mass is at the geometric center of the spacecraft launch configuration (1.1 m)

The beam bending stiffness (EI) in Eq. (22-1) is intended to be constant along the entire length. However, this is obviously not the case for any realistic spacecraft

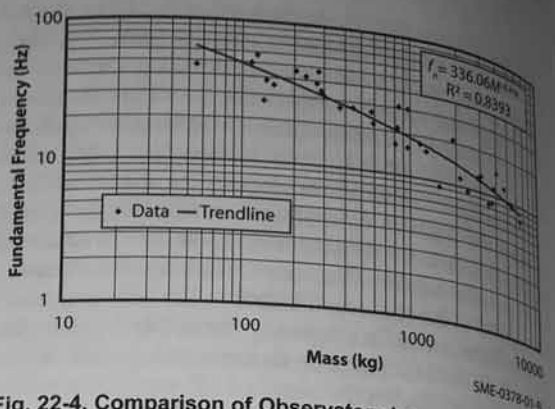


Fig. 22-4. Comparison of Observatory Launch Mass Versus Fundamental Frequency. (Graphic courtesy of NASA)

manufactured from multiple parts and materials. Therefore, we shall assume the bending stiffness based on the properties at the launch vehicle interface because this will typically be the area of the structure with the largest strain energy. FireSat II is a low-cost mission; therefore, we will first assume a simple, all metallic structure to save cost. Therefore, we will use the modulus of elasticity of Aluminum 6061 equal to 68.2 GPa. We will also assume a ring at the launch vehicle interface with an outer diameter of 1 m and a wall thickness of 1 mm. Therefore, the area moment of inertia for this beam is calculated as follows:

$$I = \frac{\pi\left(d_o^4 - d_i^4\right)}{64} = \frac{\pi\left(1.000^4 - 0.998^4\right)}{64} = 3.92\times10^{-4}\ \text{m}^4 \qquad (22\text{-}2)$$

where d_o = outer ring diameter, d_i = outer ring diameter. Using the above results in Eq. (22-1), the fundamental frequency is estimated in the following equation

$$f_n = \frac{1}{2\pi}\sqrt{\frac{3\times68.2\times10^9\times3.92\times10^{-4}}{260\times1.1^3}} = 76.6\ \text{Hz} \qquad (22\text{-}3)$$

The FireSat II fundamental frequency is estimated at approximately 77 Hz using the given assumptions. Note that this estimate is considered an upper bound because this equation assumes an ideal structural load path and doesn't account for the added flexibilities of the upper separation system.

For FireSat II, we can also estimate the spacecraft fundamental frequency using the empirical data in Fig. 22-4. This data shows that we should expect a fundamental frequency of approximately 35 Hz. In fact, it could be lower because this empirical data does not include many samples of stacked spacecraft connected via separation systems. The separation system compliance often plays a big role in the stack fundamental frequency;

Table 22-2◀, Fig. 22-4, Eq. 22-3

therefore, further analysis may be required if there is a concern about the FireSat II fundamental frequency.

Fundamental Frequency of Deployable Systems

Large deployables often have fundamental frequencies of approximately 1 Hz or less. The deployable fundamental frequency is an important parameter for the ACS engineers to consider during design of the spacecraft attitude control system. We can quickly estimate the fundamental frequency of a deployable such as solar array using simplified equations. One method to estimate the fundamental frequency of a solar array is to assume it is a cantilever beam with its root at the spacecraft interface. Assuming the spacecraft interface is fixed is a reasonable assumption if the spacecraft is significantly heavier than the solar array. We can then apply Eq. (22-1) as follows

$$f_n = \frac{1}{2\pi}\sqrt{\frac{3EI}{ML^3}} \tag{22-4}$$

In this case, we would assume a beam bending stiffness (EI) based on the properties of the deployable structure adjacent to the spacecraft interface, such as the boom. We would also apply the mass properties of the deployable.

The deployable fundamental frequency may also be estimated using an alternative method. Due to the relatively low stiffness of the actuators or hinges, these parts often drive the fundamental frequency of the entire solar array. Using this method, the solar panel is assumed to be a rigid body and the fundamental frequency is estimated using the following equation,

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k}{J}} \tag{22-5}$$

where f_n = fundamental frequency (Hz), k = the actuator or hinge torsional stiffness, and J = the mass moment of inertia at the center of rotation of the actuator or hinge. Note that this method requires some knowledge of the deployed actuator or hinge torsional stiffness.

Example: SCS Solar Array

We can use the methods described above to estimate the deployed fundamental frequency of the SCS solar arrays. SCS is a 200 kg, cube-shaped spacecraft carrying two deployable solar arrays. The power system requires that each solar array support 1.7 m^2 of cell area. The total mass of the solar panels is estimated at 9.5 kg (cells and substrate) based on the initial power system estimates. Based on this information, we make the following key assumptions,

- 1m cube-shaped spacecraft
- Solar array mass is 4.75 kg each, including cells and substrate
- Solar array panel is 1 m × 0.7 m
- Center of mass is at the geometric center of the array
- A boom length of 0.5 m is required to accommodate the field of view requirements

It should first be noted that a 1.7 m^2 deployed solar array is a relatively small array and SCS is a low-cost mission. Therefore, we will first assume a simple, all metallic solar array structure to save cost. We will assume the solar array boom is made of Aluminum 6061 and has a modulus of elasticity equal to 68.2 GPa. We will also assume the boom is a tube with an outer diameter of 50 mm and a wall thickness of 0.5 mm at the spacecraft interface. Therefore, the boom area moment of inertia is calculated as follows:

$$I = \frac{\pi\left(d_o^4 - d_i^4\right)}{64} = \frac{\pi\left(0.050^4 - 0.049^4\right)}{64} = 2.38\times10^{-8}\ \text{m}^4 \tag{22-6}$$

We then assume that the center of mass is at a distance of 1 m from the spacecraft interface (boom length plus half long dimension of panel). Using these assumptions, we estimate the solar array fundamental frequency using Eq. (22-4),

$$f_n = \frac{1}{2\pi}\sqrt{\frac{3\times68.2\times10^9\times2.38\times10^{-8}}{4.75\times1^3}} = 5.1\ \text{Hz} \tag{22-7}$$

The SCS solar array fundamental frequency is estimated at approximately 5 Hz using the given assumptions. Note that this estimate is considered an upper bound because this equation assumes an ideal structural load path and doesn't account for the added flexibilities of hinges and actuators.

We next estimate the fundamental frequency of the system assuming that the actuator stiffness is known to drive the system stiffness. We will assume an actuator torsional stiffness of 10,000 Nm/radian.

We next need to calculate the mass moment of inertia, J, at the center of rotation of the actuator. For simplicity, we will assume the elevation actuator is at the spacecraft interface. We first estimate the solar array panel mass moment of inertia J_{panel} at the center of mass assuming a rectangular, homogenous panel

$$J_{panel} = \frac{m_{panel}\left(a_{panel}^2 + b_{panel}^2\right)}{12} \tag{22-8}$$

where m_{panel} is the mass of the panel, a_{panel} is the length of side 1, and b_{panel} is the length of side 2. Using our assumptions outlined above,

$$J_{panel} = \frac{4.75\left(1^2 + 0.7^2\right)}{12} = 0.59\ \text{kg m}^2 \tag{22-9}$$

We can safely assume the boom mass moment of inertia is negligible. However, we must now use the parallel axis theorem to transform the solar array panel mass moment of inertia from the panel center of mass to the actuator center of rotation.

$$J = J_{panel} + m_{panel} d^2 \qquad (22\text{-}10)$$

where d is the length from the solar array panel center of mass to the actuator center of rotation. In this case, d is equivalent to 1 m based on our previous assumptions.

Therefore, the mass moment of inertia is calculated as follows,

$$J = 0.59 + 4.75 \times 1^2 = 5.34 \text{ kg m}^2 \qquad (22\text{-}11)$$

Finally, we calculate the fundamental frequency of the solar array system with actuator from Eq. (22-5)

$$f_n = \frac{1}{2\pi}\sqrt{\frac{10000}{5.34}} = 6.89 \text{ Hz} \qquad (22\text{-}12)$$

Therefore, the solar array panel fundamental frequency is estimated at 6.9 Hz using this method.

These examples show that both the structure and mechanisms can drive the fundamental frequency of a deployable. Therefore, care must be taken during the design and analysis process of both items to ensure that the deployed system meets the ACS frequency requirements.

Stress

Spacecraft stress analysis is often performed using both *Finite Element Analysis* (FEA) and "hand" calculations. Once a *Finite Element Model* (FEM) is created, the stress analysis should be performed using the driving loads and environments uncovered from the assessment described above. As mentioned above, the driving loads for the primary structure are typically the spacecraft launch limit loads. The analysis is performed in multiple load cases by applying the axial load factor and clocking the lateral load factors around in all lateral directions (every 22.5 deg, typically).

The analysis results should provide stresses that should be evaluated against material allowables using a margin of safety calculation and appropriate factors of safety. Material properties are most often found in well-established sources such as MMPDS-01. However, GFRP composites and other non-traditional materials have properties that depend on the manufacturing process and the specific layup. Properties for these items should be attained from material characterization testing of coupons that represent the design materials and manufacturing process.

The specific stress components or combinations required from the analysis depend on the type of material and material properties. Von mises stresses are used to evaluate isotropic and elastic materials, such as aluminum. Brittle and non-isotropic materials require assessment of the margins of safety using the stress components and interaction criteria. GFRP composites often require margins of safety to be evaluated at the ply-level.

Forces are recovered from the FEM to evaluate failures of specific items separate from the FEA. This is because the FEM doesn't have the fidelity to accurately calculate the stresses in those areas. These include inserts/fittings in honeycomb panels, fasteners, and bonded joints. It is wise to compare the recovered forces in primary structure joints to allowables recovered from a detailed joint FEM or previous test data. These joints are often the critical part of the design and can add significant mass.

The stress analysis results should show positive margins of safety for the closing design. All critical margins of safety should be reported. Stress contour plots should also be reported to demonstrate the locations of the critical margins.

Mass Estimate

The mass of the structure should be updated once a preliminary design has been established using the procedure outlined above. The estimate should include all structural entities modeled using physical elements. It should also estimate the mass of fittings, inserts, and other members not represented with physical elements. A maturity factor should also be presented with the estimate.

At this point, there should be two independent mass estimates: one from the CAD model and one from the FEM. In working with the FEM to perform preliminary sizing, we should now have a good approximation for the mass of the mechanical system. As a sanity check, the mechanical system, including the primary structure, secondary structures, and fasteners, should be approximately 15% of the dry mass. Dry mass is the mass of the spacecraft less any propellant (wet or dry) or any gases (expendables) used in the propulsion or attitude control system. An efficient primary structure will be from 8% (very efficient) to 12% (still efficient). The secondary structures will be around 2–5%, and fasteners are usually around 2%. These percentages should include the proper margin at any stage of the mission life. As with any guidelines, these are approximations and should be used as goals, not absolute requirements.

22.1.4 Structural Design Verification Philosophy and Criteria

The number one credo for verification is "Test like you fly, and fly like you test." While this credo has some variations, following this one tenet will give you a fairly high success rate. Having said that, let's try to provide a little more detail and guidance.

Philosophy: Trust but verify. Everything must be verified. There are three typical ways to accomplish this, relative to the mechanical subsystem: by *test*, by *analysis*, or by *inspection*. How a verification method is utilized can depend on the mission phase, the hardware

concept, and the management philosophy. In the following section, I will try to provide a high level overview of some philosophies [Sarafin, 1995; Pisacane, 2005].

Analysis: During the mission design phase, analysis will be the primary tool to verify designs and concepts. Analysis is used to determine preliminary loads and environments that will be used in the design and testing phases. FEM are used to assist with preliminary sizing and more detailed analysis for strength and stiffness verifications. CAD models are used to analytically verify mass properties, volumes, interfaces, and possible deployments paths.

Some will argue that one can also verify by similarity. Similarity is a red flag to a lot of review teams and systems engineers. It is becoming widely accepted that there is no such thing as "previously flown" hardware or similarity. See text box below. As most system engineers know, one major technical risk is using existing hardware or designs in a new but similar context. There will be a burden of proof that the projected environment is identical to the previous one. Remember that you will need to look over the product lifecycle, including the manufacturing processes, methods and controls, not just the operational environment. It is rare that an existing design will be able to actually claim similarity. Rarely will it be worth the expense to meet the burden of proof.

> An excerpt from **A Project Manager's Lessons Learned,** [Madden, 2005]
>
> *Item 1.* There is no such thing as previously-flown hardware, i.e., the people who build the next unit probably never saw the previous unit; there are probably minor changes; *the operational environment has probably changed*; and the people who check the unit out will in most cases not understand the unit or the test equipment.

Qualification by analysis is just that—analytically verifying a structure that meets its design requirements. I have seen some hardware with large production runs that use statistical analysis for verification. This is a unique case that merits more writing space than we have here. Since your analysis is only as good as your model, you will either have high factors of safety on your analysis to account for uncertainties (this typically results in an over-designed solution), or else you will need some testing to verify your models. The testing is performed to obtain data to correlate model predictions with actual test results. For smaller spacecraft, if you have to test to verify the model and go through the rigors of setting up the test, you may as well just qualify by testing to begin with. The design efficiencies you gain from designing to much lower *design loads* (predicted load times the factor of safety) should pay benefits in the area of overall mass and general risk reduction from a failure in flight. Some large spacecraft may be too large to test as a system, so final verification may need to be done by analysis.

Another time to qualify by analysis can be for small tertiary structures. They are typically low mass, low stress parts where a lack of elegance in the design is a minimal penalty on mass and risk. One should be able to simply and conservatively analyze the parts. Also, any testing of these components would generally be to high test loads (40+ gs') where any instrumentation (accelerometers) of the test article could be of a significant enough mass to influence the test and results.

Test: All flight hardware should see some type of testing. This follows the "test as you fly" mantra. There are four different types of hardware classifications (*development, qualification, proto-flight and flight*) you can use for testing. Those are addressed in the next section. Those types of hardware are used to support three types of testing: *development*, *qualification*, and *acceptance*.

Development testing. This test is not part of the verification process but can be an important part of the design lifecycle. Development testing is used to provide proof of concept or design in areas where we may not be extremely confident in our analysis. Development testing is also used to retire risk on new materials, manufacturing methods or processes and pathfinder operations. In these situations, development testing usually happens early in the design process as we are making trades on concepts.

At this early time in the lifecycle, the final mission design loads will not be accurately known since the mission design will not be that mature. So, extra margin is also carried in the form of a model/load uncertainty factor when determining the testing loads. Development tests are usually to twice the expected mission load and at times, hardware can be tested to failure. The test is used to gather data used to evaluate the design or process. It is not used to qualify a design.

On LRO, we designed a new method for integration of our flight structural panels. The bonded joint between the primary spacecraft deck and the bus walls had never been used before. The integration was planned to take place very late in the integration flow, after hardware was already mated to the flight panels. In a case like this, a development test to retire the risk from using a new process was recommended. Had there been an issue with integration, there would have been a significant set back in cost and schedule if we had been forced to recover an integration issue.

Qualification testing: Qualification testing is a preferred step in the verification process. Qualification testing typically tests primary structures and load bearing components to design loads above the predicted flight loads. See Table 22-3 for nominal test factors of safety. Qualification tests are also taken to destructive levels some times. The tests are usually performed on test articles (known as Qualification or Engineering Units) and not the actual flight components. So, there is the expense of building an extra set of hardware that needs to be accounted for if you decide to use a qualification unit. These tests typically occur when your manufacturing methods and processes are repeatable and there is reasonable proof that subsequent units will perform

Table 22-3. Test Factors and Durations from GEVS.

Test	Protoype Qualification	Proto-Flight Qualification	Acceptance
Structural Loads[1] Level	1.25 × Limit Load	1.25 × Limit Load	1.00 × Limit Load
Duration Centrifuge/Static Load Sine Burst	1 minute 5 cycles @ full level per axis	30 seconds 5 cycles @ full level per axis	30 seconds 5 cycles @ full level per axis
Acoustics Level[2] Duration	Limit Level + 3 dB 2 minutes	Limit Level + 3 dB 1 minute	Limit Level 1 minute
Random Vibration Level[2] Duration	Limit Level + 3 dB 2 minutes/axis	Limit Level + 3 dB 1 minute/axis	Limit Level 1 minute/axis
Sine Vibration[3] Level Sweep Rate	1.25 × Limit Level 2 oct/min	1.25 × Limit Level 4 oct/min	Limit Level 4 oct/min
Mechanical Shock Actual Device Simulated	2 actuations 1.4 × Limit Level 2 × Each Axis	2 actuations 1.4 × Limit Level 1 × Each Axis	1 actuation Limit Level 1 × Each Axis
Thermal-Vacuum	Max./min. predict ±10 C	Max./min. predict ±10 C	Max./min. predict ±5 C
Thermal Cycling[4] Level Duration	Max./min. predict ±25 C	Max./min. predict ±25 C	Max./min. predict ±20 C
EMC & Magnetics	As Specified for Mission	Same	Same

1 If qualified by analysis only, positive margins must be shown for factors of safety of 2.0 on yield and 2.6 on ultimate. Beryllium and composite materials cannot be qualified by an analysis alone.

Note: Test levels for beryllium structure are 1.4 × Limit Level for both qualification and acceptance testing. Also composite structure, including metal matrix, requires acceptance testing to 1.25 × Limit Level.

2 As a minimum, the test level shall be equal to or greater than the workmanship level.

3 The sweep direction should be evaluated and chosen to minimize the risk of damage to the hardware. If a sine sweep is used to satisfy the loads or other requirements, rather than to simulate an oscillatory mission environment, a faster sweep rate may be considered, e.g., 6*8 oct/min to reduce the potential for over stress.

4 It is recommended that the number of thermal cycles and dwell times be increased by 50% for thermal cycle (ambient pressure) testing.

similarly. *Inspection* plays a vital role in this proof and will be addressed later. Qualification tests are performed at higher than predicted environments and provide design margin as the mission and design matures.

A special type of Qualification testing can be done by using proto-flight hardware. Sometimes, due to schedule, cost, or risk tolerance, there is not adequate resources for a Qualification Unit. The design will still need to be qualified. Proto-flight testing allows you to qualify a design but allows you to do so are lower test levels than Qualification Units. Most important, the unit is still tested to levels that provide design margin. In some cases where material fatigue can be an issue, fewer cycles are used during proto-flight testing to ensure adequate material life remains in the flight article. Proto-flight testing also provides testing to adequate levels to verify *workmanship*, the third test type.

Proto-flight testing is pretty standard in the industry and most often used in areas where risk mitigation is not a priority. Proto-flight is also typically used for composite structures. Composite are unique in that they are heavily susceptible to workmanship issues compared to metallic structures (the brittleness of beryllium makes it another outlier among metallic materials). So even if you test a composite qualification unit to qualification levels, you will still need to perform a qualification-level test on the flight unit when it comes to composite structures. As mentioned above, proto-flight levels can cover both qualification and workmanship verifications.

GEVS, Section

Acceptance testing: So once a design has been qualified, the flight unit would then only need to be tested to acceptance levels, also known as *workmanship testing*. This is the third level of testing used for verification. It is strongly recommended to at least perform some kind of test on all hardware. Testing is a final chance to demonstrate that workmanship was of an acceptable level and that all manufacturing processes were adequately controlled. It does not verify the design is adequate. Only the Qualification or Proto-flight test accomplishes the design verification. Workmanship tests are designed to catch more of the human error in building and integrating spacecraft. Acceptance or workmanship tests are typically tested to

Table 22-3, Fig. 22-4◀, Eq. 22-12◀

Table 22-4. GEVS Factors of Safety. Flight Hardware Design/Analysis Factors of safety Applied to Limit Loads.[1, 2]

Type	Static	Sine	Random/Acoustic[4]
Metallic Yield	1.25[3]	1.25	1.6
Metallic Ultimate	1.4[3]	1.4	1.6
Stability Ultimate	1.4	1.4	1.6
Beryllium Yield	1.4	1.4	1.6
Beryllium Ultimate	1.6	1.6	2.0
Composite Ultimate	1.5	1.5	1.9
Bonded Inserts/Joints Ultimate	1.5	1.5	1.9

1 Factors of safety for pressurized systems to be compliant with EWR-127 (range safety)
2 Factors of safety for glass and structural glass bonds specified in NASA-STD-5001
3 If qualified by analysis only, positive margin must be shown for factors of safety of 2.0 on yield and 2.6 on ultimate. (See Sec. 22.2.4.)
4 Factors shown should be applied to statistically derived peak response based on RMS level. As a minimum, the peak response shall be calculated as a 3-sigma value.

the design loads. These tests are usually performed at higher levels of assembly (and final assemblies) provided there are adequate facilities.

Inspection: Inspection is typically used in conjunction with other tests to provide verification. Inspection is performed after testing to ensure structural integrity remains. Inspections are also used during manufacturing and integration processes to ensure methods are followed and repeatability can be verified. Inspections can be critical when used with qualification testing to verify there were not deformations, yielding, or any other type of structural integrity violation.

GEVS Factors of Safety Testing

At this point, it is worth discussing test factors, factors of safety, and limit loads. (See Table 22-4.) As mentioned in Sec. 22.1.3, programmatic requirements can affect many areas of the design. GSFC uses GEVS as its guide for general verification. GEVS specifies the test factors and the design factors of safety for a project. It is important to note that different stakeholders (sponsors) can have different guidelines. There are even differences between NASA Centers. It is important to know those guidelines.

At GSFC, structural and mechanical verification testing shall be supported by structural analysis to provide confidence that the hardware will not experience failure or detrimental permanent deformation under test or launch conditions.

Factors of Safety are applied to *limit loads* in order to calculate margins of safety. Factors of Safety include both factors on ultimate strength and yield strength. The yield factor of safety ensures that a prototype or protoflight test can be conducted with low risk of the hardware experiencing detrimental yielding. The ultimate factor of safety provides adequate separation between yield and ultimate failure modes and ensures that the hardware will not experience an ultimate failure under expected loading conditions.

Limit loads is the assumed maximum load the structure will experience. The design load is the limit load multiplied by the factor of safety. The test load is the limit load multiplied by the test factor.

There is also one more factor to keep in mind. In determining the limit load, sometimes the expected loads are derived from models (coupled loads—see Sec. 26.5.1). Depending on the fidelity and accuracy of the models, others factors can be applied to the derived load like a model uncertainty factor (MUF) to determine the limit load.

Hopefully, one can see this is an area where factors can get nested inside other factors and conservatism can get out of hand. Different sponsors have different levels of conservatism and that's how you get different guidelines.

Test Articles

In general, there are four types of test articles or hardware to discuss: development, qualification, proto-flight, and flight. Each type of hardware has specific purposes and associated test criteria. These classifications can range from piece parts all the way up to major assemblies. In general, this test article discussion is for primary structural components or assemblies.

Development Hardware: This type of hardware has many names including development or engineering units. In general, these units are not intended to fly. Engineering or development models are typically rudimentary models (but sufficiently representative) used early in a program development to gather evaluation data. They are used to retire risk and they help verify proof of concepts. The units are typically components and lower level assemblies.

Qualification Hardware: This type of hardware is also known as proto-type hardware. These units typically follow flight manufacturing and handling processes so they are flight-like, but they are never intended to fly. Like development hardware, these units are subject to higher test loads. Qualification hardware is critical to verification of your flight hardware. These units are used to confirm designs are adequate to meet performance requirements (survive predicted flight and ground environments).

Proto-Flight Hardware: Proto-flight hardware is a cross between a qualification unit and a flight unit. Proto-flight units are typically the first article of a production. Unlike Qualification units, Proto-flight units are expected to fly. Proto-flight units are typically used instead of Qualification Units. Testing of these units serves as both a qualification test and it also verifies flight worthiness (workmanship).

Flight Hardware: Flight hardware is just that; hardware that is expected to fly. The testing of these units are more to confirm workmanship. As any material, manufacturing, or integration process is subject to variations, workmanship (or acceptance) tests on flight units intended to verify that all units meet some minimal expected performance level. It is a good practice to test all hardware to at least workmanship (or acceptance) test levels.

The above text is meant as an overview. More details can be found in two books that I constantly reference: Sarafin [1995] and Pisacane [2005].

Mechanical Verification and Environmental Testing

Typically, mechanical tests used for verification are as follows: strength, stiffness, acoustic, shock, articulation/deployment, and mass properties.

Strength: It is recommended to perform at least some kind of strength test on all hardware. Qualification testing is typically performed on qualification units or protoflight units. Test levels are typically above design load levels to provide design margin. Pre-test analysis will determine what areas are critical or have the highest stresses. Testing can be accomplished at any level from component to assembly. Any structural part that is made from a composite material will need to undergo strength testing to at least 1.25 times the design load. Static load testing or sinusoidal burst testing are the norms. For static load testing, actuators or load cells apply loads and displacements and strains are measured with gauges. This test is probably the most effective test, but set-up and execution of numerous configurations can be time consuming. Sine burst tests are performed on a vibration table which imparts high level, low cycle accelerations. Accelerometers measure the responses. Large systems can have issues finding a table that can adequately drive the test article to desired levels.

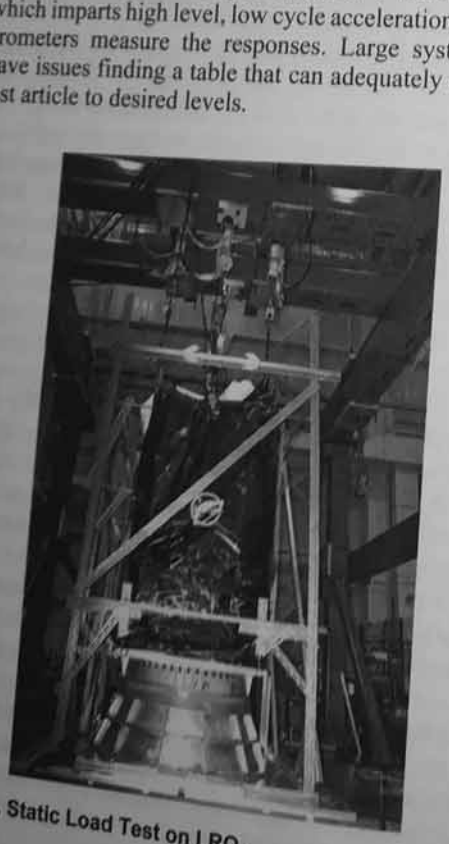


Fig. 22-5. Static Load Test on LRO.

Table 22-4◀, Fig. 22-5, Eq. 22-12◀

Stiffness: Stiffness tests are typically done via a sinusoidal survey vibration test or modal survey. Accelerometers are used to capture structural responses that are correlated to the input vibration frequencies. The response accelerometers will show resonance occurring in the structure. This data is what is used to correlate predicted responses from FEM. Any discrepancies in the model are tuned to match the test results. Programs will define how accurate the models need to be, relative to the test result (within 2 Hz up to 100 Hz input and assuming a model mass participation of greater than 5–10%). The critical point is to verify that the predicted/measured responses are not near major launch vehicle response frequencies [specified in the Launch Vehicle (LV) Spacecraft Interface Control Document (ICD)]. Spacecraft frequencies that are close to major LV frequencies can easily increase the input loads orders of magnitude above the design load due to *dynamic coupling*. This will end any mission very quickly. Another key area to keep an eye on is the ACS system response. Again, the control frequency of the ACS system should be an octave away from any resultant mode of the spacecraft or deployed appendage. See Sec. 19.1.5 and note box below. This can be tricky sometimes for deployed or articulating antenna systems or solar arrays. However, if the appendage frequency couples with the ACS control system, the resultant responses can overpower the control system and the mission could be lost or severely degraded.

Acoustic: At final assembly, the spacecraft is typically tested for acoustic loads that are generated during launch. The acoustic environment is a random vibration on the spacecraft from noise pressure inside the fairing. Acoustic loading can be amplified within a LV fairing if there is lots of free volume in the fairing. Acoustic loads can only be applied in air. So this issue only applies to the stowed or launch configuration of the spacecraft. LVs can add blankets or use other tricks to minimize acoustic power transmission on a spacecraft in the fairing. Work with the LV representatives if you have any concerns. Special accommodations come at a price. Make sure you really need one before you ask. Typically, one is concerned with acoustic noise when a spacecraft has any large, lightweight, flat surfaces/panels. Imagine a large ship's sail. The bigger the sail, the more wind (or acoustic energy) it will catch. So, large flat surfaces will capture more acoustic energy and probably should be tested. Niche analysts can help analyze acoustic energy imparted on a spacecraft. However, this is an area for specialists. If there is any doubt, test. I think most specialists will tend to err on the side of testing too.

Shock: Shock testing is another area for specialists. Years ago, shock was a large concern due to the pyrotechnics that were used to split bolts or nuts for deployment systems and the high g-loads those pyros created. These days, Non-Explosive Actuators (NEAs) are fairly typical and have flight history. The shock response from NEAs is magnitudes less than previous pyro devices. However, it is still recommended to test and measure responses from explosive device actuations. Remember that the

Transmissibility, T, is the ratio of input to output ($T = x_o/x_i$). It is unitless and is sometimes referred to as the "Q" factor.

Using Newton's second law, the equation of motion, and assuming a sinusoidal input, transmissibility can be derived into

$$T = \left\{ \left[1 - \left(\frac{\omega}{\omega_n} \right)^2 \right]^2 + \left[2\xi \left(\frac{\omega}{\omega_n} \right) \right]^2 \right\}^{-1/2} \quad (22\text{-}13)$$

Where T is transmissibility, ω is the component frequency, ω_n is the natural frequency, and ξ is the structural damping.

Assuming forcing at resonance $(\omega = \omega_n)$, and $\xi = 0.05$ (a typical structural damping value), then

$$T = \left[(2\xi)^2 \right]^{-1/2} = 1/(2\xi) = 10 \quad (22\text{-}14)$$

When the component frequency is twice the mounting structure's natural frequency (octave rule) or $\omega/\omega_n = 2$, then

$$T = \left\{ \left[(1-4)^2 + \left[2(0.05)(2) \right]^2 \right] \right\}^{-1/2} \quad (22\text{-}15)$$

$$= \left[9 + 0.04 \right]^{-1/2} = 0.33$$

Keeping the component frequency an octave above the mounting structure's natural frequency results in a reduction of transmissibility by a factor of 30. This is enough to assume that any input loading will not amplify the component.

shock test is to verify that components near the shock devices are not susceptible to the high impulsive loads from the device. This test should be done at higher or final levels of assembly. Qualifying the shock device is typically



Fig. 22-6. LRO Acoustic Test.

performed at low levels of component assembly and is done at temperature extremes. Close scrutiny should take place if the explosive device is near sensitive electronics or RF equipment. As with acoustic testing, the analysis of the shock response and how it is propagated across interfaces is not easily analyzed or quantified (modeling structural damping). Testing and measuring responses are always recommended if there is any concern.

Deployment/Articulation: The next mechanical test we will touch on is related to deployment of a system (e.g., a solar array, high gain antenna, boom or mast) Some of these systems also articulate through a gimbal or equivalent motion system. This is actually a good lead into the next section on mechanisms. However, this section will focus on testing the system at the assembly or spacecraft level. All deployment and articulation systems need to be tested both as a verification (range of motion) and workmanship. In following the "test as you fly" mantra, the test needs to carefully consider two items: temperature and gravity. The final flight assembly needs to be tested to verify range of motion and show

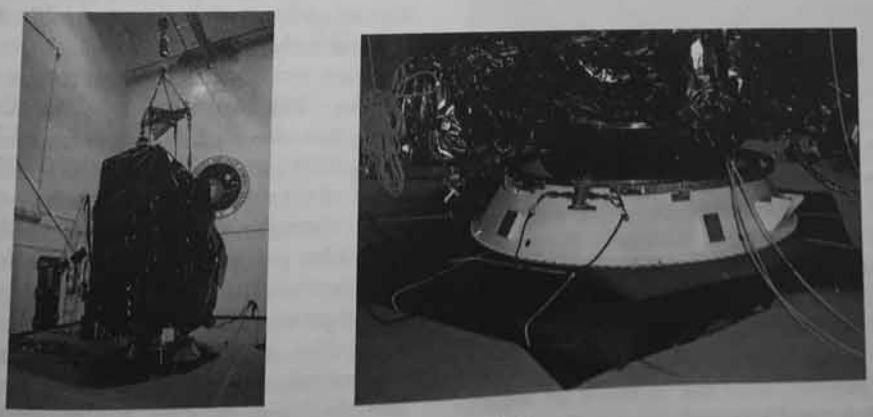


Fig. 22-7. LRO Shock Test/Launch Vehicle Separation Test.

there are no interferences that can bind up the actuation. Pay special attention to blankets, harnesses, launch lock systems or anything else that is near the articulation/deployment path. To assist with the deployment or articulation, you may need to use air tables (simple g-negation test where the article "floats" over the table during a motion in a single plan). More complex systems are needed for multiple plane motions. Simple deployments should verify workmanship test as late as possible. It is recommended to perform a simple "*Pop and Catch*" test at the launch site. The "pop and catch" tests are simply verifying the release mechanisms work. The launch restraint is released and the deployed article is stopped (caught) prior to full deployment. It tests both the final copper connections (electrical configuration and commanding) and the mechanical integration of the release mechanism. Pop and Catch tests are almost always performed if any of the following actions are performed at the launch site: installation of flight pyro or NEAs, integration of appendages like Solar Arrays (sometimes, due to packaging concerns, arrays are shipped separately and integrated at the launch site), blanket installation or adjustments around the mechanisms, any mate/demating of the deployment or mechanism harness.



Fig. 22-8. LRO Mass Properties Test Prior to Environmental Testing.

For all tests, measuring response times and full deployment/articulation times are good ways to verify the system is performing nominally. Trending this data is a recommended practice. It is recommended to track deployment metrics like power, voltage, current, resistance, duration of motion, torque margin, and alignments to name a few. Metrics are unit dependent, but it is rare to keep too much information. These trending/verification tests should be performed prior to each environmental test the unit experiences to ensure the unit entered the test in working order. The units should undergo the same test after each environmental test to verify the deployment system successfully passed the test. It is recommended to perform the deployment test after each environmental test (random, acoustic, thermal vacuum, shock, long transports to the launch site). Depending on project risk tolerance, schedules, and budgets, some projects test only prior to a suite of environmental tests and post-environmental tests. I advise against it because if there are any issues post-testing, it will be extremely difficult in determining the cause of the issue between all the tests the unit experienced. And realistically, you will not be able to prove the mechanism survived any of the environmental test. The project could be required to perform the full suite of tests again. The risk is too high in my opinion and deployment/articulation tests should be added to the comprehensive functional tests used to show post-environmental test acceptance after each environmental test.

Mass Properties: Mass properties are typically measured during a spin balance test. These are crucial for any spacecraft that are spin stabilized either on their own or as part of a spin stabilized launch vehicle. Final mass properties are also critical to attitude control systems to maintain proper control of the spacecraft during the mission. These tests are usually performed just prior to a spacecraft undergoing fine environmental testing and also at the launch site, prior to launch vehicle mating or fairing encapsulation.

The discussion above is a quick overview of environmental testing, more details can be found in two books that I constantly reference: Sarafin [1995] and Pisacane [2005].

22.1.5 Mechanisms and Deployables

Matt Ashmore, ***ATK Space***
Chris Lashley, ***ATK Space***

By definition, space mechanisms must move to satisfy their required function. Mechanism development includes all the required considerations of structures, such as mass, volume, stiffness, strength and material selection. However, the functional requirements add extra failure modes and another layer of criticality. The high launch vibration environments and harsh thermal/vacuum environments of space, as well as the requirement for high reliability and the impracticality of service and repair add to the difficulty.

Space mechanisms development is governed by the AIAA [2005] Standard S-114-2005. Excellent guidance can also be found in NASA-STD-5017 [NASA, 2006] and GSFC-STD-7000 (GEVS). In response to the relatively high incidence of mechanisms on-orbit failure, the NASA [2011] Glenn Space Mechanisms Project has performed research to capture lessons learned from mechanisms failures and to document best practices.

The process for mechanisms development is shown in Table 22-5. In reality, the process is not as linear as the table implies. Good requirements require some idea of a feasible design solution. The requirements may then be iterated as the design matures and the performance and

Table 22-4◀, Fig. 22-8, Eq. 22-15◀

Table 22-5. Process for Mechanisms Development. See text for discussion.

Requirements Definition	Identify requirements and document in specification Identify design drivers
Preliminary Design and Analysis	Identify design concepts capable of satisfying requirements Perform trade studies to select "best" design concept Select and size major components. Characterize force/torque margins. Create preliminary design model and functional analysis Ends with PDR
Detailed Design and Analysis	Detailed models and fab/assembly drawings Strength, dynamic, and thermal analysis Final torque margin estimates Assembly and test plans and procedures Ends with CDR
Engineering Testing	Preliminary testing of risky elements in the design Can be components test or full assembly Engineering model Can involve life testing of bearings or gears
Design Qualification	Manufacture and test of a flight like mechanism Tested to qualification levels and durations which exceed flight predictions Can be Qualification Unit which is consumed in testing or Proto-flight Unit which is qualified and then used as the operational unit
Manufacture, Assembly and Test of Operational Units	Final manufacture, assembly and test of operational unit Must be identical or very similar to Qualification Unit Controlled by documented processes and verified by QA

Table 22-6. Examples of the Three Principal Types of Space Mechanisms.

Deployables	Drives	Instrument Mechanisms
• Solar arrays • Deployable antennas (e.g., mesh, linear, quadrifilar, rigid dish) • Deployable booms and trusses • Deployable thermal radiators • Separation systems	• Solar array drives • Antenna pointing gimbals • Reaction and momentum wheels	• Mirror Adjustment mechanisms • Scan/Pointing Mechanisms • Shutter/Selector Mechanisms • Instrument covers

cost constraints become clearer. Requirements should be essentially finalized by the end of the Preliminary Design phase. Verification activities to confirm that the actual performance meets requirements are ongoing throughout the program. Conformance to requirements can be verified by analysis, inspection, demonstration, or test.

22.1.5.1 Classification and Examples

Table 22-6 shows the 3 main types of space mechanisms: deployables, drives, and instrument mechanisms.

Deployables are generally used to create spacecraft on-orbit configurations which are larger than can be packaged in the launch vehicle fairing. These devices generally deploy only once on orbit, but some applications require automatic retraction capability, sometimes for tens or hundreds of cycles. The primary challenge in most deployables' design is to achieve tight packaging efficiency without compromising the strength needed to survive launch or the reliability to successfully deploy on orbit.

Conventional planar solar arrays are typically deployed using lug/clevis hinges and non-explosive release devices. There are also advanced concepts which use motor/cable system to deploy lightweight flexible arrays, such as the Ultraflex design shown in Fig. 22-9. Antenna deployment systems may use rigid tube sections

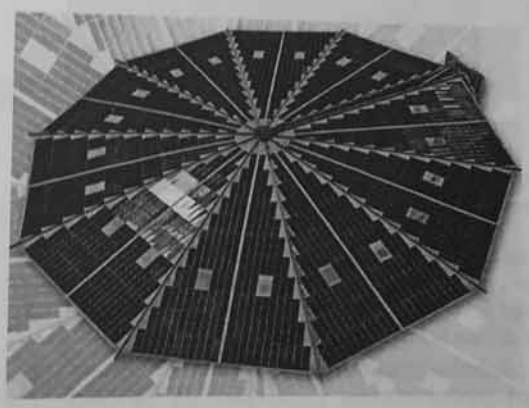


Fig. 22-9. Ultraflex Solar Array. (Photo courtesy of ATK-Goleta)

and rigid hinges. For applications requiring larger and lighter weight structures, deployable truss booms, shown in Fig. 22-10, can be used. Pyrotechnically actuated marman clampbands, Fig. 22-11, are the classical method of separating spacecraft from launch vehicles.

Drives typically require continuous or frequent operation for the life of the mission. Therefore wear, lubrication and life are critical considerations. Solar array drives

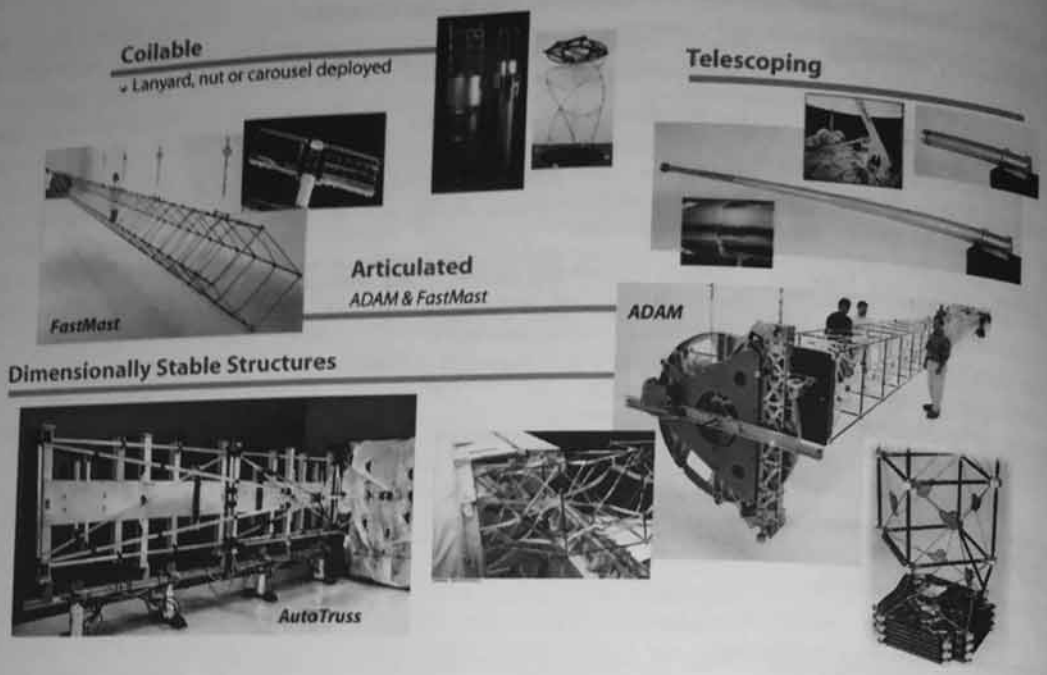


Fig. 22-10. Deployable Structures Examples. (Graphic courtesy of ATK-Goleta)



Fig. 22-11. Clamp Band. (Photo courtesy of Sierra Nevada Corporation)

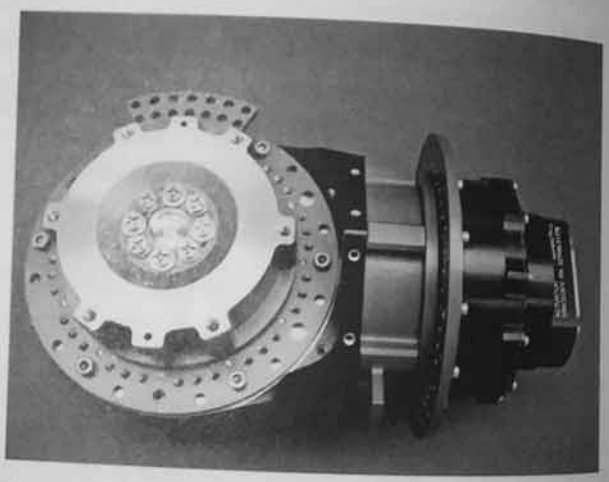


Fig. 22-12. Two-Axis Gimbal. (Photo courtesy of Sierra Nevada Corporation)

and antenna gimbals typically use stepper motors acting through harmonic drive gear reductions to reorient their load as needed. An example of a two axis antenna gimbal is shown in Fig. 22-12.

Reaction wheels (and less common momentum wheels and control moment gyroscopes) are universally employed for spacecraft attitude control. These actuators must function with high reliability over extremely long, high cycle life times. These wheels are typically supported by ball bearings and are driven by brushless DC motors.

Instrument Mechanisms encompass a wide range of devices used to provide optical adjustments for space instruments. High precision and stringent contamination control are drivers here. These mechanisms are highly specialized to meet the requirements and available envelope of the host instrument. Moderate life requirements are typical. Precision adjustment and scanning mechanisms usually incorporate brushless DC motors, precision feedback and closed loop control. Shutter / selector mechanisms are generally simpler and may use open-loop stepper motors. A precision scan mechanism is shown in Fig. 22-13.

22.1.5.2 Requirements

The first step in the mechanisms development process is the definition of requirements. If trade study and design activities proceed too far without well defined requirements, rework and accompanying cost impact usually results. Table 22-7 is a listing of requirements that generally apply to mechanisms.

22.1.5.3 Space Mechanisms Design

To start the preliminary design phase, the development team gets together to brainstorm design approaches

Table 22-6◀, Fig. 22-12, Eq. 22-15◀

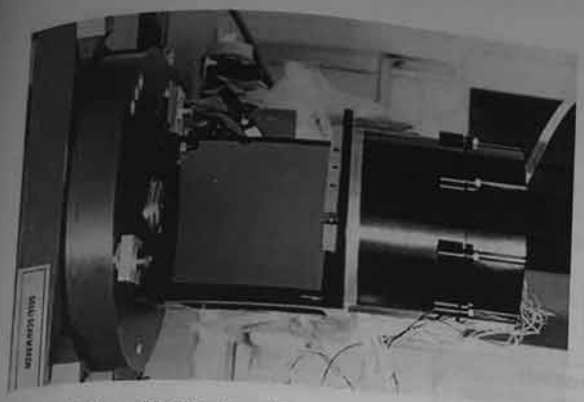


Fig. 22-13. Scan Mechanism. (Photo courtesy of ATK)

that can satisfy the requirements. Trade studies are conducted to rank the design concepts and downselect to one or more superior options. Some general principles are applicable to successful mechanisms design.

The simplest approach possible will usually be the most reliable, lowest cost and easiest to verify. Unnecessary components create more opportunities for failure. Redundancy is a very effective approach to increased reliability. However, 100% mechanical redundancy may lead to a very complex mechanism with extra features, such as differentials and mechanical couplings. In contrast, electrical redundancy with selected mechanical redundancy usually gives the highest overall reliability. Redundant rotation surfaces are an example of selected mechanical redundancy.

Rotary mechanisms are usually more reliable than linear mechanisms. Linear (sliding) mechanisms are prone to cocking and jamming. *Kinematic* techniques should be used for mechanism interfaces and moving parts. A kinematic approach constrains only the *6 degrees of freedom* (DOF) needed to locate a rigid body and produces determinate loads. All 6 DOF are constrained in this approach, but each DOF is constrained no more than once. Additional non-kinematic constraints can produce unpredictable loads and distortions, that can also reduce reliability.

Mechanical parts should be designed and toleranced for efficient manufacture. Some features such as bearing bores and optical alignment features required tight tolerances. Others such as exterior dimensions and structural ribbing can be relaxed to reduce manufacturing cost. Parts should be self jigging to minimize the need for assembly fixtures and designed to be assembled only one way to reduce the risk of assembly error. Reducing parts count also facilitates assembly. Mechanisms should also be designed to be disassembled and refurbished. "I don't anticipate this needing to come apart" are famous last words.

Material should be selected for acceptable outgassing properties in vacuum. NASA [2011] maintains an online database of approved materials and test results. Material combinations in contact should be hard, dissimilar and coated with a suitable surface layer. Hardcoat anodize surfaces impregnated with Teflon are used for aluminum and titanium substrates. Steels may be nitride or carbonized to improve hardness and reduce possibility of galling. Molybdium Disulphide (MoS_2) coatings are also effective in reducing friction and wear. Note: All contacting surfaces that must move in operation must be lubricated. Redundant coatings of dry (MoS_2) and liquid (space-rated) grease are recommended. Dissimilar materials in contact need to be evaluated for corrosion and protected with an appropriate coating if necessary.

Table 22-7. Key Performance Parameters for Mechanisms.

Requirement	Discussion
Performance	Specific description of what the mechanism needs to accomplish
Motion Profile	Position, velocity and/or acceleration requirements Can be simple point-to-point or complex programmable profiles
Accuracy, Resolution and Stability	Allowable errors. Usually stated in deg/arcsec for position
Mass and Power	Allowable assembly mass and peak/average power draw
Life	Operational, ground test and storage life
Reliability and Redundancy	Numerical probability of success for specified life
Temperature Range	Operational, non-op and survival temperatures
Vacuum Environment	All space devices require operation in hard vacuum of space Drives selection of materials for low outgassing
Stiffness	First resonant mode of assembly Must be high enough to avoid resonance with launch vehicle vibration environment Deployed frequency must be high enough not to impact ACS
Design Loads and Random Environment	Quasi-static acceleration loads, random vibration, and shock environments Safety factors used for design & derivation of qualification / acceptance test levels
Interface Definition	Mechanical and electrical interfaces and allowable envelope
Contamination Control	Proximity to sensitive space instruments levies specific requirements on cleaning and assembly controls

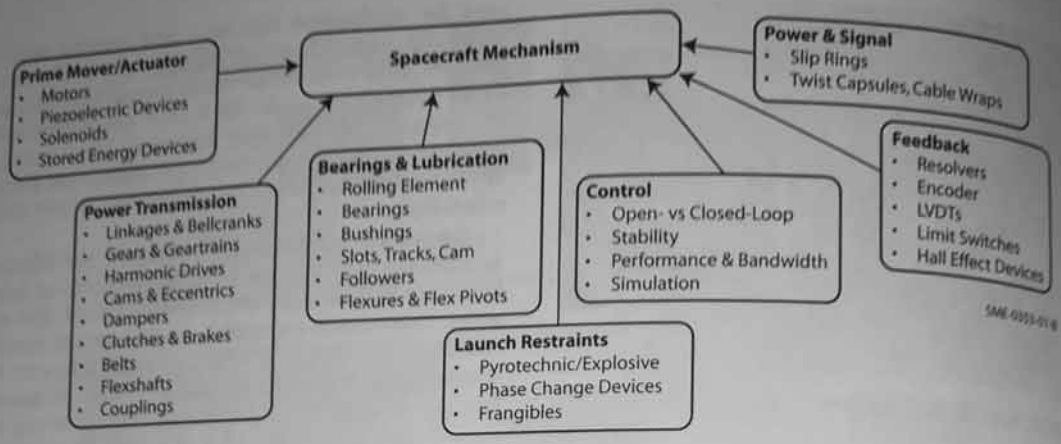


Fig. 22-14. Spacecraft Mechanism Building Blocks.

Most spacecraft mechanisms can be broken into a series of building blocks, drawn from "bins" that can be divided into unique categories. Layout of a mechanism then involves pulling the appropriate building blocks from each bin to create an assembly that meets performance requirements. For preliminary design, the job of the mechanism engineer is to establish a trade space for each bin to determine the best building block (or blocks) to draw, as shown in Fig. 22-14.

The following sections describe these building blocks and how to select and size them in performing a preliminary mechanism layout.

Actuator/Prime Mover Selection

Table 22-8 shows typical primary considerations for actuator selection. Frequently the type of prime mover chosen for the specific mechanism design is greatly narrowed by the nature of the problem itself; (i.e., whether the system must rotate or translate). However, the most obvious actuator may not always end up as the best actuator for certain design scenarios. For example, given proper design, a simple compression spring is just as capable of imparting rotary motion as a torsion spring, and in many circumstances provides a more reliable and space-efficient solution.

Additionally, selection of the prime mover is not necessarily the first step in laying out the mechanism. Factors that influence actuator selection like inertia of the load and mounting provisions are often times unknown until all of the system **ahead** of the actuator has been defined. Smart preliminary design allocates ample real estate within the system structure early on to accommodate unforeseen changes in actuator size, position, and interface needs—a black box with volume margin.

Torque and Force Margins

The objective of any prime mover is to impart motion to a system (the load) by exerting a force, torque, or some combination of the two. In order to assure the motion is imparted under worst-case environmental conditions, a

Table 22-8, Fig. 22-14, Eq. 22-16

Table 22-8. Typical Issues Considered in Actuator Selection

Type of Motion; Linear or Rotary (or combination)	Total stroke (linear) or angular excursion (rotary)
Type of Load Application; Force or Torque	Whether the actuator must bear launch loads from the remainder of the mechanism
Number of Actuation Cycles	Whether force or torque may vary over mechanism stroke
Duty Cycle	Required positioning accuracy
Required Support Electronics/Power Requirements	Available interface provisions for mounting
Whether There is a Necessity to Return-to-Zero or Reset	Whether there is a necessity to hold position at beginning-of-travel, end-of-travel, & in between

robust mechanism incorporates a prime mover that possesses more available force/torque than is required by the load. This is analogous to a structural factor of safety and ensures that the prime mover will be able to overcome any number of unexpected perturbations to motion, such as stiction, localized cold-welding or galling, or loss of lubrication. Like a structural factor of safety, the torque or force margin accounts for the accretion of "human factors" that may affect mechanism performance, such as buyoffs on mis-toleranced parts, undetected contamination, or cut vs. rolled threads.

A torque or force margin is a convenient metric by which to demonstrate to an outside reviewer that the selected prime mover is appropriately sized for its intended application. Generally, the margin can be calculated according to the following equations:

Torque Margin:

$$T_{margin} = \left[\frac{T_{avail}}{FS_{known}\left(\Sigma T_{known}\right) + FS_{unknown}\left(\Sigma T_{unknown}\right)} \right]^{-1} \quad (22\text{-}16)$$

Force Margin:

$$F_{margin} = \left[\frac{F_{avail}}{FS_{known}\left(\Sigma F_{known}\right) + FS_{unknown}\left(\Sigma F_{unknown}\right)} \right]^{-1} \quad (22\text{-}17)$$

T_{avail}, F_{avail} represent the available torque/force from the prime mover. T_{known}, F_{known} represent readily quantifiable resistance loads, such as inertial accelerations, position-based-forces from return springs, system running torque. $T_{unknown}$, $F_{unknown}$ represent resistance loads that, though calculable, are difficult to characterize as they be time- and temperature-variant or may not be able to be measured during the test program. This category generally covers all resistances that are frictional in nature, including forces and torques due to thread preload, as well as parasitic drag from grease or shaft seals.

FS_{known}, $FS_{unknown}$ represent the factors of safety over which the available force or torque must demonstrate positive margin. The factors of safety are usually chosen as part of the requirements development process. In general, the "unknown" factor of safety is larger than the "known" in order to account for the uncertainty in determination of "unknown" resistances. Typical values are 1.25–1.5 for FS_{known} and 3.0–4.0 for $FS_{unknown}$.

In some texts, the subscripts "known" and "unknown" are replaced with "fixed" and "variable," respectively.

A robust mechanism will demonstrate a torque or force margin greater than zero. While high margins are desirable, the margin in itself does not equate to a well-designed mechanism. A system which possesses extraordinarily high torque margins must provide adequate protection against being over-driven at high loads. This negatively impacts the ancillary structure around the prime mover—shafts, for example, must be able to withstand the increased drive torque, increasing the system mass. A motor capable of outputting 50 ft-lbf at stall, driving a #10-32 lead screw with a required load of 20 lbf, indeed shows very high Torque Margin, but would not be classified as a well-designed mechanism. As for this design, the motor would consume far more mass, volume, and power than is necessary. **Good mechanism design efficiently balances force and torque margins with mass and size optimization.**

22.1.6 Typical Space Mechanisms

The book website discusses:



- Motors
- Stored energy devices
- Gearing/power transmission
- Bearings, lubrication, and life.

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22.2 Spacecraft Thermal Control*

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The Thermal Control subsystem of a spacecraft sets and maintains the temperature range of the spacecraft and the temperatures of all its components. The desired temperatures are achieved by balancing the flow of heat energy across spacecraft interfaces. All Spacecraft components are designed to operate over a defined temperature range. While most components operate at or near room temperatures, others have narrow temperature requirements or are exposed to a wide range of heating environments which demand operation over a wide temperature range.

Therefore, to begin the design process, we need to know the temperature requirements of the components and the heating environment to which the spacecraft will be exposed.

The basic approach to thermal control begins with requirements definition followed by an iterative process of analysis, design and re-analysis and ends with a physical test of the design to verify acceptable performance. In its simplest form the steps are as follows:

- **Define Problem:** Identify temperature requirements and heating environments.
- **Analysis:** Compute maximum and minimum orbit average temperature for the spacecraft using surface property estimates and average power dissipation when subjected to the maximum and minimum heating conditions.
- **Design:** Adjust thermal coatings to see if passive design is possible.
- **Analysis:** Develop multi-node thermal model with individual node power levels, surface properties and network of node connections.
- **Design:** Determine conduction and radiation couplings which combined with external properties and power dissipations will provide acceptable temperatures. If necessary, recommend component relocations and define active thermal control elements.
- **Analysis:** Analytically test the design resiliency by varying couplings, surface properties, power dissipations and environment loads.
- **Test:** Physically perform a thermal balance test to verify design and performance.

See a more complete and detailed process description in Sec. 22.2.4.

* This section was prepared with the help and assistance of Wes Ousley, for thermal control hardware information, and Rob Chalmers, for the thermal flux tables.

22.2.1 Requirements

Mission requirements, constraints and boundaries should be set in concert with all other system and subsystem considerations. Thermal considerations must also be included early in the mission concept studies and physical baselines. This leads to a well integrated overall spacecraft system design. Trying to force a rigid physical design to operate at the required temperature is a poor approach to thermal control.

Table 22-9. Typical Temperature Requirements.

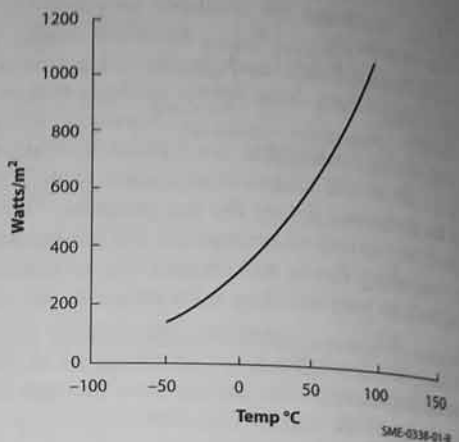
Equipment	Operational	Survival
Avionics Baseplates	–20 to 60	–40 to 75
Batteries	10 to 30	0 to 40
Hydrazine Fuel	15 to 40	5 to 50
Solar Arrays	–150 to 110	–200 to 130
Antennas	–100 to 100	–120 to 120
Reaction Wheels	–10 to 40	–20 to 50
Typical Temperature Requirements °C		

Examples of typical spacecraft element operating temperature requirements are shown in Table 22-9. Components that are sensitive to temperature, (e.g., batteries, star trackers) can be located in positions on the spacecraft that provide thermal stability and heaters can be allocated to provide additional control. Components that must be exposed directly to the space environment to perform their functions (antennas, solar arrays) have to be designed to withstand a very wide range of temperatures.

The spacecraft and its elements are required to perform various functions during the mission. These can include pointing instruments, collecting energy from solar arrays, or communicating with ground stations. These functional requirements usually dictate a general configuration and a spacecraft location and orientation in space that will enable the functions. The physical configuration, its location, and orientation will define the radiant energy that falls on the spacecraft surfaces. This energy can be absorbed or reflected as necessary to control the flow of heat into the spacecraft and set the heat balance to achieve the required temperatures. The spacecraft external surfaces become the boundary condition interfaces to control the heat flow between the environment and the spacecraft. The balance of the heat in, out, and through the spacecraft sets temperatures to meet requirements during the mission.

22.2.2 Radiant Energy Heat Balance in Space

The external surfaces of a spacecraft are the boundary conditions which ultimately provide the energy balance with the environment and therefore control the average temperature. The thermal properties of these surfaces are chosen to balance the energy transfer over the required spacecraft temperature range. This is possible because the radiant energy absorbed from the environment is independent of the surface temperature, but the energy radiated from the surface is a strong function of surface temperature (T^4). Figure 22-15 shows how much energy is radiated from a surface with an emissivity of 1 as a function the surface temperature.



Live Calc

Fig. 22-15. Heat Energy Radiated from a Perfect Emitting Surface (Black Body) as a Function of the Surface Temperature.

Consider a simple flat plate with one side normal to the Sun at 1 AU, and the other side insulated (adiabatic). We can compute the temperature of this object by writing the heat energy balance Eq. (22-18) and solving for T. At equilibrium the heat energy in, Q_{in}, equals the heat energy out, Q_{out}:

$$Q_{in} = Q_{out} \tag{22-18a}$$

$$Q_{in} = SA_p\alpha \tag{22-18b}$$

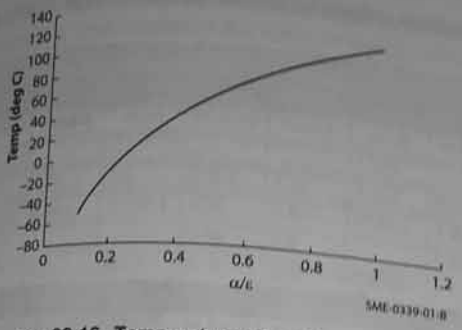
$$Q_{out} = A_r\varepsilon\sigma T^4 \tag{22-18c}$$

where S is the solar constant (1,366 W/m^2 at 1 AU), A_P is the projected area of the plate toward the Sun, α is the absorptivity of the surface, A_r is the radiating surface area of the plate, ε is the emissivity of the surface, σ is the Stefan-Boltzmann constant (5.67051×10^{-8} W/m^{2}K^4, and T is the absolute temperature in Kelvin. Therefore,

$$T = \sqrt[4]{S\left(\frac{\alpha}{\varepsilon}\right)\left(\frac{A_p}{A_r}\right)/\sigma} \tag{22-19}$$

For the insulated flat plate geometry, the projected area, A_p equals the radiating area, A_r and the temperature for the plate becomes a function of the α/ε as shown in Fig. 22-16.

From Eq. (22-19) the ratio of the projected area receiving energy to the radiating area dissipating energy is also an important factor in setting the temperature. Geometric factors determine how the average surface area exposed to heat sources compares to the surface area available to reradiate heat. This area ratio is an important element of heat balance and can be used to adjust the desired temperature of an object in space. Some simple



Live Calc

Fig. 22-16. Temperature of a Flat Plate Normal to the Sun at 1 AU with an Insulated Back as a Function of the Surface Properties (α/ε ratio).

geometries with various ratios of absorbing area to radiating area are shown in Table 22-10. The temperatures are based on surface properties where $\alpha/\varepsilon = 1$ with solar illumination at 1 AU. These six geometries can often be used to estimate average temperatures for various proposed spacecraft configurations or specific surfaces on a spacecraft. They include:

- Flat plate facing Sun with insulated back
- Flat plate facing Sun with radiating back
- Spinning flat plate with both sides active
- Sphere
- Cylinder with insulated ends
- Cube with 1 side facing Sun

Note: A sphere at 1 AU from the Sun with high α and ε will provide an average surface temperature that is close to ideal for life (6 C). If the sphere's surface is coated such that the absorptivity (α) is reduced to 0.2 the temperature drops to –87 C.

Therefore, the average sphere temperature can be controlled if the surface thermal properties can be controlled.

22.2.3 Sources of Radiant Energy in Space

Direct solar energy flux is a function of the distance from the Sun. The radiant energy from the Sun is usually the most significant heat source in spacecraft thermal control. The Sun radiates its energy equally in all directions. However, from the Earth, the Sun appears as a small disc subtending an angle of only ½ deg. This geometry causes the Suns energy to impinge on a surface normal to the Sun as parallel rays. Therefore, one surface can cast a shadow on another and a specular surface can reflect energy to illuminate another surface.

There are two important characteristics of the Sun's radiant energy for spacecraft thermal control. These are

Table 22-10. Temperatures for Various Geometries of Absorbing to Radiating Area ($\alpha = \varepsilon = 1$).

Geometry	Figure	Absorbing Area	Radiating Area	Area Ratio	Resulting Temp, °C Sun (1,366 W/m²) ($\alpha = \varepsilon = 1$)
Plate Facing Sun with Insulated Back		1	1	1	121
Plate Facing Sun with Radiating Back		1	2	1/2	58
Plate Rotating in Sun both Sides Active		$2/\pi$	2	$1/\pi$	23
Sphere		πr^2	$4\pi r^2$	4	6
Cylinder Normal to Sun-Insulated Ends		$D \times H$	$\pi \times D \times H$	$1/\pi$	23
Cube (Isothermal) One Side Facing Sun		1	6	1/6	–21